TEJAS200

BITSAT

Chemistry

Chapter-wise Question Book

50 MCQs per chapter / One answer-key page after every chapter

32 chapter units

1,600 questions

Revised practice edition V2 / October 2026

Using this book

Each chapter contains 50 single-correct, four-option questions. Select one option, A–D, for each question. The next page after the chapter contains its complete answer key. Solutions are not included. Questions 1–10 retain the strongest ten questions selected from the previous middle block, 11–40 are newly written middle-to-hard replacements, and 41–50 preserve the previous hard block. All previous easy questions and ten weaker middle questions have been removed. These are intended practice levels, not official difficulty grades or empirical calibration. Work through the chapter before opening its answer key. Use the questions to check understanding, selection of the first step, calculation, and the stated conditions. Constants and idealisations given in a question take priority for that question. Chapter units follow the updated Tejas200 chapter engine. They are study groupings, rather than a claim that the official syllabus uses the same chapter headings. This is original practice material, not an official BITS question paper. **Reference conventions.** Explicit question data take priority. Otherwise use atomic masses: H 1, C 12, N 14, O 16, F 19, Na 23, Mg 24, Al 27, Si 28, P 31, S 32, Cl 35.5, K 39, Ca 40, Cr 52, Mn 55, Fe 56, Co 59, Ni 59, Cu 64, Zn 65, Br 80, Ag 108, I 127, Ba 137. Use $N_A = 6.0 \times 10^{23}\,\mathrm{mol}^{-1}$ and $F = 96500\,\mathrm{C\,mol}^{-1}$. M denotes $\mathrm{mol\,L}^{-1}$. The logarithms in pH, pOH and the stated base-10 Nernst relation have base 10; ln denotes the natural logarithm. At 25°C, use $K_w = 10^{-14}$ and $\mathrm{pH} + \mathrm{pOH} = 14$. Conventional STP in this book means 0°C, 1 atm and $22.4\,\mathrm{L\,mol}^{-1}$. Use 0.0591 V for the base-10 Nernst factor at 25°C unless specified.

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Chapter 1

Mole Concept

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Two liquids have densities 1.00 and 1.50 g mL^{-1}. They mix without volume change to density 1.20 g mL^{-1}. What fraction of the mixture mass came from the denser liquid?

(A) 0.6 (B) 0.75

(C) 0.5 (D) 0.4

02. A mixture contains only CO and CO_2. Its mean molar mass is 36 g mol^{-1}. How many moles of oxygen atoms are present in 18 g of the mixture?

(A) 1.5 (B) 1

(C) 0.5 (D) 0.75

03. A compound contains only C, H, O. Complete combustion of 0.60 g gives 0.88 g CO_2 and 0.36 g water. If its molar mass is 90 g mol^{-1}, the number of hydrogen atoms per molecule is

(A) 4 (B) 6

(C) 2 (D) 12

04. A 10.0 g sample contains $CaCO_3$ and inert material. It is treated with 0.200 mol HCl; the unreacted acid needs 0.0800 mol NaOH. Percentage by mass of $CaCO_3$ is (molar mass 100)

(A) 60 (B) 80

(C) 100 (D) 40

05. A hydrate $MSO_4 \cdot xH_2O$ has 40.0% water by mass. Its anhydrous salt has molar mass 162 g mol^{-1}. The number of oxygen atoms per formula unit of the hydrate is

(A) 10 (B) 8

(C) 4 (D) 6

06. A gaseous hydrocarbon consumes five times its own volume of oxygen on complete combustion and forms three times its volume of CO_2. All gas volumes are at the same T, P. Hydrogen atoms per molecule are

(A) 8 (B) 4

(C) 6 (D) 10

07. 2.00 g of a mixture of NaOH and Na_2CO_3 needs 0.04509434 mol HCl for complete neutralization. Mass of NaOH in g is (molar masses 40, 106)

(A) 1.60 (B) 0.40

(C) 0.80 (D) 1.20

08. A reaction $2A + 3B \rightarrow 4C$ is run with 0.60 mol A and 0.60 mol B. The isolated C contains 0.48 mol. What is the percentage yield?

(A) 60 (B) 75

(C) 40 (D) 80

09. A 20.0 g mixture of $CaCO_3$ and an inert solid loses 4.40 g on complete heating. What percentage of the final solid mass is CaO? ($CaCO_3$ =100, CaO =56)

(A) 56 (B) 28

(C) 35.897436 (D) 64.102564

10. A mixture contains equal numbers of CH_4 and C_2H_6 molecules. What is the mass percentage of carbon in the mixture? C=12, H=1.

(A) 78.26087 (B) 80

(C) 75 (D) 85.714286

11–40 / New middle-to-hard practice

11. A mixture contains only Na_2CO_3 and $NaHCO_3$. One aliquot consumes 25.0 mL of 0.100 M HCl to the phenolphthalein end point and 65.0 mL in total to the methyl-orange end point. The mole fraction of Na_2CO_3 in the original mixture is

(A) $\frac{3}{8}$ (B) $\frac{10}{13}$

(C) $\frac{5}{13}$ (D) $\frac{5}{8}$

12. An 8.20 g mixture of MgO and $MgCO_3$ consumes 0.300 mol HCl completely. On strong heating a fresh identical sample to constant mass, the mass lost is (Mg=24, C=12, O=16)

(A) 4.40 g (B) 2.20 g

(C) 1.10 g (D) 3.30 g

13. A pure hydrate $CuSO_4 \cdot xH_2O$ weighs 25.00 g. Complete dehydration leaves 16.00 g, and all sulfate in that residue gives 23.30 g $BaSO_4$. With $M(BaSO_4) = 233$, the number of water molecules per copper ion is

(A) 10 (B) 4

(C) 6 (D) 5

14. A 0.890 g compound containing C, H, N and O gives 1.320 g CO_2 and 0.630 g water on complete combustion. A separate identical sample gives 112 mL N_2 at STP. Its molar mass is 89. The molecular formula is (STP molar volume 22.4 L)

(A) $C_3H_9N_2O$ (B) $C_3H_7NO_2$

(C) $C_2H_5NO_2$ (D) C_3H_7NO

15. A 1.00 L mixture of ethene and ethyne is hydrogenated completely using 1.60 L H_2. All volumes refer to the same temperature and pressure. The original gas mixture is then considered before hydrogenation: the oxygen volume needed for its complete combustion is

(A) 2.40 L (B) 2.60 L

(C) 3.00 L (D) 2.70 L

16. A gaseous hydrocarbon has density 1.25 $g\,L^{-1}$ at STP. A 0.10 mol sample consumes 0.30 mol O_2 completely. The number of hydrogen atoms in one molecule is (STP molar volume 22.4 L)

(A) 2 (B) 4

(C) 6 (D) 8

17. A 9.88 g mixture of $CaCO_3$ and $MgCO_3$ is heated completely. It loses 4.84 g. The mole fraction of $CaCO_3$ in the original carbonate mixture is (molar masses 100 and 84)

(A) $\frac{4}{11}$ (B) $\frac{2}{5}$

(C) $\frac{7}{11}$ (D) $\frac{3}{5}$

18. A mixture of FeO and Fe_2O_3 contains 0.150 mol Fe atoms. Complete reduction gives 8.40 g Fe and removes 2.80 g oxygen. The mole fraction of FeO among the two oxides is (Fe=56, O=16)

(A) $\frac{4}{5}$ (B) $\frac{1}{5}$

(C) $\frac{3}{5}$ (D) $\frac{2}{3}$

19. One mole of an element contains isotopes of mass numbers 20, 21 and 22. Their number ratio is $x : 1 : 2$, and the mean mass number is 21.0. The fraction of total mass contributed by isotope 22 is

(A) $\frac{22}{105}$ (B) $\frac{44}{105}$

(C) $\frac{11}{25}$ (D) $\frac{2}{5}$

20. A_2B_3 contains 40% A by mass. AB_2 containing the same isotopes has molar mass 180. The atomic molar mass of B is

(A) 90 (B) 45

(C) 120 (D) 60

21. A 0.500 mol feed of A and 0.600 mol B undergoes $2A + 3B \rightarrow 4C$. All limiting reagent reacts, but only 80.0% of the C formed is recovered. The recovered C is used in $C + 2D \rightarrow E$ with 0.500 mol D. Maximum E obtained is

(A) 0.640 mol (B) 0.500 mol

(C) 0.250 mol (D) 0.320 mol

22. A solution is 20.0% by mass in H_2SO_4 and has density 1.225 g mL^{-1}. A 20.0 mL aliquot is diluted and fully neutralised. Moles of NaOH needed are ($M(H_2SO_4) = 98$)

(A) $\frac{1}{20}$ (B) $\frac{1}{40}$

(C) $\frac{1}{10}$ (D) $\frac{1}{5}$

23. A 9.20 g alloy contains zinc and aluminium only. It gives 5.60 L hydrogen at STP with excess dilute HCl. A fresh identical sample is completely converted to ZnO and Al_2O_3 without metal loss. The ZnO mass fraction in the oxide mixture is (Zn=65, Al=27, O=16)

(A) $\frac{27}{40}$ (B) $\frac{17}{44}$

(C) $\frac{27}{44}$ (D) $\frac{13}{22}$

24. A 24.0 g mixture contains only methane and oxygen. On complete ignition followed by cooling, exactly 9.00 g liquid water is collected and oxygen remains. The mass of unreacted oxygen is

(A) 4.00 g (B) 2.00 g

(C) 0 (D) 8.00 g

25. A solid mixture contains NaCl and KCl only. A 6.650 g sample gives 14.35 g AgCl with excess $AgNO_3$. The mole fraction of KCl is (Na=23, K=39, Cl=35.5, Ag=108)

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

26. A 1.000 g sample contains anhydrous Na_2CO_3 and water only. It consumes 15.0 mL of 1.00 M HCl for complete neutralisation. The sample is dried without decomposing carbonate. Percentage mass loss is ($M(Na_2CO_3) = 106$)

(A) 20.5% (B) 79.5%

(C) 10.25% (D) 41.0%

27. A solution initially contains 0.100 mol $AgNO_3$. It is treated with 0.0400 mol $CaCl_2$, and the precipitate is removed completely. The filtrate is then treated with excess NaBr. Mass of the second precipitate is (Ag=108, Br=80)

(A) 18.80 g (B) 7.52 g

(C) 3.76 g (D) 15.04 g

28. A 0.200 mol mixture of Na_2SO_4 and Na_3PO_4 contains 0.500 mol Na atoms. The mixture is treated with excess $BaCl_2$. Total moles of barium ions incorporated in both precipitates are

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$

(C) $\frac{1}{2}$ (D) $\frac{3}{10}$

29. A hydrocarbon mixture contains 0.200 mol methane and an unknown amount of ethane. Complete combustion yields 0.800 mol CO_2. The amount of water formed is

(A) 1.00 mol (B) 1.60 mol

(C) 1.30 mol (D) 0.800 mol

30. A 2.00 mol mixture of N_2 and H_2 has mean molar mass 8.00. It is passed over an ammonia catalyst with complete conversion of the limiting reactant, then cooled to remove all ammonia. Moles of gas left are (N=14, H=1)

(A) $\frac{8}{13}$ (B) $\frac{2}{13}$

(C) $\frac{6}{13}$ (D) $\frac{4}{13}$

31. A salt M_2X_3 contains metal M in the $+3$ state and X in the -2 state. Its metal mass fraction is 0.400. If the equivalent mass of M is 18.0, the atomic molar mass of X is

(A) 36 (B) 54

(C) 27 (D) 81

32. A gaseous mixture of CO and H_2 contains equal masses of the two gases. Its 15.0 mol sample is converted completely by $CO + 2H_2 \rightarrow CH_3OH$. Moles of hydrogen remaining are

(A) 10 (B) 11

(C) 13 (D) 12

33. 0.100 mol $Ca_3(PO_4)_2$ is reacted with excess H_2SO_4 to form only H_3PO_4 and $CaSO_4$. The phosphoric acid is then completely neutralised with NaOH. Total moles of NaOH required are

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{3}{10}$ (D) $\frac{9}{10}$

34. A 27.8 g sample of $FeSO_4 \cdot xH_2O$ leaves 15.2 g anhydrous $FeSO_4$ on heating without redox change. The fraction of all original oxygen atoms that remain in the solid is ($M(FeSO_4) = 152$)

(A) $\frac{7}{11}$ (B) $\frac{4}{11}$

(C) $\frac{4}{7}$ (D) $\frac{8}{11}$

35. In an oxygen sample, 80.0% of atoms are ^{16}O and the rest are ^{18}O. Atoms pair randomly into molecules. The mass fraction of $^{16}O^{18}O$ molecules in the gas is

(A) $\frac{17}{50}$ (B) $\frac{68}{205}$

(C) $\frac{8}{25}$ (D) $\frac{4}{25}$

36. A salt contains only Fe^{2+}, Fe^{3+} and oxide ions. A sample contains 0.300 mol Fe and 0.400 mol O. The fraction of iron atoms in the $+3$ state is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{3}{4}$ (D) $\frac{1}{3}$

37. A gas mixture contains 0.100 mol N_2O_4 and 0.100 mol NO_2. Exactly half the initial N_2O_4 dissociates into NO_2. The ratio of final gas-molecule count to final nitrogen-atom count is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{3}{5}$ (D) $\frac{5}{6}$

38. An alloy contains copper and zinc only. A 12.8 g sample evolves 2.24 L H_2 at STP with excess dilute HCl; copper does not react. The residual copper is oxidised to CuO. Mass of CuO formed is (Cu=63, Zn=65, O=16)

(A) 15.80 g (B) 6.30 g

(C) 7.90 g (D) 8.10 g

39. A mixture contains $Fe_2(SO_4)_3$ and $MgSO_4$ only, with 0.500 mol sulfate ions and 0.400 mol metal ions. Both metals are precipitated completely as hydroxides. The mass fraction of $Fe(OH)_3$ in the precipitate is (Fe=56, Mg=24, O=16, H=1)

(A) $\frac{107}{223}$ (B) $\frac{1}{2}$

(C) $\frac{58}{165}$ (D) $\frac{107}{165}$

40. A 5.60 g mixture of CO and CO_2 occupies 3.36 L at STP. It is passed through hot carbon so that all CO_2 becomes CO by $CO_2 + C \rightarrow 2CO$. The mass of carbon consumed is

(A) 1.20 g (B) 0.900 g

(C) 1.05 g (D) 0.600 g

41–50 / Preserved hard practice

41. A primary acyclic saturated monoamine produces 0.0100 mol N_2 in Dumas analysis from a 1.180 g sample. Assuming quantitative nitrogen recovery and formula $C_nH_{2n+3}N$, the number of constitutional primary amines with this formula is

(A) 3 (B) 4

(C) 2 (D) 1

42. A mixture contains ethanol and propan-1-ol only. Its mass is 5.80 g and excess sodium liberates 0.0500 mol H_2. Complete combustion of the original mixture gives CO_2 in amount (mol)

(A) 0.28571429 (B) 0.25

(C) 0.3 (D) 0.2

43. A 0.100 mol mixture of propane and ethene consumes 0.380 mol oxygen on complete combustion. When a fresh equal sample is hydrogenated quantitatively, moles of H_2 required are

(A) 0.06 (B) 0.08

(C) 0.02 (D) 0.04

44. A 0.0200 mol sample of a chromium complex has empirical composition $CrCl_3 \cdot 6H_2O$. Excess $AgNO_3$ immediately precipitates 0.0400 mol AgCl. The complex is octahedral with only water and chloride ligands. How many moles of coordinated water are in the sample?

(A) 0.06 (B) 0.1

(C) 0.08 (D) 0.12

45. A 2.060 g sample containing NaOH, Na_2CO_3 and an inert solid is titrated with HCl. The phenolphthalein endpoint needs 0.0300 mol HCl; complete neutralization needs 0.0400 mol. The mass percentage of Na_2CO_3 is (NaOH =40, Na_2CO_3 =106)

(A) 25.728155 (B) 77.184466

(C) 51.456311 (D) 48.543689

46. A 4.22 g mixture of $CaCO_3$ and $MgCO_3$ is fully decomposed. It releases 0.0450 mol CO_2. The final oxide residue is completely neutralized by HCl. What mass of MgO was present in that residue? Molar masses: $CaCO_3$ =100, $MgCO_3$ =84, MgO =40.

(A) 0.4 (B) 0.7

(C) 1.8 (D) 1.1

47. A mixture of Fe^{2+} and Fe^{3+} contains 0.0200 mol total iron. Oxidizing all Fe^{2+} to Fe^{3+} needs 0.00300 mol dichromate in acid. Reducing all iron in a fresh identical sample to metal requires moles of electrons equal to

(A) 0.04 (B) 0.042

(C) 0.036 (D) 0.06

48. A liquid containing C, H, O only gives C: H atom ratio 1: 2 by combustion. Its oxygen mass fraction is 8/29. Its vapour molar mass is 58 g mol^{-1}. It gives a silver mirror but no iodoform precipitate. Number of carbon atoms in its molecule is

(A) 2 (B) 6

(C) 4 (D) 3

49. A 0.100 mol mixture of $FeSO_4$ and $Fe_2(SO_4)_3$ is treated with excess $BaCl_2$ to precipitate sulfate. A separate identical sample consumes 0.0100 mol acidic dichromate to oxidize Fe^{2+}. Moles of $BaSO_4$ formed are

(A) 0.12 (B) 0.1

(C) 0.24 (D) 0.18

50. A 0.100 mol mixture of an alkane C_nH_{2n+2} and the corresponding alkene C_nH_{2n} has mass 5.70 g and consumes 0.0500 mol H_2 on complete hydrogenation. Total oxygen required for complete combustion of the original sample is (mol)

(A) 0.6 (B) 0.65

(C) 0.675 (D) 0.625

Answer key

Mole Concept

C-MOCO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**D**	21	**C**	31	**B**	41	**C**
02	**D**	12	**B**	22	**C**	32	**D**	42	**A**
03	**B**	13	**D**	23	**C**	33	**A**	43	**A**
04	**A**	14	**B**	24	**A**	34	**B**	44	**B**
05	**A**	15	**D**	25	**C**	35	**B**	45	**C**
06	**A**	16	**B**	26	**A**	36	**B**	46	**B**
07	**D**	17	**A**	27	**C**	37	**D**	47	**B**
08	**A**	18	**A**	28	**A**	38	**C**	48	**D**
09	**C**	19	**B**	29	**C**	39	**D**	49	**D**
10	**A**	20	**D**	30	**B**	40	**C**	50	**D**

Chapter 2

States of Matter

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For a van der Waals gas, $a = 1.20\,\mathrm{L^2\,bar\,mol^{-2}}$, $b = 0.0400\,\mathrm{L\,mol^{-1}}$, $T = 300\,\mathrm{K}$, $V_m = 1.00\,\mathrm{L\,mol^{-1}}$. Using $R = 0.0800\,\mathrm{L\,bar\,mol^{-1}K^{-1}}$, pressure in bar is

(A) 25 (B) 24

(C) 26.2 (D) 23.8

02. At low pressure, a gas has $Z \simeq 1 + B/V_m$ with $B = b - a/(RT)$. If $a = 2.40$ and $b = 0.0400$ in mutually consistent L, bar, mol units, and R=0.0800, the Boyle temperature in K is

(A) 750 (B) 1500

(C) 375 (D) 600

03. Two capillaries contain the same liquid at the same temperature. Their radii are in ratio 1: 2 and contact angles are 0 and 60 degrees. Their rises are in ratio h_1/h_2 equal to

(A) 2 (B) 4

(C) 1 (D) 8

04. A water droplet has radius 1 mm and surface tension 0.072 N/m. It divides into 8 equal spherical droplets at unchanged temperature. The excess pressure inside each new droplet in Pa is

(A) 144 (B) 576

(C) 288 (D) 72

05. A vessel holds 0.25 mol O_2 and 0.75 mol He at total pressure 4 bar. Relative to He, the oxygen partial pressure divided by its own molar mass is smaller by what factor? Use He=4, O_2 =32.

(A) 24 (B) 12

(C) 8 (D) 3

06. At fixed temperature, 1 mol of a gas occupies V initially. Its amount is doubled and volume halved. In the van der Waals equation, the attractive pressure correction an^2/V^2 becomes how many times its original value?

(A) 8 (B) 4

(C) 16 (D) 2

07. A van der Waals gas has critical constants $P_c = 50\,\text{bar}$ and $V_{m,c} = 0.120\,\text{L mol}^{-1}$. Its parameter a, in L ^{2} bar mol $^{-2}$, is

(A) 2.16 (B) 1.08

(C) 6.48 (D) 0.72

08. At 300 K an ideal gas has pressure 2 bar and molar density c. At 600 K its compressibility factor is 1.20 and pressure remains 2 bar. Its new molar density divided by c is

(A) 0.41666667 (B) 0.5

(C) 1.2 (D) 0.6

09. Under laminar flow through a capillary, volumetric flow is proportional to $r^4\Delta P/(\eta l)$. A liquid's viscosity doubles and the tube radius increases by $\sqrt{2}$. Other conditions unchanged, flow changes by factor

(A) 2 (B) 1

(C) 0.5 (D) 4

10. The same amount of a gas has identical P and V in states 1 and 2. Temperatures are 300 K and 360 K. If $Z_1 = 1.20$, then Z_2 is

(A) 1.44 (B) 0.833333

(C) 1.2 (D) 1

11–40 / New middle-to-hard practice

11. A rigid vessel at 300 K contains equal masses of methane and oxygen at total pressure 3.00 bar. It is heated to 450 K while a selective membrane removes exactly half the methane and no oxygen. The final oxygen partial pressure is

(A) 1.50 bar (B) 1.00 bar

(C) 2.25 bar (D) 3.00 bar

12. Equal volumes of H_2 and O_2, initially at the same temperature and pressure, are mixed in a rigid vessel and ignited. After cooling to the initial temperature, all water condenses. Neglect its vapour pressure. The final pressure divided by the pressure immediately before ignition is

(A) $\frac{3}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{1}{4}$

13. A vessel contains N_2O_4 at a pressure of 0.800 bar before dissociation at fixed temperature and volume. The equilibrium pressure is 1.20 bar. The equilibrium mole fraction of NO_2 is

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

14. At 300 K, gas A effuses through an aperture at half the molecular-number rate of gas B at 600 K. The gases have equal pressures and the same aperture area. Their ideal-gas density ratio ρ_A/ρ_B under these respective conditions is

(A) 4 (B) 8

(C) 2 (D) 16

15. An ideal He–oxygen mixture has mean molar mass 11.0 g mol^{-1}. A tiny fraction effuses into vacuum at constant temperature. The earliest effusate has helium mole fraction (He=4, oxygen=32)

(A) $9 - 6\sqrt{2}$ (B) $\frac{3}{4}$

(C) $\frac{8}{7} - \frac{2\sqrt{2}}{7}$ (D) $\frac{72}{71} - \frac{6\sqrt{2}}{71}$

16. An ideal He–nitrogen mixture has vapour density 6.00 relative to hydrogen under identical conditions. A tiny amount effuses through a hole into vacuum. The earliest effusate nitrogen mole fraction is (He=4, nitrogen=28, hydrogen=2)

(A) $\frac{1}{3}$ (B) $-\frac{1}{6} + \frac{\sqrt{7}}{6}$

(C) $-\frac{1}{27} + \frac{2\sqrt{7}}{27}$ (D) $\frac{7}{3} - \frac{2\sqrt{7}}{3}$

17. One mole of a van der Waals gas has $b = 0.0500$ L mol^{-1} and Boyle temperature 600 K. At 300 K and $V_m = 1.00$ L mol^{-1}, its pressure is (use $R = 0.0800$ L bar mol^{-1}K^{-1})

(A) 2526/95 bar (B) 2286/95 bar

(C) 2172/95 bar (D) 24 bar

18. A van der Waals gas has $P_c = 40.0$ bar and $V_{m,c} = 0.150$ L mol^{-1}. At $T = 2T_c$ and $V_m = 3V_{m,c}$, the reduced pressure P/P_c is

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$

(C) $\frac{1}{3}$ (D) $\frac{4}{3}$

19. A gas obeys the supplied low-density model $Z = 1 + [b - a/(RT)]/V_m$. At $V_m = 20.0$ L mol^{-1}, Z is 0.980 at 300 K and 1.010 at 600 K. Its Boyle temperature is

(A) 450 K (B) 300 K

(C) 600 K (D) 900 K

20. A sealed gas initially contains pure A_2 at 300 K, with pressure 1.00 bar and Z=0.900. At 600 K, some $A_2 \rightarrow 2A$ dissociation occurs; the same vessel has pressure 3.20 bar and Z=1.20. The final mole fraction of undissociated A_2 is

(A) $\frac{2}{3}$ (B) $\frac{1}{5}$

(C) $\frac{1}{3}$ (D) $\frac{4}{5}$

21. An ideal He–Ne–Ar mixture has mean molar mass 20.0 and contains 10.0% helium by mass. At total pressure 8.00 bar, its neon partial pressure is (He=4, Ne=20, Ar=40)

(A) 4.00 bar (B) 1.60 bar

(C) 0.800 bar (D) 3.20 bar

22. Two equal rigid bulbs contain ideal gases. Bulb A has N_2 at 2.00 bar and 300 K; bulb B has He at 3.00 bar and 600 K. The bulbs are connected, and the whole system is held at 400 K. The final nitrogen mole fraction is

(A) $\frac{4}{7}$ (B) $\frac{3}{7}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

23. A He–Ne mixture is collected over water at 300 K and total pressure 1.00 bar. Water vapour pressure is 0.0400 bar, and the measured mean molar mass of the wet gas is 12.24. After drying, it is heated to 400 K in the same volume. Its helium partial pressure is (He=4, Ne=20, water=18)

(A) 0.480 bar (B) 1.28 bar

(C) 0.640 bar (D) 0.960 bar

24. A gas mixture contains A and C of molar masses 40 and 90, and has mean molar mass 60.0. In a rigid vessel they react completely by $A \to 2B$ and $C \to 3D$. At the original temperature all D condenses and B remains gaseous. Neglect vapour pressure of D. Final pressure divided by initial pressure is

(A) $\frac{9}{5}$ (B) $\frac{6}{5}$

(C) $\frac{12}{5}$ (D) $\frac{3}{5}$

25. An ideal mixture contains gases A and B of molar masses 16 and 64. At the same temperature, the total translational kinetic energies carried by A and B are in the ratio 3 : 2. The fraction of mixture mass contributed by A is

(A) $\frac{1}{3}$ (B) $\frac{2}{11}$

(C) $\frac{3}{5}$ (D) $\frac{3}{11}$

26. An equimolar ideal mixture of He and oxygen is initially at temperature T. Exactly one-fourth of the He escapes, while all oxygen remains. The mixture is then heated until its total translational kinetic energy equals the original value. The final He rms speed divided by its initial rms speed is

(A) $\frac{2\sqrt{14}}{7}$ (B) $\frac{8}{7}$

(C) $\frac{2\sqrt{3}}{3}$ (D) $\frac{\sqrt{14}}{4}$

27. A liquid rises 6.00 cm in a capillary of radius r with zero contact angle. The same liquid at a new temperature has surface tension 80.0% and density 96.0% of their original values. In a tube of radius $2r/3$ with contact angle 60°, its rise is

(A) 2.50 cm (B) 7.50 cm

(C) 5.00 cm (D) 3.75 cm

28. Two capillaries contain the same liquid. It rises 8.00 cm in tube A at contact angle zero and 2.00 cm in tube B at 60°. In a later laminar-flow experiment the pressure drops are equal and B has twice A length. The volume-flow ratio Q_B/Q_A is

(A) 2 (B) 4

(C) 16 (D) 8

29. A liquid forms droplets with radii r and 2r only. Their combined volume is 32 times that of one radius-r drop and combined area is 20 times its area. The fraction of all initial droplets with radius 2r is

(A) $\frac{1}{4}$ (B) $\frac{3}{11}$

(C) $\frac{8}{11}$ (D) $\frac{3}{5}$

30. One liquid drop divides into equal smaller drops without volume loss. Their total area is four times the original. After a surfactant changes the surface tension, each small drop has twice the original drop excess pressure. New surface tension divided by original surface tension is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) 4 (D) 2

31. Two identical soap bubbles of radius r merge isothermally into one, retaining all their ideal gas. Surface tension is γ and external pressure is $12\gamma/(5r)$. Neglect film volume. The final radius is

(A) $\sqrt{2}r$ (B) $3r/2$

(C) $4r/3$ (D) $2^{1/3}r$

32. A soap bubble of radius r is inflated slowly to 2r at constant surface tension in external pressure P_0. Initially its excess internal pressure equals P_0. The work against atmospheric pressure and surface tension, divided by $\pi P_0 r^3$, is

(A) $\frac{70}{3}$ (B) $\frac{46}{3}$

(C) $\frac{28}{3}$ (D) 6

33. Two unequal-radius capillaries of equal length carry the same liquid laminarly. At the same total pressure drop, their combined flow in parallel is 25/4 times their combined flow in series. Tube A has the smaller radius. In the parallel arrangement, A carries what fraction of total flow?

(A) $\frac{4}{25}$ (B) $\frac{1}{4}$

(C) $\frac{1}{5}$ (D) $\frac{4}{5}$

34. Two capillaries A and B carry the same liquid in parallel at a common pressure drop. Together their flow is $5Q_0$. Blocking A leaves flow Q_0 through B. If A has twice B length, the radius ratio r_A/r_B is

(A) $\sqrt{2}$ (B) $2^{\frac{3}{4}}$

(C) 2 (D) $\sqrt[4]{2}$

35. Two equal-length capillaries have radii r and 2r. They run in parallel under the same pressure drop. First liquid A fills the narrow tube and B the wide one; then the liquids are interchanged. The first total flow divided by the second is 6/11. The viscosity ratio η_B/η_A is

(A) 4 (B) $\frac{1}{2}$

(C) 2 (D) $\frac{11}{6}$

36. For a gas, the second virial coefficient is $B = b - a/(RT)$. At T_1, B is $-b$; at T_2, B is $b/2$. The temperature ratio T_2/T_1 is

(A) 2 (B) 4

(C) 3 (D) $\frac{1}{4}$

37. Two van der Waals gases A and B have $T_{c,A}/T_{c,B} = 2$ and $P_{c,A}/P_{c,B} = 4$. The ratios a_A/a_B and b_A/b_B, respectively, are

(A) 4, 2 (B) 1, 1/2

(C) 1/2, 1 (D) 2, 1

38. A 4.00 g gas sample occupies 1.00 L at 300 K and 3.84 bar. It is expanded without loss to 4.00 L at 600 K, where its pressure is 2.40 bar and its compressibility factor is 1.00. With $R = 0.0800$ L bar mol^{-1}K^{-1}, its molar mass and initial Z are

(A) 25.0, 1.00 (B) 20.0, 0.800

(C) 10.0, 0.400 (D) 20.0, 1.25

39. A rigid vessel initially contains 1.00 mol He and 1.00 mol Ne at 300 K. After selective leakage, its pressure at 400 K is 80.0% of the original, and the remaining gas mass is 12.8 g. The fraction of the initial helium that escaped is (He=4, Ne=20)

(A) $\frac{1}{2}$ (B) $\frac{7}{10}$

(C) $\frac{2}{5}$ (D) $\frac{3}{10}$

40. An ideal mixture contains He, nitrogen and oxygen. Helium and oxygen contribute equal masses, and mean molar mass is 12.0. The oxygen mole fraction is (He=4, nitrogen=28, oxygen=32)

(A) $\frac{4}{47}$ (B) $\frac{8}{47}$

(C) $\frac{1}{12}$ (D) $\frac{1}{9}$

41–50 / Preserved hard practice

41. A vessel contains 0.10 mol N_2O_4 and 0.10 mol NO_2 at 300 K. On heating to 600 K, 40% of N_2O_4 dissociates to 2 NO_2. The volume is fixed and all gases are ideal. Final-to-initial pressure ratio is

(A) 2 (B) 2.4

(C) 2.2 (D) 2.8

42. A hydrocarbon gas has density 1.25 g/L at 300 K and 1 bar. Taking R=0.0800 L bar mol $^{-1}$ K $^{-1}$, a 5.00 g sample consumes how many moles of O_2 if it is an acyclic saturated hydrocarbon?

(A) 0.625 (B) 0.5

(C) 0.58333333 (D) 0.75

43. A gaseous mixture of He and N_2 has total mass 6.00 g and occupies 6.00 L at 300 K and 1 bar. Using R=.0800 L bar mol $^{-1}$ K $^{-1}$, helium partial pressure in bar is

(A) 0.083333333 (B) 0.83333333

(C) 0.16666667 (D) 0.33333333

44. A solution exerts water vapour pressure 0.030 bar while pure water at the same temperature has 0.040 bar. Treat the solution as ideal and solute nonvolatile. A 0.200 mol vapour-free liquid sample contains water in amount(mol)

(A) 0.15 (B) 0.2

(C) 0.05 (D) 0.1

45. An ideal gas initially has 2 mol at 300 K. A reaction at constant pressure and without temperature change doubles the total gas amount. Assuming only PV work, work done by the system in kJ is(R=8.314 J mol $^{-1}$ K $^{-1}$)

(A) 2.4942 (B) 4.9884

(C) 0 (D) 9.9768

46. A wet gas occupies 2.50 L at 300 K and 1.00 bar; water vapour pressure is 0.040 bar. Dry gas is a mixture of CO and CO_2 with average molar mass 36.0 g/mol. Use R=.0800 L bar mol $^{-1}$ K $^{-1}$. Mass of CO_2 in the vessel in g is

(A) 1.4 (B) 3.6

(C) 4.4 (D) 2.2

47. A gaseous compound has relative effusion rate $1/\sqrt{2}$ compared with O_2 under identical conditions. It contains 50.0% sulfur by mass and the rest oxygen. Its empirical formula is(S=32, O=16)

(A) SO_3 (B) SO

(C) S_2O_3 (D) SO_2

48. A capillary experiment is used to compare an unknown liquid with water. At equal tube radius and zero contact angles, rises are 3.0 cm and 6.0 cm respectively. The unknown density is 0.80 times water density. Surface tension of unknown relative to water is

(A) 1.6 (B) 0.5

(C) 0.625 (D) 0.4

49. Initially 0.100 mol PCl_5 is placed in a vessel at T. It partly dissociates into PCl_3 and Cl_2. If the gas mean molar mass at equilibrium is 139 g/mol, the dissociation fraction is(PCl_5 =208.5 g/mol)

(A) 0.25 (B) 1

(C) 0.5 (D) 0.75

50. A vessel contains 0.10 mol H_2 and 0.10 mol O_2 at 300 K. A spark converts all H_2 to water vapour; the final gas remains at 600 K and unchanged volume, with no water condensation. Final-to-initial pressure ratio is

(A) 2 (B) 1.5

(C) 3 (D) 1

Answer key

States of Matter

C-STMA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**A**	21	**C**	31	**C**	41	**B**
02	**A**	12	**D**	22	**A**	32	**B**	42	**C**
03	**B**	13	**A**	23	**C**	33	**C**	43	**C**
04	**C**	14	**D**	24	**B**	34	**B**	44	**A**
05	**A**	15	**D**	25	**D**	35	**C**	45	**B**
06	**C**	16	**C**	26	**A**	36	**B**	46	**D**
07	**A**	17	**C**	27	**D**	37	**B**	47	**D**
08	**A**	18	**B**	28	**D**	38	**B**	48	**D**
09	**A**	19	**A**	29	**B**	39	**D**	49	**C**
10	**D**	20	**A**	30	**B**	40	**A**	50	**B**

Chapter 3

Solid State

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. An fcc element has atomic molar mass $60\,\mathrm{g\,mol^{-1}}$ and edge $400\,\mathrm{pm}$. Taking $N_A = 6.0 \times 10^{23}\,\mathrm{mol^{-1}}$, density in $\mathrm{g\,cm^{-3}}$ is

(A) 6.25 (B) 3.125

(C) 25 (D) 12.5

02. A metal changes from bcc to fcc while retaining the same atomic radius. Ratio of new to old cell edge is

(A) 1.4142136 (B) 2

(C) 1.2247449 (D) 0.81649658

03. In an fcc anion cell, each tetrahedral site at $(1/4, 1/4, 1/4)$ has distance from the corner anion equal to

(A) $a/4$ (B) $\sqrt{3}a/4$

(C) $a/2$ (D) $\sqrt{2}a/4$

04. A crystal has $\mathrm{Fe_{0.95}O}$ composition with iron only as $\mathrm{Fe^{2+}}$ and $\mathrm{Fe^{3+}}$. Per mole of oxide ions, moles of $\mathrm{Fe^{3+}}$ are

(A) 0.1 (B) 0.2

(C) 0.05 (D) 0.15

05. An fcc array of monovalent anions has half its octahedral sites occupied by cations. The cation charge required for neutrality is

(A) 4 (B) 1

(C) 3 (D) 2

06. An fcc array of divalent anions has all tetrahedral sites occupied by identical cations. Cation charge is

(A) 2 (B) 3

(C) 4 (D) 1

07. An fcc element has atomic radius r. If its atoms per cell are counted correctly but $a = 2r$ is incorrectly used, the computed density is how many times the true density?

(A) 1.4142136 (B) 4

(C) 2.8284271 (D) 2

08. A crystal of $M_{0.90}O$ contains only M^{2+} and M^{3+}. What fraction of metal ions is trivalent?

(A) 0.22222222 (B) 0.11111111

(C) 0.1 (D) 0.2

09. An fcc crystal has 4 atoms per cell, each with 12 nearest neighbours. Number of distinct nearest-neighbour contacts assigned per cell is

(A) 36 (B) 12

(C) 24 (D) 48

10. In a zinc-blende-type AB crystal, anions are fcc and cations occupy half tetrahedral holes. The number of nearest-neighbour A-B contacts per conventional cell is

(A) 8 (B) 16

(C) 4 (D) 32

11–40 / New middle-to-hard practice

11. An fcc metal consists of mass-58 and mass-62 isotopes in equal numbers. Its cell edge is 400 pm and density 5.625 $\mathrm{g\,cm^{-3}}$. Only random metal-site vacancies occur. With $N_A = 6.00 \times 10^{23}$, the vacant-site fraction is

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$

(C) $\frac{9}{10}$ (D) $\frac{1}{5}$

12. An fcc metal has cell edge 400 pm. Its density is initially 6.00 $\mathrm{g\,cm^{-3}}$; annealing fills all vacancies and raises density to 6.25 without changing the edge. No other defects occur. Its atomic molar mass and original vacancy fraction are, respectively ($N_A = 6.00 \times 10^{23}$)

(A) 60, 0.0400 (B) 62.5, 0.0400

(C) 60, 0.0800 (D) 57.6, 0.0400

13. An fcc array contains oxide ions. A chemical assay finds 0.800 mol metal ions per mole oxide, all either A^{2+} in octahedral holes or B^{3+} in tetrahedral holes. No other ions or defects occur. Fraction of tetrahedral holes empty is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

14. A close-packed array of monovalent X anions has monovalent A in octahedral holes and divalent B in tetrahedral holes only. An assay gives A:B ion-number ratio 2 : 1. With no other defects, the fraction of tetrahedral holes filled by B is

(A) $\frac{3}{8}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{1}{8}$

15. In $M_{1-x}O$, M occurs only as M^{2+} and M^{3+}. One-fourth of the metal ions are trivalent. The metal-site vacancy fraction x, relative to ideal MO, is

(A) $\frac{1}{8}$ (B) $\frac{1}{9}$

(C) $\frac{2}{9}$ (D) $\frac{1}{4}$

16. A nonstoichiometric oxide has formula $M_{0.96}O$. All metal ions are either +2 or +3. The ratio of divalent to trivalent metal ions is

(A) 12 (B) 11

(C) $\frac{1}{11}$ (D) 23

17. An initially ideal 111.000 g $CaCl_2$ crystal is doped by replacing some Ca^{2+} with Na^+. Charge compensation occurs only by chloride vacancies. Total cell count is unchanged; final crystal mass is 110.475 g. Fraction of original chloride sites now vacant is (Ca=40, Na=23, Cl=35.5)

(A) $\frac{1}{400}$ (B) $\frac{1}{100}$

(C) $\frac{1}{50}$ (D) $\frac{1}{200}$

18. An initially ideal 58.500 g NaCl crystal is doped with Ca^{2+} on sodium sites. Compensation occurs only through vacant sodium sites. With total anion-site count unchanged, final mass is 58.440 g. Fraction of original sodium sites still occupied by sodium is (Na=23, Ca=40, Cl=35.5)

(A) $\frac{99}{100}$ (B) $\frac{199}{200}$

(C) $\frac{49}{50}$ (D) $\frac{24}{25}$

19. A metal changes from bcc to fcc with unchanged atomic radius and atomic mass. The final density divided by its initial density is

(A) $\frac{\sqrt{6}}{2}$ (B) 2

(C) $\frac{4\sqrt{6}}{9}$ (D) $\frac{3\sqrt{6}}{8}$

20. For one metal, bcc and fcc forms have identical measured density and no vacancies. If a_f and a_b denote their cell edges, the ratio of the hard-sphere radii inferred for fcc and bcc is

(A) $\frac{2^{\frac{5}{6}}\sqrt{3}}{3}$ (B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt[3]{2}$ (D) $\frac{2^{\frac{5}{6}}\sqrt{3}}{6}$

21. A fully occupied cubic 1:1 salt MX is known to adopt either rock-salt or CsCl structure. Its edge is 400 pm; diffraction gives nearest unlike-ion distance $200\sqrt{3}$ pm. Density is 3.00 g cm^{-3}. Formula molar mass is ($N_A = 6.00 \times 10^{23}$)

(A) $\frac{1152}{5}$ (B) $\frac{576}{5}$

(C) $\frac{288}{5}$ (D) $\frac{144}{5}$

22. A fully occupied cubic 1:1 salt is either rock-salt or CsCl. Its edge is 600 pm, nearest unlike-ion separation is 300 pm, and density is 2.00 $\mathrm{g\,cm^{-3}}$. Formula molar mass is ($N_A = 6.00 \times 10^{23}$)

(A) $\frac{324}{5}$ (B) $\frac{648}{5}$

(C) $\frac{1296}{5}$ (D) $\frac{162}{5}$

23. A fluorite-type crystal has 4 metal ions and 8 anion sites per cell. Half its divalent anions are replaced by monovalent ions. No vacancies occur, and metal ions are only +2 or +4. The fraction of metal ions in the +4 state is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{1}{4}$

24. An antifluorite M_2X initially has monovalent M and divalent X. One-fourth of M ions are removed, with no anion vacancies. The remaining M ions become a mixture of +1 and +2 only. The fraction of remaining M ions that must be divalent is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

25. An fcc oxide-ion array has 80.0% of its octahedral holes occupied by metal ions. Only divalent and trivalent metal ions are present; no other defects occur. The fraction of occupied metal sites that are trivalent is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) $\frac{2}{5}$ (D) $\frac{3}{5}$

26. A close-packed anion array has all tetrahedral holes filled by monovalent A and one-fourth of octahedral holes filled by divalent B. Anions are either 2− or 3−, with no other defects. The fraction of anions that are 3− is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

27. A bcc crystal has randomly vacant sites but unchanged atomic mass and cell edge. Its density is 7/8 of the fully occupied value. Relative to the perfect crystal, the expected fraction of nearest-neighbour pairs with both endpoints occupied is

(A) $\frac{7}{8}$ (B) $\frac{49}{64}$

(C) $\frac{1}{8}$ (D) $\frac{3}{4}$

28. A fully occupied fcc model crystal contains N atoms. Atoms A and B have masses 10 and 40 u, with mean atomic mass 30 u. Labels are independently distributed. The expected number of distinct unlike nearest-neighbour pairs is

(A) $16N/3$ (B) $4N/3$

(C) $4N$ (D) $8N/3$

29. A 1 : 1 ionic crystal develops a Schottky vacancy fraction 1/20 on each sublattice. Simultaneously, its unit-cell edge shrinks to 99% of the initial value. Its final-to-initial density ratio is

(A) $\frac{970299}{950000}$ (B) $\frac{100000}{107811}$

(C) $\frac{950000}{970299}$ (D) $\frac{19}{20}$

30. An MX crystal initially has 10^{20} sites on each monovalent sublattice. Cell dimensions stay fixed. It develops Schottky pairs plus cation Frenkel pairs; density is 98.0% of its initial value and cation interstitials number 10^{18}. Total normal-site vacancies on both sublattices are

(A) 4000000000000000000 (B) 3000000000000000000

(C) 5000000000000000000 (D) 6000000000000000000

31. A silver-halide crystal is heated reversibly. Its mass, cell volume and stoichiometry remain constant, but mobile silver interstitials and an equal number of silver vacancies appear. Which combination best follows?

(A) Anion vacancies only; electron conductivity is necessarily the only current

(B) Frenkel defects; silver content must decrease; density increases

(C) Schottky defects; density must decrease; conductivity cannot change

(D) Frenkel defects; ideal density unchanged; ionic conductivity may increase

32. An MX crystal has fixed cell dimensions and develops only Schottky pairs and cation Frenkel pairs. Density falls to 97.0% of its initial value. Per 10^{20} original cation sites, there are 2×10^{18} cation interstitials. Fraction of all normal cation vacancies belonging to Frenkel pairs is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{2}{3}$

33. A silicon sample contains 1.10×10^{16} fully ionised donor plus acceptor atoms. Intrinsic carriers are negligible, and the net mobile carriers are 5.00×10^{15} electrons. The donor fraction among all dopant atoms is

(A) $\frac{8}{11}$ (B) $\frac{1}{2}$

(C) $\frac{3}{11}$ (D) $\frac{5}{11}$

34. A silicon crystal has 1.40×10^{16} donor plus acceptor atoms. At the stated temperature only half the donors and all acceptors are ionised. Intrinsic carriers are negligible; net mobile electrons number 1.00×10^{15}. The donor fraction among all dopants is

(A) $\frac{1}{2}$ (B) $\frac{6}{7}$

(C) $\frac{5}{7}$ (D) $\frac{2}{7}$

35. An fcc anion cell has edge a. A void on the diagonal has position (x, x, x), $0 < x < 1/2$, and is equidistant from the corner anion $(0, 0, 0)$ and face-centre anions such as $(0, 1/2, 1/2)$. Its distance to the cell-centre octahedral void, divided by the anion nearest-neighbour distance, is

(A) $\frac{\sqrt{6}}{2}$ (B) $\frac{\sqrt{3}}{4}$

(C) $\frac{1}{2}$ (D) $\frac{\sqrt{6}}{4}$

36. In an fcc anion lattice, the nearest-neighbour anion-centre distance is $200\sqrt{2}$ pm. A tetrahedral-hole centre is $100\sqrt{3}$ pm from each surrounding anion centre. A small sphere fits exactly, touching those anions. Its radius is

(A) $100(\sqrt{2}-1)$ pm (B) $100(\sqrt{3}-\sqrt{2})$ pm

(C) $100(\sqrt{3}-1)$ pm (D) $100\sqrt{3}$ pm

37. A close-packed array contains oxide ions. One-sixth of its tetrahedral holes contain D^{3+}; some octahedral holes contain M^{2+}. No other defects or ions occur. Charge neutrality gives which simplest formula?

(A) $M_3D_2O_6$ (B) M_3DO_6

(C) $M_2D_3O_6$ (D) MDO_3

38. An fcc array consists of divalent X anions. All octahedral holes contain monovalent A; some tetrahedral holes contain divalent B. With no other defects, the simplest formula and fraction of tetrahedral holes occupied are

(A) A_4BX_4; 1/8 (B) A_2BX_2; 1/4

(C) ABX; 1/2 (D) AB_2X_2; 1/2

39. A mixed oxide–chloride solid contains only a divalent metal, oxide ions and chloride ions. A sample contains 0.300 mol anions in total and 1.60 g oxygen. Its cation-to-anion number ratio is (O=16)

(A) $\frac{2}{3}$ (B) 1

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

40. A solid contains only M^{3+}, oxide and chloride ions. A sample has 0.500 mol anions; after complete dissolution its chloride gives 43.05 g AgCl quantitatively. The amount of metal ions is ($M(\mathrm{AgCl}) = 143.5$)

(A) 1/6 mol (B) 1/5 mol

(C) 7/30 mol (D) 3/10 mol

41–50 / Preserved hard practice

41. In a hypothetical cubic oxide, O^{2-} forms an fcc array and M^{3+} occupies only octahedral holes, with just enough M ions for neutrality. The edge is 400 pm, M=52, O=16, and $N_A = 6.0 \times 10^{23}\,\mathrm{mol}^{-1}$. The ideal density in $\mathrm{g\,cm}^{-3}$ is

(A) 6.25 (B) 7.5

(C) 3.3333333 (D) 5.2777778

42. A cubic ionic solid has edge 400 pm, density 5.0 g/ cm^3 and formula mass 96 g/mol. Using N_A =6.0×10 ^{23}, number of formula units per cell is

(A) 2 (B) 3

(C) 1 (D) 4

43. A metal crystallizes in fcc with atomic radius $100\sqrt{2}$ pm. Its density is 6.25 g/ cm^3. Taking N_A =6.0×10 ^{23}, atomic molar mass in g/mol is

(A) 30 (B) 90

(C) 120 (D) 60

44. A 6.976 g sample of $Fe_{0.96}O$ contains only Fe(II) and Fe(III). Use Fe=56, O=16. After acid dissolution, Fe(II) is selectively oxidized to Fe(III) by acidic permanganate. Required MnO_4^- amount, in mol, is

(A) 0.0192 (B) 0.0176

(C) 0.02 (D) 0.0088

45. In a hypothetical cubic ionic crystal, X^{3-} occupies all corners and the body centre, while M^+ occupies all edge centres and face centres. If atomic masses are M=23 and X=31, the empirical-formula mass in g/mol is

(A) 123 (B) 100

(C) 77 (D) 54

46. An fcc array of Cl^- contains Na^+ in all octahedral sites. A small fraction 0.01 of Na^+ is replaced by Ca^{2+}. To preserve neutrality using only cation vacancies, the number of vacant Na sites per 10000 original Na sites is

(A) 200 (B) 50

(C) 1000 (D) 100

47. A stoichiometric 1: 1 ionic crystal loses 1.00% of each ion species by Schottky -defect formation. Its number of unit cells remains fixed, but heating also increases every cell edge by 1.00%. Relative to the original crystal, the density decreases by what percentage?

(A) 3.9115754 (B) 1

(C) 4 (D) 2.9409852

48. A monatomic bcc metal has edge 300 pm. A 1.00 cm^3 specimen dissolves quantitatively as M^{2+} in excess HCl, with hydrogen as the only reduced product. Use $N_A = 6.0 \times 10^{23}$ mol^{-1}. The hydrogen amount formed, in mol, is

(A) 0.061728395 (B) 0.012345679

(C) 0.24691358 (D) 0.12345679

49. An ionic compound has 4 formula units per cell and molar mass 120 g/mol. Its edge shrinks by 20% at constant crystal composition and number of cells. Density ratio final/initial is

(A) 0.512 (B) 1.953125

(C) 1.5625 (D) 1.25

50. A 6.640 g sample of $Fe_{0.90}O$ contains Fe(II) and Fe(III) only. Use Fe=56, O=16. After acid dissolution, 0.0100 mol permanganate oxidizes Fe(II) selectively to Fe(III). All permanganate is consumed. The final iron mixture is then reduced quantitatively to metal. Required electron amount, in mol, is

(A) 0.27 (B) 0.18

(C) 0.25 (D) 0.2

Answer key

Solid State

C-SOST / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**B**	31	**D**	41	**D**
02	**C**	12	**A**	22	**A**	32	**C**	42	**A**
03	**B**	13	**D**	23	**B**	33	**A**	43	**D**
04	**A**	14	**D**	24	**C**	34	**C**	44	**B**
05	**D**	15	**B**	25	**B**	35	**D**	45	**B**
06	**D**	16	**B**	26	**A**	36	**B**	46	**D**
07	**C**	17	**D**	27	**B**	37	**A**	47	**A**
08	**A**	18	**C**	28	**D**	38	**B**	48	**D**
09	**C**	19	**C**	29	**C**	39	**A**	49	**B**
10	**B**	20	**A**	30	**C**	40	**C**	50	**C**

Chapter 4

Atomic Structure

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. An orbital has two radial nodes and two angular nodes. Its principal quantum number is

(A) 3 (B) 4

(C) 6 (D) 5

02. The energy of a hydrogen-like ion in $n = 2$ is $-30.6\,\text{eV}$. Its nuclear charge Z is

(A) 4 (B) 2

(C) 3 (D) 1

03. The number of unpaired electrons in ground-state Fe^{3+} is (Fe atomic number 26)

(A) 3 (B) 4

(C) 1 (D) 5

04. How many electrons in a neutral ground-state argon atom have $l = 1$?

(A) 18 (B) 6

(C) 8 (D) 12

05. An orbital has principal quantum number 5 and a total of 4 nodes. Exactly one node is radial. Its azimuthal quantum number is

(A) 0 (B) 3

(C) 2 (D) 1

06. In a hydrogen-like ion, orbit A has n=2 and orbit B has n=4. Ratio of ionization energy from A to B, multiplied by ratio of radius A to B, is

(A) 16 (B) 0.25

(C) 1 (D) 4

07. How many electrons can simultaneously have $n = 4$, $l \leq 1$ and $m_s = -1/2$?

(A) 16 (B) 4

(C) 8 (D) 2

08. A hydrogen atom is ionized from an excited state by 1.51 eV approximately. Maximum distinct cascade lines from that state are

(A) 6 (B) 3

(C) 1 (D) 10

09. For a hydrogen-like ion, the radius of its $n = 2$ orbit equals the radius of the first hydrogen orbit. Its ground-state ionization energy in eV is

(A) 54.4 (B) 108.8

(C) 435.2 (D) 217.6

10. An atom has completely filled shells $n = 1$ and $n = 2$, followed by $3s^2 3p^4$. Number of orbitals that are occupied by at least one electron is

(A) 7 (B) 8

(C) 9 (D) 10

11–40 / New middle-to-hard practice

11. An electron in He^+ has the same energy as an electron in hydrogen $2p$ and the same orbital angular-momentum magnitude. Which orbital of He^+ meets both requirements?

(A) $3d$ (B) $2p$

(C) $4p$ (D) $4s$

12. A hydrogen-like ion has the same energy in its $n = 6$ state as hydrogen in $n = 2$. Its $n = 6$ orbital has two angular nodes. The number of radial nodes of that orbital and the nuclear charge are, respectively

(A) 3, 2 (B) 4, 9

(C) 2, 3 (D) 3, 3

13. A hydrogen-like ion has nuclear charge 4. A level has energy -13.6 eV. Among the orbitals at that energy, how many have exactly two angular nodes?

(A) 5 (B) 9

(C) 7 (D) 16

14. For hydrogen, a level has energy $-13.6/25$ eV. Count the orbitals at this energy having at least three angular nodes; ignore spin.

(A) 9 (B) 16

(C) 7 (D) 25

15. An orbital has four total nodes. Its angular-node count is three times its radial-node count. The number of allowed m_l values for that orbital is

(A) 5 (B) 3

(C) 9 (D) 7

16. An orbital has principal quantum number 6. Its radial-node count exceeds its angular-node count by one. The number of orbitals in this subshell is

(A) 9 (B) 5

(C) 3 (D) 7

17. Hydrogen atoms initially occupy an excited state with ionisation energy 0.544 eV. They undergo all possible downward cascades. How many distinct emitted frequencies have photon energy greater than 3.00 eV? Use $E_n = -13.6/n^2$ eV.

(A) 6 (B) 4

(C) 5 (D) 10

18. A sample of hydrogen atoms is excited to $n = 4$ and emits every allowed downward cascade line. A detector records only photons of energy above 2.00 eV. How many distinct frequencies can it record? ($E_n = -13.6/n^2$ eV)

(A) 3 (B) 4

(C) 6 (D) 5

19. A hydrogen atom initially at $n = 2$ absorbs a photon of 2.55 eV. It is then ionised from the new level. The minimum additional photon energy needed is ($E_n = -13.6/n^2$ eV)

(A) 2.55 eV (B) 0.850 eV

(C) 1.51 eV (D) 3.40 eV

20. A He^+ ion initially at $n = 2$ absorbs 10.2 eV without ionising. After relaxation from the resulting level, the maximum possible number of distinct cascade frequencies is

(A) 3 (B) 4

(C) 10 (D) 6

21. Ion A has charge $+2e$, ion B has charge $+e$, and A is 12 u heavier than B. Starting from rest, A is accelerated through V and B through 6 V. Their nonrelativistic de Broglie wavelengths are equal. Mass of A is

(A) 36 u (B) 6 u

(C) 24 u (D) 18 u

22. An oxygen ion has 8 protons, 6 electrons and mass 16 u. A helium ion has 2 protons, 1 electron and mass 4 u. Starting from rest, their accelerating potentials are chosen for equal de Broglie wavelengths. The helium-to-oxygen potential ratio is

(A) 8 (B) $\frac{1}{8}$

(C) 4 (D) 2

23. Photons of energies E and 2E eject electrons from the same metal. The maximum electron de Broglie wavelength for E photons is $\sqrt{3}$ times that for 2E photons. If $E = 4.00$ eV, the maximum kinetic energy for 6.00 eV incident photons is

(A) 3.00 eV (B) 6.00 eV

(C) 2.00 eV (D) 4.00 eV

24. Two electron beams have wavelengths λ and $\lambda/2$. A mixed pulse contains $3N$ electrons and deposits total kinetic energy $6NK$, where K is the energy per electron in the longer-wavelength beam. All electrons stop. The fraction from the shorter-wavelength beam is

(A) $\frac{1}{4}$ (B) $\frac{2}{3}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

25. Two minimum-uncertainty wavepackets A and B satisfy $\Delta x_A = 2\Delta x_B$ and $\Delta v_A = (\Delta v_B)/4$. Their particle masses sum to 9 u. The mass of A is

(A) 6 u (B) 3 u

(C) 9/2 u (D) 12 u

26. Two minimum-uncertainty particles have total mass 5 u. Particle A has twice particle B position uncertainty and one-third its velocity uncertainty. The mass of A is

(A) 4 u (B) 5/2 u

(C) 3 u (D) 2 u

27. How many distinct one-electron states satisfy $n = 4$, even l, nonnegative m_l, and $m_s = +1/2$?

(A) 8 (B) 6

(C) 4 (D) 10

28. How many electrons can simultaneously have $n = 5$, $l \leq 2$, $m_l = 0$, and either allowed spin?

(A) 6 (B) 3

(C) 9 (D) 18

29. A hydrogen-like ion has ground-state ionisation energy 122.4 eV. An electron in its $n = 3$ shell has one angular node. The orbital and its radius relative to hydrogen $1s$ are

(A) $2p$; 4/3 (B) $3s$; 3

(C) $3p$; 3 (D) $3p$; 1

30. A hydrogen-like ion has a state with binding-energy magnitude 13.6/9 eV and characteristic radius $18a_0$, where a_0 is the hydrogen ground-orbit radius. Its ground-state ionisation energy is

(A) 217.6 eV (B) 13.6 eV

(C) 122.4 eV (D) 54.4 eV

31. Two hydrogen orbitals have the same l. The second has two more radial nodes, and the first-to-second Bohr-speed ratio is 3/2. Photon energy for transition from the second state to the first is ($E_n = -13.6/n^2$ eV)

(A) 17/16 eV (B) 17/36 eV

(C) 17/9 eV (D) 17/144 eV

32. A state of an unknown hydrogen-like ion has the same binding energy as hydrogen $n = 2$, but its characteristic radius is three times as large. The unknown ion nuclear charge is

(A) 2 (B) 9

(C) 3 (D) 6

33. A two-colour pulse contains 10^{15} photons of energies 4.00 and 8.00 eV only, and total energy 9.60×10^{-4} J. All photons strike a metal with work function 5.00 eV. Assuming one electron per above-threshold photon and maximum kinetic energy for each, the total electron kinetic energy is ($1\text{ eV} = 1.60 \times 10^{-19}$ J)

(A) 2.40×10^{-4} J (B) 1.60×10^{-4} J

(C) 3.20×10^{-4} J (D) 4.80×10^{-4} J

34. A pulse has 2.00 eV and 4.00 eV photons only, with total energy 9.60×10^{-4} J. A detector yields 0.25 electron per 2 eV photon and 0.75 per 4 eV photon; it emits 10^{15} electrons. The fraction of incident photons having energy 4.00 eV is ($1\text{ eV} = 1.60 \times 10^{-19}$ J)

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{2}{3}$ (D) $\frac{1}{4}$

35. A hydrogen-like ion has ground ionisation energy 54.4 eV. The wavelength of its $3 \to 2$ line divided by that of its $n \to 2$ line is 27/20. The values of n and nuclear charge, respectively, are

(A) 4, 2 (B) 5, 2

(C) 4, 4 (D) 3, 2

36. An unknown hydrogen-like ion has ground-state ionisation energy 54.4 eV. A line from an unknown upper level to $n = 2$ has photon energy 10.2 eV. If λ_0 is the hydrogen Lyman-limit wavelength, the wavelength of the line from that upper level to the ion ground state is

(A) $4\lambda_0/15$ (B) $\lambda_0/4$

(C) $15\lambda_0/4$ (D) $4\lambda_0/3$

37. Electrons accelerated from rest initially through an unknown potential have wavelength λ. Adding 150 V to the accelerating potential halves the wavelength. A further 250 V is then added. The final wavelength divided by the original wavelength is

(A) $\frac{\sqrt{6}}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

38. For a nonrelativistic electron beam, adding 96 V to the accelerating potential changes its wavelength to 5/7 of its original value. What further potential increase makes the new wavelength $\sqrt{7/8}$ times the wavelength after the first increase?

(A) 28 V (B) 32 V

(C) 24 V (D) 14 V

39. Two orbitals of a hydrogen atom have the same l. The higher-energy orbital has one more radial node. The ratio of lower-state to higher-state binding-energy magnitudes is 25/16. Maximum distinct cascade frequencies from the higher state are

(A) 15 (B) 5

(C) 10 (D) 6

40. Two hydrogen orbitals A and B each have one angular node. Their radial-node counts sum to three, and $r_B/r_A = 9/16$ for their characteristic radii. Minimum ionising energy from A is ($E_n = -13.6/n^2$ eV)

(A) 1.511 eV (B) 0.544 eV

(C) 0.850 eV (D) 3.40 eV

41–50 / Preserved hard practice

41. A dipositive ion contains 24 electrons. In the free-ion ground state it loses its $4s$ electrons before $3d$ electrons. Its spin-only magnetic moment in Bohr magnetons is closest to

(A) 3.87 (B) 4.9

(C) 5.92 (D) 2.83

42. A hydrogen-like ion emits a photon of energy 40.8 eV for a transition from n=2 to n=1. Its bare nucleus subsequently captures one electron directly into the ground state. Energy released in eV is

(A) 54.4 (B) 40.8

(C) 13.6 (D) 122.4

43. A third-period element X forms a stable monatomic ion with charge 2- and 18 electrons. In neutral X, the number of unpaired electrons in its ground-state valence shell is

(A) 2 (B) 0

(C) 4 (D) 1

44. A metal ion M^{3+} has 23 electrons. In an octahedral strong-field complex with electron pairing, its number of unpaired electrons is

(A) 3 (B) 1

(C) 0 (D) 5

45. An ionic compound contains X^+ and Y^{2-}. X^+ and Y^{2-} are isoelectronic with Ne. Sum of the atomic numbers of all atoms in one empirical formula unit is

(A) 19 (B) 27

(C) 30 (D) 38

46. A 0.0200 mol mixture contains free high-spin M^{2+} and M^{3+} ions. The M^{3+} ion has 23 electrons. Oxidizing all M^{2+} to M^{3+} removes 0.0140 mol electrons. Before oxidation, the amount of unpaired electrons, counting each unpaired electron once, is (mol)

(A) 0.1 (B) 0.094

(C) 0.086 (D) 0.08

47. One mole of hydrogen atoms initially in n=3 is ionized completely. Taking 1 eV per particle=96.5 kJ /mol, minimum energy needed in kJ is

(A) 145.82222 (B) 1312.4

(C) 328.1 (D) 36.455556

48. A species has nuclear charge 13 and 10 electrons. A compound made from this cation and oxide ions contains how many total electrons per empirical formula unit?

(A) 30 (B) 40

(C) 60 (D) 50

49. A 0.0200 mol ideal mixture contains free high-spin Fe^{2+} and Fe^{3+}. Its effective moment obeys $\mu_{\text{eff}}^2 = x\mu_{\text{Fe(II)}}^2 + (1-x)\mu_{\text{Fe(III)}}^2$ and equals $30.6\,\mu_B^2$. Use spin-only moments $\mu^2 = n(n+2)\mu_B^2$. The amount of electrons removed when all Fe(II) becomes Fe(III) is (mol)

(A) 0.008 (B) 0.012

(C) 0.016 (D) 0.02

50. An excited hydrogen-like one-electron ion occupies an orbital with two radial nodes and one angular node. Direct emission to $n = 1$ releases 51.0 eV. Use $E_n = -13.6Z^2/n^2$ eV and 1 eV per particle $= 96.5\,\text{kJ mol}^{-1}$. Energy needed to ionize 0.0100 mol of these ions after emission, in kJ, is

(A) 52.496 (B) 3.281

(C) 49.215 (D) 13.124

Answer key

Atomic Structure

C-ATST / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**D**	31	**B**	41	**B**
02	**C**	12	**D**	22	**A**	32	**C**	42	**A**
03	**D**	13	**A**	23	**D**	33	**A**	43	**A**
04	**D**	14	**B**	24	**D**	34	**B**	44	**B**
05	**B**	15	**D**	25	**A**	35	**A**	45	**C**
06	**C**	16	**B**	26	**C**	36	**A**	46	**C**
07	**B**	17	**B**	27	**C**	37	**B**	47	**A**
08	**B**	18	**B**	28	**A**	38	**A**	48	**D**
09	**D**	19	**B**	29	**C**	39	**C**	49	**A**
10	**C**	20	**D**	30	**D**	40	**C**	50	**A**

Chapter 5

Periodic Properties

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Which descending-radius order is correct for O^{2-}, F^-, Na^+, Mg^{2+}?

(A) $F^- > O^{2-} > Mg^{2+} > Na^+$

(B) $Na^+ > Mg^{2+} > O^{2-} > F^-$

(C) $O^{2-} > F^- > Na^+ > Mg^{2+}$

(D) $Mg^{2+} > Na^+ > F^- > O^{2-}$

02. Which cation should be harder to oxidize by removing one further electron from the gas-phase ion?

(A) Li

(B) Na

(C) Na^+

(D) K

03. Which statement about the first electron-gain enthalpies of O and S is correct?

(A) Both are zero because both are chalcogens

(B) S cannot accept an electron

(C) S is generally more negative because O has stronger compact-orbital repulsion

(D) O must be more negative solely because it is above S

04. Among NaCl, $MgCl_2$, $AlCl_3$, which anhydrous chloride is expected to show the greatest covalent character by Fajans' reasoning?

(A) All three exactly equal

(B) $MgCl_2$

(C) NaCl

(D) $AlCl_3$

05. A Period 3 main-group atom shows its first very large successive-ionization jump after removal of 3 electrons. Its neutral atom has how many $3p$ electrons?

(A) 1

(B) 3

(C) 2

(D) 0

06. A Period 3 element forms a monatomic $2-$ ion with an argon configuration. Its neutral atom has how many unpaired electrons in the ground state?

(A) 0 (B) 4

(C) 2 (D) 1

07. Two species are isoelectronic with Ne. One has charge$+1$ and the other$+3$. Which comparison follows?

(A) The$+1$ ion has a smaller radius and larger nuclear charge (B) Their nuclear charges are equal

(C) The$+3$ ion has a smaller radius and larger nuclear charge (D) They have identical radii because electron counts match

08. A main-group element in Period 2 forms a monoxide XO in which X has oxidation state$+2$. Its first large ionization jump is after removing 2 electrons. Atomic number of X is

(A) 3 (B) 5

(C) 6 (D) 4

09. An element X has atomic number 13. If its stable cation is isoelectronic with Ne, how many oxide ions are required per 2 cations for a neutral oxide?

(A) 3 (B) 2

(C) 6 (D) 1

10. A Group 17 anion and a Group 1 cation are isoelectronic with Ar. The halogen is in Period 3 and the alkali metal in Period 4. Sum of their atomic numbers is

(A) 36 (B) 38

(C) 34 (D) 28

11–40 / New middle-to-hard practice

11. A Period-3 element has its first very large successive-ionisation jump between the third and fourth electrons. Its highest oxide is tested with dilute HCl and concentrated NaOH. Which prediction is consistent with both clues?

(A) The element is P; its oxide reacts only with HCl (B) The element is Si; its oxide reacts readily with both reagents

(C) The element is Mg; its oxide reacts only with HCl (D) The element is Al; its oxide reacts with both reagents

12. An element in Period 2 forms a stable divalent cation and shows a large jump after its second ionisation. Which combination of oxide behaviour and chloride character is most consistent?

(A) BeO is amphoteric; anhydrous $BeCl_2$ has appreciable covalent character

(B) CaO is amphoteric; $CaCl_2$ has the greatest covalent character in Group 2

(C) BeO is strongly acidic; $BeCl_2$ has negligible covalent character

(D) MgO is amphoteric; $MgCl_2$ is exclusively molecular

13. Ions X^{2-} and Y^{+} each have 18 electrons. X and Y are in their normal main-group states. Which pair correctly gives the ground-state unpaired-electron count of neutral X and the ionic-radius comparison?

(A) 2; $r(X^{2-}) > r(Y^{+})$

(B) 1; equal ionic radii

(C) 2; $r(X^{2-}) < r(Y^{+})$

(D) 0; $r(X^{2-}) > r(Y^{+})$

14. Ions A^{3+} and B^{-} are isoelectronic with neon. The neutral atoms A and B have how many valence electrons in total, and which ion is smaller?

(A) 8; B^{-}

(B) 12; A^{3+}

(C) 10; B^{-}

(D) 10; A^{3+}

15. Consider $O(g) + e^{-} \rightarrow O^{-}(g)$ and $O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$. Why can a solid oxide be stable even though the second process is endothermic?

(A) The second addition is always exothermic after the octet is completed; lattice formation destabilises the solid

(B) Both gas-phase additions must be exothermic because oxide solids exist

(C) The second gas-phase addition faces electron repulsion; lattice formation can more than compensate energetically

(D) Oxide stability requires the isolated O^{2-} ion to have the lowest gas-phase energy

16. Chlorine has a more negative first electron-gain enthalpy than fluorine, while fluoride is smaller than chloride. Which explanation accommodates both observations?

(A) Adding an electron to F removes a $2p$ electron before attachment

(B) F is smaller, but its compact $2p$ shell gives stronger repulsion to the added electron

(C) Cl has the smaller anion and therefore always the stronger attraction

(D) Electron-gain enthalpy and anion radius must vary in the same direction without exceptions

17. A Period-3 oxide is insoluble in water, dissolves in strong NaOH, and does not behave as a basic oxide with dilute HCl. Its element forms a tetrahalide. Which oxide is most consistent?

(A) Na_2O

(B) Al_2O_3

(C) MgO

(D) SiO_2

18. An unknown Period-3 oxide dissolves in both dilute acid and concentrated alkali. Its element has one unpaired p electron in the neutral ground state. Which highest chloride is expected?

(A) $MgCl_2$

(B) $SiCl_4$

(C) $AlCl_3$

(D) PCl_5

19. Two anhydrous halides have the same anion: LiCl and KCl. A second pair has the same cation: LiF and LiI. Which pair correctly identifies the more covalent member of each comparison?

(A) KCl; LiI

(B) LiCl; LiI

(C) KCl; LiF

(D) LiCl; LiF

20. Compare NaCl with $MgCl_2$ and MgF_2 with MgI_2. Which choice is consistent with Fajans reasoning for greater covalent character in each pair?

(A) $MgCl_2$; MgF_2
(B) $MgCl_2$; MgI_2
(C) NaCl; MgF_2
(D) NaCl; MgI_2

21. Two neutral Period-3 elements X and Y have configurations $3s^2$ and $3s^2 3p^1$ outside a Ne core. Which comparison of first and second ionisation enthalpies is expected?

(A) Both are equal because both contain a $3s^2$ pair
(B) Both first and second ionisation enthalpies of X are larger
(C) $IE_1(X) < IE_1(Y)$ and $IE_2(Y) < IE_2(X)$
(D) $IE_1(X) > IE_1(Y)$ and $IE_2(Y) > IE_2(X)$

22. For N and O, the first ionisation enthalpy of N is higher. After one electron is removed, which ion has the half-filled p subshell, and how does this affect the next ionisation comparison?

(A) O^+; its second ionisation enthalpy exceeds that of N
(B) Both ions have half-filled p subshells and identical second ionisation enthalpies
(C) O^+; its second ionisation enthalpy is lower than that of N
(D) N^+; its second ionisation enthalpy exceeds that of O

23. Three isoelectronic ions have radii 0.95, 0.65 and 0.50 in the same arbitrary unit. They are Na^+, Mg^{2+} and Al^{3+}. What is the sum of neutral atomic numbers of the ions assigned the largest and smallest radii?

(A) 24
(B) 22
(C) 23
(D) 25

24. Three ions with 10 electrons have charges $-3, -2, -1$ and radii 1.71, 1.40, 1.36 in matching units. Assign the radii by the usual isoelectronic trend. The neutral atom corresponding to radius 1.40 has how many unpaired electrons?

(A) 3
(B) 2
(C) 0
(D) 1

25. For a hypothetical atom M, $IE_1 = 500$, $IE_2 = 1500$ and $IE_3 = 7000$ kJ mol^{-1}. One mole gaseous M atoms forms one mole gaseous M^{2+}, then half those ions become M^{3+}. Total energy absorbed is

(A) 2000 kJ
(B) 9000 kJ
(C) 3500 kJ
(D) 5500 kJ

26. Gaseous M atoms absorb 800 kJ to form a mixture containing 0.500 mol M^+ and 0.250 mol M^{2+}. No atoms remain and $IE_1 = 400$ kJ mol^{-1}. The second ionisation enthalpy is

(A) 1600 kJ mol^{-1}
(B) 2400 kJ mol^{-1}
(C) 800 kJ mol^{-1}
(D) 2000 kJ mol^{-1}

27. A cation M^{2+} has 18 electrons. Its neutral atom belongs to Period 4. Which combination is correct for its highest oxide and its ionic radius relative to K^+?

(A) M_2O_3 is amphoteric; the ions have equal radii

(B) MO is acidic; M^{2+} is larger than K^+

(C) MO is basic; M^{2+} is smaller than K^+

(D) MO_2 is acidic; M^{2+} has fewer protons than K^+

28. A main-group element X forms X^{3-} with 18 electrons. Which pair gives the highest oxide formula and neutral-atom valence configuration?

(A) X_2O_3; $3s^2 3p^1$

(B) XO_3; $3s^2 3p^4$

(C) X_2O_5; $3s^2 3p^3$

(D) XO_2; $3s^2 3p^2$

29. Four atoms have valence configurations $2s^2 2p^3$, $2s^2 2p^4$, $3s^2$, $3s^2 3p^1$. Among the two same-period adjacent pairs, in how many does the atom to the right have the lower first ionisation enthalpy?

(A) 3

(B) 2

(C) 1

(D) 0

30. Consider B, C, N and O. Which ordered pair has a higher first ionisation enthalpy followed by an atom of larger covalent radius, respectively, when comparing N with O and B with C?

(A) N; C

(B) N; B

(C) O; C

(D) O; B

31. The gaseous process $X^+ \rightarrow X^{2+} + e^-$ needs far more energy than the corresponding second ionisation of its neighbour Y. X and Y are adjacent Period-3 metals, with X on the left. Which oxide-formula pair is expected?

(A) X_2O and YO

(B) XO_2 and Y_2O_5

(C) X_2O_3 and YO_2

(D) XO and Y_2O_3

32. Two adjacent Period-2 atoms have neutral p-electron counts differing by one. The atom on the left has a higher first ionisation enthalpy despite lower nuclear charge, and both have at least three p electrons. Their most common simple hydrides contain how many H atoms per central atom, left then right?

(A) 2, 1

(B) 4, 3

(C) 3, 1

(D) 3, 2

33. A Group-1 atom and a Group-17 atom form ions isoelectronic with argon. Their neutral atomic numbers sum to 36. Which comparison of the neutral atoms is correct?

(A) The Group-1 atom has larger radius and lower first ionisation enthalpy

(B) The atoms have equal radius because their ions are isoelectronic

(C) The Group-1 atom has smaller radius but greater electron affinity magnitude

(D) The Group-17 atom has larger radius and lower first ionisation enthalpy

34. Neutral atoms A and B form A^+ and B^{2-} with 10 electrons each. Which comparison simultaneously follows for neutral-atom radius and the ion radius?

(A) Neutral radii are equal; only charges determine ionic radius

(B) Both neutral and ionic radii of A are smaller

(C) $r(A) > r(B)$ but $r(A^+) < r(B^{2-})$

(D) Both neutral and ionic radii of A are greater

35. For hypothetical monovalent metal M, sublimation is 100 and first ionisation I is unknown. MX formation is -420; X atomisation is 80 and MX lattice formation is -700. Separately, M_2O formation is -150; O atomisation is 250, its two electron-gain steps sum to $+600$, and lattice formation is -2000. All data are $kJ\,mol^{-1}$ per formula reaction. First electron-gain enthalpy of X is

(A) $-300\ kJ\,mol^{-1}$

(B) $-100\ kJ\,mol^{-1}$

(C) $-500\ kJ\,mol^{-1}$

(D) $+300\ kJ\,mol^{-1}$

36. A hypothetical divalent metal has sublimation 100, $IE_1 = 500$ and unknown IE_2. MO formation is -550; O atomisation plus two electron-gain steps is $+850$, and MO lattice formation is -3000. MX_2 formation is -800; each X atom costs 100 to atomise and gains an electron with enthalpy -300. All data are $kJ\,mol^{-1}$ per formula reaction. MX_2 lattice dissociation enthalpy is

(A) $2000\ kJ\,mol^{-1}$

(B) $1500\ kJ\,mol^{-1}$

(C) $1000\ kJ\,mol^{-1}$

(D) $2500\ kJ\,mol^{-1}$

37. A Period-3 element has three unpaired electrons in its neutral ground state. Another in the same period has a closed s subshell and no p electrons. Which pair orders their highest oxides as acidic then basic?

(A) Al_2O_3; P_4O_{10}

(B) SiO_2; Na_2O

(C) SO_3; Al_2O_3

(D) P_4O_{10}; MgO

38. An element X in Period 3 has two unpaired p electrons and forms an acidic highest oxide. Another element Y has one p electron and an amphoteric oxide. The pair of highest oxidation states, X then Y, is

(A) +2 for X; +1 for Y

(B) +3 for X; +4 for Y

(C) +7 for X; +2 for Y

(D) +4 or +6 for X; +3 for Y

39. Atoms A and B are adjacent main-group elements in Period 3. A forms A^{2-} and B forms B^-, both with 18 electrons. Of their gaseous neutral atoms, which has the smaller radius and more negative first electron-gain enthalpy?

(A) B for both comparisons

(B) B has smaller radius; A has more negative electron-gain enthalpy

(C) A for both comparisons

(D) A has smaller radius; B has more negative electron-gain enthalpy

40. X and Y are alkali metals in adjacent periods. X^+ has a Ne configuration, and Y^+ has an Ar configuration. Which simultaneous comparison is correct?

(A) Both first ionisation enthalpies and hydration tendencies are equal

(B) $IE_1(X) < IE_1(Y)$; gaseous Y^+ has stronger hydration tendency

(C) $IE_1(X) > IE_1(Y)$; Y^+ has the smaller ionic radius

(D) $IE_1(X) > IE_1(Y)$; gaseous X^+ has stronger hydration tendency

41–50 / Preserved hard practice

41. A Period-3 metal M has its first very large successive-ionization jump after two electrons. A 1.760 g mixture contains M and MO only and consumes 0.1000 mol HCl completely. Use M=24 and O=16. Both components react quantitatively. The hydrogen amount released, in mol, is

(A) 0.05 (B) 0.015

(C) 0.025 (D) 0.035

42. A Period-3 element X forms a 3+ ion with ten electrons. A 5.010 g anhydrous mixture contains XCl_3 and NaCl only. Complete chloride analysis finds 0.1000 mol Cl atoms. Use X=27, Na=23, Cl=35.5. Formal hydrolysis of only XCl_3 releases three HCl per formula unit. Moles of NaOH needed to neutralize only that released HCl are

(A) 0.1 (B) 0.06

(C) 0.04 (D) 0.02

43. Two Period 3 elements X and Y form X^+ and Y^{2-} ions. X has its first large ionization jump after one electron; Y^{2-} has 18 electrons. With X=23, Y=32, mass percentage of Y in X_2Y is

(A) 41.818182 (B) 41.025641

(C) 58.974359 (D) 50

44. A Group-15 Period-3 element X forms a 0.100 mol mixture of XCl_3 and XCl_5. The mixture contains 0.400 mol Cl atoms. Hydrolysis is complete. Treat the resulting H_3XO_3 as di basic and H_3XO_4 as tri basic, and include all liberated HCl. Total NaOH required for complete neutralization, in mol, is

(A) 0.7 (B) 0.5

(C) 0.65 (D) 0.4

45. An element X has configuration [Ne] $3s^2\ 3p^1$. Its oxide is amphoteric. In formal reduction of one mole of X_2O_3 to metal, moles of electrons gained by X are

(A) 3 (B) 2

(C) 6 (D) 12

46. A metal M forms M^{2+} with the Ar configuration. A 4.080 g mixture of M and MH_2 reacts completely with water, giving 0.1400 mol hydrogen. Use M=40, H=1. The mass fraction of MH_2 is

(A) 0.4 (B) 0.41176471

(C) 0.6 (D) 0.58823529

47. A Group 17 element X lies in Period 3. In disproportionation $3\ X_2 + 6\ OH^- \rightarrow 5\ X^- + XO_3^- + 3\ H_2O$, moles of electrons transferred internally per mole of X 2 consumed are

(A) 1 (B) 1.6666667

(C) 5 (D) 2

48. A Group 15 Period 2 element combines with a Group 1 Period 2 element to form the normal ionic nitride. Given atomic masses 7 and 14, mass of nitride from 0.300 mol metal with nitrogen in excess in g is

(A) 4.2 (B) 3.5

(C) 2.1 (D) 10.5

49. X^+ and Y^- are isoelectronic with Ar. A mixture contains 0.100 mol XY and 0.0500 mol Y_2. Total moles of electrons in the mixture are

(A) 5.3 (B) 4.5

(C) 3.6 (D) 7.2

50. A Period 3 main-group atom forms a monatomic 3− ion with the Ar electron configuration. Its ordinary trichloride has one central lone pair. The number of chlorine atoms in 0.250 mol of that trichloride, expressed as a multiple of N_A, is

(A) 0.5 (B) 0.75

(C) 0.25 (D) 1

Answer key

Periodic Properties

C-PEPR / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**D**	21	**D**	31	**A**	41	**B**
02	**C**	12	**A**	22	**A**	32	**D**	42	**B**
03	**C**	13	**A**	23	**A**	33	**A**	43	**B**
04	**D**	14	**D**	24	**B**	34	**C**	44	**C**
05	**A**	15	**C**	25	**D**	35	**A**	45	**C**
06	**C**	16	**B**	26	**D**	36	**A**	46	**B**
07	**C**	17	**D**	27	**C**	37	**D**	47	**B**
08	**D**	18	**C**	28	**C**	38	**D**	48	**B**
09	**A**	19	**B**	29	**B**	39	**A**	49	**A**
10	**A**	20	**B**	30	**B**	40	**D**	50	**B**

Chapter 6

Chemical Bonding

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Which pair is isoelectronic and has the same elementary bond order?

(A) N_2 and N_2^+

(B) F_2 and Ne_2

(C) N_2 and CO

(D) O_2 and O_2^+

02. In a trigonal bipyramid, how many axial-equatorial bond-angle pairs are 90 degrees?

(A) 3

(B) 6

(C) 10

(D) 4

03. An ion has central atom X with 7 valence electrons, four X-F bonds and net charge-1. Assuming closed-shell Lewis bonding, the number of central lone pairs is

(A) 0

(B) 3

(C) 1

(D) 2

04. An ion XF_4^- has a central atom X with seven valence electrons and four single X–F bonds. Use the usual six-domain VSEPR model, with the lone pairs placed to minimise repulsion. The number of distinct 90° F–X–F bond-angle pairs is

(A) 6

(B) 2

(C) 4

(D) 8

05. A neutral molecule has central X with 6 valence electrons, four single bonds to F and one lone pair. In the school VSEPR model, number of axial F atoms in its seesaw structure is

(A) 2

(B) 4

(C) 1

(D) 0

06. Number of species with zero permanent dipole moment among CO_2, BF_3, CH_4, NH_3, H_2O is

(A) 3 (B) 1

(C) 4 (D) 2

07. In a mixture containing equal numbers of O_2 and O_2^+ ions, mean bond order per diatomic species is

(A) 2.5 (B) 2

(C) 1.75 (D) 2.25

08. A diatomic species has 10 bonding and 7 antibonding electrons. Number of electrons that must be removed exclusively from antibonding orbitals to raise bond order to 2.5 is

(A) 4 (B) 3

(C) 1 (D) 2

09. Number of sigma and pi bonds, respectively, in $HC \equiv C - CH = CH_2$ is

(A) $7, 2$ (B) $7, 3$

(C) $9, 4$ (D) $8, 3$

10. The resonance-average bond order of each $C - O$ bond in carbonate in the elementary equivalent-contributor model is

(A) 1.5 (B) 1

(C) 1.3333333 (D) 2

11–40 / New middle-to-hard practice

11. An ion XF_4^- has central atom X with seven valence electrons. In the usual VSEPR model, determine both the number of central lone pairs and the number of distinct 90° F–X–F angle pairs.

(A) 2, 6 (B) 1, 4

(C) 3, 4 (D) 2, 4

12. A neutral XF_4 molecule has X with six valence electrons. Use elementary VSEPR with the lone pair in its preferred position. The numbers of central lone pairs and axial–equatorial F–X–F pairs are, respectively

(A) 1, 4 (B) 0, 6

(C) 2, 4 (D) 1, 6

13. For I_3^-, count total valence electrons before drawing its standard linear Lewis structure. The numbers of total lone pairs on all atoms and central lone pairs are

(A) 10, 4 (B) 9, 2

(C) 3, 3 (D) 9, 3

14. In standard Lewis/VSEPR models, compare I_3^- and I_3^+. Which pair gives the respective central lone-pair counts and predicts whether both ions are linear?

(A) 2, 3; only I_3^+ is linear

(B) 3, 2; only I_3^- is linear

(C) 3, 3; both are linear

(D) 2, 2; neither can be bent

15. An equimolar mixture contains N_2 and O_2^+. Under elementary molecular-orbital theory, the mean bond order per diatomic species and mean unpaired-electron count per species are

(A) 11/4, 1/2

(B) 9/4, 1/2

(C) 5/2, 1

(D) 11/4, 0

16. Two diatomic species are N_2^{2+} and O_2^{2-}. In their elementary ground-state MO configurations, which comparison of bond order and magnetism is correct?

(A) Bond orders are 1 and 2; both are diamagnetic

(B) Both have bond order 2; both are paramagnetic

(C) Bond orders are 2 and 1, respectively; both are diamagnetic

(D) Bond orders are 3 and 2; only the peroxide ion is paramagnetic

17. A mixture contains only O_2 and O_2^+. Its mean bond order is 2.30. The fraction of diatomic species with two unpaired electrons is

(A) $\frac{2}{5}$

(B) $\frac{3}{5}$

(C) $\frac{4}{5}$

(D) $\frac{1}{5}$

18. A mixture contains O_2^- and O_2^{2-} only. Mean unpaired-electron count per species is 0.40. Its mean bond order is

(A) $\frac{7}{5}$

(B) 1

(C) $\frac{3}{2}$

(D) $\frac{6}{5}$

19. A diatomic ion initially contains 15 electrons occupying bonding or antibonding molecular orbitals. Exactly one antibonding electron is removed; the resulting bond order is 3. The original antibonding-electron count is

(A) 5

(B) 6

(C) 7

(D) 4

20. A model diatomic species contains 14 bonding plus antibonding electrons in total. Promoting one electron from a bonding to an antibonding orbital changes its bond order to 1. Its original bonding-electron count was

(A) 10

(B) 9

(C) 8

(D) 7

21. An ideal octahedral MA_4B_2 has its B ligands cis. Among all 15 unordered ligand pairs, how many A–B pairs subtend 90°?

(A) 6

(B) 4

(C) 2

(D) 8

22. An ideal octahedral MA_3B_3 has a facial arrangement. The number of A–B pairs at 180° and A–B pairs at 90°, respectively, is

(A) 2, 7 (B) 1, 8

(C) 3, 6 (D) 3, 3

23. An ion YH_3^+ has eight valence electrons in total; Y is a Period-2 main-group atom. In its usual octet Lewis structure, identify Y and predict the molecular shape.

(A) O; trigonal pyramidal (B) O; trigonal planar

(C) N; trigonal planar (D) C; trigonal pyramidal

24. An ion XH_3^- has eight valence electrons in total and X is a Period-2 atom. It binds one proton without electron loss. The central atom and shape of the resulting neutral molecule are

(A) C; trigonal planar (B) C; tetrahedral

(C) O; bent (D) N; trigonal pyramidal

25. An acyclic hydrocarbon has formula C_4H_6 and consumes two moles H_2 per mole on complete hydrogenation. Its molecule has two sp carbons and no sp^2 carbon. Total sigma and pi bonds are

(A) 8, 2 (B) 9, 1

(C) 9, 2 (D) 10, 2

26. An acyclic hydrocarbon C_5H_8 contains four sp^2 carbons and no sp carbon. Its sigma- and pi-bond totals are

(A) 12, 2 (B) 11, 2

(C) 13, 1 (D) 12, 3

27. A 0.0100 mol mixture contains ortho- and para-nitrophenol. A calibrated steam-distillation step recovers 80% of the predominantly intramolecular-H-bonded isomer and 20% of the other, yielding 0.00600 mol distillate. Ortho-isomer mole fraction in the feed is

(A) $\frac{2}{3}$ (B) $\frac{3}{5}$

(C) $\frac{4}{5}$ (D) $\frac{1}{3}$

28. A 0.0200 mol mixture contains ortho- and para-hydroxybenzaldehyde. A calibrated steam step recovers 75% of the intramolecular-H-bonded isomer and 25% of the other, yielding 0.0125 mol distillate. From a fresh equal feed, all para isomer is separated. The remaining aldehyde deposits how much Ag in quantitative Tollens reaction? (Ag=108; two Ag per aldehyde)

(A) 4.32 g (B) 1.08 g

(C) 2.70 g (D) 3.24 g

29. Two fixed bond-dipole magnitudes p and q are carried unchanged between idealised arrangements. Their parallel resultant is 6.00 D and antiparallel resultant is 2.00 D, with $p > q$. If the bond vectors meet at 90°, the resultant magnitude is

(A) 6 D (B) 4 D

(C) $2\sqrt{2}$ D (D) $2\sqrt{5}$ D

30. In a fixed-bond-moment model, two positive bond-dipole magnitudes sum to 5.00 D. When the vectors are perpendicular their resultant is $\sqrt{13}$ D. Their antiparallel resultant magnitude is

(A) 2.00 D (B) 1.00 D

(C) $\sqrt{5}$ D (D) 3.00 D

31. A carbonate–nitrate mixture contains 0.300 mol oxygen atoms and 0.140 mol units of negative ionic charge. In the usual one-double-bond Lewis contributors for the two ions, the mean formal charge on the central atom is

(A) $\frac{2}{5}$ (B) $\frac{7}{5}$

(C) $\frac{3}{5}$ (D) $\frac{4}{5}$

32. A 0.200 mol carbonate–nitrate mixture has total negative charge equivalent to 0.250 mol electrons. Its mean central-atom oxidation state is

(A) $\frac{9}{2}$ (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{19}{4}$

33. A gas mixture contains 0.200 mol BF_3 plus NH_3 and weighs 8.50 g (molar masses 68 and 17). They form a 1:1 Lewis adduct quantitatively. The amount of adduct and boron geometry in it are

(A) 0.200 mol; tetrahedral (B) 0.100 mol; trigonal planar

(C) 0.100 mol; tetrahedral (D) 0.0500 mol; trigonal planar

34. A 0.300 mol mixture of ammonia and water weighs 5.30 g (molar masses 17 and 18). Add 0.150 mol protons. Use ideal complete stoichiometric protonation, with ammonia protonated before water. Among the resulting protonated molecules, the fraction having trigonal-pyramidal shape is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

35. A mixture of O_2^+ and O_2^- has mean bond order 2.10. One mole of the mixture gains enough electrons to convert every species into O_2^{2-}. The electron amount required is

(A) 2 mol (B) 9/5 mol

(C) 3 mol (D) 11/5 mol

36. A mixture contains equal molecular amounts of O_2, O_2^- and O_2^{2-}. Remove exactly one electron from every diatomic species, with all products remaining in their elementary ground states. Mean bond order increases by

(A) $\frac{1}{2}$ (B) 1

(C) $\frac{1}{6}$ (D) $\frac{1}{3}$

37. A 0.100 mol mixture of CN^- and NO^+ weighs 2.84 g (molar masses 26 and 30). Exactly one electron is added to every NO^+ while CN^- remains unchanged. Under elementary MO theory, the final mean bond order is

(A) $\frac{5}{2}$ (B) $\frac{27}{10}$

(C) 3 (D) $\frac{14}{5}$

38. A 0.100 mol mixture of CN^- and NO^+ has net charge equivalent to 0.0200 mol excess electrons. Remove one electron from each CN^- and add one to each NO^+, making neutral species. The final mean bond order and net electron amount released are

(A) 5/2; 0.0800 mol (B) 3; 0.100 mol

(C) 5/2; 0.0200 mol (D) 3; 0.0200 mol

39. An idealised hydrogen-bonding mixture contains 0.300 mol water plus ammonia molecules and 0.800 mol O–H plus N–H bonds. Count each O–H/N–H as one donor site, each water O as two acceptor sites and each ammonia N as one. Neglect geometry and allow one H bond per site. Maximum H bonds are

(A) 0.600 mol (B) 0.400 mol

(C) 0.300 mol (D) 0.800 mol

40. A 0.200 mol mixture contains ethanol and ammonia only, with 0.400 mol hydrogen-bond donor sites. Use one donor and two acceptor sites per ethanol, three donors and one acceptor per ammonia. In the ideal one-bond-per-site model, with geometry ignored, maximum H bonds are

(A) 0.600 mol (B) 0.200 mol

(C) 0.400 mol (D) 0.300 mol

41–50 / Preserved hard practice

41. A gas sample initially contains equal numbers of O_2 and O_2^- species. Every superoxide ion is oxidized by one electron to O_2, with original O_2 unchanged. The increase in mean O–O bond order per initial diatomic species is

(A) 1 (B) 0.25

(C) 0.5 (D) 0.125

42. A central atom in Period 3 has valence configuration $3s^2\ 3p^3$. Its pentachloride is treated with one equivalent of F^- to form $[XCl_5F]^-$. In the ideal octahedral product, number of geometrically distinct cis/trans arrangements for its single F ligand is

(A) 3 (B) 6

(C) 1 (D) 2

43. A metal ion M^{2+} has d^8 configuration. Four identical strong-field ligands form an ideal square-planar complex. Number of unpaired electrons in the standard low-spin square-planar description is

(A) 4 (B) 1

(C) 0 (D) 2

44. A 0.100 mol mixture contains ethene and ethyne only. Complete hydrogenation consumes 0.160 mol hydrogen. The original sample contains how many moles of C–H sigma bonds?

(A) 0.4 (B) 0.28

(C) 0.32 (D) 0.16

45. A 0.0500 mol mixture contains but-1-y ne and but a-1, 3-diene only. Under a quantitative terminal-alkyne assay, one Ag atom replaces the terminal hydrogen of each but-1-y ne molecule; the diene does not react. The isolated acetylide contains 0.0200 mol Ag. The original sample contains sp^2 carbon atoms in amount (mol)

(A) 0.04 (B) 0.2

(C) 0.06 (D) 0.12

46. A compound C_2H_6O is either ethanol or dimethyl ether. A 0.100 mol sample liberates 0.0500 mol H_2 with excess sodium. Number of intermolecular $O-H$ hydrogen-bond donor sites per molecule in the identified pure compound is

(A) 1 (B) 3

(C) 0 (D) 2

47. An isolated atom has 5 valence electrons. Its trihydride is pyramidal and neutral. After complete protonation of one mole of hydride, formal charge summed over central atoms is (in elementary-charge mole units)

(A) 1 (B) 3

(C) 0 (D) 2

48. A 1.00 mol mixture of N_2 and O_2 has total elementary MO bond-order sum 2.60 mol. Moles of unpaired electrons associated with the mixture are

(A) 0.4 (B) 1.2

(C) 0.8 (D) 1.6

49. A 0.200 mol gas mixture contains only a neutral heteronuclear 14-electron diatomic species containing carbon and ordinary O_2. Its mass is 6.00 g. Use C=12, O=16. Each oxygen molecule then accepts exactly one electron. Using elementary MO bond orders, the final mean bond order per diatomic species is

(A) 2 (B) 2.25

(C) 2.5 (D) 1.75

50. A 0.100 mol mixture contains SF_6 and XeF_4 only, and has 0.480 mol fluorine atoms. Use ideal octahedral and square-planar shapes, respectively. The amount of distinct 90° F–X –F angle pairs in the sample, in mol, is

(A) 1.2 (B) 0.72

(C) 0.6 (D) 0.48

Answer key

Chemical Bonding

C-CHBO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**D**	21	**A**	31	**C**	41	**B**
02	**B**	12	**A**	22	**C**	32	**D**	42	**C**
03	**D**	13	**D**	23	**A**	33	**C**	43	**C**
04	**C**	14	**B**	24	**B**	34	**B**	44	**B**
05	**A**	15	**A**	25	**C**	35	**D**	45	**D**
06	**A**	16	**C**	26	**A**	36	**A**	46	**A**
07	**D**	17	**A**	27	**A**	37	**B**	47	**A**
08	**D**	18	**D**	28	**D**	38	**C**	48	**C**
09	**B**	19	**A**	29	**D**	39	**B**	49	**B**
10	**C**	20	**B**	30	**B**	40	**D**	50	**B**

Chapter 7

Chemical Thermodynamics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Standard formation enthalpies of $CO_2(g)$, $H_2O(l)$ and $CH_4(g)$ are respectively -394, -286 and $-75\,\mathrm{kJ\,mol^{-1}}$. The standard combustion enthalpy of methane, in $\mathrm{kJ\,mol^{-1}}$, is

(A) -680

(B) -891

(C) -745

(D) 891

02. A reaction has $\Delta C_P = 20\,\mathrm{J\,mol^{-1}K^{-1}}$ and $\Delta H = -50\,\mathrm{kJ\,mol^{-1}}$ at 300 K. Assume constant ΔC_P. Its ΔH at 400 K, in $\mathrm{kJ\,mol^{-1}}$, is

(A) -48

(B) -30

(C) -70

(D) -52

03. Two moles of an ideal diatomic gas, with vibrations inactive, are heated by 100 K at constant pressure. With $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$, their internal-energy change, in kJ, is

(A) 1.6628

(B) 2.4942

(C) 4.157

(D) 5.8198

04. A two-mole ideal-gas sample is heated by 100 K at constant pressure while absorbing 5.00 kJ. With $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$, its ΔU, in kJ, is

(A) 3.3372

(B) 6.6628

(C) 5

(D) 1.6628

05. A reaction has $\Delta H = 30\,\mathrm{kJ\,mol^{-1}}$ and $\Delta S = 100\,\mathrm{J\,mol^{-1}K^{-1}}$. At 400 K, the maximum reversible non- PV work done by the system per mole, in kJ, is

(A) 10

(B) 0

(C) 30

(D) 70

06. A substance melts at 300 K with fusion enthalpy 6.0 kJ mol^{-1}. Neglect heat-capacity differences. Its Gibbs energy of melting at 290 K, in kJ mol^{-1}, is

(A) -0.2 (B) 5.8

(C) 6.2 (D) 0.2

07. At 300 K, $A \to B$ has $\Delta H = -20$ and $\Delta G = -5$ kJ mol^{-1}. Its entropy change, in J mol^{-1}K^{-1}, is

(A) -50 (B) 83.3333

(C) 50 (D) -83.3333

08. An exothermic process releases 30 kJ to surroundings at 300 K. Its system entropy change is -80 J K^{-1}. The combined entropy change, in J K^{-1}, is

(A) 20 (B) 100

(C) -20 (D) 180

09. One mole of $A(g)$ forms three moles of $B(g)$ at 300 K. The constant-volume heat for this conversion is -10.0 kJ. With $R = 8.314$ J mol^{-1}K^{-1}, the constant-pressure heat, in kJ, is

(A) -5.0116 (B) -14.9884

(C) -7.5058 (D) -2.5174

10. $A \to B$ has $\Delta H = +40$ and $B \to C$ has $\Delta H = -70$ kJ mol^{-1}. If ΔG for $A \to C$ is -10 kJ mol^{-1} at 400 K, its ΔS, in J mol^{-1}K^{-1}, is

(A) -50 (B) -75

(C) 75 (D) 50

11–40 / New middle-to-hard practice

11. At a temperature where $RT = 2.50$ kJ mol^{-1}, standard combustion enthalpies of C(s) to CO_2(g) and CO(g) to CO_2(g) are -393.0 and -283.0 kJ mol^{-1}. For C(s) + $\frac{1}{2}O_2$(g) $\to$ CO(g), constant-volume heat per mole CO is

(A) -108.75 kJ (B) -110.00 kJ

(C) -111.25 kJ (D) -112.50 kJ

12. Liquid ethanol burns as $C_2H_5OH(l) + 3O_2(g) \to 2CO_2(g) + 3H_2O(l)$. Bomb heat is -1364.0 kJ mol^{-1} where $RT = 2.50$ kJ mol^{-1}. Given formation enthalpies of CO_2(g) and H_2O(l) as -393.5 and -285.8, ethanol formation enthalpy is

(A) $+277.9$ kJ mol^{-1} (B) -277.9 kJ mol^{-1}

(C) -280.4 kJ mol^{-1} (D) -275.4 kJ mol^{-1}

13. A calorimeter containing 100 g water rises 5.00 K when a standard releases 10.0 kJ. In a second run it contains 200 g water; burning 0.500 g unknown fuel raises temperature 3.00 K. Neglect losses and use water heat capacity 4.00 J g^{-1}K^{-1}. Heat released per gram unknown fuel is

(A) 28.8 kJ g^{-1} (B) 12.0 kJ g^{-1}

(C) 7.20 kJ g^{-1} (D) 14.4 kJ g^{-1}

14. A calorimeter with 200 g water rises 4.00 K when a standard releases 12.0 kJ. With 100 g water in a later constant-volume reaction, 0.0500 mol reactant produces a 5.00 K rise. Water heat capacity is 4.00 $\mathrm{J\,g^{-1}K^{-1}}$; neglect losses. Reaction internal-energy change per mole reactant is

(A) $-300\ \mathrm{kJ\,mol^{-1}}$ (B) $-220\ \mathrm{kJ\,mol^{-1}}$

(C) $-260\ \mathrm{kJ\,mol^{-1}}$ (D) $+260\ \mathrm{kJ\,mol^{-1}}$

15. A rigid reactor initially contains a total of 0.200 mol of A and inert gas, with unknown composition. Reaction $2A(g) \rightarrow B(g)$ consumes all A. After restoration to the initial temperature, pressure is 75.0% of the original. The constant-volume reaction releases 5.00 kJ. Internal-energy change per mole of reaction as written is

(A) $-200\ \mathrm{kJ\,mol^{-1}}$ (B) $-50.0\ \mathrm{kJ\,mol^{-1}}$

(C) $-100\ \mathrm{kJ\,mol^{-1}}$ (D) $-25.0\ \mathrm{kJ\,mol^{-1}}$

16. A rigid reactor initially contains a total of 0.200 mol of A and inert gas, with unknown composition. Reaction $A(g) \rightarrow 2B(g)$ consumes all A; at the restored initial temperature pressure is 1.40 times the original. Heat absorbed is 8.00 kJ at constant volume. With $RT = 2.50\ \mathrm{kJ\,mol^{-1}}$, reaction enthalpy per mole A is

(A) $105.0\ \mathrm{kJ\,mol^{-1}}$ (B) $100.0\ \mathrm{kJ\,mol^{-1}}$

(C) $97.5\ \mathrm{kJ\,mol^{-1}}$ (D) $102.5\ \mathrm{kJ\,mol^{-1}}$

17. For a reaction with temperature-independent standard enthalpy and entropy, $\ln K = 4$ at 400 K and $\ln K = 6$ at 800 K. At 600 K, $\Delta G^\circ/(RT)$ is

(A) $\frac{16}{3}$ (B) -4

(C) $-\frac{16}{3}$ (D) -6

18. A reaction has $\ln K = 5$ at 300 K and $\ln K = 3$ at 600 K, with constant standard enthalpy and entropy. Which conclusion follows?

(A) $\Delta H^\circ > 0$, $\Delta S^\circ = R$; crossover is 1200 K

(B) $\Delta H^\circ = 0$; the different K values arise only from a catalyst

(C) $\Delta H^\circ < 0$, $\Delta S^\circ = -R$; crossover is 1200 K

(D) $\Delta H^\circ < 0$, $\Delta S^\circ = R$; standard Gibbs energy is negative at every positive temperature

19. A reaction has $\Delta G^\circ = +10.0$ kJ/mol at 300 K and -10.0 kJ/mol at 500 K. Treat enthalpy and entropy as constant. Among temperatures $200, 250, 300, 350, 400, 450, 500, 550, 600$ K, how many give strictly negative standard Gibbs energy?

(A) 6 (B) 4

(C) 3 (D) 5

20. A reaction has standard Gibbs energies -10.0 kJ/mol at 300 K and $+20.0$ kJ/mol at 600 K. With constant enthalpy and entropy, how many temperatures in $\{250, 300, 350, 400, 450, 500\}$ K give $\Delta G^\circ < 0$?

(A) 5 (B) 3

(C) 2 (D) 4

21. At 300 K a reaction has $\Delta H = -20.0$ kJ/mol and $\Delta S = -50.0$ J/mol/K. Its constant ΔC_p is 40.0 J/mol/K. Use $\Delta H_2 = \Delta H_1 + \Delta C_p(T_2 - T_1)$ and $\Delta S_2 = \Delta S_1 + \Delta C_p \ln(T_2/T_1)$, with $\ln 2 = 0.693$. At 600 K, ΔG is

(A) $-5.368\ \text{kJ mol}^{-1}$ (B) $22.0\ \text{kJ mol}^{-1}$

(C) $5.368\ \text{kJ mol}^{-1}$ (D) $-8.00\ \text{kJ mol}^{-1}$

22. At 300 K, $\Delta H = 30.0$ and $\Delta G = 15.0$ kJ/mol. Constant $\Delta C_p = 20.0$ J/mol/K. Using $\Delta S_2 = \Delta S_1 + \Delta C_p \ln(T_2/T_1)$, $\ln 2 = 0.693$, and the usual enthalpy correction, ΔG at 600 K is

(A) $-2.316\ \text{kJ mol}^{-1}$ (B) $+2.316\ \text{kJ mol}^{-1}$

(C) $6.00\ \text{kJ mol}^{-1}$ (D) $-6.00\ \text{kJ mol}^{-1}$

23. An ideal gas expands isothermally from 10.0 L to 20.0 L in two irreversible stages. External pressure is 1.00 bar until an unknown intermediate volume, then 0.500 bar. It absorbs 750 J heat overall. With 1 L bar = 100 J, the intermediate volume is

(A) 17.5 L (B) 15.0 L

(C) 10.0 L (D) 12.5 L

24. One mole ideal monatomic gas absorbs 3000 J while its temperature rises 100 K. It first expands by 10.0 L against 1.00 bar, then by an unknown volume against 0.500 bar. Use $R = 8.00\ \text{J mol}^{-1}\text{K}^{-1}$ and 1 L bar = 100 J. The second volume increase is

(A) 20.0 L (B) 16.0 L

(C) 8.00 L (D) 12.0 L

25. A closed working fluid completes the clockwise triangle $A : (V = 1\ \text{L}, P = 1\ \text{bar})$, $B : (1\ \text{L}, 3\ \text{bar})$, $C : (5\ \text{L}, 1\ \text{bar})$, then A. Only PV work occurs. Heat on B–C is +1100 J and on C–A is −1000 J. Heat on A–B is (1 L bar = 100 J)

(A) +100 J (B) −300 J

(C) +400 J (D) +300 J

26. A closed fluid follows rectangle $A : (1\ \text{L}, 1\ \text{bar})$, $B : (1\ \text{L}, 3\ \text{bar})$, $C : (4\ \text{L}, 3\ \text{bar})$, $D : (4\ \text{L}, 1\ \text{bar})$, then A. Only PV work occurs. Heats on A–B, B–C, C–D are +400, +2700, −1600 J. Heat on D–A is (1 L bar = 100 J)

(A) +900 J (B) −300 J

(C) −1500 J (D) −900 J

27. A solid melts at 0°C. Ten grams heated from −10°C to liquid at 0°C absorb 1200 J; 20 g heated from −5°C to liquid at 5°C absorb 2600 J. Liquid specific heat is $4.00\ \text{J g}^{-1}\text{K}^{-1}$, and solid specific heat and latent heat are constant. Heat for 10 g from −20°C to liquid at 20°C is

(A) 2400 J (B) 2000 J

(C) 2200 J (D) 1800 J

28. A material melts at 0°C. Ten grams from −10°C to liquid at 0°C absorb 1200 J; 20 g from −5°C to liquid at 5°C absorb 2600 J. Liquid specific heat is 4.00 $\mathrm{J\,g^{-1}K^{-1}}$. If 10 g solid initially at −10°C absorb only 700 J, the final liquid mass fraction is

(A) $\frac{7}{10}$ (B) $\frac{1}{2}$

(C) $\frac{7}{12}$ (D) $\frac{2}{5}$

29. A 0.200 mol ideal mixture contains monatomic and rigid diatomic gases. Its measured constant-volume heat capacity is $0.400R$. At constant pressure it absorbs 200 J. The monatomic mole fraction and expansion work by the gas are

(A) 1/2; 100 J (B) 1/4; 200/3 J

(C) 1/2; 200/3 J (D) 3/4; 200 J

30. An 18.0 g ideal He–nitrogen mixture has $C_p/C_v = 3/2$ with nitrogen vibrations inactive. With He=4 and nitrogen=28, its total constant-volume heat capacity is

(A) $3R/2$ (B) $9R/4$

(C) $2R$ (D) $3R$

31. For gaseous X and Y, $D(X-X) : D(Y-Y) = 1:2$. The X–Y bond enthalpy is 400 kJ/mol, and $\frac{1}{2}X_2 + \frac{1}{2}Y_2 \to XY$ has enthalpy −100 kJ/mol. The Y–Y bond enthalpy is

(A) 600 $\mathrm{kJ\,mol^{-1}}$ (B) 200 $\mathrm{kJ\,mol^{-1}}$

(C) 400 $\mathrm{kJ\,mol^{-1}}$ (D) 800 $\mathrm{kJ\,mol^{-1}}$

32. Diatomic X_2 and Y_2 have equal bond enthalpy A, while XY has bond enthalpy C. Reaction $X_2 + Y_2 \to 2XY$ releases 200 kJ per mole of reaction. Atomising 0.100 mol X_2 plus 0.200 mol XY requires 80.0 kJ. C is

(A) 300 $\mathrm{kJ\,mol^{-1}}$ (B) 400 $\mathrm{kJ\,mol^{-1}}$

(C) 200 $\mathrm{kJ\,mol^{-1}}$ (D) 500 $\mathrm{kJ\,mol^{-1}}$

33. At a temperature where $RT = 2.50$ kJ/mol, a target reaction has $K = e^4$ but current $Q = e^6$. A coupled process has standard Gibbs change -8.00 kJ per mole and is maintained at unit reaction quotient $Q_c = 1$, so its actual Gibbs change is also -8.00 kJ per mole; it has no other constraint. If its extent per mole target is chosen from $0, 0.1, \ldots, 1.0$, the smallest extent making the combined Gibbs change strictly negative is

(A) $\frac{7}{10}$ (B) $\frac{3}{5}$

(C) $\frac{1}{2}$ (D) $\frac{4}{5}$

34. At one temperature a reaction has $\Delta G = +5.00$ kJ/mol when $Q = e^2$ and $\Delta G = -15.0$ kJ/mol when $Q = e^{-2}$. Its equilibrium constant at that temperature is

(A) e^2 (B) e^4

(C) e^{-1} (D) e

35. Two bodies with constant heat capacities 200 and 300 J/K start at 600 and 300 K. They are placed in an isolated enclosure and reach a common temperature without work. With $\ln 0.7 = -0.357$ and $\ln 1.4 = 0.336$, their total entropy change is

(A) $29.4\ \mathrm{J\,K^{-1}}$ (B) $171.6\ \mathrm{J\,K^{-1}}$

(C) 0 (D) $-29.4\ \mathrm{J\,K^{-1}}$

36. Two bodies have total heat capacity 600 J/K. One starts at 500 K and the other at 300 K; adiabatic contact gives final 350 K without work. Their separate heat capacities are constant. Using $\ln 0.7 = -0.357$ and $\ln(7/6) = 0.154$, the total entropy change is

(A) $-15.75\ \mathrm{J\,K^{-1}}$ (B) $15.75\ \mathrm{J\,K^{-1}}$

(C) $23.1\ \mathrm{J\,K^{-1}}$ (D) 0

37. An adiabatic constant-pressure reactor receives 0.100 mol A and 0.150 mol B at 300 K. Reaction $A + 2B \to C$ is complete for the limiting reactant and has $\Delta H = -80.0$ kJ per mole reaction. Effective total heat capacity is 200 J/K, constant. Final temperature is

(A) 330 K (B) 320 K

(C) 360 K (D) 340 K

38. A constant-pressure adiabatic reactor starts with 0.200 mol A and 0.500 mol B for $A + B \to C$. It warms from 300 to 330 K with effective heat capacity 400 J/K. Isolated product is 0.150 mol at 75.0% recovery; no side reaction occurs. Reaction enthalpy per mole C formed is

(A) $-45.0\ \mathrm{kJ\,mol^{-1}}$ (B) $+60.0\ \mathrm{kJ\,mol^{-1}}$

(C) $-80.0\ \mathrm{kJ\,mol^{-1}}$ (D) $-60.0\ \mathrm{kJ\,mol^{-1}}$

39. A reversible two-electron cell at 300 K has $E^\circ = 0.500$ V. Its reaction is $A(g) + B(g) \to C(g)$; constant-volume reaction heat is -120.0 kJ/mol. Use $F = 96500\ \mathrm{C\,mol^{-1}}$ and $RT = 2.50$ kJ/mol. Standard reaction entropy is

(A) $-235/3\ \mathrm{J\,mol^{-1}K^{-1}}$ (B) $+260/3\ \mathrm{J\,mol^{-1}K^{-1}}$

(C) $-100\ \mathrm{J\,mol^{-1}K^{-1}}$ (D) $-260/3\ \mathrm{J\,mol^{-1}K^{-1}}$

40. A reversible two-electron cell has $E^\circ = 0.500$ V at 300 K and 0.450 V at 400 K. Assume its voltage varies linearly over this interval. With $F = 96500\ \mathrm{C\,mol^{-1}}$, standard reaction enthalpy at 350 K is

(A) $-91.675\ \mathrm{kJ\,mol^{-1}}$ (B) $-57.90\ \mathrm{kJ\,mol^{-1}}$

(C) $-125.45\ \mathrm{kJ\,mol^{-1}}$ (D) $+125.45\ \mathrm{kJ\,mol^{-1}}$

41–50 / Preserved hard practice

41. A 10.0 g mixture of carbon and sulfur burns completely to CO_2 and SO_2, releasing 300.0 kJ. Take combustion enthalpies of C and S as -400 and $-300\,\mathrm{kJ\,mol^{-1}}$, and atomic masses C=12, S=32. The carbon mass, in g, is

(A) 5
(B) 1.3913043
(C) 7.5
(D) 8.6086957

42. A 2.00 g sample containing NaOH and an inert solid is neutralized by excess acid. The released heat alone raises 100.0 g of water by 2.00 K. Take water heat capacity $4.00\,\mathrm{J\,g^{-1}K^{-1}}$ and neutralization enthalpy $-40.0\,\mathrm{kJ\,mol^{-1}}$. The NaOH mass percentage is

(A) 20
(B) 80
(C) 40
(D) 60

43. A 0.0500 mol reaction raises a bomb calorimeter temperature by 2.00 K. Its total heat capacity is $5.00\,\mathrm{kJ\,K^{-1}}$. The reaction has $\Delta n_g = -2$ at 300 K. With $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$, the molar ΔH, in $\mathrm{kJ\,mol^{-1}}$, is

(A) -100
(B) -195.0116
(C) -200
(D) -204.9884

44. $A(g) \rightarrow 2B(g)$ has $\Delta G^\circ = -11.416\,\mathrm{kJ\,mol^{-1}}$ at 298 K. Take $2.303RT = 5.708\,\mathrm{kJ\,mol^{-1}}$. For activities $a_A = 1$ and $a_B = 5$, its actual ΔG, in $\mathrm{kJ\,mol^{-1}}$, is. Use $\log_{10} 25 = 1.39794$.

(A) 3.4365585
(B) -3.4365585
(C) -19.395442
(D) -11.416

45. A reaction has $\Delta H^\circ = +40.0\,\mathrm{kJ\,mol^{-1}}$. Its equilibrium constant doubles between temperatures T_1 and T_2. With $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ and $\ln 2 = 0.693$, $1/T_1 - 1/T_2$, in $\mathrm{K^{-1}}$, is

(A) 0.0014404005
(B) 0.00014404005
(C) 6942.5136
(D) -0.00014404005

46. A cell reaction transfers two moles of electrons per mole of reaction and has $E^\circ = 1.00\,\mathrm{V}$. With $F = 96500\,\mathrm{C\,mol^{-1}}$ and $\Delta S^\circ = +100\,\mathrm{J\,mol^{-1}K^{-1}}$ at 300 K, its ΔH°, in $\mathrm{kJ\,mol^{-1}}$, is

(A) -193
(B) -163
(C) -223
(D) 163

47. Quantitative hydrogenation of a molecule containing one C = C bond has $\Delta H = -120\,\mathrm{kJ\,mol^{-1}}$. A 5.60 g sample releases 12.0 kJ on complete hydrogenation. Its molar mass, in $\mathrm{g\,mol^{-1}}$, is

(A) 56
(B) 28
(C) 84
(D) 112

48. A closed vessel contains 0.100 mol each of H_2 and Cl_2. If 80.0% of the limiting reactant reacts by $H_2 + Cl_2 \rightarrow 2HCl$ with $\Delta H = -180\,\text{kJ}$ per mole of reaction, the heat released, in kJ, is

(A) 28.8 (B) 14.4

(C) 7.2 (D) 18

49. One mole each of A and B mix to form an ideal solution at constant T, P with zero enthalpy change. At 300 K, with $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$ and $\ln 2 = 0.693$, the maximum reversible useful work obtainable from mixing, in kJ, is

(A) -3.4569612 (B) 1.7284806

(C) 0 (D) 3.4569612

50. A one-step reversible reaction has forward activation energy $80.0\,\text{kJ mol}^{-1}$ and forward reaction enthalpy $-30.0\,\text{kJ mol}^{-1}$. Both directions use the same transition state and obey Arrhenius laws with constant barriers and prefactors. With $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$, the value of $\ln[k_r(330\,\text{K})/k_r(300\,\text{K})]$ is closest to

(A) 1.8224098 (B) -4.0093016

(C) 4.0093016 (D) 2.9158557

Answer key

Chemical Thermodynamics

C-CHTH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**C**	31	**C**	41	**D**
02	**A**	12	**B**	22	**A**	32	**A**	42	**C**
03	**C**	13	**D**	23	**B**	33	**A**	43	**D**
04	**A**	14	**C**	24	**B**	34	**D**	44	**B**
05	**A**	15	**C**	25	**D**	35	**A**	45	**B**
06	**D**	16	**D**	26	**D**	36	**B**	46	**B**
07	**A**	17	**C**	27	**C**	37	**A**	47	**A**
08	**A**	18	**D**	28	**B**	38	**D**	48	**B**
09	**A**	19	**B**	29	**C**	39	**D**	49	**D**
10	**A**	20	**B**	30	**B**	40	**C**	50	**C**

Chapter 8

Solutions

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 20% by-mass aqueous solution has density 1.20 g/ mL and solute molar mass 60 g/mol. Its molarity is

(A) 2 (B) 3

(C) 4 (D) 5

02. A 1.0 g polymer sample in 100 mL solution has osmotic pressure 0.0246 atm at 300 K. R=.082. Polymer molar mass in g/mol is

(A) 2460 (B) 1000

(C) 10000 (D) 100000

03. A solute dimerizes with observed molar mass 180 g/mol; its true molar mass is 120 g/mol. Dimerization fraction is

(A) 0.33333333 (B) 1

(C) 0.66666667 (D) 0.5

04. For the ideal mixture with x_A =.25, p_A° =100 torr and p_B° =200 torr, the vapour-phase mole fraction of A is

(A) 0.25 (B) 0.14285714

(C) 0.33333333 (D) 0.57142857

05. A 0.20 mol/kg solution with K_f =2.0 shows freezing depression 0.60 K. If solute dissociates as AB $\rightarrow$ $A^+ + B^-$, dissociation fraction is

(A) 0.5 (B) 0.75

(C) 0.25 (D) 1

06. A solute with true M=100 has apparent M=125 in osmometry. If it only dimerizes, fraction associated is

(A) 0.4 (B) 0.8

(C) 0.5 (D) 0.2

07. An ideal binary liquid solution has pure-component vapour pressures $p_A^\circ = 300\,\text{torr}$ and $p_B^\circ = 100\,\text{torr}$. Its total equilibrium vapour pressure is 180 torr. The mole fraction of A in the equilibrium vapour is

(A) 0.5 (B) 0.66666667

(C) 0.75 (D) 0.4

08. An ideal solution has 1 mol solute in 9 mol solvent. Doubling only the solvent amount changes relative vapour-pressure lowering by factor

(A) 1 (B) 0.52631579

(C) 2 (D) 0.5

09. An osmosis experiment at 300 K has M=.10 and observed i=1.5. At fixed T, a second solution of the same solute with M=.20 and i=1.25 has osmotic pressure ratio second/first of

(A) 1.25 (B) 1.5

(C) 2 (D) 1.6666667

10. A solution has freezing depression 0.40 K and boiling elevation 0.10 K. For the same effective particles, K_f / K_b is

(A) 4 (B) 0.25

(C) 5 (D) 2

11–40 / New middle-to-hard practice

11. A 12.0 g glucose–urea mixture is dissolved in 1.00 kg water. Its freezing depression is 0.186 K with $K_f = 1.86\ \text{K kg mol}^{-1}$. Both are non-electrolytes; glucose is $C_6H_{12}O_6$ (180) and urea CH_4N_2O (60). Carbon mass in the original solute mixture is

(A) 3.60 g (B) 2.40 g

(C) 6.00 g (D) 4.20 g

12. A 19.35 g NaCl–glucose mixture in 1.00 kg water gives freezing depression 0.5115 K, with $K_f = 1.86$. Assume complete NaCl dissociation and no glucose association. With molar masses 58.5 and 180, the NaCl mass fraction is

(A) $\frac{30}{43}$ (B) $\frac{1}{3}$

(C) $\frac{13}{43}$ (D) $\frac{1}{2}$

13. A 3.00 g hypothetical salt AB_2 in 1.00 L solution gives 3.76 g AgBr after complete destruction and precipitation of all B as bromide. Osmotic pressure before destruction is 0.576 bar at 300 K. Assume only $AB_2 \rightleftharpoons A^{2+} + 2B^-$, ideal particles, $R = 0.0800\ \text{L bar mol}^{-1}\text{K}^{-1}$, and AgBr=188. Salt molar mass and dissociation fraction are

(A) 300, 1.00 (B) 300, 0.400

(C) 150, 0.700 (D) 300, 0.700

14. A 14.2 g hypothetical M_2SO_4 sample gives 23.3 g $BaSO_4$ after complete sulfate recovery. Its 1.00 L solution at 300 K has osmotic pressure 6.00 bar. Use ideal dissociation into three ions, $R = 0.0800\ \text{L bar mol}^{-1}\text{K}^{-1}$, $BaSO_4$=233 and SO_4=96. Atomic mass of M and dissociation fraction are

(A) 23, 0.750 (B) 46, 0.750

(C) 23, 1.00 (D) 23, 0.500

15. An ideal binary liquid has total vapour pressure 160 torr; vapour is 75.0 mol% A. Pure B vapour pressure is 80.0 torr. For a fresh solution containing one mole A and three moles B, total vapour pressure at the same temperature is

(A) 120 torr (B) 140 torr

(C) 100 torr (D) 160 torr

16. At a fixed temperature an ideal binary solution has P=180 torr, vapour mole fraction of A 2/3, and pure-A pressure 300 torr. A fresh mixture contains 1 mol A and 3 mol B. Its total vapour pressure is

(A) 200 torr (B) 120 torr

(C) 180 torr (D) 150 torr

17. A He–CO_2 gas mixture has mean molar mass 14.0 and total pressure 4.00 bar. It contacts 20.0 mol water at fixed temperature; K_H for CO_2 is 2000 bar and He is insoluble. Use $p = K_H x$ and the dilute approximation $x \simeq n_{gas}/n_{water}$. Dissolved CO_2 mass is (He=4, CO_2=44)

(A) 0.440 g (B) 0.880 g

(C) 0.110 g (D) 1.76 g

18. An ideal nitrogen–oxygen mixture has mean molar mass 28.8 at total pressure 5.00 bar. At fixed temperature K_H for oxygen is 50000 bar. Using $x \simeq n_{O_2}/n_{water}$, oxygen mass dissolved in 100 mol water is (nitrogen=28, oxygen=32)

(A) 0.128 g (B) 0.320 g

(C) 0.0128 g (D) 0.0640 g

19. An aqueous non-electrolyte solution has density 1.20 g mL^{-1}, molarity 2.00 M and molality 2.50 mol kg^{-1}. Solute molar mass is

(A) 100 g mol^{-1} (B) 160 g mol^{-1}

(C) 200 g mol^{-1} (D) 240 g mol^{-1}

20. An aqueous solution is 3.00 M and 4.00 mol kg^{-1}. Add 120 g water to a 20.0 mL aliquot without loss of solute. Final molality is

(A) 1/3 mol kg^{-1} (B) 4/9 mol kg^{-1}

(C) 1/2 mol kg^{-1} (D) 2/5 mol kg^{-1}

21. A mixture of $MgCl_2$ and KCl in 1.00 kg water gives freezing depression 0.930 K ($K_f = 1.86$). Complete chloride assay gives 0.300 mol Cl. Assume complete dissociation. The original solute mass is ($MgCl_2$=95, KCl=74.5)

(A) 16.95 g (B) 19.00 g

(C) 22.35 g (D) 14.25 g

22. A $CaCl_2$–NaCl mixture in 1.00 kg water gives freezing depression 1.116 K with $K_f = 1.86$. Total chloride is 0.350 mol, and both salts dissociate completely. Fraction of cations that are divalent is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$

(C) $\frac{1}{3}$ (D) $\frac{2}{7}$

23. Two compartments initially have 1.00 L solution each and are separated by a flexible membrane permeable only to water. A contains 2.00 g unknown non-electrolyte polymer; B contains 0.00100 mol glucose. At equilibrium A volume is 0.400 L, total volume remains 2.00 L, and hydrostatic effects are negligible. Polymer molar mass is

(A) 3200 g mol^{-1} (B) 8000 g mol^{-1}

(C) 5000 g mol^{-1} (D) 2000 g mol^{-1}

24. Two 1.00 L compartments contain respectively 0.100 and 0.300 mol glucose; the first also has 0.200 mol urea. Their membrane passes water and urea but not glucose. At equilibrium total volume is 2.00 L, urea concentration is equal, and hydrostatic effects are negligible. Urea amount in the second compartment is

(A) 0.200 mol (B) 0.150 mol

(C) 0.0500 mol (D) 0.100 mol

25. An ideal liquid contains 1.00 mol A and 1.00 mol B; pure vapour pressures are 300 and 100 torr. A controlled separation removes 0.200 mol material with composition equal to the initial equilibrium vapour, after which the residual liquid re-equilibrates. Residual total vapour pressure is

(A) 190 torr (B) 1750/9 torr

(C) 200 torr (D) 1850/9 torr

26. For an ideal binary liquid, pure vapour pressures are 300 torr for A and 100 torr for B. A 1.50 mol liquid sample has equilibrium vapour with A mole fraction 0.600. Add 0.500 mol A without evaporation. New total vapour pressure is

(A) 200 torr (B) 220 torr

(C) 180 torr (D) 240 torr

27. A 2.44 g sample of an unknown monobasic carboxylic acid consumes 0.0200 mol NaOH. In a nonpolar solvent its apparent osmotic molar mass is 183. Assume dimerisation only. Fraction of original acid molecules associated is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

28. A monobasic acid sample of mass 1.20 g consumes 0.0200 mol NaOH. In a nonpolar solvent it has apparent molar mass 100 owing only to dimerisation. A 0.100 mol/kg solution at that association degree has $K_f = 2.00$. Its freezing depression is

(A) 0.200 K (B) 0.160 K

(C) 0.120 K (D) 0.0800 K

29. A 1.50 g polymer blend contains one-third its mass as polymer A of molar mass 10000; the rest is polymer B of unknown molar mass. Its 1.00 L ideal solution has $\pi/(RT) = 1.00 \times 10^{-4}$ mol L^{-1}. Molar mass of B is

(A) 15000 g mol^{-1} (B) 20000 g mol^{-1}

(C) 25000 g mol^{-1} (D) 10000 g mol^{-1}

30. A 1.00 L solution contains polymers A and B of molar masses 10000 and 20000. Initially $\pi/(RT) = 2.00 \times 10^{-4}$ mol L^{-1}. A separation removes half the B chains and none of A; solution is restored to 1.00 L, giving $\pi/(RT) = 1.50 \times 10^{-4}$. Original polymer mass is

(A) 2.00 g (B) 4.00 g

(C) 3.00 g (D) 2.50 g

31. A hypothetical salt contains only A^{3+} and B^{2-}. Its 0.100 mol/kg solution shows freezing depression 0.372 K ($K_f = 1.86$). Under the ideal undissociated-formula-unit/fully-separated-ion model, dissociation fraction is

(A) $\frac{3}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

32. A hypothetical AB_2 salt has 9.50 g solute per 1.00 kg water. A calibrated instantaneous probe reports 0.0400 mol free A^{2+} without perturbing equilibrium; freezing depression is 0.3348 K ($K_f = 1.86$). Assume ideal dissociation into three ions only. True salt molar mass is

(A) 95.0 g mol^{-1} (B) 190 g mol^{-1}

(C) 52.8 g mol^{-1} (D) 118.75 g mol^{-1}

33. An ideal 0.100 mol/kg solution of a salt dissociating into five ions gives freezing depression 0.372 K with $K_f = 1.86$ and boiling elevation 0.200 K with $K_b = 0.500$. Allow different dissociation degrees at the two phase temperatures. The ratio $\alpha_{boiling}/\alpha_{freezing}$ is

(A) 2 (B) $\frac{1}{3}$

(C) 4 (D) 3

34. A nonpolar-solvent solution contains 0.100 mol/kg monobasic acid that only dimerises. Freezing depression is 0.120 K with $K_f = 2.00$; boiling elevation is 0.0800 K with $K_b = 1.00$. The ratio of fractions associated at freezing to boiling is

(A) 4 (B) $\frac{3}{2}$

(C) $\frac{1}{2}$ (D) 2

35. A 100 mL non-electrolyte solution is isotonic with 0.100 M glucose at the same temperature. Then 25.0% of its solute molecules decompose by $X \rightarrow 2Y$ and the solution is diluted to 250 mL. It is isotonic with what molarity of fully dissociated NaCl?

(A) 0.0125 M (B) 0.0250 M

(C) 0.0200 M (D) 0.0500 M

36. A 0.100 M solution of a salt with five ions on full dissociation is isotonic with 0.200 M glucose. It is diluted with an equal volume of water, and an independent measurement then finds dissociation fraction 0.500. The final-to-initial osmotic-pressure ratio at unchanged temperature is

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$

(C) 1 (D) $\frac{1}{2}$

37. An ideal solution of 30.0 g nonvolatile non-electrolyte in 441 g water has vapour pressure 98.0 torr when pure water pressure is 100 torr. All solute is then dissolved to make 2.50 L solution at 300 K. With $R = 0.0800\ \text{L bar mol}^{-1}\text{K}^{-1}$, true molar mass and osmotic pressure are

(A) 60.0, 2.40 bar (B) 120, 4.80 bar

(C) 60.0, 9.60 bar (D) 60.0, 4.80 bar

38. A nonvolatile ideal solution has vapour pressure 98.0 torr when pure solvent pressure is 100 torr. Adding exactly one mole solvent raises solution pressure to 99.0 torr at the same temperature. Initial solvent amount is

(A) 0.0200 mol (B) 0.490 mol

(C) 1.98 mol (D) 0.980 mol

39. A closed vessel at 300 K contains 0.100 mol gas in 1.20 L gas space over 100 mol solvent. The gas obeys $p = K_H x$, $K_H = 10000$ bar; use $x \simeq n_{dissolved}/n_{solvent}$ and $R = 0.0800\ \text{L bar mol}^{-1}\text{K}^{-1}$. Gas-space volume and solvent amount remain fixed. Fraction of all gas dissolved at equilibrium is

(A) $\frac{1}{10}$ (B) $\frac{1}{2}$

(C) $\frac{1}{6}$ (D) $\frac{1}{5}$

40. A closed vessel initially has 0.200 mol He–CO_2 gas of mean molar mass 24.0 at 300 K in 2.40 L gas space over 100 mol water. He is insoluble; CO_2 has $K_H = 10000$ bar. Use ideal gas law, dilute Henry model $x \simeq n_{dissolved}/100$, and $R = 0.0800\ \text{L bar mol}^{-1}\text{K}^{-1}$. Final gas-phase CO_2 mole fraction is (He=4, CO_2=44)

(A) $\frac{1}{2}$ (B) $\frac{11}{21}$

(C) $\frac{10}{21}$ (D) $\frac{5}{11}$

41–50 / Preserved hard practice

41. A 1.800 g mixture of glucose(M=180) and urea(M=60) is dissolved in 100.0 g water. Freezing depression is 0.372 K with K_f =1.86; both solutes are nonelectrolytes. Glucose mass fraction is

(A) 0.75 (B) 0.66666667

(C) 0.25 (D) 0.5

42. A monoprotic acid HA has molality 0.100 and observed van't Hoff factor 1.20. Approximating molarity by 0.100 M and using ideal activities, K_a is

(A) 0.025 (B) 0.004

(C) 0.005 (D) 0.0005

43. A solute dimerizes in benzene. Its true molar mass is 120 and observed mass is 150 g/mol. If initial formal solute concentration is 0.100 M, dimer concentration at equilibrium in M is

(A) 0.02 (B) 0.08

(C) 0.01 (D) 0.04

44. A solution contains 0.0100 mol $CaCl_2$ and 0.0200 mol Na_2SO_4 in one kg water. Assume $CaSO_4$ precipitates completely and others stay fully dissociated. Final total moles of dissolved solute ions are

(A) 0.08 (B) 0.07

(C) 0.09 (D) 0.11

45. Two aqueous non electrolyte solutions have equal osmotic pressures at 300 K. Solution A contains 2.00 g solute(M=100) per 200 mL. Solution B contains 3.00 g unknown solute per 100 mL. Unknown molar mass in g/mol is

(A) 150 (B) 300

(C) 600 (D) 75

46. Liquid A and B form an ideal solution with pure pressures 100 and 300 torr. The equilibrium vapour has y_A =.25. Liquid mole fraction x_A is

(A) 0.25 (B) 0.33333333

(C) 0.5 (D) 0.75

47. A 0.100 mol solute partially dissociates as $AB_2 \rightarrow$A 2++2 B^-. Freezing measurement indicates 0.220 mol of particles. Total moles of B^- are

(A) 0.1 (B) 0.12

(C) 0.06 (D) 0.24

48. A 0.100 M weak acid HA is 20% dissociated. Its osmotic pressure is measured at 300 K. Then a nonreacting isotonic solution of glucose is prepared at 360 K. Required glucose molarity in M is

(A) 0.144 (B) 0.12

(C) 0.08 (D) 0.1

49. For an ideal solution, x_A =.40, p_A° =200 torr, p_B° =100 torr. A closed gas vessel filled with its equilibrium vapour contains 0.700 mol gas. Moles of A in vapour are

(A) 0.4 (B) 0.56

(C) 0.3 (D) 0.28

50. A 1.000 g mixture of NaCl (M=58.5) and KCl (M=74.5) is dissolved in water. Chloride precipitation gives 0.01500 mol AgCl. Both salts fully dissociate. Effective osmotic particles in mol are

(A) 0.06 (B) 0.03

(C) 0.02 (D) 0.015

Answer key

Solutions

C-SOLU / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**D**	21	**A**	31	**C**	41	**D**
02	**C**	12	**C**	22	**A**	32	**A**	42	**C**
03	**C**	13	**D**	23	**B**	33	**D**	43	**A**
04	**B**	14	**A**	24	**B**	34	**D**	44	**B**
05	**A**	15	**A**	25	**B**	35	**B**	45	**B**
06	**A**	16	**D**	26	**A**	36	**A**	46	**C**
07	**B**	17	**A**	27	**D**	37	**D**	47	**B**
08	**B**	18	**D**	28	**C**	38	**D**	48	**D**
09	**D**	19	**C**	29	**B**	39	**C**	49	**A**
10	**A**	20	**B**	30	**C**	40	**C**	50	**B**

Chapter 9

Chemical Equilibrium

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A⇌B has K=4. Starting with 2.0 mol A and 3.0 mol B in one vessel, net moles of A converted to B are

(A) 1.5 (B) 1

(C) 0.5 (D) 2

02. For A+B⇌C, K_c =4 L/mol. In 1 L, start with 1 mol of each reactant and no C. If equilibrium C=x mol, the physical root is closest to

(A) 0.6096118 (B) 0.25

(C) 0.3903882 (D) 1.6403882

03. For N_2O_4 (g)⇌2 NO_2 (g), dissociation fraction is.50 and total pressure 3.0 bar. Numerical K_p using bar pressures is

(A) 2 (B) 4

(C) 1 (D) 3

04. For A(g)⇌2 B(g), K_p =4 bar and total pressure 3 bar. Starting from pure A, equilibrium dissociation fraction is

(A) 0.66666667 (B) 0.5

(C) 0.25 (D) 0.75

05. For A+B⇌C+D, K_c =4. Equal initial reactant amounts and no products give conversion fraction of each reactant of

(A) 0.8 (B) 0.33333333

(C) 0.66666667 (D) 0.5

06. For 2 A⇌B, K_c =8 and $[A]$ =.50 M at initial equilibrium. One adds B instantaneously until $[B]$ =3.0 M without volume change. Q/K immediately after addition is

(A) 1 (B) 1.5

(C) 3 (D) 0.5

07. For A⇌B, K=3. After equilibration, the total analytical concentration $[A] + [B]$ is.80 M. Equilibrium $[B]$ in M is

(A) 0.6 (B) 0.75

(C) 0.4 (D) 0.2

08. At fixed temperature, two reactions A⇌Band 2 B⇌C have K_1 =2 and K_2 =3. K for 2 A⇌C is

(A) 6 (B) 18

(C) 9 (D) 12

09. For $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, pure PCl_5 initial pressure is 2 bar at fixed TV. Equilibrium total pressure is 3 bar. PCl_5 partial pressure in bar is

(A) 2 (B) 0.5

(C) 1.5 (D) 1

10. For A(g)+B(g)⇌C(g), an instantaneous isothermal compression triples all partial pressures. Q/K from an initial equilibrium becomes

(A) 1 (B) 0.33333333

(C) 9 (D) 3

11–40 / New middle-to-hard practice

11. Initially a 1.00 L vessel contains a total of 0.300 mol of A and B, with unknown composition and total mass 8.00 g. Molar masses of A and B are 20 and 40. For $A + B \rightleftharpoons C$, equilibrium C is 0.0500 mol, with no C initially. Numerical K_c using molar concentrations is

(A) $\frac{20}{3}$ (B) 20

(C) $\frac{40}{3}$ (D) 10

12. A 1.00 L vessel initially holds a total of 0.500 mol of A and B, with unknown composition and total mass 6.00 g. Molar masses are 20 for A and 10 for B. Reaction $A + 2B \rightleftharpoons C$ gives 0.0500 mol C at equilibrium, from none initially. Numerical K_c is

(A) $\frac{100}{9}$ (B) $\frac{10}{3}$

(C) $\frac{20}{3}$ (D) $\frac{200}{9}$

13. A rigid vessel initially has a total of 0.200 mol of A_2 and He, with unknown composition and total mass 4.40 g (A_2=40, He=4). At fixed T, $A_2 \rightleftharpoons 2A$ raises pressure to 1.25 times its original value; final total pressure is 3.00 bar. Numerical K_p in bar pressures is

(A) 4 (B) $\frac{12}{5}$

(C) $\frac{6}{5}$ (D) $\frac{24}{5}$

14. A rigid vessel initially contains a total of 0.300 mol of A_2 and He, with unknown composition and total mass 13.2 g (A_2=64, He=4). Dissociation $A_2 \rightleftharpoons 2A$ at unchanged T raises pressure by a factor 1.50, to 9.00 bar. Numerical K_p is

(A) 72 (B) 27

(C) 18 (D) 36

15. For $A+B \rightleftharpoons C$ in 1.00 L, $K_c = 20$ numerically. Initial A=0.100 mol, B=0.200 mol, C=0. Equilibrium C amount is

(A) $(7-\sqrt{17})/40$ mol (B) $(7+\sqrt{17})/40$ mol

(C) $(7-\sqrt{17})/20$ mol (D) 0.100 mol

16. A 1.00 L vessel has A=B=C=0.100 mol for $A+B \rightleftharpoons C$, with numerical $K_c = 1.00$. After equilibration, B has increased by

(A) $(7-3\sqrt{5})/10$ mol (B) $(3\sqrt{5}+6)/10$ mol

(C) $(3\sqrt{5}-6)/10$ mol (D) 0.100 mol

17. At fixed T, a gaseous equilibrium $A \rightleftharpoons 2B$ contains 1 mol A and 2 mol B at total pressure 3 bar. One mole He is added while total pressure is maintained at 3 bar, then equilibrium is restored. Additional A converted is

(A) 0 mol (B) $(\sqrt{97}+9)/8$ mol

(C) $(\sqrt{5}-2)$ mol (D) $(\sqrt{97}-9)/8$ mol

18. At fixed T, $A(g) \rightleftharpoons 2B(g)$ is at equilibrium in 1.00 L with A=1.00 mol and B=2.00 mol. Volume is suddenly doubled and then held at 2.00 L. Additional A converted by the time equilibrium returns is

(A) 0 mol (B) 1/2 mol

(C) $(\sqrt{5}+2)$ mol (D) $(\sqrt{5}-2)$ mol

19. At one T, equilibrium I has $[A] = 0.200$ M, $[B] = 0.400$ M for $A \rightleftharpoons B$. Equilibrium II has B=0.100 M, C=0.200 M, D=0.300 M for $B+C \rightleftharpoons 2D$. Numerical K_c for $2A+2C \rightleftharpoons 4D$ is

(A) 36 (B) 18

(C) 81 (D) 9

20. At one T, $2A \rightleftharpoons B$ has equilibrium A=0.200 M and B=0.200 M. Separately, $B \rightleftharpoons C$ has B=0.100 M and C=0.400 M. Numerical constant for $\frac{1}{2}C \rightleftharpoons A$ is

(A) $\frac{1}{20}$ (B) $2\sqrt{5}$

(C) $\frac{\sqrt{5}}{10}$ (D) $\frac{1}{10}$

21. For $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 1.00$ bar. A vessel has $V/(RT) = 1.00\,\text{mol}\,\text{bar}^{-1}$, initially 0.500 mol carbonate and 0.200 mol CO_2, with CaO seed. It equilibrates; then 0.500 mol carbonate is added and re-equilibrated. Total carbonate consumed across both stages is

(A) 1.00 mol (B) 0.800 mol

(C) 0.300 mol (D) 0.500 mol

22. For carbonate decomposition with $K_p = 2.00$ bar, a vessel initially has 0.600 mol carbonate, 0.100 mol CO_2, and CaO seed. Its initial $V/(RT) = 0.250\ \text{mol bar}^{-1}$. After equilibrium, volume is increased fourfold at the same T. Increase in CO_2 amount during this expansion stage is

(A) 0.200 mol (B) 1.50 mol

(C) 0.400 mol (D) 0.600 mol

23. Two solid equilibria share gas G: $A(s) \rightleftharpoons B(s) + G(g)$ has $K_p = 1.00$ bar; $C(s) \rightleftharpoons D(s) + G(g)$ has $K_p = 2.00$ bar. A and D are ample, B=0.200 mol, C=0.250 mol. Initially gas pressure is 1.00 bar and $V/(RT) = 0.100\ \text{mol bar}^{-1}$. Final pressure is

(A) 2.50 bar (B) 2.00 bar

(C) 1.50 bar (D) 1.00 bar

24. In the two solid equilibria $A \rightleftharpoons B + G$ ($K_p = 1$ bar) and $C \rightleftharpoons D + G$ ($K_p = 2$ bar), A and D are ample. Initially B=0.200 mol, C=0.500 mol, G pressure=1.00 bar, and $V/(RT) = 0.100\ \text{mol bar}^{-1}$. Final amount of C is

(A) 0.200 mol (B) 0.300 mol

(C) 0 mol (D) 0.100 mol

25. In a fixed-volume solution, $A \rightleftharpoons B$ has $[B]/[A] = 2$ and $2B \rightleftharpoons C$ has $[C]/[B]^2 = 1.00\ \text{L mol}^{-1}$. Analytical monomer-unit concentration is $[A] + [B] + 2[C] = 1.25$ M. Equilibrium C concentration is

(A) 0.125 M (B) 0.250 M

(C) 0.400 M (D) 0.500 M

26. For $A \rightleftharpoons B$, $[B]/[A] = 2$; for $2B \rightleftharpoons C$, $[C]/[B]^2 = 0.500\ \text{L mol}^{-1}$. An equilibrium has A=0.500 M, B=1.00 M, C=0.500 M. Remove all C instantaneously without changing volume, then re-equilibrate. New C concentration is

(A) $(21 + 3\sqrt{33})/16$ M (B) 0.500 M

(C) $(9 - \sqrt{33})/8$ M (D) $(21 - 3\sqrt{33})/16$ M

27. A pure dimer gas has molar mass 60 before dissociation $A_2 \rightleftharpoons 2A$. At equilibrium its mean molar mass is 40.0 and total pressure 3.00 bar. It is compressed isothermally to equilibrium at 15.0 bar. New mean molar mass is

(A) $40.0\ \text{g mol}^{-1}$ (B) $50.0\ \text{g mol}^{-1}$

(C) $48.0\ \text{g mol}^{-1}$ (D) $30.0\ \text{g mol}^{-1}$

28. A monomer gas A has molar mass 20.0. At equilibrium for $2A \rightleftharpoons A_2$, mean molar mass is 25.0 at 4.00 bar. The gas is compressed isothermally to equilibrium at 9.00 bar. Final mean molar mass is

(A) 30 (B) 25

(C) $50 - 10\sqrt{5}$ (D) $25 - 5\sqrt{5}$

29. Two separate 1.00 L vessels are at the same T for $A + B \rightleftharpoons C$. Vessel I has A=B=0.500 mol, C=1.00 mol. Vessel II has A=1.00 mol, B=0.250 mol, C=1.00 mol. They are mixed into 2.00 L. Forward reaction extent before equilibrium returns is

(A) $(11 + \sqrt{113})/8$ mol (B) $(11 - \sqrt{113})/8$ mol

(C) $(\sqrt{17} - 4)$ mol (D) 0 mol

30. At the same T for $A + B \rightleftharpoons C$, one 1.00 L vessel contains A=B=C=1.00 mol. A second contains A=B=2.00 mol and C=4.00 mol. They are mixed into 2.00 L. Reverse reaction extent on re-equilibration is

(A) 0 mol (B) $(\sqrt{17} + 4)$ mol

(C) $(11 - \sqrt{113})/8$ mol (D) $(\sqrt{17} - 4)$ mol

31. For $A(g) \rightleftharpoons 2B(g)$, $K_p = 4.00$ bar at 300 K and 1.00 bar at 600 K. Assume $\ln K_p$ is linear in $1/T$. Starting from pure A, equilibrium total pressure at 400 K is 6.00 bar. Dissociation fraction is

(A) $\frac{\sqrt{13}}{13}$ (B) $\frac{1}{4}$

(C) $\frac{\sqrt{7}}{7}$ (D) $\frac{1}{2}$

32. A reaction R has $\ln K = 4$ at 300 K and $\ln K = 2$ at 600 K, with $\ln K$ linear in $1/T$. At 400 K, reverse R and divide every coefficient by two to make reaction S. If current $Q_S = e^{-2}$, which direction is favoured?

(A) Forward S; $K_S = e^{-3/2}$ (B) Reverse S; $K_S = e^{3/2}$

(C) Reverse S; $K_S = e^{-3/2}$ (D) Forward S; $K_S = e^{-3}$

33. For $A \rightleftharpoons 2B$ in 1.00 L, starting from A=1.00 mol and B=0 gives equilibrium A=0.500 mol and B=1.00 mol. Remove 0.500 mol B without changing volume and re-equilibrate. Removed B plus final B amount is

(A) 1.50 mol (B) $(\sqrt{7} + 1)/2$ mol

(C) $(\sqrt{7} - 1)/2$ mol (D) $\sqrt{7}/2$ mol

34. The 1.00 L equilibrium $A \rightleftharpoons 2B$ has A=0.500 mol and B=1.00 mol. Add 1.00 mol B without volume change and re-equilibrate. Final A amount is

(A) $(7 + \sqrt{13})/4$ mol (B) 1.00 mol

(C) $(5 - \sqrt{13})/4$ mol (D) $(7 - \sqrt{13})/4$ mol

35. A 1.00 L equilibrium for $A + B \rightleftharpoons C$ contains 0.300 mol gas in total, with all three species present in positive amounts. A selective probe consumes two reagent equivalents per A, one per B and none per C; total uptake is 0.300 mol equivalents. Numerical $K_c = 10.0$. Equilibrium C amount is

(A) 0.200 mol (B) 0.0500 mol

(C) 0.100 mol (D) 0.150 mol

36. A 1.00 L gas mixture at equilibrium for $A_2 \rightleftharpoons 2A$ also contains inert He. Total amount is 0.400 mol and mass 8.40 g; molar masses are A_2=40, A=20, He=4. Numerical $K_c = 0.400$. Equilibrium He amount is

(A) 0.100 mol (B) 0.150 mol

(C) 0.200 mol (D) 0.0500 mol

37. A constant-pressure adiabatic reactor starts with 1.00 mol A at 300 K for $A \rightleftharpoons B$. Reaction enthalpy is -12000 J/mol and total effective heat capacity 100 J/K. Use $R = 8.00$ and $K(T) = e^{5/6}\exp[1500(1/T - 1/300)]$. The equilibrium fraction converted to B is

(A) $\frac{2}{3}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

38. A constant-pressure adiabatic reactor starts with 1.00 mol A at 420 K for $A \rightleftharpoons B$. Reaction enthalpy is $+12000$ J/mol and effective total heat capacity 100 J/K. Use $K(T) = e^{25/42}\exp[-1500(1/T - 1/420)]$ and ideal A/B activity ratio. Final equilibrium temperature is

(A) 420 K (B) 360 K

(C) 400 K (D) 300 K

39. An elementary reversible first-order reaction $A \rightleftharpoons B$ has equilibrium B fraction 0.750 and measured relaxation constant $k_f + k_r = 0.400\ \text{min}^{-1}$. Starting with pure A, use $f_B(t) = f_{eq}[1 - e^{-(k_f+k_r)t}]$. B fraction after $5\ln 2$ min is

(A) $\frac{1}{2}$ (B) $\frac{9}{16}$

(C) $\frac{3}{4}$ (D) $\frac{3}{8}$

40. For reversible first-order $A \rightleftharpoons B$, pure A approaches equilibrium B fraction 0.600. After 10.0 min B fraction is 0.300. Use $f_B = f_{eq}[1 - e^{-(k_f+k_r)t}]$, $f_{eq} = k_f/(k_f + k_r)$, and $\ln 2 = 0.693$. Forward rate constant is

(A) 0.04158 min^{-1} (B) 0.02079 min^{-1}

(C) 0.0693 min^{-1} (D) 0.02772 min^{-1}

41–50 / Preserved hard practice

41. Pure N_2O_4 has initial pressure 1.00 bar in a fixed volume at T. Equilibrium total pressure is 1.40 bar. Numerical K_p using bar pressures is

(A) 1.0666667 (B) 1.4

(C) 0.8 (D) 0.4

42. A 0.100 mol PCl_5 sample of mass 20.85 g partly dissociates in a gas vessel. Equilibrium gas mean molar mass is 139 g/mol. Numerical K_p at a total pressure 3.00 bar is

(A) 0.5 (B) 1

(C) 3 (D) 2

43. A(g)⇌2 B(g)has K_p =4 bar. An equilibrium mixture at 3 bar starts with pure A. It is compressed isothermally to final total pressure 8 bar. New dissociation fraction is

(A) 0.5 (B) 0.66666667

(C) 0.25 (D) 0.33333333

44. A⇌B has K=100 at 298 K. For an initial mixture with Q=10, ΔG in units RT is

(A) 2.3025851 (B) -4.6051702

(C) -2.3025851 (D) 4.6051702

45. For N_2 +3 H_2 ⇌2 NH_3, an initial equilibrium has partial pressures 1 bar, 3 bar, and 2 bar. Nitrogen alone is added at fixed TV until its partial pressure is 4 bar. Q/K immediately after addition is

(A) 4 (B) 0.25

(C) 2 (D) 0.5

46. In 1.00 L, A+B⇌C+D has K=1. Start with 1.00 mol A, 2.00 mol Band 1.00 mol C, but no D. Equilibrium D amount in mol is

(A) 0.25 (B) 0.75

(C) 0.5 (D) 0.66666667

47. A reaction has ΔH° =+40 kJ /mol. At T 1, its K is 4.0. At T 2, van't Hoff relation gives ln(K_2 / K_1)=ln 2. K for the reverse reaction at T 2 is

(A) 8 (B) 0.5

(C) 0.25 (D) 0.125

48. A 0.100 mol N_2O_4 sample is dissociated by 50%. The NO_2 formed is then quantitatively absorbed without further N_2O_4 dissociation. Absorbed nitrogen atom amount in mol is

(A) 0.1 (B) 0.15

(C) 0.2 (D) 0.05

49. One mole of ideal gas A undergoes A⇌2 B with K_p =4 bar. Initially equilibrium total pressure is 3 bar. Moles of B present are

(A) 2 (B) 0.5

(C) 1 (D) 1.5

50. A(g)⇌B(g)+C(g)starts from pure A. Equilibrium mean molar mass is 60 g/mol while initial A has M 90 g/mol. If equilibrium total pressure is 6 bar, numerical K_p using bar pressures is

(A) 1 (B) 4

(C) 3 (D) 2

Answer key

Chemical Equilibrium

C-CHEQ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**A**	21	**B**	31	**A**	41	**A**
02	**A**	12	**A**	22	**A**	32	**A**	42	**B**
03	**B**	13	**B**	23	**C**	33	**D**	43	**D**
04	**B**	14	**D**	24	**A**	34	**D**	44	**C**
05	**C**	15	**A**	25	**B**	35	**C**	45	**B**
06	**B**	16	**C**	26	**D**	36	**A**	46	**C**
07	**A**	17	**D**	27	**C**	37	**C**	47	**D**
08	**D**	18	**D**	28	**C**	38	**B**	48	**A**
09	**D**	19	**C**	29	**B**	39	**B**	49	**C**
10	**B**	20	**C**	30	**D**	40	**A**	50	**D**

Chapter 10

Ionic Equilibrium

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Equal volumes of 0.10 M HCl and 0.050 M NaOH are mixed.At 25°C, final H^+ concentration in M is

(A) 0.05 (B) 0.15

(C) 0.075 (D) 0.025

02. A 0.10 M acetate solution has conjugate-acid $K_a = 10^{-5}$. At 25°C, using weak-hydrolysis approximation, pH is

(A) 9 (B) 7

(C) 11 (D) 5

03. A 0.10 M NH_4Cl solution has NH_3 $K_b = 10^{-5}$. At 25°C using weak-hydrolysis approximation, pH is

(A) 5 (B) 9

(C) 3 (D) 7

04. For CaF_2, K_{sp} =4× 10^{-12}. Neglect hydrolysis and complexation.Molar solubility in pure water in M is

(A) 1×10^{-6} (B) 0.001

(C) 0.0001 (D) 0.0002

05. A buffer initially contains 0.10 mol HA and 0.20 mol A^-. Add 0.050 mol HCl with negligible volume change. Final $[A^-]$ / [HA] is

(A) 0.5 (B) 2

(C) 1 (D) 3

06. A buffer initially contains 0.20 mol HA and 0.10 mol A^-. Add 0.050 mol NaOH. Final $[A^-]$ / [HA] is

(A) 1 (B) 0.5

(C) 3 (D) 2

07. A salt $M(OH)_2$ has K_{sp} =4× 10^{-15}. If $[M^{2+}] = 10^{-3}$ M, precipitation onset OH^- concentration in M is

(A) 2×10^{-6} (B) 1×10^{-6}

(C) 2×10^{-5} (D) 4×10^{-6}

08. A weak base B has $K_b = 10^{-5}$. In buffer $[B]$ =0.20 M, $[BH^+]$ =0.020 M. At 25°C, pH is

(A) 9 (B) 11

(C) 10 (D) 4

09. At 25°C, indicator HIn has pK_a =7. At pH 8, the fraction of indicator present as In^- is

(A) 0.99 (B) 0.90909091

(C) 0.5 (D) 0.090909091

10. At 25°C, HCl concentration is 10^{-8} M. Including water ionization, exact H^+ concentration in units 10^{-7} M is

(A) 0.1 (B) 1

(C) 1.1 (D) 1.0512492

11–40 / New middle-to-hard practice

11. A 25.0 mL weak monoprotic acid sample needs 40.0 mL 0.100 M NaOH for equivalence. At half-equivalence pH is 4.80. Under the usual buffer approximation, pH after adding only 10.0 mL base is $(\log_{10} 3 = 0.477)$

(A) $\frac{5277}{1000}$ (B) $\frac{4323}{1000}$

(C) $\frac{24}{5}$ (D) $\frac{4499}{1000}$

12. A 100 mL weak monoprotic acid is titrated with 0.100 M NaOH. Before equivalence, pH is 4.000 after 10.0 mL and $4.000 + \log_{10} 6$ after 30.0 mL. Assume Henderson–Hasselbalch is valid. Acid K_a is

(A) $\frac{1}{40000}$ (B) $\frac{1}{20000}$

(C) $\frac{1}{10000}$ (D) $\frac{1}{100000}$

13. A buffer of HA/NaA has pH equal to pK_a. Adding 0.0500 mol HCl, without exhausting the buffer, lowers pH to $pK_a - \log_{10} 3$. Neglect volume change and free-ion contributions. Total initial amount HA plus A^- was

(A) 0.300 mol (B) 0.100 mol

(C) 0.200 mol (D) 0.150 mol

14. A buffer with $pK_a = 5$ initially has pH $5 + \log_{10} 2$. Adding 0.0300 mol NaOH makes pH $5 + \log_{10} 5$, without exhausting HA. Neglect volume change. Initial HA amount was

(A) 0.0900 mol (B) 0.0600 mol

(C) 0.120 mol (D) 0.0300 mol

15. Mix 100 mL 10^{-7} M HCl with 100 mL 4.00×10^{-8} M NaOH at 25°C. Include water ionisation, $K_w = 10^{-14}$. Final hydrogen-ion concentration is

(A) $(\sqrt{409} - 3) \times 10^{-8}/2$ M (B) 1.00×10^{-7} M

(C) $(3 + \sqrt{409}) \times 10^{-8}/2$ M (D) 3.00×10^{-8} M

16. At 25°C, 100 mL unknown NaOH is mixed with 100 mL 10^{-7} M HCl. Measured final $[H^+] = 5.00 \times 10^{-8}$ M and $K_w = 10^{-14}$. Original NaOH concentration is

(A) 4.00×10^{-7} M (B) 2.00×10^{-7} M

(C) 1.00×10^{-7} M (D) 3.00×10^{-7} M

17. A solution has $[Ag^+] = [Pb^{2+}] = 0.100$ M. Add chloride slowly with negligible volume change. $K_{sp}(AgCl) = 10^{-10}$ and $K_{sp}(PbCl_2) = 10^{-5}$. Ignore complexes. The fraction of original silver remaining dissolved just as $PbCl_2$ starts precipitating is

(A) $\frac{1}{100000000}$ (B) $\frac{1}{10000000}$

(C) $\frac{1}{100000}$ (D) $\frac{1}{1000000}$

18. A solution initially has Ba^{2+}=0.0100 M and Sr^{2+}=0.100 M. Sulfate is added slowly. $K_{sp}(BaSO_4) = 10^{-10}$ and $K_{sp}(SrSO_4) = 10^{-7}$. At $SrSO_4$ precipitation onset, percentage of original Ba already precipitated is

(A) $\frac{999}{10}$ (B) 90

(C) $\frac{9999}{100}$ (D) 99

19. At 25°C an ammonia buffer has pH 9.00, $pK_b = 5.00$, and $[NH_3] + [NH_4^+] = 0.200$ M. A large reservoir keeps these concentrations fixed during AgCl dissolution. With $K_{sp} = 10^{-10}$, $K_f[Ag(NH_3)_2^+] = 10^8$, assume complex dominance. AgCl solubility is

(A) 0.0200 M (B) 0.0100 M

(C) 0.100 M (D) 0.00100 M

20. AgCl dissolves in a closed 0.100 M NH_3 solution. Assume all dissolved Ag is $Ag(NH_3)_2^+$; ignore ammonia acid-base reaction and other complexes. $K_{sp} = 10^{-10}$, $K_f = 10^8$. Ligand depletion must be included. AgCl molar solubility is

(A) 0.0100 M (B) 1/120 M

(C) 1/1000 M (D) 0.00500 M

21. A hypothetical $M(OH)_2$ has molar solubility 10^{-5} M in pure water at 25°C. Neglect water autoionisation in this calibration. In a large buffer fixed at pH 8.00, its molar solubility, ignoring complexes, is

(A) 0.0400 M
(B) 10^{-5} M
(C) 0.000400 M
(D) 0.00400 M

22. A hypothetical $M(OH)_3$ dissolves to 10^{-3} M in pure water at 25°C; ignore water autoionisation. A large buffer then fixes pH at 12.0. Ignoring all complexation, the metal-ion equilibrium concentration is

(A) 2.70×10^{-8} M
(B) 9.00×10^{-5} M
(C) 2.70×10^{-5} M
(D) 10^{-3} M

23. A diprotic acid is titrated with NaOH. First and second equivalence volumes are 20.0 and 40.0 mL. At 10.0 mL base, pH is 3.00; at 30.0 mL it is 7.00. Use well-separated dissociations and the usual amphiprotic-salt approximation. pH at first equivalence is

(A) 4
(B) 3
(C) 7
(D) 5

24. A pure amphiprotic HA^- salt has pH 6.00 under the usual approximation. Titration halfway from HA^- to A^{2-} gives pH 9.00. For a fresh 0.100 mol H_2A sample partly neutralised with 0.0750 mol NaOH, buffer pH is ($\log_{10} 3 = 0.477$)

(A) $\frac{9477}{1000}$
(B) $\frac{2523}{1000}$
(C) $\frac{3477}{1000}$
(D) $\frac{6477}{1000}$

25. A weak monoprotic acid stock has pH 3.00 and degree of dissociation 0.0100. Under the usual weak-acid/buffer approximation, 25.0 mL stock is neutralised until A^-/HA is 4.00. Amount NaOH added is

(A) 0.000500 mol
(B) 0.00100 mol
(C) 0.00200 mol
(D) 0.00250 mol

26. Weak acid HA is 0.100 M with $[H^+] = 10^{-4}$ M; weak acid HB is 0.200 M with $[H^+] = 2.00 \times 10^{-4}$ M. Equal volumes are mixed. Use weak-ionisation approximations, neglect water and mutual complexation. Mixture $[H^+]$ is

(A) $\sqrt{5}/10000$ M
(B) 2.00×10^{-4} M
(C) 1.50×10^{-4} M
(D) $\sqrt{10}/20000$ M

27. A carbonate–bicarbonate sample needs 0.0100 mol acid to phenolphthalein and 0.0300 mol in total to methyl orange. A fresh identical sample receives only 0.00500 mol HCl. With $pK_{a2} = 10.00$, final buffer pH is ($\log_{10} 3 = 0.477$)

(A) $\frac{9699}{1000}$
(B) 10
(C) $\frac{10477}{1000}$
(D) $\frac{9523}{1000}$

28. A NaOH–Na_2CO_3 sample consumes 0.0300 mol acid to phenolphthalein and 0.0400 mol in total to methyl orange. A fresh equal sample receives 0.0250 mol HCl. Use $pK_{a2} = 10.00$ and neglect minor species. Final pH is

(A) 12 (B) 10

(C) $\frac{9699}{1000}$ (D) $\frac{10301}{1000}$

29. An indicator HIn has $pK_a = 6.00$ and its basic-form fraction in a solution is 0.800. A 0.100 mol weak-acid HA sample with $pK_a = 5.00$ is partly neutralised to this same pH. Under ideal buffer assumptions, NaOH amount needed is

(A) 1/41 mol (B) 0.0800 mol

(C) 0.0900 mol (D) 4/41 mol

30. In one calibration solution, indicator A ($pK_a = 5.00$) has basic/acidic ratio 10.0, while indicator B has basic fraction 0.500. A separate HA/A^- buffer ($pK_a = 5.00$) initially has 0.100 mol of each; add 0.0500 mol NaOH. Indicator B basic fraction in the final buffer is

(A) $\frac{1}{4}$ (B) $\frac{3}{10}$

(C) $\frac{3}{4}$ (D) $\frac{3}{13}$

31. A 100 mL mixture contains HCl and weak HA ($pK_a = 5.00$). Its initial pH is 2.00 and full neutralisation takes 0.00300 mol NaOH. Neglect the small HA contribution to initial hydrogen concentration. After adding only 0.00150 mol NaOH, pH is ($\log_{10} 3 = 0.477$)

(A) 5 (B) $\frac{4523}{1000}$

(C) $\frac{2301}{1000}$ (D) $\frac{5477}{1000}$

32. A 100 mL solution contains NaOH and NH_3 ($pK_b = 5.00$). Initial pOH is 2.00 and complete acid titration consumes 0.00400 mol HCl. Neglect ammonia contribution to initial OH. After adding only 0.00200 mol HCl, pH at 25°C is ($\log_{10} 2 = 0.301$)

(A) $\frac{9301}{1000}$ (B) $\frac{8699}{1000}$

(C) 9 (D) 12

33. A solution with NH_3=0.0200 M and KOH=0.0100 M has measured $[NH_4^+] = 3.60 \times 10^{-5}$ M. Use the common-ion approximation. A second solution of the same base has NH_3=0.0300 M and KOH=0.0200 M. Its $[NH_4^+]$ is

(A) 2.70×10^{-5} M (B) 3.60×10^{-5} M

(C) 1.80×10^{-5} M (D) 5.40×10^{-5} M

34. A HA/A^- solution with concentrations 0.0200 and 0.0100 M has $[H^+] = 2.00 \times 10^{-5}$ M. Use buffer approximations. A second 1.00 L buffer initially has 0.0300 mol HA and 0.0600 mol A^-. Add 0.0100 mol HCl with negligible volume change. Final $[H^+]$ is

(A) 8.00×10^{-6} M (B) 1.00×10^{-5} M

(C) 5.00×10^{-6} M (D) 2.00×10^{-5} M

35. A mixture has M^{2+} and N^{2+} each at 0.100 M, with hydroxide K_{sp} values 10^{-16} and 10^{-10}. At 25°C, select a pH that precipitates at least 99.9% of M while leaving N unprecipitated. Ignore complexes and volume change.

(A) $\frac{15}{2}$ (B) $\frac{17}{2}$

(C) $\frac{49}{5}$ (D) $\frac{21}{2}$

36. A solution contains M^{3+}=0.0100 M and N^{2+}=0.100 M. Their hydroxide K_{sp} values are 10^{-30} and 10^{-16}. At 25°C choose pH giving at least 99.9% M precipitation and no N precipitation. Neglect complexes.

(A) $\frac{15}{2}$ (B) 6

(C) 5 (D) $\frac{34}{5}$

37. For a diprotic acid at pH 5.00, spectroscopy shows HA^- fraction 50/51, with equal amounts H_2A and A^{2-}. Concentration ratios obey the two usual dissociation equilibria. At pH 6.00, A^{2-} fraction is

(A) $\frac{100}{1101}$ (B) $\frac{1}{101}$

(C) $\frac{100}{1001}$ (D) $\frac{1}{11}$

38. An ampholyte has states H_2A^+, HA and A^-. At pH 6.00, HA fraction is 5/6 and the other two are equal. At pH 7.00, mean net molecular charge in elementary-charge units is

(A) $\frac{33}{67}$ (B) $-\frac{1}{2}$

(C) $-\frac{33}{67}$ (D) $-\frac{1}{3}$

39. A salt of weak acid HA and weak base B has pH 8.00 and hydrolysis fraction $h = 0.0100$ at 25°C. Use $pH = 7 + \frac{1}{2}\log(K_b/K_a)$ and $h^2 \simeq K_w/(K_aK_b)$, $K_w = 10^{-14}$. A separate B/BH^+ buffer initially has 0.100 mol each; add 0.0500 mol HCl. Final pH is ($\log 3 = 0.477$)

(A) 10 (B) $\frac{8477}{1000}$

(C) $\frac{9523}{1000}$ (D) $\frac{10477}{1000}$

40. A weak-acid/weak-base salt has pH 6.00 and hydrolysis fraction 0.0200 at 25°C. Use $pH = 7 + \frac{1}{2}\log(K_b/K_a)$ and $h^2 \simeq K_w/(K_aK_b)$ with $K_w = 10^{-14}$. A separate HA/A^- buffer has 0.100 mol each, then receives 0.0500 mol NaOH. Final hydrogen concentration is

(A) 1/20000 M (B) 1/60000 M

(C) 1/120000 M (D) 1/30000 M

41–50 / Preserved hard practice

41. A 50.0 mL sample contains 0.100 M HA and 0.200 M NaA, pK_a =5.00. Add 10.0 mL 0.500 M HCl. Neglect activity effects.The final pH is

(A) 6 (B) 5

(C) 4.69897 (D) 4

42. AgCl and AgBr have K_{sp} 10^{-10} and 10^{-13}, respectively. A solution contains $[Cl^-]$ =0.010 M and $[Br^-]$ =0.0010 M. As Ag^+ is added slowly, AgBr precipitates first. At AgCl onset, the fraction of initial Br^- remaining is

(A) 1 (B) 0.1

(C) 0.01 (D) 0.001

43. A mixture contains 0.0100 mol NaOH and 0.0100 mol NH_3 in 100 mL solution. Add 0.0150 mol HCl without significant volume change. With pK_b NH_3 =5.00 and pK_w 14, final pH is

(A) 5 (B) 10

(C) 7 (D) 9

44. A 0.100 M monoprotic acid HA is 20.0% dissociated. Its conjugate base is prepared at 0.100 M. At 25°C, using weak-hydrolysis approximation, the OH^- concentration in M is

(A) 4.472136×10^{-7} (B) 5×10^{-7}

(C) 1.4142136×10^{-6} (D) 1×10^{-7}

45. A 0.0100 M solution of MCl_2 is buffered at pH 10.0. At 25°C and K_{sp} $[M(OH)_2] = 10^{-14}$, the fraction of initial M^{2+} remaining dissolved at equilibrium is

(A) 0.01 (B) 0.1

(C) 0.001 (D) 0.0001

46. A 0.100 mol weak acid with pK_a 5.00 is neutralized with 0.0750 mol NaOH. Using Henderson and log 3=.4771, the pH is

(A) 5.4771 (B) 5

(C) 4.5229 (D) 6

47. A solution contains 0.0100 mol Ca^{2+} and 0.0200 mol F^- in 1.00 L. If CaF_2 K_{sp} =4× 10^{-12}, neglecting hydrolysis, the total moles of free ions left after precipitation are approximately

(A) 0.0002 (B) 0.0003

(C) 0.0001 (D) 0.003

48. A diprotic acid H_2A has K_a 1= 10^{-3}, K_a 2= 10^{-8}. At pH 5, the equilibrium ratio[A 2-]/ $[H_2A]$ is

(A) 0.01 (B) 10

(C) 0.001 (D) 0.1

49. An acetate buffer contains 0.100 mol HA and 0.100 mol A^-. A carbonate impurity consumes 0.0200 mol H^+ from the buffer, with CO_2 escaping. Remaining $[A^-]$ / [HA] is

(A) 1.5 (B) 0.66666667

(C) 2 (D) 1

50. An ideal model solution contains only a monoprotic acid HA (M=100 g/mol) and glucose(M=180 g/mol). At the stated temperature, $K_a = 10^{-3}$ and the solution has pH 2.00; neglect water autoionization. Total analytical solute mass is $27.2\,\mathrm{g\,L^{-1}}$. At 300 K with $R = 0.0800\,\mathrm{L\,bar\,mol^{-1}K^{-1}}$, its ideal osmotic pressure in bar is

(A) 4.8 (B) 5.04

(C) 5.28 (D) 2.64

Answer key

Ionic Equilibrium

C-IOEQ / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**B**	21	**D**	31	**B**	41	**C**
02	**A**	12	**A**	22	**C**	32	**A**	42	**C**
03	**A**	13	**C**	23	**D**	33	**A**	43	**D**
04	**C**	14	**B**	24	**C**	34	**A**	44	**A**
05	**C**	15	**C**	25	**C**	35	**B**	45	**D**
06	**A**	16	**A**	26	**D**	36	**B**	46	**A**
07	**A**	17	**B**	27	**D**	37	**A**	47	**B**
08	**C**	18	**D**	28	**B**	38	**C**	48	**D**
09	**B**	19	**B**	29	**D**	39	**C**	49	**A**
10	**D**	20	**B**	30	**D**	40	**B**	50	**B**

Chapter 11

Redox Reactions

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 0.0200 mol permanganate sample in acid oxidizes Fe^{2+} to Fe^{3+}. Moles of Fe^{2+} consumed are

(A) 0.1 (B) 0.02

(C) 0.004 (D) 0.04

02. A 0.0100 mol dichromate sample in acid oxidizes Fe^{2+} to Fe^{3+}. Moles of Fe^{2+} consumed are

(A) 0.01 (B) 0.02

(C) 0.03 (D) 0.06

03. A 0.0500 mol oxalate sample is completely oxidized to CO_2 by acidified permanganate. Moles of permanganate required are

(A) 0.02 (B) 0.025

(C) 0.01 (D) 0.05

04. A 25.0 mL Fe^{2+} sample requires 20.0 mL 0.0200 M $KMnO_4$ in acid. Fe^{2+} concentration in mol/L is

(A) 0.016 (B) 0.08

(C) 0.1 (D) 0.04

05. In 3 Cl_2 +6 OH^- →5 Cl^- + ClO_3^- +3 H_2O, the fraction of reactant chlorine atoms that are oxidized is

(A) 0.33333333 (B) 0.83333333

(C) 0.5 (D) 0.16666667

06. For Fe_3O_4 treated as FeO. Fe_2O_3, the fraction of iron atoms that are Fe^{3+} is

(A) 0.66666667 (B) 1

(C) 0.33333333 (D) 0.5

07. A 0.0300 mol H_2O_2 sample is oxidized completely by MnO_4^- in acid. Permanganate amount in mol is

(A) 0.03 (B) 0.006

(C) 0.012 (D) 0.015

08. One mole NO_3^- is reduced to NH_4^+ in acid. Electrons accepted in mol are

(A) 5 (B) 6

(C) 3 (D) 8

09. For Zn+ $Cu^{2+} \rightarrow Zn^{2+}$ +Cu, a 0.0500 mol reaction extent transfers electrons in mol of

(A) 0.1 (B) 0.05

(C) 0.2 (D) 0.025

10. A 0.0200 mol mixture of Fe^{2+} and Fe^{3+} needs 0.0120 mol electrons to reduce all irons to Fe^{2+}. Initial Fe^{2+} amount in mol is

(A) 0.012 (B) 0.008

(C) 0.004 (D) 0.02

11–40 / New middle-to-hard practice

11. A mixture contains $FeSO_4$ and FeC_2O_4 only, with 0.100 mol Fe in total. Complete acidic permanganate oxidation to Fe^{3+} and CO_2 takes 0.180 mol electron equivalents. CO_2 formed is

(A) 0.120 mol (B) 0.0400 mol

(C) 0.180 mol (D) 0.0800 mol

12. A mixture contains FeC_2O_4 and $Fe_2(SO_4)_3$ with 0.100 mol Fe atoms. Oxidation of all Fe to +3 and all oxalate to CO_2 consumes 0.0120 mol acidic permanganate. Fraction of initial iron atoms in the +2 state is

(A) $\frac{2}{5}$ (B) $\frac{1}{5}$

(C) $\frac{1}{10}$ (D) $\frac{3}{5}$

13. A mixture contains 0.0500 mol total $KMnO_4$ and $K_2Cr_2O_7$. Complete reduction in acid accepts 0.260 mol electrons. Amount of Cr^{3+} formed is

(A) 0.0400 mol (B) 0.0100 mol

(C) 0.0500 mol (D) 0.0200 mol

14. A 9.26 g $KMnO_4$–$K_2Cr_2O_7$ mixture accepts 0.260 mol electrons on complete acidic reduction. Molar masses are 158 and 294. Total K^+ amount in the sample is

(A) 0.0600 mol (B) 0.0800 mol

(C) 0.0500 mol (D) 0.100 mol

15. A 1.00 g impure MnO_2 sample is treated with 0.0400 mol Fe^{2+} in acid. Remaining Fe^{2+} consumes 0.00400 mol acidic permanganate. MnO_2 reduces to Mn^{2+}; impurities are inert. With MnO_2=87, purity by mass is

(A) 43.5% (B) 87.0%

(C) 100% (D) 69.6%

16. A pure 2.39 g oxide MO_2 converts M(IV) to M(II) on reacting with Fe^{2+} in acid. Start with 0.0400 mol Fe^{2+}; the remainder needs 0.00400 mol acidic permanganate. Atomic molar mass of M is (O=16)

(A) 119 (B) 207

(C) 87 (D) 55

17. A FeO–Fe_2O_3 mixture contains 0.100 mol Fe atoms. It needs 0.0400 mol electrons to convert all iron to Fe^{2+}. Mole fraction of FeO among the oxide formula units is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{3}{5}$ (D) $\frac{2}{5}$

18. A CuO–Cu_2O mixture contains 0.150 mol Cu atoms. Complete reduction to copper consumes 0.100 mol H_2. Fraction of original copper atoms in the +1 state is

(A) $\frac{3}{4}$ (B) $\frac{2}{3}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

19. A 0.100 mol Cl_2 sample disproportionates in alkali, forming only chloride, hypochlorite and chlorate. Assay finds 0.0300 mol chlorate. Fraction of all chlorine atoms that were oxidised is

(A) $\frac{3}{20}$ (B) $\frac{1}{5}$

(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

20. Initially 0.0900 mol permanganate accepts 0.150 mol electrons in strong alkali, forming only manganate(VI) and MnO_2. On acidification, only manganate disproportionates by $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$. Permanganate regenerated is

(A) 0.0400 mol (B) 0.0600 mol

(C) 0.0200 mol (D) 0.0500 mol

21. An acidic sample contains Cu^{2+} and dichromate. Total Cu plus Cr atoms are 0.100 mol. Excess iodide converts Cu to CuI and Cr(VI) to Cr(III), liberating 0.100 mol I_2. Initial copper mole fraction among metal atoms is

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

22. A 250 mL Cu^{2+}–Fe^{3+} sample liberates iodine with excess iodide; quantitative iodine titration uses 0.100 mol thiosulfate. A separate validated assay gives 0.0250 mol Cu. Assume $Cu^{2+} \rightarrow$CuI and $Fe^{3+} \rightarrow Fe^{2+}$. Initial Fe^{3+} molarity is

(A) 0.150 M (B) 0.100 M

(C) 0.300 M (D) 0.400 M

23. A thiosulfate–sulfite mixture consumes 0.0300 mol I_2 on oxidation to tetrathionate and sulfate, respectively. A separate equal sample on acidification liberates 0.0500 mol SO_2. Elemental sulfur formed in the acid test is

(A) 0.0200 mol (B) 0.0400 mol

(C) 0.0500 mol (D) 0.0300 mol

24. A thiosulfate–sulfite sample contains 0.100 mol sulfur atoms and consumes 0.0400 mol iodine when thiosulfate forms tetrathionate and sulfite forms sulfate. On separate acidification, elemental sulfur amount divided by SO_2 amount is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) 1

25. A 0.0100 mol permanganate sample accepts 0.0400 mol electrons, ending only as Mn^{2+} and MnO_2. Additional Fe^{2+} needed in acid to reduce all resulting MnO_2 to Mn^{2+} is

(A) 0.0200 mol (B) 0.0150 mol

(C) 0.0100 mol (D) 0.00500 mol

26. A 0.0200 mol permanganate sample is reduced in a controlled process to Mn(VI) and Mn(II) only, accepting 0.0600 mol electrons. A later selective oxidation converts all Mn(VI), and none of Mn(II), back to Mn(VII). Electron amount released in this second step is

(A) 0.0100 mol (B) 0.00500 mol

(C) 0.0300 mol (D) 0.0200 mol

27. In acid, permanganate oxidises sulfite to sulfate and becomes Mn^{2+}. Feed contains 0.0100 mol permanganate and 0.0200 mol sulfite, with ample acid. Amount of H^+ consumed by the net balanced redox equation is

(A) 0.0240 mol (B) 0.0300 mol

(C) 0.0400 mol (D) 0.0160 mol

28. In acid, 0.0100 mol dichromate reacts with 0.0300 mol Fe^{2+}, all acid being ample. Cr(VI) ends as Cr^{3+}. A later controlled precipitation recovers 80.0% of formed chromium as $Cr(OH)_3$. With Cr=52, O=16, H=1, recovered precipitate mass is

(A) 0.412 g (B) 1.648 g

(C) 0.824 g (D) 1.030 g

29. A 0.100 mol mixture of ethanol and ethanal releases 0.300 mol electrons when both are completely oxidised to ethanoic acid. Before that oxidation, an equal mixture sample undergoes quantitative Tollens reaction. Silver mass formed is (Ag=108; ethanal gives two Ag)

(A) 5.40 g (B) 10.8 g

(C) 16.2 g (D) 21.6 g

30. A 3.10 g mixture contains methanol and methanal. Complete oxidation to CO_2 releases 0.500 mol electrons. Molar masses are 32 and 30. Initial methanol mole fraction is

(A) $\frac{3}{5}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

31. A dry mixture contains $Ca(OCl)_2$ and $CaCl_2$ only, with 0.0100 mol Ca. On complete acidic reduction of hypochlorite to chloride it accepts 0.0200 mol electrons. Molar masses are 143 and 111. $Ca(OCl)_2$ mass fraction is

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$

(C) $\frac{143}{254}$ (D) $\frac{111}{254}$

32. A NaClO–$NaClO_3$ mixture contains 0.100 mol Na and accepts 0.400 mol electrons on complete chlorine reduction to chloride. Molar masses are 74.5 and 106.5. Sodium chlorate mass fraction is

(A) $\frac{149}{362}$ (B) $\frac{3}{4}$

(C) $\frac{1}{2}$ (D) $\frac{213}{362}$

33. A 0.100 mol gas mixture contains NO and NO_2. It requires 0.0200 mol O_2 to convert all NO into NO_2. Initial mean molar mass is (NO=30, NO_2=46)

(A) 36.4 $g\,mol^{-1}$ (B) 39.6 $g\,mol^{-1}$

(C) 42.8 $g\,mol^{-1}$ (D) 38.0 $g\,mol^{-1}$

34. A 0.100 mol mixture contains NO_2 and N_2O_4. Complete oxidation of all nitrogen to nitrate releases 0.180 mol electrons. Fraction of the initial nitrogen atoms present in N_2O_4 is

(A) $\frac{4}{5}$ (B) $\frac{1}{5}$

(C) $\frac{8}{9}$ (D) $\frac{2}{3}$

35. An unknown Fe(II)/Fe(III) sample releases 0.0400 mol electrons when all iron is formally oxidised to Fe(III). Complete reduction of an equal sample to Fe(0) requires 0.260 mol electrons. Initial Fe(III) fraction among iron atoms is

(A) $\frac{2}{5}$ (B) $\frac{4}{5}$

(C) $\frac{1}{3}$ (D) $\frac{3}{5}$

36. A copper-compound mixture contains Cu(I) and Cu(II) only. Oxidising all Cu to Cu(II) releases 0.0300 mol electrons. Reducing an equal sample completely to Cu(0) requires 0.130 mol electrons. Cu(I) fraction among copper atoms is

(A) $\frac{5}{8}$ (B) $\frac{1}{4}$

(C) $\frac{3}{5}$ (D) $\frac{3}{8}$

37. A dry Na_2O–Na_2O_2 mixture has 0.200 mol Na atoms and accepts 0.0600 mol electrons when all oxygen is converted to oxide. On initial hydrolysis use $Na_2O + H_2O \rightarrow 2NaOH$ and $Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$. Water mass consumed is

(A) 1.08 g (B) 3.60 g

(C) 1.80 g (D) 2.34 g

38. A dry KO_2–K_2O_2 mixture contains 0.100 mol K atoms. Reducing all oxygen to oxide requires 0.200 mol electrons. With KO_2=71 and K_2O_2=110, superoxide mass fraction is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{55}{126}$ (D) $\frac{71}{126}$

39. In ample acid, 0.0100 mol iodate reacts with 0.0300 mol iodide by $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$. All liberated iodine is quantitatively separated from the unreacted iodate and transferred to a suitable titration medium. Then 0.0200 mol thiosulfate reduces part of that isolated iodine to iodide, forming tetrathionate. I_2 remaining is

(A) 0.00400 mol (B) 0.0100 mol

(C) 0.00800 mol (D) 0.0180 mol

40. In ample acid, 0.0100 mol bromate reacts with 0.100 mol bromide by $BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$. The bromine oxidises 0.0400 mol Fe^{2+} completely to Fe^{3+}. Remaining Br_2 is quantitatively converted to I_2 with excess KI. Thiosulfate required for that iodine is

(A) 0.0100 mol (B) 0.0400 mol

(C) 0.0200 mol (D) 0.0600 mol

41–50 / Preserved hard practice

41. A 0.100 mol mixture of Fe^{2+} and Fe^{3+} requires 0.0120 mol acidic dichromate to oxidize all Fe^{2+} to Fe^{3+}. Total moles of electrons needed to reduce afresh equal sample to metal are

(A) 0.244 (B) 0.3

(C) 0.228 (D) 0.2

42. A 3.000 g mixture of FeC_2O_4 (M=144) and inert matter consumes 0.0100 mol acidic permanganate. Both Fe^{2+} and oxalate oxidize fully to Fe^{3+} and CO_2. Mass percent FeC_2O_4 is

(A) 60 (B) 80

(C) 72 (D) 90

43. A mixture of oxalate and H_2O_2 consumes 0.0400 mol acidic permanganate. On separately decomposing the H_2O_2 quantitatively, 0.0300 mol O_2 is released. Oxalate amount in mol is

(A) 0.08 (B) 0.06

(C) 0.02 (D) 0.04

44. A 0.0200 mol chlorate sample oxidizes I^- to I_2 in acid while ClO_3^- is reduced to Cl^-. I_2 formed is titrated with $S_2O_3^{2-}$ by $I_2 + 2\ S_2O_3^{2-} \rightarrow 2\ I^- + S_4O_6^{2-}$. Thiosulfate amount in mol is

(A) 0.24 (B) 0.12

(C) 0.04 (D) 0.06

45. A mixture contains 0.0100 mol MnO_4^- and 0.0100 mol $Cr_2O_7^{2-}$. In acid, it oxidizes Fe^{2+} to Fe^{3+}. If each Fe^{2+} sample contains 0.0200 mol Fe^{2+}, number of complete samples that can be oxidized is

(A) 5 (B) 5.5

(C) 4 (D) 6

46. A 0.0200 mol MnO_4^- sample is reduced to a mixture of Mn^{2+} and MnO_2 and accepts 0.0800 mol electrons in total. Moles of Mn^{2+} formed are

(A) 0.01 (B) 0.02

(C) 0.005 (D) 0.015

47. A 0.0100 mol NO_3^- sample is reduced to NO and NH_4^+ and accepts 0.0550 mol electrons. Fraction of nitrogen forming NH_4^+ is

(A) 0.25 (B) 0.6

(C) 0.5 (D) 0.75

48. A 2.000 g impure H_2O_2 sample is reacted with 0.0200 mol acidic MnO_4^-. Excess MnO_4^- is back titrated by 0.0100 mol Fe^{2+}. Mass percent H_2O_2 is

(A) 34 (B) 76.5

(C) 68 (D) 51

49. A compound contains one S atom per molecule with oxidation number-2. On complete oxidation to the+6 state, 0.500 g sample transfers 0.0400 mol electrons. Its molar mass in g/mol is

(A) 100 (B) 80

(C) 50 (D) 200

50. A mixture of Cl_2 and Br_2 oxidizes 0.200 mol I^- to I_2. If the halogens had total mass 12.10 g(Cl_2 =71, Br_2 =160), moles of Cl 2 were

(A) 0.025 (B) 0.075

(C) 0.05 (D) 0.043820225

Answer key

Redox Reactions

C-RERE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**C**	31	**C**	41	**C**
02	**D**	12	**B**	22	**C**	32	**D**	42	**B**
03	**A**	13	**D**	23	**B**	33	**B**	43	**D**
04	**B**	14	**A**	24	**A**	34	**C**	44	**B**
05	**D**	15	**B**	25	**C**	35	**D**	45	**A**
06	**A**	16	**B**	26	**A**	36	**D**	46	**A**
07	**C**	17	**A**	27	**A**	37	**D**	47	**C**
08	**D**	18	**B**	28	**C**	38	**D**	48	**B**
09	**A**	19	**B**	29	**B**	39	**C**	49	**A**
10	**B**	20	**A**	30	**C**	40	**C**	50	**D**

Chapter 12

Electrochemistry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For a two-electron cell at 298 K with $E^\circ = 1.10\,\text{V}$ and reaction quotient Q=100, E in V is

(A) 1.0409 (B) 1.1

(C) 0.9818 (D) 1.1591

02. A concentration cell has identical Zn electrodes, $[Zn^{2+}]$ values 0.010 and 1.00 M. At 298 K its emf in V is

(A) 0 (B) 0.1182

(C) 0.0591 (D) 0.02955

03. For a cell with n=2 and E° =.1773 Vat 298 K, log 10 K is

(A) 4 (B) 12

(C) 3 (D) 6

04. Same charge passes through Ag^+ and Cu^{2+} cells in series. Deposited mass ratio Ag/Cu is(Ag=108, Cu=64)

(A) 1.6875 (B) 2

(C) 6.75 (D) 3.375

05. A dilute weak acid hasα=.10 and formal concentration.010 M. Approximate K_a using the exact mass balance expression is

(A) 0.00011111111 (B) 0.011111111

(C) 1×10^{-5} (D) 0.001

06. Limiting conductivities of HCl, NaCl, and NaA are 426, 126, and 91 respectively. Limiting conductivity of HA is

(A) 461 (B) 209

(C) 391 (D) 643

07. A copper electrolysis experiment deposits 2.56 g using 9650 C. Current efficiency in % is(Cu=64, F=96500)

(A) 40 (B) 60

(C) 80 (D) 100

08. A conductivity cell has constant.50 cm-1 and conductance.004 S. Its 0.020 M solution has molar conductivity in S cm^2 /mol of

(A) 200 (B) 100

(C) 10 (D) 50

09. For a one-electron concentration cell at 298 K, emf.1182 V requires activity ratio high/low of

(A) 1000 (B) 10

(C) 10000 (D) 100

10. A two-electron cell has E° =.0591 Vat 298 K. At what Q does its voltage become zero?

(A) 100 (B) 0.01

(C) 10 (D) 1

11–40 / New middle-to-hard practice

11. At a cathode, only $Cu^{2+} \rightarrow$Cu and hydrogen evolution occur: 0.384 g Cu and 0.00200 mol H_2 are collected. At the anode, only O_2 and Cl_2 form, including 0.00300 mol O_2. With Cu=64, Cl_2 formed is

(A) 0.00100 mol (B) 0.00200 mol

(C) 0.00800 mol (D) 0.00400 mol

12. A cathode deposits 0.320 g Cu and evolves 112 mL dry H_2 at STP; no other cathodic process occurs. With Cu=64, molar gas volume 22.4 L and $F = 96500$, time at a constant 2.00 A is

(A) 1447.5 s (B) 965 s

(C) 1930 s (D) 482.5 s

13. During electrolysis of 0.500 L $CuSO_4$, a pure Cu anode loses 1.28 g and the cathode gains 0.960 g Cu. Only Cu dissolution occurs at the anode; the remaining cathodic current makes H_2. With Cu=64 and unchanged volume, the pair (H_2 amount, increase in $[Cu^{2+}]$) is

(A) $(0.00500 \text{ mol}, 0.0100 \text{ M})$ (B) $(0.00500 \text{ mol}, 0.0200 \text{ M})$

(C) $(0.00250 \text{ mol}, 0.0100 \text{ M})$ (D) $(0.0100 \text{ mol}, 0.00500 \text{ M})$

14. A 100 mL, 0.200 M $CuSO_4$ sample is electrolysed with a pure copper anode. Anode loss is 0.640 g; cathodic Cu gain is 0.480 g. Other cathodic current evolves H_2 and no Cu is otherwise lost. The solution is then diluted to 250 mL. With Cu=64, final $[Cu^{2+}]$ is

(A) 0.0900 M (B) 0.100 M

(C) 0.0800 M (D) 0.0225 M

15. The same charge deposits 10.8 g Ag from Ag^+ and 0.900 g metal M from M^{n+} at full efficiency. The molar mass of its chloride MCl_n is 133.5. With Ag=108, Cl=35.5, atomic molar mass of M is

(A) 54 (B) 40

(C) 27 (D) 9

16. In validated quantitative electrodeposition experiments, one charge deposits 21.6 g Ag from Ag^+ and 2.40 g M from M^{n+}. The chloride MCl_n has molar mass 95.0. With Ag=108, Cl=35.5, O=16, molar mass of the neutral oxide M_2O_n in its simplest formula is

(A) 24 (B) 56

(C) 40 (D) 80

17. A 100 mL sample has 0.0200 mol Ag^+ and 0.0300 mol Cu^{2+}. In an ideal controlled deposition, Ag deposits completely before Cu; hydrogen starts only after both metal ions are exhausted. Charge $0.0600F$ passes. With Ag=108 and Cu=64, metal mass deposited is

(A) 2.80 g (B) 4.08 g

(C) 2.16 g (D) 3.44 g

18. A controlled cathode initially has 0.0100 mol Ag^+ and 0.0200 mol Cu^{2+} available. Ag deposits first, then Cu, then only H_2 evolves; all steps are quantitative. Charge $0.0800F$ passes. Dry H_2 volume at STP is (22.4 L mol^{-1})

(A) 896 mL (B) 672 mL

(C) 224 mL (D) 336 mL

19. For a cell, $E = E^\circ - (0.059/n)\log Q$ at 298 K. Measured E is 0.177 V at $Q = 1$ and 0.118 V at $Q = 100$. At $Q = 0.0100$, E is

(A) 0.236 V (B) 0.295 V

(C) 0.2065 V (D) 0.118 V

20. At 298 K a cell obeys $E = E^\circ - (0.059/n)\log Q$. It gives 0.118 V at $Q = 1$ and 0.059 V at $Q = 10^3$. Equilibrium constant for the reaction as written is

(A) 100 (B) 1000000000

(C) 1000 (D) 1000000

21. In a hydrogen concentration cell with both gas pressures 1 bar, $E_{right} - E_{left} = 0.059(\text{pH}_{left} - \text{pH}_{right})$. The right acid has pH 2.000 and measured emf is 0.118 V. Equal 100 mL portions of the two strong acids are mixed. Final pH is (use $\log 5.05 = 0.7033$)

(A) 3 (B) $\frac{22967}{10000}$

(C) 4 (D) 2

22. For two hydrogen electrodes at 298 K, use $E_{right} - E_{left} = 0.059(\text{pH}_{left} - \text{pH}_{right}) - 0.0295\log(P_{right}/P_{left})$. Measured emf is 0.0885 V, right pH is 2.00 and pressures are 100 and 1 bar on right and left. After tenfold dilution of the left strong acid, its pH is

(A) $\frac{13}{2}$ (B) $\frac{9}{2}$

(C) $\frac{7}{2}$ (D) $\frac{11}{2}$

23. At 298 K an Ag^+/Ag electrode has E=0.682 V at $[Ag^+] = 0.0100$ M. An AgCl(s)/Ag electrode has E=0.269 V at $[Cl^-] = 0.100$ M. With $E = E^\circ_{Ag^+/Ag} + 0.059\log[Ag^+]$ and ideal ions, $K_{sp}(AgCl)$ is

(A) $\frac{1}{100000000}$ (B) $\frac{1}{1000000000}$

(C) $\frac{1}{10000000000}$ (D) $\frac{1}{100000000000}$

24. For M^{2+}/M at 298 K, $E = E^\circ + 0.0295\log[M^{2+}]$. E is -0.4885 V at $[M^{2+}] = 10^{-3}$ M. In equilibrium with $M(OH)_2$(s) and $[OH^-] = 10^{-2}$ M, E is -0.5770 V. For an initially 0.0100 M M^{2+} solution, precipitation begins at pH ($pK_w = 14$)

(A) 10 (B) 12

(C) 9 (D) 11

25. Standard reductions $Fe^{3+} \to Fe$ and $Fe^{2+} \to Fe$ have $E^\circ = -0.0300$ V and -0.440 V. For Ag^+/Ag, $E^\circ = 0.849$ V. At 298 K, equilibrium K for $Fe^{2+} + Ag^+ \to Fe^{3+} + Ag$ is (use 0.059 V)

(A) 100 (B) 10

(C) 1000 (D) $\frac{1}{10}$

26. Standard $E^\circ(Sn^{4+}/Sn)$=0.00500 V and $E^\circ(Sn^{4+}/Sn^{2+})$=0.150 V. $E^\circ(Zn^{2+}/Zn)$=-0.760 V. For $Sn^{2+} + Zn \to Sn + Zn^{2+}$ at 298 K, $\log K$ is (use $2.303RT/F = 0.059$ V)

(A) $\frac{1520}{59}$ (B) $\frac{620}{59}$

(C) $\frac{1820}{59}$ (D) $\frac{1240}{59}$

27. A conductance cell gives R=10.0 ohm for a standard with $\kappa = 0.0100$ S cm^{-1}. The same cell gives R=100 ohm for 0.0100 M HA. Limiting molar conductivities of HCl, NaA, NaCl are 400, 100, 100 S cm^2 mol^{-1}. Using $\alpha = \Lambda/\Lambda^\circ$, $K_a = C\alpha^2/(1-\alpha)$ is

(A) $\frac{1}{1200}$ (B) $\frac{1}{1600}$

(C) $\frac{1}{300}$ (D) $\frac{1}{400}$

28. A cell gives R=25.0 ohm for $\kappa = 0.0200$ S cm^{-1} and R=625 ohm for 0.0100 M HA. Its $\Lambda^\circ = 400$ S cm^2 mol^{-1}. Assume only HA dissociation and constant K_a. At 0.00200 M, degree of dissociation is

(A) $-\frac{1}{8} + \frac{\sqrt{17}}{8}$ (B) $\frac{1}{5}$

(C) $\frac{1}{2}$ (D) $\frac{1}{8} + \frac{\sqrt{17}}{8}$

29. Limiting molar conductivities of NaCl, KCl and KNO_3 are 126, 150 and 110 S cm^2 mol^{-1}. The Na^+ transport fraction in NaCl is 0.400. Assuming independent ionic migration, Na^+ transport fraction in $NaNO_3$ is

(A) $\frac{126}{275}$ (B) $\frac{89}{215}$

(C) $\frac{126}{215}$ (D) $\frac{2}{5}$

30. At infinite dilution, $\Lambda(KCl) = 150$, $\Lambda(CaCl_2) = 276$ and $\Lambda(Ca(NO_3)_2) = 260$ in S cm^2 mol^{-1}. K^+ carries 0.480 of current in KCl. Nitrate transport fraction in KNO_3 is

(A) $\frac{36}{71}$ (B) $\frac{35}{71}$

(C) $\frac{7}{15}$ (D) $\frac{1}{2}$

31. A 100 mL mixture contains only NaCl and KCl. Chloride assay gives 0.00200 mol Cl^-. Its conductivity is 0.00276 S cm^{-1}. Treat ionic molar conductivities as concentration-independent, with $\Lambda(NaCl) = 126$, $\Lambda(KCl) = 150$. KCl molarity is

(A) 0.00500 M (B) 0.0150 M

(C) 0.0200 M (D) 0.0100 M

32. A mixture contains only HCl and NaCl. Chloride concentration is 0.0100 M and conductivity is 0.002142 S cm^{-1}. Use constant molar conductivities 420 and 126 S cm^2 mol^{-1}. After twofold dilution its pH is ($\log 1.5 = 0.1761$)

(A) 2 (B) $\frac{28239}{10000}$

(C) $\frac{25229}{10000}$ (D) 3

33. A hydrogen fuel cell receives 0.100 mol H_2 and leaves 0.0400 mol unused. All consumed H_2 supplies two electrons per molecule at a constant 0.800 V. Half of the electrical energy lifts a 20.0 kg load. With $F = 96500$, $g = 10.0$, lift height is

(A) 38.60 m (B) 11.58 m

(C) 46.32 m (D) 23.16 m

34. A methanol cell is fed 3.20 g CH_3OH (M=32); 20% remains unused. Consumed methanol transfers six electrons per molecule at 0.750 V. A mechanism uses 40% of electrical energy to expand against 100 kPa. With $F = 96500$, volume change is

(A) 0.06948 m^3 (B) 0.34740 m^3

(C) 0.17370 m^3 (D) 0.13896 m^3

35. A membrane-separated brine cell sends 5.00 A for 193 s through a cathode compartment containing 0.0200 mol HA and 0.0100 mol A^-. Each electron generates one OH^-; no anodic product enters. With $F = 96500$, $pK_a = 5.00$ and $\log 2 = 0.3010$, final buffer pH is

(A) 5 (B) $\frac{5301}{1000}$

(C) $\frac{4699}{1000}$ (D) 12

36. A separated cathode compartment initially contains 0.0200 mol HCl and ample inert electrolyte. It receives 3.00 A for 965 s; all cathodic charge reduces H^+ or water, and no anodic product enters. Final volume is 0.500 L. With $F = 96500$, $pK_w = 14$, $\log 2 = 0.3010$, pH is

(A) $\frac{1699}{1000}$ (B) $\frac{12301}{1000}$

(C) $\frac{63891}{5000}$ (D) 13

37. A 0.200 mol dry feed contains H_2 and N_2 only, with mean molar mass 15.0 (H_2=2, N_2=28). It meets 0.0800 mol O_2 in a fuel cell. 90.0% of the stoichiometrically consumable hydrogen is actually used. Water mass formed is

(A) 2.592 g (B) 1.80 g

(C) 1.62 g (D) 3.24 g

38. A 3.00 g, 0.200 mol fuel sample contains H_2 and CO only (M=2,28). In an ideal cell both have the same fractional conversion; H_2 forms water and CO forms CO_2, each transferring two electrons per molecule. The delivered charge is $0.300F$. Water formed is

(A) 1.35 g (B) 0.900 g

(C) 1.80 g (D) 2.70 g

39. In a protected iron sample, the only oxidation is $Fe \rightarrow Fe^{2+} + 2e^-$ and the only reduction is oxygen to OH^- (four electrons per O_2). Measured O_2 uptake is 0.00400 mol. An external protection supply delivers $0.0100F$ of cathodic charge. With Fe=56, iron mass lost is

(A) 0.168 g (B) 0.280 g

(C) 0.448 g (D) 0.0560 g

40. Iron corrodes only to Fe^{2+}; the cathodic reactions are H^+ to H_2 and oxygen reduction, with no external current. Iron mass loss is 0.560 g and 0.00300 mol H_2 is collected. With Fe=56 and STP molar volume 22.4 L, O_2 uptake is

(A) 44.8 mL (B) 112 mL

(C) 78.4 mL (D) 156.8 mL

41–50 / Preserved hard practice

41. A 0.100 M weak acid has conductivity $1.00\times\ 10^{-3}$ S/cm. Its limiting molar conductivity is 400 S cm^2 /mol. Usingα=$\Lambda/\Lambda\infty$and K_a =Cα^2/(1-α), K_a is

(A) 6.25×10^{-5}

(B) 0.00064102564

(C) 6.4102564×10^{-5}

(D) 0.000625

42. AgCl $K_{sp} = 10^{-10}$. In 0.010 M Cl^-, Ag^+ at saturation is determined only by K_{sp}. With Ag^+ /Ag E° =.80 V, find reduction potential at 298 K in V.

(A) 0.3863

(B) 0.209

(C) 0.3272

(D) 0.8

43. A one-electron cell has E° =.0591 Vat 298 K. For reaction A⇌B with activities a A=.10, a B=1.0, the actual Gibbs energy per mole in units of RT is

(A) 2.3025851

(B) 0

(C) -2.3025851

(D) 1

44. Charge 0.100 F passes through a mixture containing 0.0200 mol Ag^+ and 0.100 mol Cu^{2+}. Assume Ag^+ is reduced fully before any Cu^{2+} and no other reaction. Total deposited metal mass in g is(Ag=108, Cu=64)

(A) 4.72

(B) 5.36

(C) 3.2

(D) 2.16

45. An inert anode electrolyzes water with 0.200 F charge and 100% efficiency. O_2 is collected at 300 K and 1 bar. With R=.0800 L bar mol^{-1} K^{-1}, gas volume in L is

(A) 1.2

(B) 0.6

(C) 2.4

(D) 4.8

46. A copper cell cathode gains 1.92 g while its copper anode loses 2.56 g. If anodic dissolution is 100% efficient and Cu=64, the cathodic current efficiency in % is

(A) 50

(B) 80

(C) 60

(D) 75

47. A cell reaction transfers 2 electrons and has E° 1.00 Vat 300 K. Its entropy change is-100 J/mol/K. With F=96500, standard enthalpy in kJ /mol is

(A) 223

(B) -223

(C) -193

(D) -163

48. A 0.100 mol mixture of Fe^{2+} and Fe^{3+} requires 0.250 F charge to reduce all iron to metal with 100% efficiency. Moles of Fe^{3+} initially were

(A) 0.1

(B) 0.05

(C) 0.025

(D) 0.075

49. A cell has standard potential.1182 V and n=2 at 298 K. At its current state, Q= 10^3. With F=96500, actual Gibbs energy in kJ /mol is

(A) -11.4063 (B) -5.70315

(C) 5.70315 (D) -22.8126

50. A concentration cell with identical Ag electrodes has one half-cell free Ag^+ activity 10^{-2} and the other 10^{-6}. If complexation reduces only the first free activity by factor 100 without changing the second, emf at 298 K in V is

(A) 0.2364 (B) 0.0591

(C) 0.1182 (D) 0.3546

Answer key

Electrochemistry

C-ELEC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**B**	31	**D**	41	**C**
02	**C**	12	**B**	22	**D**	32	**B**	42	**C**
03	**D**	13	**A**	23	**C**	33	**D**	43	**B**
04	**D**	14	**A**	24	**A**	34	**D**	44	**A**
05	**A**	15	**C**	25	**B**	35	**B**	45	**A**
06	**C**	16	**C**	26	**D**	36	**B**	46	**D**
07	**C**	17	**D**	27	**A**	37	**C**	47	**B**
08	**B**	18	**D**	28	**A**	38	**A**	48	**B**
09	**D**	19	**A**	29	**C**	39	**A**	49	**B**
10	**A**	20	**D**	30	**B**	40	**C**	50	**C**

Chapter 13

Chemical Kinetics

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A first-order reaction has half-life 5 min. Time for 87.5% completion in min is

(A) 40 (B) 15

(C) 20 (D) 10

02. A first-order reaction is 25% complete in 10 min. With ln(4/3)=.2877, rate constant in min^{-1} is

(A) 0.2877 (B) 0.014385

(C) 0.02877 (D) 0.05754

03. For a second-order reaction in A only, 1/ $[A]$ versus th as slope 2.0 L/mol/min. At $[A]$ 0=.50 M, time until $[A]$ =.25 M in min is

(A) 2 (B) 1

(C) 0.5 (D) 4

04. For rate=k $[A]^m$ $[B]^n$, doubling A quadruples rate and doubling B halves rate. Overall order m+n is

(A) 2 (B) 0

(C) 3 (D) 1

05. For A→B, the initial rates are.020 M/sat $[A]$ =.10 M and.080 M/sat $[A]$ =.20 M. The rate constant in L/mol/sis

(A) 2 (B) 0.2

(C) 8 (D) 4

06. A first-order reaction has A_0 =.80 M. It is half-complete after 10 min. After another 20 min, the concentration in M is

(A) 0.2 (B) 0.05

(C) 0.4 (D) 0.1

07. A first-order reaction has initial rate.040 M/min and A_0 =.20 M. Its half-life in min is(ln 2=.693)

(A) 3.465 (B) 6.93

(C) 5 (D) 1.7325

08. For rate=k $[A]^2$ $[B]$, dilution instantaneously doubles the solution volume without changing amounts. Initial rate after dilution is what fraction of the previous rate?

(A) 2 (B) 0.5

(C) 0.25 (D) 0.125

09. For a zero-order reaction, the concentration falls from.50 to.25 M during 5 min. Time from.25 to.125 M in min is

(A) 2.5 (B) 10

(C) 7.5 (D) 5

10. A first-order reaction has 90% completion time t 90 and half-life t Half. Using ln 10=2.303, ln 2=.693, t 90/t Half is

(A) 3.3232323 (B) 4

(C) 2 (D) 0.30091185

11–40 / New middle-to-hard practice

11. At fixed T, V, first-order $A(g) \rightarrow 2P(g) + Q(g)$ occurs with an inert gas present. Initial total pressure is 0.500 bar; after completion it is 1.300 bar. At 100 s it is 0.900 bar. Total pressure at 300 s is

(A) 1.250 bar (B) 1.200 bar

(C) 1.100 bar (D) 1.300 bar

12. First-order $A(g) \rightarrow 2B(g)$ proceeds at fixed T, V with an inert gas. Initial and final total pressures are 1.000 and 1.600 bar. At 40 s total pressure is 1.150 bar. At 80 s it is

(A) 1.2625 bar (B) 1.2250 bar

(C) 1.3000 bar (D) 1.4500 bar

13. For first-order disappearance of A, a signal obeys $S = S_\infty + b[A]$ with constant b. Initial signal is 100, final signal is 20, and signal at 50 s is 60. Signal at 150 s is

(A) 35 (B) 25

(C) 15 (D) 30

14. A first-order reaction gives an optical signal $S = S_\infty + b[A]$. Readings at 0,20,40 s are 140,100,80. The constant background S_∞ is

(A) 0 (B) 60

(C) 40 (D) 70

15. A zero-order reaction has $[A]_0 = 0.100$ M and $[A]_{20s} = 0.0800$ M. At 20 s the solution is instantly diluted twofold, without changing the zero-order rate constant. After a further 20 s, [A] is

(A) 0.0300 M (B) 0.0200 M

(C) 0.0400 M (D) 0.0600 M

16. A first-order reaction has [A]=0.0800 M at 10 s and 0.0400 M at 20 s. At 20 s its volume is instantly increased fourfold. Reaction conditions and k remain unchanged. [A] at 30 s is

(A) 0.0200 M (B) 0.00250 M

(C) 0.00500 M (D) 0.0100 M

17. For $r = k[A]^m[B]^n$, changing (A, B) from (a, b) to (2a,4b) multiplies r by 32; changing to (4a,2b) multiplies it by 16. The rate at (3a,2b), divided by that at (a, b), is

(A) 18 (B) 6

(C) 12 (D) 24

18. For $r = k[A]^m[B]^n$, doubling A and tripling B multiplies r by 4/3; tripling A and doubling B multiplies it by 9/2. If both concentrations are halved, the new-to-old rate ratio is

(A) 1 (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) 2

19. An irreversible reaction follows a single power-law order. Half-lives are 40 s at initial 0.100 M and 20 s at initial 0.200 M. Using the corresponding integrated law, time to decrease from 0.400 to 0.100 M is

(A) 20 s (B) 120 s

(C) 60 s (D) 30 s

20. A single power-law reaction has half-lives 20 s at initial 0.100 M and 40 s at initial 0.200 M. Time to fall from 0.300 M to 0.0500 M is

(A) 100 s (B) 60 s

(C) 120 s (D) 200 s

21. A first-order reaction obeys one Arrhenius law over the range. Its half-lives are 80 s at 300 K and 20 s at 400 K. Temperature at which half-life is 10 s is

(A) 600 K (B) 450 K

(C) 480 K (D) 500 K

22. For one Arrhenius reaction, $\ln k = -10$ at 300 K and $\ln k = -6$ at 400 K. At 350 K, $\ln k$ is

(A) $-\frac{62}{7}$ (B) -8

(C) $-\frac{50}{7}$ (D) $-\frac{54}{7}$

23. A has independent first-order paths $A \to B$ and $A \to C$. Total A half-life is 20 s; the final B:C ratio is 3:1. Starting with A only, B formed at 40 s divided by initial A is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{3}{16}$ (D) $\frac{9}{16}$

24. A has parallel first-order paths to B and C. Starting with 1.00 mol A, at 10 s 0.800 mol A and 0.0500 mol B remain/form. No product reacts further. Amount of C at 30 s is

(A) 0.488 mol (B) 0.366 mol

(C) 0.150 mol (D) 0.384 mol

25. A sample has 100 arbitrary equal-count units of radioactive atoms A and B combined. Half-lives are 10 and 20 days, respectively. Its initial activity is $140k_B$ in the corresponding count units. At 20 days, fraction of total activity due to B is

(A) $\frac{2}{5}$ (B) $\frac{1}{2}$

(C) $\frac{3}{5}$ (D) $\frac{3}{4}$

26. Two independently decaying nuclides have half-lives 10 and 20 days. Total activity is 120 units initially and 45 units at 20 days. At 40 days, fraction of surviving radioactive atoms belonging to the shorter-lived nuclide is

(A) $\frac{1}{17}$ (B) $\frac{1}{5}$

(C) $\frac{1}{3}$ (D) $\frac{1}{9}$

27. A decays by a background first-order path and a B-assisted path: $k_{obs} = k_0 + k[B]$, with B maintained constant. Observed k values are 0.0300 s^{-1} at B=0.100 M and 0.0700 s^{-1} at B=0.300 M. At B=0.200 M, half-life is ($\ln 2 = 0.693$)

(A) 17.325 s (B) 9.90 s

(C) 23.10 s (D) 13.86 s

28. For independent first-order disappearance channels of A, $k_{obs} = k_0 + k[B]$. Observed k values are 0.00800 s^{-1} at B=0.0200 M and 0.0160 s^{-1} at B=0.0600 M. At B=0.0400 M, fraction of A consumption through the B-assisted path is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

29. A reacts simultaneously by $A \to 2P$ and $A \to Q$. Measured instantaneous formation rates are $d[P]/dt = 0.0600$ and $d[Q]/dt = 0.0100$ M s^{-1}. Fraction of A consumption passing through the P path is

(A) $\frac{6}{7}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) $\frac{2}{3}$

30. Two simultaneous reactions are $A + B \to 2C$ and $2A \to D$. At one instant, $-d[B]/dt = 0.0200$ and $-d[A]/dt = 0.0800$ M s^{-1}. The ratio $(d[C]/dt)/(d[D]/dt)$ is

(A) $\frac{2}{3}$ (B) 2

(C) $\frac{4}{3}$ (D) $\frac{8}{3}$

31. For reversible first-order $A \rightleftharpoons B$, equilibrium has $[B]/[A] = 3$. Departure from equilibrium halves in 20 s. Use $k_f/k_b = [B]_{eq}/[A]_{eq}$ and relaxation constant $k_f + k_b$. With $\ln 2 = 0.693$, k_b is

(A) 0.0086625 s^{-1} (B) 0.034650 s^{-1}

(C) 0.017325 s^{-1} (D) 0.0259875 s^{-1}

32. For reversible first-order $A \rightleftharpoons B$, start with [A]=1.00 M, [B]=0. Equilibrium [B] is 0.600 M; at 10 s [B]=0.300 M. Given $[B]_t = [B]_{eq}(1 - e^{-(k_f+k_b)t})$, [B] at 30 s is

(A) 0.900 M (B) 0.450 M

(C) 0.525 M (D) 0.0750 M

33. A first-order reaction has rate constant $3k$ at T_2 and k at T_1. A sample spends 10 s at T_1 and 20 s at T_2; one quarter remains. Half-life at T_1 is

(A) 70 s (B) 25 s

(C) 15 s (D) 35 s

34. For a first-order reaction, $k(T_2) = 2k(T_1)$. Ten seconds at each temperature leaves half the sample. It is then kept at T_1. Additional time needed to leave one eighth of the original amount is

(A) 30 s (B) 40 s

(C) 60 s (D) 90 s

35. Parallel first-order paths to B and C each obey Arrhenius laws. The ratio k_B/k_C is 2 at 300 K and 8 at 400 K. At the temperature satisfying $1/T = (1/300 + 1/400)/2$, fraction of reacted A becoming B is

(A) $\frac{2}{3}$ (B) $\frac{4}{5}$

(C) $\frac{1}{2}$ (D) $\frac{8}{9}$

36. Catalysed and uncatalysed first-order rates each follow one Arrhenius law with temperature-independent prefactors. Their k ratio is 4 at 300 K and 2 at 400 K. At what temperature are their rate constants equal?

(A) 800 K (B) 450 K

(C) 600 K (D) 500 K

37. An empirical rate law is $v = kC/(1 + KC)$. Rates at C=0.100 and 0.300 M are 0.0100 and 0.0200 M s^{-1}. With unchanged k, K, rate at C=0.600 M is

(A) 0.0400 M s^{-1} (B) 2/75 M s^{-1}

(C) 0.0300 M s^{-1} (D) 0.0600 M s^{-1}

38. An empirical rate law is $v = kC/(1 + KC)$. Measured rates are 0.0200 M s^{-1} at C=0.100 M and 0.0500 at C=0.400 M. Concentration required for v=0.0750 M s^{-1} is

(A) 0.600 M (B) 1.50 M

(C) 1.20 M (D) 0.800 M

39. Radioactive P decays to retained stable D with half-life 100 years; S is a separate stable isotope. Initially P:S=3:2 and no D is present. A current assay gives D:S=9:8. Sample age is

(A) 100 years (B) 150 years

(C) 200 years (D) 300 years

40. P decays first-order into retained stable D. Initially $D_0/P_0 = 0.200$. After 30 years $D/P = 1.40$. At what age is $D/P = 3.80$?

(A) 60 years (B) 120 years

(C) 45 years (D) 90 years

41–50 / Preserved hard practice

41. A(g)→2 B(g) is first order in A in a fixed vessel at constant T. Initial pure A pressure is 1.00 bar. At t=10 min, total pressure is 1.50 bar. At what later time in min is total pressure 1.875 bar?

(A) 40 (B) 30

(C) 20 (D) 25

42. A(g)→B(g)+C(g) is first order. Initial pure A pressure is 2.00 bar. At 5 min, total pressure 3.00 bar. Remaining A partial pressure at 15 minin bar is

(A) 0.75 (B) 0.125

(C) 0.5 (D) 0.25

43. A compound A decays by parallel first-order paths A→B and A→C with k B=.030, k C=.010 min^{-1}. Starting with 1.00 mol A and no products, after half of A has reacted, moles of B are

(A) 0.375 (B) 0.125

(C) 0.25 (D) 0.5

44. For a reaction, ln(k 2/k 1)=ln 2 when T 1=300 K and T 2=330 K. With R=8.314 and ln 2=.693, activation energy in kJ /mol is

(A) 1.9013287 (B) 19.013287

(C) 38.02572 (D) 57.03858

45. A first-order decomposition of A produces 2 B. A 0.200 mol A sample releases 10.0 kJ after 75% conversion. Molar reaction enthalpy in kJ per mol A is

(A) -66.666667 (B) -200

(C) -50 (D) -100

46. A first-order reaction has concentration.400 M initially and.100 M after 20 min. At 30 min, a 25.0 mL aliquot requires strong base for 1: 1 neutralization of remaining monoprotic A. Required 0.100 M base volume in mL is

(A) 25 (B) 12.5

(C) 6.25 (D) 50

47. A reversible one-step reaction shares one transition state. Its forward activation energy is $80.0\,\mathrm{kJ\,mol^{-1}}$ and forward enthalpy is $-20.0\,\mathrm{kJ\,mol^{-1}}$. Both activation barriers and Arrhenius prefactors are temperature independent. The reverse rate constant at 300 K is $0.0100\,\mathrm{s^{-1}}$. Use $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$. The reverse rate constant at 330 K, in $\mathrm{s^{-1}}$, is closest to

(A) 0.38275868 (B) 0.18464606

(C) 0.089074835 (D) 0.00026126122

48. A closed vessel initially contains 0.100 mol A and 0.100 mol inert I. A→2 B is first order with 10 min half-life. At 20 min and constant T, V, total pressure divided by initial pressure is

(A) 1.375 (B) 1.75

(C) 1.25 (D) 1.5

49. A first-order reaction has k=.0693 $\mathrm{min^{-1}}$. A reactant sample initially contains.0800 mol A. At 20 min, all B formed by A→B is quantitatively oxidized by acid MnO_4^-, each B donating 2 e. Required MnO_4^- amount in mol is

(A) 0.032 (B) 0.016

(C) 0.008 (D) 0.024

50. A reactant A is consumed by parallel first-order paths with k 1=.01 and k 2=.03 $\mathrm{s^{-1}}$. Pure A initially has 1 mol. Total A surviving after time t is.25 mol. If only path 1 releases one gas molecule per A consumed, moles of released gas are

(A) 0.0625 (B) 0.1875

(C) 0.25 (D) 0.5625

Answer key

Chemical Kinetics

C-CHKI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**C**	31	**A**	41	**B**
02	**C**	12	**A**	22	**D**	32	**C**	42	**D**
03	**B**	13	**D**	23	**D**	33	**D**	43	**A**
04	**D**	14	**B**	24	**B**	34	**C**	44	**B**
05	**A**	15	**B**	25	**C**	35	**B**	45	**A**
06	**D**	16	**C**	26	**D**	36	**C**	46	**B**
07	**A**	17	**C**	27	**D**	37	**B**	47	**A**
08	**D**	18	**C**	28	**A**	38	**C**	48	**A**
09	**A**	19	**D**	29	**B**	39	**C**	49	**D**
10	**A**	20	**A**	30	**C**	40	**A**	50	**B**

Chapter 14

Surface Chemistry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A log-log Freundlich plot has slope 0.25. Parameter n equals

(A) 2 (B) 0.25

(C) 4 (D) 8

02. A gas adsorbent has accessible area 100 m^2 /g. A monolayer contains 10^{-3} mol/g. Using N_A =6.0× 10^{23}, area per adsorbed molecule in nm 2 is

(A) 6 (B) 0.0016666667

(C) 0.16666667 (D) 0.016666667

03. Two equal spherical droplets coalesce into one. New radius divided by old radius is

(A) 1.4142136 (B) 2

(C) 1.259921 (D) 0.5

04. A membrane retains all colloid particles while half the small-ion concentration is removed in each ideal dialysis exchange. After three exchanges, the small-ion fraction remaining is

(A) 0.5 (B) 0.25

(C) 0.75 (D) 0.125

05. For the same model, coverage 0.80 requires K_p of

(A) 0.2 (B) 0.8

(C) 4 (D) 5

06. For $x/m = kp^{1/2}$, loading is 8 mg/gat 4 bar. At 1 bar, it is in mg/g

(A) 4 (B) 2

(C) 8 (D) 16

07. A solid has area 50 m^2 /g and mass 4 g. On sintering, area falls to 30 m^2 /g with mass unchanged. Accessible area lost in m^2 is

(A) 20 (B) 120

(C) 200 (D) 80

08. A spherical droplet divides into 8 equal droplets. Total oil-water interface area changes by factor

(A) 0.5 (B) 4

(C) 2 (D) 8

09. At fixed total catalyst mass, spherical particles of radius 10 nm are replaced by equal-density particles of radius 5 nm. Total accessible external surface area changes by factor

(A) 0.5 (B) 8

(C) 4 (D) 2

10. A reaction is catalyzed on a surface. Initial rate is proportional to coverageθ. If K_p changes from 1 to 3 in$\theta = K_p\ /(1+\ K_p)$, rate ratio later/initial is

(A) 0.5 (B) 2

(C) 1.5 (D) 3

11–40 / New middle-to-hard practice

11. An adsorbent follows $q = kp^{1/n}$ in the stated range, with q in mg g^{-1}. Measurements are q=2 at p=1 and q=4 at p=16. At p=81, adsorbent mass needed to bind 18 mg gas is

(A) 6.00 g (B) 3.00 g

(C) 1.50 g (D) 4.50 g

12. For $q = kp^{\alpha}$, q is 3 mg g^{-1} at p=4 and 9 at p=36. At p=16, mass of adsorbent needed for 12.0 mg uptake is

(A) 1.00 g (B) 2.00 g

(C) 4.00 g (D) 3.00 g

13. At fixed T, equilibrium adsorption from solution obeys $q = \sqrt{C}$ g solute per g adsorbent, with C in g L^{-1}. Initially 1.00 L contains 1.00 g solute; 0.500 g adsorbent is added. Volume is unchanged. Equilibrium C is

(A) $(9 + \sqrt{17})/8$ g L^{-1} (B) $(9 - \sqrt{17})/8$ g L^{-1}

(C) 1/2 g L^{-1} (D) 3/4 g L^{-1}

14. An adsorption law is $q = K\sqrt{C}$, q in g g^{-1} and C in g L^{-1}. A 1.00 L solution initially at 0.750 g L^{-1} equilibrates at 0.250 after adding 1.00 g adsorbent. For an equal initial solution with 2.00 g adsorbent, final C is

(A) 0.250 g L^{-1} (B) $(11 + 4\sqrt{7})/4$ g L^{-1}

(C) 0.125 g L^{-1} (D) $(11 - 4\sqrt{7})/4$ g L^{-1}

15. Two adsorbents obey power laws in pressure. A has uptake 2 and 4 mg g^{-1} at p=1 and 16; B has uptake 1 and 4 at those pressures. At p=256, the mass ratio A:B giving equal gas uptake is

(A) 2 (B) 4

(C) $\frac{1}{2}$ (D) 1

16. At p=1 and 9, adsorbent A has q=1 and 3 mg g^{-1}; B has q=2 at both. Within this range both obey $q = kp^{\alpha}$. Pressure where their capacities per gram are equal is

(A) 2 (B) 4

(C) 6 (D) 3

17. An isotherm obeys $q = q_m Kp/(1+Kp)$. Measured q values are 2 at p=1 and 3 at p=3, in consistent units. At p=0.500, q is

(A) 1 (B) $\frac{4}{3}$

(C) $\frac{8}{3}$ (D) $\frac{2}{3}$

18. For $q = q_m Kp/(1+Kp)$, q=0.400 at p=1 and 0.640 at p=4. A treatment blocks half the adsorption sites without changing K. After treatment, q at p=2 is

(A) $\frac{8}{15}$ (B) $\frac{2}{5}$

(C) $\frac{1}{5}$ (D) $\frac{4}{15}$

19. A closed vessel contains 1.00 mol gas at a temperature with RT=1.00 L bar mol^{-1}. Free-gas volume is 1.00 L. Adsorption is $q = Kp$ mol per g adsorbent. With 1.00 g adsorbent equilibrium p is 0.250 bar. With 2.00 g and the same total gas amount, equilibrium p is

(A) 1/7 bar (B) 1/8 bar

(C) 1/4 bar (D) 1/5 bar

20. For a closed adsorption vessel, $N = pV/(RT) + mKp$, with N=1.00 mol and RT=1.00 L bar mol^{-1}. Unknown V and K are unchanged. At m=1.00 g, p=0.250 bar; at m=3.00 g, p=0.125 bar. At m=2.00 g, p is

(A) 1/8 bar (B) 1/6 bar

(C) 3/16 bar (D) 1/5 bar

21. AgI sol is made by mixing 10.0 mL 0.0200 M $AgNO_3$ with 30.0 mL 0.0100 M KI; the lattice ion remaining in excess is preferentially adsorbed. In an electric field its particles move toward which electrode, and among equal-molar NaCl, $BaCl_2$, $AlCl_3$ which is the most effective coagulant by counter-ion valency?

(A) Positive electrode; $AlCl_3$ (B) Negative electrode; $AlCl_3$

(C) Positive electrode; NaCl (D) Negative electrode; $BaCl_2$

22. AgI sols X and Y are prepared separately. X uses 20.0 mL 0.0200 M $AgNO_3$ and 10.0 mL 0.0100 M KI; Y uses 10.0 mL 0.0100 M $AgNO_3$ and 20.0 mL 0.0200 M KI. Excess lattice ions set particle charge. From KCl, $CaCl_2$, Na_3PO_4, the strongest equal-molar coagulants for X and Y, respectively, are

(A) Na_3PO_4; $CaCl_2$ (B) KCl; KCl

(C) $CaCl_2$; Na_3PO_4 (D) Na_3PO_4; Na_3PO_4

23. For a negative sol, assume critical counter-ion concentration follows $C_{crit}(z) = 0.100/z^6$ M. What volume of 0.0100 M $AlCl_3$ just coagulates 100 mL sol when dilution by the added solution is included? Treat only Al^{3+} as the coagulant.

(A) 100/729 mL (B) 1000/719 mL

(C) 1000/71 mL (D) 1000/729 mL

24. For a negative sol, let $C_{crit}(z) = 0.100/z^6$ M and assume mixture contributions add as $\sum c_i/C_{crit,i} = 1$ at coagulation. Final Na^+ and Ba^{2+} concentrations are 0.0300 and 0.000500 M. Required final Al^{3+} concentration is

(A) 19/364500 M (B) 19/72900 M

(C) 1/7290 M (D) 31/364500 M

25. A 100 mL sol is dialysed twice against fresh identical salt-free water baths. Colloid is fully retained; small salt equilibrates to equal concentration inside and outside, and bath removal leaves sol volume unchanged. Final salt amount is 1/9 of the original. Total fresh water used is

(A) 200 mL (B) 300 mL

(C) 400 mL (D) 800 mL

26. A 100 mL sol undergoes two complete dialysis equilibrations with fresh salt-free baths whose volumes total 400 mL. Salt freely diffuses; colloid and sol volume are retained. What is the greatest possible fraction of salt removed by optimally dividing the water?

(A) $\frac{8}{9}$ (B) $\frac{7}{8}$

(C) $\frac{3}{4}$ (D) $\frac{4}{5}$

27. A 1.00 L dispersion contains monodisperse neutral polymer and NaCl. Before dialysis, $\pi/(RT) = 0.0150$ mol L^{-1}; after complete salt removal at unchanged volume it is 0.00100. Polymer mass is 4.00 g. Treat NaCl as two ions and polymers as separate molecules. The pair (polymer molar mass, original NaCl concentration) is

(A) $(4000\ \mathrm{g\,mol^{-1}}, 0.00700\ \mathrm{M})$ (B) $(266.7\ \mathrm{g\,mol^{-1}}, 0.00700\ \mathrm{M})$

(C) $(2000\ \mathrm{g\,mol^{-1}}, 0.00700\ \mathrm{M})$ (D) $(4000\ \mathrm{g\,mol^{-1}}, 0.0140\ \mathrm{M})$

28. A neutral polymer–NaCl dispersion initially has $\pi/(RT) = 0.0100$ M. After complete dialysis, exactly half of the original polymer molecules associate into dimers; final $\pi/(RT) = 0.00150$ M at unchanged volume. NaCl is fully dissociated. Initial NaCl amount divided by initial polymer-molecule amount is

(A) 2 (B) $\frac{7}{3}$

(C) 1 (D) 4

29. Identical spherical colloid particles adsorb dye in proportion to total surface area. Seventy-five percent of the original particles coalesce into compact spheres, each made from eight originals; the rest remain single. Initial dye capacity was 2.00 mg. New capacity is

(A) 1.75 mg (B) 1.00 mg

(C) 0.6875 mg (D) 1.25 mg

30. Initially there are 288N equal spherical particles. After compact coalescence there are 17N clusters, each containing either 8 or 27 original particles; volume is conserved. Surface-controlled adsorption capacity after/before is

(A) $\frac{1}{2}$ (B) $\frac{3}{8}$

(C) $\frac{17}{288}$ (D) $\frac{9}{16}$

31. A catalyst rate is proportional to unblocked sites. Initially rate is 100 units. An irreversible poison dose of 0.0200 mmol lowers rate to 60; each poison molecule blocks two sites. A further 0.0150 mmol poison is added, all adsorbing on remaining sites. New rate is

(A) 15 (B) 30

(C) 45 (D) 20

32. A supplied inhibition model is $v = v_0/(1 + KC)$, where C is poison concentration. Rates are 40 at C=0.100 M and 20 at C=0.300 M. At C=0.200 M rate is

(A) $\frac{80}{3}$ (B) $\frac{40}{3}$

(C) 30 (D) 20

33. For a nonionic surfactant, free monomer concentration above CMC stays 0.00200 M. At total concentration 0.0200 M, measured total osmotic particle concentration is 0.00236 M. Assume every micelle has the same aggregation number and ideal particles. Aggregation number is

(A) 45 (B) 9

(C) 50 (D) 100

34. A nonionic micellar solution obeys $C_{particles} = C_0 + (C - C_0)/N$ above CMC C_0. Measured particle concentrations are 0.00220 M at total C=0.0120 M and 0.00240 M at C=0.0220 M. Pair (aggregation number N, CMC) is

(A) $(50, 0.00196\ \text{M})$ (B) $(100, 0.00210\ \text{M})$

(C) $(50, 0.00200\ \text{M})$ (D) $(25, 0.00200\ \text{M})$

35. Emulsion X can be diluted substantially with water without breaking and has a continuous blue background with a dye confined to water. Y has a continuous red background with a dye confined to oil. Which emulsifier assignment matches the standard tendency of sodium versus calcium oleate?

(A) Both: calcium oleate (B) X: sodium oleate; Y: calcium oleate

(C) X: calcium oleate; Y: sodium oleate (D) Both: sodium oleate

36. An emulsion containing aqueous NaCl initially conducts well. After a controlled phase inversion its conductivity drops sharply, microscopy shows aqueous droplets within continuous oil, and total chloride assay is unchanged. The joint interpretation best supported is

(A) Water changed from continuous to dispersed phase; most ions remain in isolated aqueous droplets.

(B) NaCl was removed by dialysis; oil remained the continuous phase throughout.

(C) Water remained continuous; the conductivity drop proves complete ion-pair formation.

(D) Chloride was adsorbed irreversibly on the container; water stayed continuous.

37. Gold numbers of protective colloids A and B are 0.0200 and 0.0500 mg per standard test. Assume their protective capacities add as $m_A/g_A + m_B/g_B$. A blend is 40% A and 60% B by mass. Minimum blend mass to protect ten standard test quantities is

(A) 0.200 mg (B) 0.380 mg

(C) 5/16 mg (D) 0.500 mg

38. A and B protect one standard gold-sol test with 0.0200 and 0.0500 mg respectively; capacities add. Costs are 3 and 1 units per mg. At most 0.300 mg B is available and A stock is ample. Minimum cost for ten tests is

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$

(C) $\frac{33}{50}$ (D) $\frac{27}{50}$

39. Initially 100 mL of a dye solution is 0.0100 M. After contacting charcoal and filtering, the unchanged-volume filtrate is 0.00400 M in intact dye. Washing the charcoal quantitatively recovers 0.000300 mol intact dye. No dye reacts or precipitates. Fraction of initial dye still retained on charcoal after washing is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$

(C) $\frac{3}{10}$ (D) $\frac{1}{2}$

40. Two colour-removal experiments use the same organic dye. With charcoal, filtrate organic carbon decreases and unchanged dye is recovered from the solid by solvent extraction. With a dissolved oxidant, colour disappears while organic carbon stays in the liquid and extraction yields no unchanged dye. The best joint assignment is

(A) Charcoal: bulk dilution only; oxidant: reversible indicator ionisation only.

(B) Both: intact dye removed from solution by the same physical adsorption process.

(C) Charcoal: chemical oxidation; oxidant: adsorption on a new solid.

(D) Charcoal: adsorption of intact dye; oxidant: chemical conversion of the chromophore.

41–50 / Preserved hard practice

41. An adsorbent removes a gaseous solute X from an ideal mixture. Initially vessel holds 0.100 mol total gas and 40% mole fraction X. After selectively adsorbing 0.0200 mol X at fixed T, V, final total pressure divided by initial pressure is

(A) 0.6 (B) 1.2

(C) 0.5 (D) 0.8

42. A 1.00 g adsorbent takes up 40.0 mg CO_2 and 20.0 mg CO at an equilibrium mixture pressure. Total adsorbed oxygen atom amount in mol is(CO_2 =44, CO =28)

(A) 0.00375 (B) 0.0016233766

(C) 0.0032467532 (D) 0.0025324675

43. A negative sol is coagulated by $AlCl_3$. If 0.0010 mol $AlCl_3$ is added and fully dissociates, moles of counter-ion charge equivalents added are

(A) 0.001 (B) 0.002

(C) 0.004 (D) 0.003

44. Adsorption of HA from a buffer removes 0.0200 mol HA without adsorbing A^-. Initially HA and A^- amounts are 0.100 mol each, pK_a 5.00. Using log 1.25=.09691, final buffer pH is

(A) 4.90309 (B) 5.09691

(C) 5 (D) 5.25

45. A closed gas vessel contains 0.100 mol A which adsorbs on to catalyst. Adsorption removes 20% from gas phase. If gas-phase A then undergoes 50% conversion by A→2 B without further adsorption, final total gas moles are

(A) 0.08 (B) 0.15

(C) 0.12 (D) 0.1

46. An adsorbent removes 0.0100 mol I_2 from an initial solution containing 0.0400 mol I_2. Remaining iodine is titrated by I_2 +2 $S_2O_3{}^{2-}$ →2 I^- + $S_4O_6{}^{2-}$. Required 0.100 M thiosulfate volume in mL is

(A) 800 (B) 300

(C) 200 (D) 600

47. For a provided two-site competitive adsorption model,θA= KA p A/(1+ KA p A+ KB p B). Take KA =2 bar^{-1}, KB =1 bar^{-1}, p A=1 bar and p B=1 bar. Fraction of sites covered by A is

(A) 0.66666667 (B) 0.5

(C) 0.25 (D) 0.75

48. Adsorption releases 30.0 kJ /mol and desorption returns the same amount. A calorimeter with heat capacity 2.00 kJ /K rises 0.300 K from adsorption only. If adsorbate M=40 g/mol, mass adsorbed in g is

(A) 2.4 (B) 0.8

(C) 0.4 (D) 1.2

49. A colloid particle has 1000 fixed negative charges. Complete charge neutralization using only Ca^{2+} requires how many calcium ions per particle?

(A) 500 (B) 2000

(C) 250 (D) 1000

50. A gas adsorbs according to q=kp^a, where q is mg/g and p is in bar. Measured loadings are q=200 at 1 bar and q=400 at 4 bar. A closed 24.0 L vessel at 300 K initially contains 2.00 mol gas and 220 g dry adsorbent. Gas molar mass is 44 g/mol and $R = 0.0800\,\mathrm{L\,bar\,mol^{-1}K^{-1}}$. With no other loss, the final equilibrium pressure in bar is

(A) 1 (B) 1.5

(C) 0.25 (D) 0.5

Answer key

Surface Chemistry

C-SUCH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**A**	31	**B**	41	**D**
02	**C**	12	**B**	22	**A**	32	**A**	42	**D**
03	**C**	13	**B**	23	**B**	33	**C**	43	**D**
04	**D**	14	**D**	24	**A**	34	**C**	44	**B**
05	**C**	15	**A**	25	**C**	35	**B**	45	**C**
06	**A**	16	**B**	26	**A**	36	**A**	46	**D**
07	**D**	17	**B**	27	**A**	37	**C**	47	**B**
08	**C**	18	**D**	28	**A**	38	**D**	48	**B**
09	**D**	19	**A**	29	**D**	39	**C**	49	**A**
10	**C**	20	**B**	30	**B**	40	**D**	50	**A**

Chapter 15

Hydrogen

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 3.40 g H_2O_2 sample decomposes completely. Oxygen volume at 22.4 L/mol, in L, is

(A) 2.24 (B) 0.56

(C) 3.4 (D) 1.12

02. A water sample contains 40 mg Ca^{2+} per litre. With Ca=40, $CaCO_3$ =100, hardness expressed as mg/L $CaCO_3$ is

(A) 40 (B) 100

(C) 200 (D) 80

03. A water sample contains 24 mg Mg^{2+} per litre. With Mg=24, $CaCO_3$ =100, hardness as mg/L $CaCO_3$ is

(A) 24 (B) 200

(C) 100 (D) 50

04. A mixture contains H_2 and D_2 with mean molar mass 2.50 g/mol. Mole fraction of D_2 is

(A) 0.5 (B) 0.25

(C) 0.75 (D) 0.125

05. A water sample has 100 mg/L hardness as $CaCO_3$. In 2.00 L, moles of divalent hardness cations are

(A) 0.02 (B) 0.004

(C) 0.002 (D) 0.001

06. A 10.0 mL H_2O_2 solution needs 20.0 mL 0.0200 M $KMnO_4$ in acid. H_2O_2 molarity is

(A) 0.1 (B) 0.2

(C) 0.02 (D) 0.04

07. A peroxide solution produces 5.60 L oxygen per litre solution on complete decomposition at 22.4 L/mol. Its H_2O_2 molarity is

(A) 1 (B) 0.5

(C) 2 (D) 0.25

08. A 1.00 g sample contains 50% NaH by mass. Excess water liberates hydrogen in mol of(NaH =24)

(A) 0.083333333 (B) 0.020833333

(C) 0.010416667 (D) 0.041666667

09. A 1.00 mol H_2 and 1.00 mol D 2 mixture undergoes complete isotopic recombination to random paired atoms. Assuming statistical pairing only, expected HD amount in mol is

(A) 1 (B) 0.5

(C) 1.5 (D) 2

10. A 0.0500 mol CaH_2 sample hydrolyzes completely. Hydrogen is then burned with oxygen. Water formed in mol is

(A) 0.025 (B) 0.05

(C) 0.2 (D) 0.1

11–40 / New middle-to-hard practice

11. A 100 mL sample contains H_2O_2 and methanal. Selective catalytic decomposition of peroxide gives 0.0100 mol O_2. A separate equal sample releases 0.0800 mol electrons on complete oxidation of peroxide to O_2 and methanal to CO_2. Initial methanal amount is

(A) 0.0100 mol (B) 0.00500 mol

(C) 0.0200 mol (D) 0.0300 mol

12. A mixture contains H_2O_2 and ethanal. Selective complete decomposition of H_2O_2 produces 224 mL dry O_2 at STP. In a separate equal sample, oxidation to O_2 and ethanoic acid consumes 0.0120 mol acidic permanganate. Initial ethanal amount is (22.4 L mol^{-1})

(A) 0.00500 mol (B) 0.0300 mol

(C) 0.0100 mol (D) 0.0200 mol

13. A 10.0 mL peroxide solution is assayed by complete decomposition and yields 112 mL O_2 at STP. A 40.0 mL portion of the same solution is decomposed; its oxygen reacts with 0.0500 mol SO_2 by $2SO_2 + O_2 \rightarrow 2SO_3$. At full conversion, SO_2 remaining is (22.4 L mol^{-1})

(A) 0.0100 mol (B) 0

(C) 0.0200 mol (D) 0.0300 mol

14. One peroxide sample releases 0.0300 mol O_2 on decomposition. Only 80% of that gas is recovered and mixed with 0.0500 mol H_2. After ignition, water vapour is condensed. Dry oxygen remaining is

(A) $\frac{1}{200}$ (B) $\frac{1}{1000}$

(C) $\frac{3}{125}$ (D) 0

15. A 0.0500 mol H_2O_2 portion first reduces 0.00400 mol acidic permanganate, forming O_2. The remaining peroxide oxidises excess I^- to I_2 in acid. I_2 formed is

(A) 0.0400 mol (B) 0.0300 mol

(C) 0.0100 mol (D) 0.0200 mol

16. A peroxide sample is split into two equal portions. One reduces 0.00600 mol acidic permanganate completely. The other oxidises excess iodide; all liberated iodine is isolated and titrated. Thiosulfate needed for that iodine is

(A) 0.0300 mol (B) 0.0150 mol

(C) 0.0600 mol (D) 0.0120 mol

17. Water contains $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ only, totalling 0.0100 mol divalent ions. Complete boiling gives 0.0160 mol CO_2; use Ca bicarbonate$\rightarrow CaCO_3+CO_2+H_2O$ and Mg bicarbonate$\rightarrow Mg(OH)_2+2CO_2$. $CaCO_3$ mass formed is ($CaCO_3$=100)

(A) 0.800 g (B) 1.00 g

(C) 0.400 g (D) 0.600 g

18. A water sample contains 0.0100 mol total Ca^{2+} as $Ca(HCO_3)_2$ and $CaSO_4$ only. Complete boiling releases 0.00400 mol CO_2 and the precipitate is removed. Na_2CO_3 mass needed to remove the remaining calcium is (M=106)

(A) 0.636 g (B) 0.424 g

(C) 1.060 g (D) 0.848 g

19. A sample contains $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ with 0.0100 mol metal ions combined. Complete lime treatment needs 0.0160 mol $Ca(OH)_2$; one and two mol lime are needed per Ca and Mg bicarbonate respectively. $Mg(OH)_2$ sludge mass is (M=58)

(A) 0.174 g (B) 0.580 g

(C) 0.348 g (D) 0.232 g

20. A water sample has 0.0100 mol Ca^{2+} plus Mg^{2+}, only as chlorides. It requires 0.00300 mol $Ca(OH)_2$ to precipitate all Mg as $Mg(OH)_2$. Subsequent soda removes every Ca ion, including calcium added by lime. With $CaCO_3$=100 and $Mg(OH)_2$=58, total dry sludge is

(A) 1.174 g (B) 1.300 g

(C) 0.874 g (D) 1.000 g

21. Feed water contains only $CaCl_2$ and $MgSO_4$. Total metal concentration is 0.00400 M; chloride is 0.00600 M. A pretreatment removes half the calcium and none of the magnesium. An H^+-form exchanger then replaces all remaining metal ions and has 0.0600 mol charge-equivalent capacity. Feed volume treatable is

(A) 12.0 L (B) 7.50 L

(C) 15.0 L (D) 10.0 L

22. One litre of feed contains 0.0150 mol total NaCl and $CaCl_2$, with total salt mass 1.140 g (M=58.5,111). A selective resin binds Ca^{2+} only and has 0.0300 mol charge-equivalent capacity. Maximum litres of this feed treatable before exhaustion is

(A) 6.00 L (B) 1.00 L

(C) 3.00 L (D) 2.00 L

23. A CaH_2–CaC_2 mixture gives 0.100 mol total gas on complete hydrolysis. Complete combustion of this gas consumes 0.110 mol O_2; gases are only H_2 and C_2H_2. Initial CaH_2 amount is

(A) 0.0150 mol (B) 0.0700 mol

(C) 0.0350 mol (D) 0.0300 mol

24. A NaH–CaC_2 mixture hydrolyses to 0.0800 mol total H_2 and C_2H_2. Combustion of the gas gives 0.0600 mol CO_2. Initial NaH amount is

(A) 0.0300 mol (B) 0.0500 mol

(C) 0.100 mol (D) 0.0200 mol

25. A 2.25 g mixture of NaH and CaH_2 gives 0.100 mol H_2 on complete hydrolysis. With molar masses 24 and 42, initial CaH_2 mole fraction is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

26. A 3.30 g mixture of NaH and CaH_2 produces 0.150 mol H_2 on hydrolysis. Molar masses are 24 and 42. After complete hydrolysis, carbonate quantitatively precipitates the calcium only. $CaCO_3$ mass is (M=100)

(A) 2.50 g (B) 10.0 g

(C) 5.00 g (D) 7.50 g

27. A gas mixture contains B_2H_6 and CH_4, total 0.100 mol. On selective complete hydrolysis only diborane reacts, releasing 0.240 mol H_2 by $B_2H_6 + 6H_2O \rightarrow 2B(OH)_3 + 6H_2$. Mean molar mass of the original gas is (28,16)

(A) 18.4 $g\,mol^{-1}$ (B) 22.0 $g\,mol^{-1}$

(C) 20.8 $g\,mol^{-1}$ (D) 23.2 $g\,mol^{-1}$

28. Two separate bulbs hold B_2H_6 and NH_3, with no contact between the gases. They contain 0.100 mol molecules combined. Hydrolysis of the diborane bulb gives 0.180 mol H_2; ammonia from the other bulb is recovered unchanged. With B=11, H=1, N=14, recovered NH_3 mass is

(A) 1.19 g (B) 0.510 g

(C) 1.70 g (D) 0.850 g

29. H and D atoms pair randomly into H_2, HD and D_2. Measured HD mole fraction is 0.480 and H atoms are more abundant than D. With isotopic masses 1 and 2, mean molecular mass is

(A) 3 (B) $\frac{14}{5}$

(C) $\frac{12}{5}$ (D) $\frac{16}{5}$

30. Random pairing of H, D gives mean gas molecular mass 3.50 with isotope masses 1,2. A 1.00 mol gas sample is separated into H_2, HD and D_2. Amount of HD is

(A) 0.750 mol (B) 0.250 mol

(C) 0.375 mol (D) 0.500 mol

31. A closed liquid initially contains 1.00 mol H_2O and 3.00 mol D_2O. Complete rapid isotope exchange randomises H and D over the two equivalent water hydrogen sites. Expected HOD amount is

(A) 2.00 mol (B) 1.00 mol

(C) 3.00 mol (D) 1.50 mol

32. A 4.00 mol water sample contains H_2O, HOD, D_2O with completely random isotope placement. HOD amount is 1.50 mol and total D atoms exceed total H atoms. With O=16, H=1, D=2, sample mass is

(A) 74.0 g (B) 76.0 g

(C) 80.0 g (D) 78.0 g

33. Ortho and para hydrogen retain the same chemical formula but differ in nuclear-spin arrangement. For a supplied conversion model, each mol ortho converted to para releases Q. A 2.00 mol sample changes ortho mole fraction from 0.750 to 0.250, releasing 120 J. Heat released when a 3.00 mol sample changes from 0.600 to 0.200 is

(A) 144 J (B) 120 J

(C) 180 J (D) 72 J

34. A hydrogen sample initially has ortho fraction 0.750. A catalyst permits spin conversion without dissociation, and heat release is 45.0 J per mol total hydrogen. Calibrated release is 120 J per mol ortho converted to para. Final ortho:para molecular ratio is

(A) 1 (B) $\frac{3}{5}$

(C) $\frac{1}{3}$ (D) $\frac{5}{3}$

35. Quantitative electrolysis of a water-isotope sample gives 0.0500 mol O_2 and 0.300 g total dry $H_2/HD/D_2$ gas. All hydrogen isotopes enter that gas. With H=1, D=2, fraction of hydrogen-isotope atoms that are D is

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

36. A water sample yields 0.100 mol O_2 and 0.500 g total hydrogen-isotope gas on complete electrolysis. Assume all H, D are recovered; isotopic masses are 1,2. If those isotopes are randomly recombined into the original 0.200 mol water molecules, expected HOD amount is

(A) 0.100 mol (B) 0.150 mol

(C) 0.0500 mol (D) 0.0750 mol

37. A 0.0600 mol H_2O_2 batch is consumed through two controlled pathways: reduction to water and oxidation to O_2. Its net electron uptake, with donation counted negative, is 0.0400 mol. No oxygen-atom exchange occurs. Fraction of the original peroxide oxygen atoms ending in water is

(A) $\frac{3}{4}$ (B) $\frac{2}{3}$

(C) $\frac{1}{3}$ (D) $\frac{1}{2}$

38. Isotope-labelled peroxide is consumed by two controlled redox pathways only: oxidation to O_2 and reduction to water. No oxygen exchange occurs. Isotopic assay finds 0.0300 mol O_2 of peroxide origin and 0.0800 mol peroxide-origin oxygen atoms in water. Net electron amount accepted by peroxide, donation counted negative, is

(A) −0.0200 mol (B) 0.0200 mol

(C) 0.0400 mol (D) 0.0600 mol

39. Dry H_2 and O_2 from two hydrogen-related preparations have total amount 0.100 mol and mass 2.00 g. They are ignited quantitatively, then water is condensed. Dry gas amount remaining is

(A) 0.0300 mol (B) 0

(C) 0.0200 mol (D) 0.0400 mol

40. A dry H_2–O_2 mixture has mean molar mass 12.0 and total amount 0.150 mol. It is fully ignited and water is condensed. Mass of water formed is (H=1, O=16)

(A) 0.900 g (B) 2.70 g

(C) 3.60 g (D) 1.80 g

41–50 / Preserved hard practice

41. A 2.40 g mixture of NaH and CaH_2 reacts with water and liberates 0.110 mol H_2. Taking NaH =24, CaH_2 =42, mass fraction CaH_2 is

(A) 0.25 (B) 0.7

(C) 0.75 (D) 0.5

42. A 50.0 mL water sample contains Ca^{2+} and Mg^{2+} only as hardness ions. EDTA assay finds total divalent cations 0.100 mmol. Calcium alone precipitates as 0.00600 g $CaCO_3$. Magnesium amount in mmol is

(A) 0.04 (B) 0.06

(C) 0.02 (D) 0.1

43. A 25.0 mL peroxide solution is added to 0.0100 mol acidic MnO_4^-. Excess MnO_4^- is back titrated with 0.0200 mol Fe^{2+}. H_2O_2 concentration in M is

(A) 1 (B) 0.3

(C) 0.6 (D) 0.4

44. Separate samples of pure H_2 and pure D_2 have a combined amount of 0.100 mol and combined mass 0.300 g. Each is burned separately with oxygen. The mass fraction of D_2O in the combined collected water is

(A) 0.52631579 (B) 0.9

(C) 0.5 (D) 0.95

45. A 3.40 g H_2O_2 sample is 40% decomposed before analysis. Remaining peroxide is quantitatively oxidized by MnO_4^- in acid. Required permanganate in mol is

(A) 0.06 (B) 0.012

(C) 0.04 (D) 0.024

46. A water sample contains 0.0010 M $Ca(HCO_3)_2$ and 0.0020 M $MgSO_4$. After complete boiling removes temporary hardness alone, residual hardness as mg/L $CaCO_3$ is

(A) 100 (B) 200

(C) 400 (D) 300

47. Hydrogen formed from 0.0500 mol CaH_2 is collected over water at 300 K and 1.00 bar. Water vapour pressure is 0.040 bar. With R=.0800 L bar/mol/K, wet gas volume in L is

(A) 1.25 (B) 2.5

(C) 2.4 (D) 5

48. A 0.200 mol CO sample undergoes water-gas shift $CO + H_2O \rightleftharpoons CO_2 + H_2$. Equilibrium contains 0.0500 mol CO. The H_2 is used in a voltaic fuel cell with 2 electrons per H_2. Maximum charge in Faradays is

(A) 0.4 (B) 0.15

(C) 0.3 (D) 0.1

49. A peroxide solution contains 0.200 mol H_2O_2. First 25% decomposes, and then half the remaining peroxide oxidizes I^- to I_2 in acid. Moles of I_2 formed are

(A) 0.05 (B) 0.075

(C) 0.025 (D) 0.15

50. Two separately sealed samples, one pure H_2O and the other pure D_2O, have total amount 0.100 mol and total mass 1.90 g. Each reacts separately and quantitatively with Na. Moles of D 2 produced are

(A) 0.05 (B) 0.0125

(C) 0.075 (D) 0.025

Answer key

Hydrogen

C-HYGN / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**A**	21	**A**	31	**D**	41	**B**
02	**B**	12	**C**	22	**C**	32	**D**	42	**A**
03	**C**	13	**A**	23	**C**	33	**A**	43	**C**
04	**B**	14	**D**	24	**B**	34	**B**	44	**A**
05	**C**	15	**A**	25	**D**	35	**D**	45	**D**
06	**A**	16	**A**	26	**C**	36	**D**	46	**B**
07	**B**	17	**C**	27	**C**	37	**B**	47	**B**
08	**B**	18	**A**	28	**A**	38	**B**	48	**C**
09	**A**	19	**C**	29	**B**	39	**D**	49	**B**
10	**D**	20	**A**	30	**C**	40	**D**	50	**D**

Chapter 16

s-Block

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. In 4 KO_2 +2 CO_2 →2 K_2CO_3 +3 O_2, oxygen amount from 0.20 mol KO_2 in mol is

(A) 0.2 (B) 0.15

(C) 0.3 (D) 0.1

02. One mole gypsum converts quantitatively to one mole plaster. Water lost in mol is

(A) 1 (B) 0.5

(C) 2 (D) 1.5

03. $CaCO_3$ has molar mass 100 and CaO =56 g/mol. Complete calcination of 25.0 g pure $CaCO_3$ gives CaO in g of

(A) 28 (B) 14

(C) 11 (D) 25

04. $CaO + H_2O \rightarrow Ca(OH)_2$. Mass of slaked lime from 5.60 g pure CaO in g is(CaO =56, $Ca(OH)_2$ 74)

(A) 14.8 (B) 7.4

(C) 9.2 (D) 5.6

05. Na_2CO_3 +2 HCl →2 NaCl + CO_2 + H_2O. Neutralizing 0.0500 mol Na_2CO_3 requires 0.100 M HCl in volume L of

(A) 1 (B) 0.25

(C) 0.5 (D) 2

06. A carbonate with formula M_2CO_3 has molar mass 138 g/mol. Atomic mass of M is u(C=12, O=16)

(A) 69 (B) 23

(C) 39 (D) 40

07. A Group 2 carbonate MCO_3 weighs 10.0 g and gives 5.60 g oxide on complete heating. Atomic mass of M is u

(A) 40 (B) 88

(C) 56 (D) 24

08. On heating 1 mol $Ca(NO_3)_2$ completely, CaO +2 NO_2 +0.5 O_2 form. Total gas moles produced are

(A) 2 (B) 3

(C) 2.5 (D) 1.5

09. In CaC_2 +2 $H_2O \rightarrow Ca(OH)_2 + C_2H_2$, 0.200 mol CaC_2 gives H_2O consumption in mol of

(A) 0.8 (B) 0.4

(C) 0.1 (D) 0.2

10. If 0.100 mol Ba^{2+} is precipitated quantitatively as $BaSO_4$, mass of precipitate in g is(Ba=137, S=32, O=16)

(A) 13.7 (B) 46.6

(C) 9.6 (D) 23.3

11–40 / New middle-to-hard practice

11. A Na_2CO_3–$NaHCO_3$ sample releases 0.0800 mol CO_2 with excess acid, consuming 0.120 mol H^+. On heating an equal sample to convert bicarbonate into carbonate, CO_2 released is

(A) 0.0800 mol (B) 0.0200 mol

(C) 0.0600 mol (D) 0.0400 mol

12. A mixture of Na_2CO_3 and $NaHCO_3$ gives 0.0150 mol CO_2 on heating; an equal fresh sample gives 0.0700 mol CO_2 with excess acid. Total H^+ required for the fresh sample is

(A) 0.110 mol (B) 0.100 mol

(C) 0.0700 mol (D) 0.140 mol

13. A 0.100 mol formula-unit mixture of $NaNO_3$ and $Mg(NO_3)_2$ is heated completely. Sodium nitrate forms nitrite and O_2; magnesium nitrate forms MgO, NO_2, O_2. Measured NO_2 is 0.160 mol. Total solid residue mass is (MgO=40, $NaNO_2$=69)

(A) 3.20 g (B) 5.96 g

(C) 4.58 g (D) 6.90 g

14. A 0.100 mol formula-unit mixture of $LiNO_3$ and KNO_3 gives 0.0400 mol O_2 on complete heating. Lithium nitrate forms Li_2O, NO_2, O_2; potassium nitrate forms KNO_2 and O_2. With Li=7, K=39, N=14, O=16, total residue mass is

(A) 7.50 g (B) 4.80 g

(C) 5.70 g (D) 6.60 g

15. A Na_2O_2–KO_2 scrubber absorbs 0.0600 mol CO_2 and releases 0.0500 mol O_2. Use one CO_2 and half O_2 per Na_2O_2, and half CO_2 and three-quarter O_2 per KO_2. Fraction of released oxygen coming from KO_2 is

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

16. An 11.0 g dry Na_2O_2–KO_2 mixture completely absorbs 0.100 mol CO_2, forming normal carbonates. Use molar masses 78,71 and $2Na_2O_2 + 2CO_2 \rightarrow 2Na_2CO_3 + O_2$, $4KO_2 + 2CO_2 \rightarrow 2K_2CO_3 + 3O_2$. Oxygen released is

(A) 0.0500 mol (B) 0.100 mol

(C) 0.125 mol (D) 0.0750 mol

17. A Li–Na metal sample contains 0.100 mol metal atoms. Under controlled conditions only Li reacts with nitrogen; uptake is 0.00500 mol N_2, forming Li_3N. On adding water afterwards, all unreacted metal makes H_2 and the nitride makes NH_3. H_2 amount is

(A) 0.0150 mol (B) 0.0700 mol

(C) 0.0500 mol (D) 0.0350 mol

18. One Li–Na sample evolves 0.0500 mol H_2 with excess water. An equal fresh sample takes up0.0100 mol N_2 through lithium alone, forming Li_3N. With Li=7, Na=23, initial lithium mass fraction is

(A) $\frac{2}{5}$ (B) $\frac{7}{30}$

(C) $\frac{3}{5}$ (D) $\frac{21}{67}$

19. A Na–K metal mixture contains 0.100 mol atoms. Complete controlled oxidation gives Na_2O_2 and KO_2 only, incorporating 0.150 mol oxygen atoms. On initial hydrolysis, use $2KO_2 + 2H_2O \rightarrow 2KOH + H_2O_2 + O_2$; sodium peroxide gives H_2O_2 but no initial O_2. Initial O_2 formed is

(A) 0.0500 mol (B) 0.0250 mol

(C) 0.0750 mol (D) 0.0125 mol

20. A 6.20 g Na–K metal mixture is converted only into Na_2O_2 and KO_2, incorporating 0.300 mol oxygen atoms. All products are then converted quantitatively to normal carbonates. With Na=23, K=39, C=12, O=16, final dry carbonate mass is

(A) 12.2 g (B) 10.6 g

(C) 11.0 g (D) 13.8 g

21. A mixture of $MgCO_3$, $CaCO_3$, $BaCO_3$ gives 0.0900 mol CO_2 on complete decomposition. In controlled staged heating only Mg carbonate first decomposes, releasing 0.0200 mol CO_2; after calcium also decomposes cumulative gas is 0.0500 mol. With MgO=40, CaO=56, $BaCO_3$=197, solid mass after the calcium stage is

(A) 10.36 g (B) 8.60 g

(C) 7.88 g (D) 12.56 g

22. A 10.30 g $MgCO_3$–$CaCO_3$–$BaCO_3$ sample gives 0.0900 mol total CO_2 with acid. Controlled mild heating decomposes only $MgCO_3$, giving 0.0400 mol CO_2. Molar masses are 84,100,197. Initial $BaCO_3$ mass is

(A) 3.00 g (B) 1.97 g

(C) 5.91 g (D) 3.94 g

23. A $Be(OH)_2$–$Mg(OH)_2$ mixture consumes 0.100 mol HCl completely. An equal sample dissolves selectively in excess NaOH;0.0600 mol OH^- is consumed by $Be(OH)_2 + 2OH^- \rightarrow [Be(OH)_4]^{2-}$, while Mg hydroxide stays solid. With $Mg(OH)_2$=58, insoluble solid mass is

(A) 1.74 g (B) 2.90 g

(C) 0.580 g (D) 1.16 g

24. A mixture of BeO and Al_2O_3 consumes 0.120 mol H^+ completely. An equal sample consumes 0.0600 mol OH^- on complete conversion to $[Be(OH)_4]^{2-}$ and $[Al(OH)_4]^-$, respectively. Initial BeO formula-unit mole fraction is

(A) $\frac{3}{4}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{1}{2}$

25. A 2.46 g $MgSO_4 \cdot nH_2O$ sample gives 2.33 g $BaSO_4$ on sulfate assay. Partial controlled heating leaves 1.56 g $MgSO_4 \cdot mH_2O$. With $MgSO_4$=120, $BaSO_4$=233, H_2O=18, the pair (n, m) is

(A) $(5, 1)$ (B) $(6, 2)$

(C) $(7, 2)$ (D) $(7, 3)$

26. A 2.42 g mixture contains $CaCl_2 \cdot 2H_2O$ and anhydrous $MgCl_2$ only. Heating removes 0.360 g water without chloride loss. Molar masses are 147 and 95. Fraction of initial chloride ions belonging to calcium salt is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{3}{5}$ (D) $\frac{2}{3}$

27. CO_2 is absorbed completely by 0.100 mol $Ca(OH)_2$. At the end 0.0400 mol $CaCO_3$ remains as solid; no OH^- remains and all remaining dissolved calcium is bicarbonate. CO_2 absorbed at STP is (22.4 L mol^{-1})

(A) 2.240 L (B) 0.896 L

(C) 3.584 L (D) 4.480 L

28. After CO_2 absorption in limewater,4.00 g $CaCO_3$ is filtered off. Boiling the filtrate releases 0.0400 mol CO_2 and deposits only $CaCO_3$; no free hydroxide or dissolved CO_2 had remained. With $CaCO_3$=100, total CO_2 originally absorbed is

(A) 0.0800 mol (B) 0.120 mol

(C) 0.0400 mol (D) 0.160 mol

29. A CaC_2–Ca_3N_2 sample hydrolyses completely, and all C_2H_2 plus NH_3 is collected quantitatively: total gas amount 0.100 mol. An acid trap absorbs 0.0400 mol NH_3. After all calcium is converted to $CaCO_3$, mass obtained is (M=100)

(A) 10.0 g (B) 12.0 g

(C) 16.0 g (D) 8.00 g

30. A 4.48 g $CaCN_2$–CaC_2 sample is hydrolysed completely. Use $CaCN_2 + 3H_2O \rightarrow CaCO_3 + 2NH_3$ and the usual carbide hydrolysis. An acid trap measures 0.0800 mol NH_3. With $CaCN_2$=80, CaC_2=64, $CaCO_3$=100, mass after every calcium species is finally converted to carbonate is

(A) 8.00 g (B) 2.00 g

(C) 6.00 g (D) 4.00 g

31. A 5.115 g solid contains only $NaHCO_3$ and NaCl. Complete controlled heating gives a 3.875 g residue of Na_2CO_3 and unchanged NaCl. With molar masses 84,58.5,106, mass fraction of carbonate in the residue is

(A) $\frac{424}{775}$ (B) $\frac{4}{5}$

(C) $\frac{351}{775}$ (D) $\frac{424}{1023}$

32. An idealised Solvay precipitation uses $NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$. Feed has 0.100 mol NaCl,0.0800 mol NH_3,0.0600 mol CO_2 and ample water. Isolated $NaHCO_3$ is 80% of theoretical; heating converts it completely to Na_2CO_3. With carbonate M=106, product mass is

(A) 5.088 g (B) 4.240 g

(C) 2.544 g (D) 3.180 g

33. A 2.81 g powder contains $CaSO_4 \cdot \frac{1}{2}H_2O$ and anhydrous $CaSO_4$ only. Full setting to gypsum consumes 0.630 g water. With plaster M=145, anhydrite M=136, initial plaster mass fraction is

(A) $\frac{3}{5}$ (B) $\frac{136}{281}$

(C) $\frac{1}{2}$ (D) $\frac{145}{281}$

34. A 4.89 g gypsum–plaster mixture contains 0.0300 mol sulfate by an independent sulfate assay. With gypsum M=172 and plaster M=145, gypsum formula-unit mole fraction is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

35. Element X is an alkali metal whose nitrate gives oxide rather than nitrite on heating. Element Y is the period-3 alkaline-earth metal. Which joint prediction is correct for the neutral metals and their carbonates?

(A) Both metals can form nitrides directly; both carbonates decompose to oxides on sufficient heating.

(B) X is sodium and Y is magnesium; both sulfates are highly insoluble.

(C) Both give nitrite as the final solid on nitrate heating; both carbonates resist heating.

(D) Only Y forms a nitride directly; only X carbonate decomposes.

36. X^{3+} is Ne-like and Y^{2+} is He-like, with X in period 3 and Y in period 2. Select the correct joint statement for their neutral oxides and hydroxides.

(A) X forms X_2O_3 and Y forms YO; both hydroxides are basic but never amphoteric.

(B) X forms XO_2 and Y forms YO; the hydroxides dissolve only in acid.

(C) X forms X_2O_3 and Y forms YO; both hydroxides dissolve in acid and sufficiently strong aqueous base.

(D) X forms XO and Y forms Y_2O_3; only Y hydroxide dissolves in acid.

37. Unlabelled alkali metals A and B are identified as follows. A nitrate gives oxide, NO_2, O_2 on heating. B nitrate gives nitrite and O_2; B gives a lilac flame through cobalt glass. For equal 0.0100 mol nitrate samples heated completely together, total residue mass is (Li=7, K=39, N=14, O=16)

(A) 1.15 g (B) 0.840 g

(C) 1.00 g (D) 0.990 g

38. A 2.36 g alkaline-earth nitrate hydrate $M(NO_3)_2 \cdot xH_2O$ gives 0.0200 mol NO_2 and 0.560 g MO on complete heating. No other solid forms. With N=14, O=16, H=1, hydration number x is

(A) 4 (B) 6

(C) 8 (D) 2

39. Initially 0.100 mol Na dissolves in liquid ammonia, giving one solvated electron per Na. During controlled ageing, part reacts by $2Na + 2NH_3 \rightarrow 2NaNH_2 + H_2$, producing 0.0300 mol H_2. No other electron sink occurs. Fraction of the initial solvated electrons remaining is

(A) $\frac{2}{5}$ (B) $\frac{1}{5}$

(C) $\frac{3}{5}$ (D) $\frac{7}{10}$

40. A 1.70 g Na–K sample dissolves in liquid ammonia. Ageing gives 0.0200 mol H_2 through amide formation; subsequent complete quenching of remaining solvated electrons with water gives 0.0100 mol H_2. Each H_2 in either step consumes two originally supplied electrons. With Na=23, K=39, initial K mole fraction is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

41–50 / Preserved hard practice

41. A 5.00 g mixture of Na_2CO_3 and $NaHCO_3$ liberates 0.0500 mol CO_2 with acid excess. Molar masses are 106 and 84. Na_2CO_3 mass fraction is

(A) 0.6 (B) 0.5

(C) 0.152 (D) 0.77090909

42. A mixture of 0.100 mol $LiNO_3$ and 0.100 mol $NaNO_3$ is completely heated under the usual decomposition rules. Total O_2 formed in mol is

(A) 0.15 (B) 0.1

(C) 0.075 (D) 0.05

43. A gypsum plaster mixture contains 0.100 mol $CaSO_4 \cdot 2H_2O$ and 0.100 mol $CaSO_4 \cdot 0.5H_2O$. Complete dehydration releases water in g of

(A) 1.8 (B) 3.6

(C) 5.4 (D) 4.5

44. A 0.100 mol $Ca(OH)_2$ solution receives 0.150 mol CO_2. Assume first $CaCO_3$ precipitates and then excess CO_2 dissolves it quantitatively to bicarbonate. Final $CaCO_3$ precipitate amount in mol is

(A) 0.1 (B) 0.05

(C) 0 (D) 0.025

45. A sodium-peroxide sample contains 0.100 mol Na_2O_2. It reacts with water to give H_2O_2 quantitatively. The H_2O_2 then fully decomposes. O 2 volume at 22.4 L/mol, in L, is

(A) 4.48 (B) 0.56

(C) 1.12 (D) 2.24

46. A 0.100 mol mixture of Li_3N and Mg_3N_2 gives 0.150 mol NH_3 on complete hydrolysis. Moles of Li_3N were

(A) 0.1 (B) 0.05

(C) 0.025 (D) 0.075

47. A Mg/Ca alloy has mass 3.20 g and liberates 0.100 mol H_2 with excess acid. Taking Mg=24, Ca=40, mass fraction Ca is

(A) 0.5 (B) 0.75

(C) 0.375 (D) 0.625

48. A 0.200 mol CaC_2 sample hydrolyzes completely to acetylene. The acetylene is fully hydrogenated to ethane. Hydrogen needed in mol is

(A) 0.4 (B) 0.1

(C) 0.2 (D) 0.6

49. A water sample contains total hardness 300 mg/L as $CaCO_3$, of which 100 mg/L is permanent. Ideal boiling removes all temporary hardness. In 5.00 L, remaining divalent cation amount in mmol is

(A) 15 (B) 5

(C) 1 (D) 10

50. A 0.100 mol $BaCl_2$ sample is mixed with 0.150 mol Na_2SO_4 in sufficient water to dissolve both salts, then $BaSO_4$ precipitates completely. Total moles of dissolved ions remaining are

(A) 0.6 (B) 0.55

(C) 0.3 (D) 0.4

Answer key

s-Block

C-SBLK / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**A**	31	**A**	41	**D**
02	**D**	12	**A**	22	**D**	32	**C**	42	**C**
03	**B**	13	**C**	23	**D**	33	**D**	43	**D**
04	**B**	14	**C**	24	**D**	34	**D**	44	**B**
05	**A**	15	**A**	25	**C**	35	**A**	45	**C**
06	**C**	16	**B**	26	**A**	36	**C**	46	**B**
07	**A**	17	**D**	27	**C**	37	**C**	47	**D**
08	**C**	18	**D**	28	**B**	38	**A**	48	**A**
09	**B**	19	**B**	29	**B**	39	**A**	49	**B**
10	**D**	20	**A**	30	**C**	40	**A**	50	**B**

Chapter 17

p-Block

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. In SiO_2 network, each oxygen bridges 2 Si and each Si has 4 O neighbours. Ratio of total Si-O contacts to Si atoms is

(A) 2 (B) 1

(C) 4 (D) 8

02. P_4 has tetrahedral P-P connectivity. Number of P-P edges per molecule is

(A) 12 (B) 4

(C) 6 (D) 3

03. The number of replaceable OH hydrogens in H_3PO_3 is

(A) 3 (B) 2

(C) 1 (D) 4

04. The number of direct P-Hbonds in H_3PO_2 is

(A) 2 (B) 1

(C) 3 (D) 0

05. In alkaline disproportionation $3\ Cl_2 + 6\ OH^- \rightarrow 5\ Cl^- + ClO_3^- + 3\ H_2O$, moles of ClO_3^- from 0.300 mol Cl 2 are

(A) 0.15 (B) 0.3

(C) 0.1 (D) 0.05

06. Total number of valence electrons in XeF_4 is

(A) 32 (B) 40

(C) 28 (D) 36

07. Complete oxidation of 1 mol H_3PO_3 to H_3PO_4 transfers electrons in mol of

(A) 1 (B) 2

(C) 4 (D) 3

08. Complete oxidation of 1 mol H_3PO_2 to H_3PO_4 transfers electrons in mol of

(A) 1 (B) 2

(C) 6 (D) 4

09. A zeolite exchanges 2 Na^+ for one Ca^{2+}. Removing 0.0100 mol Ca^{2+} releases Na^+ in mol of

(A) 0.01 (B) 0.02

(C) 0.005 (D) 0.04

10. Ozone oxidizes I^- to I_2 by O_3 +2 I^- +2 H^+ → O_2 + I_2 + H_2O. One mole O 3 accepts electrons in mol of

(A) 3 (B) 6

(C) 2 (D) 1

11–40 / New middle-to-hard practice

11. A mixture contains $Na_2B_4O_7$ and H_3BO_3 only. Independent elemental assays give 0.0600 mol B and 0.0200 mol Na. With H_3BO_3=62, mass of initially free boric acid is

(A) 1.24 g (B) 3.72 g

(C) 0.620 g (D) 2.48 g

12. A borax–boric-acid sample has 0.0600 mol total B. Selective validated borax assay consumes 0.0200 mol H^+, two per $Na_2B_4O_7$. In an equal sample the initially free H_3BO_3 is separately extracted without converting borax, then esterified to $B(OC_2H_5)_3$ at 80% isolated yield. With ester M=146, ester mass is

(A) 7.008 g (B) 2.336 g

(C) 2.920 g (D) 1.168 g

13. A B_2H_6–BCl_3 sample contains 0.100 mol B atoms. Hydrolysis releases 0.120 mol H_2 from diborane only, using six H_2 per B_2H_6. BCl_3 hydrolysis gives three HCl per molecule. HCl amount formed is

(A) 0.120 mol (B) 0.180 mol

(C) 0.0600 mol (D) 0.240 mol

14. A sample contains 0.0600 mol B_2H_6 plus BCl_3 formula units combined. Complete hydrolysis gives 0.120 mol H_2; each diborane gives six H_2 and two boric acids, each BCl_3 gives one boric acid. With H_3BO_3=62, total boric acid mass is

(A) 2.48 g (B) 3.72 g

(C) 7.44 g (D) 4.96 g

15. A H_3PO_2–H_3PO_3 mixture contains 0.100 mol P atoms and loses 0.300 mol electrons on complete oxidation to H_3PO_4. Initial acids are respectively monobasic and dibasic. Total OH^- required for their complete neutralisation before oxidation is

(A) 0.300 mol (B) 0.100 mol

(C) 0.150 mol (D) 0.200 mol

16. A 7.40 g H_3PO_2–H_3PO_3 mixture loses 0.300 mol electrons on oxidation to H_3PO_4. With molar masses 66 and 82, mass fraction of H_3PO_3 is

(A) $\frac{41}{74}$ (B) $\frac{1}{2}$

(C) $\frac{33}{74}$ (D) $\frac{2}{3}$

17. A mixture of H_3PO_4 and $H_4P_2O_7$ contains 0.100 mol P and has 0.260 mol total ionisable O-H groups. Water needed for complete conversion of pyrophosphoric acid to orthophosphoric acid is

(A) 0.0200 mol (B) 0.0600 mol

(C) 0.0100 mol (D) 0.0400 mol

18. A pure linear condensed phosphoric acid has formula $H_{n+2}P_nO_{3n+1}$ and $n+2$ ionisable O-H groups per molecule. A sample contains 0.120 mol P and 0.200 mol such O-H groups. Its molecular formula is

(A) $H_6P_4O_{13}$ (B) $H_4P_2O_7$

(C) $H_5P_3O_{10}$ (D) H_3PO_4

19. A H_2SO_4–$H_2S_2O_8$ mixture has 0.100 mol S. In a controlled iodometric assay only the peroxide linkage reacts, producing 0.0200 mol I_2; one $H_2S_2O_8$ produces one I_2. Complete neutralisation of the original acids needs

(A) 0.200 mol OH^- (B) 0.160 mol OH^-

(C) 0.0800 mol OH^- (D) 0.120 mol OH^-

20. A H_2SO_5–$H_2S_2O_8$ mixture contains 0.100 mol S and releases 0.0600 mol I_2 in a selective assay, one I_2 per peroxide linkage. Each acid has exactly one such linkage. Fraction of original oxygen atoms lying in peroxide bonds is

(A) $\frac{2}{7}$ (B) $\frac{3}{7}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

21. A 0.100 mol Cl_2 feed is split between cold dilute alkali and hot concentrated alkali. The cold branch makes chloride plus 0.0400 mol hypochlorite; the hot branch makes chloride and chlorate only. Total chloride formed is

(A) 0.160 mol (B) 0.100 mol

(C) 0.140 mol (D) 0.120 mol

22. A $KClO_3$–$KClO_4$ sample gives 0.100 mol KCl and 0.180 mol O_2 on complete thermal decomposition. Molar masses are 122.5 and 138.5. Initial $KClO_4$ mass fraction is

(A) $\frac{1}{2}$ (B) $\frac{490}{1321}$

(C) $\frac{3}{5}$ (D) $\frac{831}{1321}$

23. A mixture of IF_5 and IF_7 contains 0.100 mol iodine and yields 0.600 mol HF on complete hydrolysis. Use $IF_5 + 3H_2O \rightarrow HIO_3 + 5HF$ and $IF_7 + 4H_2O \rightarrow HIO_4 + 7HF$. Fraction of iodine present as HIO_4 afterwards is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{3}$ (D) $\frac{3}{4}$

24. A validated sample contains 0.100 mol BrF_3 plus IF_5 formula units combined. Quantitative hydrolysis and fluoride assay gives 0.400 mol total F. With BrF_3=137 and IF_5=222, original sample mass is

(A) 19.65 g (B) 17.95 g

(C) 13.70 g (D) 22.20 g

25. A XeF_2–XeF_6 sample hydrolyses quantitatively: XeF_2 forms Xe, HF, O_2; XeF_6 forms XeO_3 and HF. Measured Xe gas is 0.0200 mol and total HF0.220 mol. With XeO_3=179, XeO_3 mass is

(A) 7.16 g (B) 8.95 g

(C) 3.58 g (D) 5.37 g

26. A XeF_4–XeF_6 mixture has 0.100 mol total Xe and gives 0.520 mol HF on hydrolysis. Use $6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$ and $XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$. Xe gas formed is

(A) 0.0600 mol (B) 2/75 mol

(C) 1/75 mol (D) 0.0400 mol

27. NO and NO_2 are completely absorbed in alkali. Use $NO + NO_2 + 2OH^- \rightarrow 2NO_2^- + H_2O$ and $2NO_2 + 2OH^- \rightarrow NO_2^- + NO_3^- + H_2O$. Product assays give 0.0800 mol nitrite and 0.0200 mol nitrate. Initial mean gas molar mass is (NO=30, NO_2=46)

(A) 42.8 $g\,mol^{-1}$ (B) 41.2 $g\,mol^{-1}$

(C) 39.6 $g\,mol^{-1}$ (D) 38.0 $g\,mol^{-1}$

28. A 0.120 mol NO–NO_2 gas mixture has mean molar mass 130/3 g mol^{-1} (30,46). It is fully absorbed by alkali through $NO + NO_2 \rightarrow 2NO_2^-$ and excess $2NO_2 \rightarrow NO_2^- + NO_3^-$, with water/$OH^-$ balanced. Nitrate formed is

(A) 0.0200 mol (B) 0.0800 mol

(C) 0.0400 mol (D) 0.0600 mol

29. NH_3 is oxidised only by $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$. Feed has 0.100 mol NH_3,0.100 mol O_2; 75% of formed NO is isolated. This isolated NO meets 0.0400 mol fresh O_2 by $2NO + O_2 \rightarrow 2NO_2$. O_2 remaining after the second stage is

(A) 0.0100 mol (B) 0.0200 mol

(C) 0.00500 mol (D) 0

30. First,0.0900 mol NO_2 is absorbed by $3NO_2 + H_2O \rightarrow 2HNO_3 + NO$. Recover 80% of that NO and oxidise it with 0.0100 mol fresh O_2 by $2NO + O_2 \rightarrow 2NO_2$; absorb all regenerated NO_2 once more. No further recycling occurs. Total HNO_3 formed is

(A) 0.0600 mol (B) 0.0760 mol

(C) 0.0800 mol (D) 11/150 mol

31. A 0.100 mol O_2–O_3 gas sample liberates 0.0300 mol I_2 from iodide under a validated ozone-selective assay, one I_2 per O_3. With O=16, mean molar mass of the sample is

(A) 36.8 $g\,mol^{-1}$ (B) 35.2 $g\,mol^{-1}$

(C) 38.4 $g\,mol^{-1}$ (D) 40.0 $g\,mol^{-1}$

32. A 3.68 g O_2–O_3 sample contains 0.100 mol gas molecules. Half its original O_3 then decomposes by $2O_3 \rightarrow 3O_2$. Ozone mole fraction after this step is

(A) $\frac{6}{43}$ (B) $\frac{3}{20}$

(C) $\frac{3}{23}$ (D) $\frac{3}{40}$

33. A $SOCl_2$–SO_2Cl_2 mixture hydrolyses completely. Total HCl formed is 0.200 mol and SO_2 gas from $SOCl_2$ is 0.0400 mol. SO_2 is removed before $BaCl_2$ is added; sulfate from SO_2Cl_2 precipitates quantitatively. With $BaSO_4$=233, precipitate mass is

(A) 23.30 g (B) 9.32 g

(C) 13.98 g (D) 18.64 g

34. A $SOCl_2$–SO_2Cl_2 sample gives 0.160 mol HCl on complete hydrolysis. Oxidation of the liberated SO_2 to sulfate releases 0.100 mol electrons; sulfuryl chloride directly forms sulfate. With molar masses 119 and 135, initial sample mass is

(A) 10.80 g (B) 10.0 g

(C) 8.00 g (D) 9.52 g

35. A PbO–PbO_2 sample contains 0.100 mol Pb. In acid, all Pb(IV) becomes Pb(II) with iodide and 0.0300 mol I_2 is formed; PbO is not oxidising. With PbO=223 and PbO_2=239, initial mass is

(A) 22.30 g (B) 23.90 g

(C) 22.78 g (D) 23.26 g

36. A Pb_3O_4–PbO_2 sample contains 0.100 mol Pb. Reduction of all Pb(IV) to Pb(II) accepts 0.0800 mol electrons. Treat Pb_3O_4 as $2PbO{\cdot}PbO_2$. With molar masses 685 and 239, Pb_3O_4 mass fraction is

(A) $\frac{3}{4}$ (B) $\frac{239}{2294}$

(C) $\frac{685}{924}$ (D) $\frac{2055}{2294}$

37. A 2.42 g sample contains Na_2SiO_3 and SiO_2 only. Complete acidification consumes 0.0200 mol HCl, with two HCl per silicate; original silica is acid-inert. After dehydration all silicon is isolated as SiO_2. With silicate M=122, SiO_2=60, final mass is

(A) 1.80 g (B) 0.600 g

(C) 2.42 g (D) 1.20 g

38. A 0.100 mol silicon-unit mixture contains $(CH_3)_2SiCl_2$ and CH_3SiCl_3. Hydrolysis releases 0.260 mol HCl. Complete condensation gives ideal units $(CH_3)_2SiO$ and $CH_3SiO_{3/2}$, ignoring end groups. With unit masses 74 and 67, polymer mass is

(A) 7.12 g (B) 6.98 g

(C) 7.40 g (D) 6.70 g

39. A hydrazine–hydroxylamine sample contains 0.100 mol N atoms. Complete oxidation of both to N_2 releases 0.160 mol electrons. Hydrazine has N oxidation state-2; hydroxylamine has-1. With N_2H_4=32, hydrazine mass is

(A) 0.960 g (B) 1.28 g

(C) 1.60 g (D) 0.640 g

40. An NH_3–N_2H_4 sample contains 0.120 mol N atoms and releases 0.320 mol electrons on complete oxidation to N_2. With NH_3=17 and N_2H_4=32, initial NH_3 mass fraction is

(A) $\frac{17}{25}$ (B) $\frac{8}{25}$

(C) $\frac{2}{3}$ (D) $\frac{4}{5}$

41–50 / Preserved hard practice

41. A 0.100 mol mixture of H_3PO_2 and H_3PO_3 consumes 0.300 mol electrons worth of oxidant to convert all P to H_3PO_4. Moles of H_3PO_2 were

(A) 0.1 (B) 0.025

(C) 0.05 (D) 0.075

42. A 0.100 mol mixture of PCl_3 and PCl_5 is completely hydrolyzed. The liberated HCl is quantitatively separated from the phosphorus oxo acids and then needs 0.400 mol NaOH for neutralization. Moles of PCl_5 were

(A) 0.1 (B) 0.075

(C) 0.025 (D) 0.05

43. A 0.0200 mol B_2H_6 sample hydrolyzes completely. Hydrogen is collected at 300 K and 1 bar. With R=.0800 L bar/mol/K, volume in L is

(A) 0.48 (B) 2.88

(C) 1.44 (D) 5.76

44. A 0.100 mol mixture of SO_2 and CO_2 needs 0.0200 mol acidic MnO_4^- to oxidize all SO_2 to sulfate. CO_2 does not react. Mole fraction SO_2 is

(A) 0.8 (B) 0.2

(C) 0.4 (D) 0.5

45. A 1.00 g Al sample contains 10% inert impurity. Excess aqueous NaOH reacts with all Al to form $Al(OH)_4^-$ and H_2. Taking Al=27, hydrogen amount in mol is

(A) 0.025 (B) 0.1

(C) 0.0333333 (D) 0.05

46. An equimolar mixture of NO and NO_2 contains 0.100 mol total nitrogen atoms. Upon complete oxidation to nitrate, N atoms release electrons in mol of

(A) 0.1 (B) 0.15

(C) 0.3 (D) 0.2

47. A 0.100 mol mixture of XeF_2 and XeF_4 contains 0.300 mol F atoms. In VSEPR models, average number of central Xe lone pairs per molecule is

(A) 3.5 (B) 3

(C) 2 (D) 2.5

48. A zeolite sample initially has 0.0600 mol Na^+ exchangeable at its negatively charged framework. Exhaustive exchange by Ca^{2+} without framework change makes Ca amount in mol of

(A) 0.03 (B) 0.015

(C) 0.12 (D) 0.06

49. A 0.100 mol mixture of H_3PO_3 and H_3PO_4 is fully neutralized using 0.250 mol NaOH. Moles of H_3PO_3 were

(A) 0.025 (B) 0.1

(C) 0.05 (D) 0.075

50. A 0.100 mol gas mixture of O_2 and O_3 contains 0.240 mol O atoms. On complete ozone decomposition without oxygen loss, final gas moles are

(A) 0.12 (B) 0.14

(C) 0.1 (D) 0.16

Answer key

p-Block

C-PBL / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**A**	21	**C**	31	**A**	41	**C**
02	**C**	12	**B**	22	**D**	32	**A**	42	**D**
03	**B**	13	**B**	23	**B**	33	**C**	43	**B**
04	**A**	14	**D**	24	**B**	34	**B**	44	**D**
05	**C**	15	**C**	25	**D**	35	**C**	45	**D**
06	**D**	16	**A**	26	**B**	36	**D**	46	**D**
07	**B**	17	**A**	27	**B**	37	**A**	47	**D**
08	**D**	18	**C**	28	**C**	38	**B**	48	**A**
09	**B**	19	**B**	29	**A**	39	**A**	49	**C**
10	**C**	20	**A**	30	**D**	40	**A**	50	**A**

Chapter 18

d & f Block

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The number of unpaired electrons in a high-spin free Fe^{2+} ion is

(A) 4 (B) 6

(C) 0 (D) 2

02. The number of unpaired electrons in a high-spin free Co^{2+} ion is

(A) 3 (B) 2

(C) 5 (D) 1

03. The number of unpaired electrons in free Ni^{2+} ion is

(A) 4 (B) 0

(C) 1 (D) 2

04. The spin-only magnetic moment of a high-spin d^5 ion, in Bohr magnetons, is

(A) 5.9160798 (B) 4.8989795

(C) 6.9282032 (D) 3.8729833

05. The spin-only magnetic moment of a free d^8 ion, in Bohr magnetons, is

(A) 4.8989795 (B) 3.8729833

(C) 1.7320508 (D) 2.8284271

06. A free ion has spin-only moment $\sqrt{15}\,\mu_B$. Number of unpaired electrons is

(A) 3 (B) 2

(C) 4 (D) 1

07. A free first-row metal ion M^{3+} has d^3 configuration. Atomic number of neutral M is

(A) 25 (B) 24

(C) 23 (D) 21

08. Gd^{3+} has 7 unpaired $4f$ electrons. Spin-only moment in Bohr magnetons is

(A) 5.9160798 (B) 7.9372539

(C) 9 (D) 7

09. A 0.0100 mol acidic $Cr_2O_7^{2-}$ sample oxidizes Fe^{2+} to Fe^{3+}. Amount Fe^{2+} in mol is

(A) 0.06 (B) 0.03

(C) 0.01 (D) 0.02

10. For a first-row M^{2+} ion that has d^9 configuration, the atomic number of neutral M is

(A) 30 (B) 27

(C) 29 (D) 28

11–40 / New middle-to-hard practice

11. A first-row transition metal M has high-spin aquo ions M^{2+} and M^{3+} with spin-only squared moments 24 and 35 BM^2, respectively. Use $\mu^2 = u(u+2)$ and ground-state d counts. Atomic number of M is

(A) 24 (B) 25

(C) 26 (D) 27

12. A first-row transition metal has high-spin M^{3+} and M^{2+} ions with squared spin-only moments 15 and 24 BM^2. What is its M^{6+} d-electron count? Use $\mu^2 = u(u+2)$ and ground-state ion configurations.

(A) 2 (B) 3

(C) 0 (D) 1

13. A 0.100 mol Fe(II)/Fe(III) aquo-ion sample has mean spin-only $\mu^2 = 30.6$ BM^2 per iron ion. Assume high-spin independent ions, with Fe(II)24 and Fe(III)35. Ce(IV) amount needed to oxidise all Fe(II) to Fe(III), one electron per Ce(IV), is

(A) 0.100 mol (B) 0.0600 mol

(C) 0.0200 mol (D) 0.0400 mol

14. A 0.0500 mol Cu(II)/Ni(II) aquo-ion mixture has mean spin-only $\mu^2 = 6.00$ BM^2. Use Cu(II)3, Ni(II)8 and additive independent-ion susceptibility. In a validated selective procedure only Cu is reduced to Cu(I) and isolated as CuCl. With CuCl=99, precipitate mass is

(A) 2.97 g (B) 4.95 g

(C) 1.98 g (D) 0.990 g

15. A mixture contains $KMnO_4$ and K_2MnO_4 only. Independent assays give 0.100 mol Mn and 0.160 mol K. Using Mn atomic number 25 and independent ions, mean spin-only μ^2 per Mn is

(A) 1.80 BM^2 (B) 3.00 BM^2

(C) 1.20 BM^2 (D) 0.600 BM^2

16. A mixture contains $Na_3[FeF_6]$ and $Na_4[FeF_6]$, with 0.100 mol Fe and 0.340 mol Na. Treat F^- as weak-field and both iron ions as high-spin independent ions. With Fe=26, mean spin-only squared moment per Fe is

(A) 35.0 BM^2 (B) 30.6 BM^2

(C) 28.4 BM^2 (D) 24.0 BM^2

17. A 0.100 mol Mn sample initially as Mn(VII) is reduced to Mn(VI), Mn(IV), Mn(II) only. It accepts 0.360 mol electrons; final mean spin-only $\mu^2 = 22.6$ BM^2. Use moments squared 3,15,35 for VI, IV, II. Final Mn(II) mole fraction is

(A) $\frac{3}{10}$ (B) $\frac{1}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{2}$

18. A 0.100 mol Cr sample contains Cr(II), Cr(III), Cr(VI) only, with mean oxidation number 3.60 and mean spin-only $\mu^2 = 13.2$ BM^2. Assume high-spin ions and moments squared 24,15,0 respectively. Fe(II) amount needed to reduce all Cr(VI) to Cr(III) is

(A) 0.0600 mol (B) 0.0900 mol

(C) 0.0300 mol (D) 0.120 mol

19. Initially 0.100 mol $KMnO_4$ is heated partially by $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ until 0.0200 mol O_2 forms. All manganate then disproportionates on acidification: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$. Total permanganate after acidification is

(A) 0.0600 mol (B) 0.0800 mol

(C) 1/75 mol (D) 11/150 mol

20. A $KMnO_4$–K_2MnO_4 sample contains 0.100 mol Mn. Complete decomposition of its permanganate by $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$ gives 0.0200 mol O_2. Afterwards all manganate disproportionates quantitatively in acid, two permanganates per three manganates. Final permanganate amount is

(A) 0.0800 mol (B) 4/75 mol

(C) 2/75 mol (D) 0.0400 mol

21. A K_2CrO_4–$K_2Cr_2O_7$ sample contains 0.100 mol Cr and 0.140 mol K. OH^- required to convert all dichromate into chromate by $Cr_2O_7^{2-} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$ is

(A) 0.0600 mol (B) 0.100 mol

(C) 0.140 mol (D) 0.0300 mol

22. A K_2CrO_4–$K_2Cr_2O_7$ sample contains 0.100 mol Cr. Conversion of all dichromate to chromate consumes 0.0400 mol OH^-, two per dichromate. The whole solution is then diluted to 0.500 L without potassium loss. $[K^+]$ is

(A) 0.320 M (B) 0.240 M

(C) 0.160 M (D) 0.400 M

23. A CoO–Co_3O_4 sample contains 0.100 mol Co and 0.130 mol O atoms. Use Co_3O_4 as one high-spin Co(II) plus two low-spin Co(III), and CoO as high-spin Co(II). Their squared moments are 15 and 0; ignore orbital and exchange effects. Mean μ^2 per Co is

(A) 4.50 BM^2 (B) 6.00 BM^2

(C) 15.0 BM^2 (D) 10.5 BM^2

24. A Fe_3O_4–Fe_2O_3 sample contains 0.100 mol Fe and 0.140 mol O atoms. Treat all ions as independent high-spin ions, ignoring exchange, with squared moments 24 for Fe(II),35 for Fe(III). Mean squared moment per Fe is

(A) 30.6 BM^2 (B) 32.8 BM^2

(C) 35.0 BM^2 (D) 28.4 BM^2

25. A lanthanoid Ln^{3+} has squared spin-only moment 35 BM^2 and fewer than seven 4f electrons. Ignore orbital contributions and assume Ln^{3+} has $[Xe]4f^{Z-57}$. If Ln^{2+} adds one 4f electron, its squared spin-only moment is

(A) 63 BM^2 (B) 35 BM^2

(C) 48 BM^2 (D) 24 BM^2

26. A lanthanoid has Ln^{3+} with six unpaired 4f electrons and Ln^{4+} with seven. Assume ionisation removes one 4f electron and $Ln^{3+} = [Xe]4f^{Z-57}$; ignore orbital moments. Atomic number Z is

(A) 63 (B) 64

(C) 65 (D) 66

27. $E^\circ(Cu^{2+}/Cu^+)$=0.150 V and $E^\circ(Cu^+/Cu)$=0.520 V. An aqueous sample contains 0.100 mol Cu as Cu(I)/Cu(II), and the favoured disproportionation is assumed quantitative. It deposits 0.0200 mol Cu metal. Final dissolved Cu(II) amount is

(A) 0.100 mol (B) 0.0400 mol

(C) 0.0800 mol (D) 0.0600 mol

28. For manganese, $E^\circ(Mn^{3+}/Mn^{2+})$=1.51 V and $E^\circ(MnO_2/Mn^{3+})$=0.95 V under the stated acid conditions. A 0.100 mol Mn(II)/Mn(III) sample disproportionates quantitatively to Mn(II) and MnO_2, yielding 0.0300 mol MnO_2. Initial mean Mn oxidation number is

(A) $\frac{27}{10}$ (B) $\frac{23}{10}$

(C) $\frac{13}{5}$ (D) $\frac{29}{10}$

29. A V_2O_5 catalyst initially has 0.100 mol V atoms, all at+5. In a controlled oxygen-transfer step $V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3$,0.0400 mol SO_3 forms. No other reaction occurs. Fraction of V still at+5 is

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

30. A catalyst has 0.120 mol V at mean oxidation state 4.50, only V(IV)/V(V). First,0.00500 mol O_2 is fully used to oxidise V(IV) to V(V), four V per O_2. Then 0.0200 mol SO_2 becomes SO_3, each reducing two V(V) to V(IV). Final mean V oxidation number is

(A) $\frac{9}{2}$ (B) 5

(C) $\frac{14}{3}$ (D) $\frac{13}{3}$

31. Self-reduction uses $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$. Feed has 0.0600 mol Cu_2S and 0.0800 mol Cu_2O. Isolated copper is 90% of theoretical. With Cu=64, metal mass obtained is

(A) 15.36 g (B) 6.912 g

(C) 20.736 g (D) 13.824 g

32. A Cu_2S–Cu_2O sample contains 0.200 mol Cu atoms and 0.0600 mol S. It self-reduces completely through $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$ until a feed is exhausted. The remaining sulfide is separately roasted quantitatively to CuO. With CuO=80, this later oxide mass is

(A) 6.40 g (B) 3.20 g

(C) 12.8 g (D) 9.60 g

33. A FeO–Fe_3O_4 mixture has 0.100 mol Fe atoms. Complete oxidation to Fe_2O_3 takes 0.0150 mol O_2. Initial FeO formula-unit mole fraction is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{2}{5}$ (D) $\frac{1}{3}$

34. A 9.26 g nickel-battery active mixture contains NiOOH (M=92) and $Ni(OH)_2$ (M=93). Complete discharge reduces every Ni(III) to Ni(II), accepting $0.0400F$; $Ni(OH)_2$ is already discharged. Initial NiOOH formula-unit mole fraction is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{2}$

35. An intensely coloured tetraoxoanion of a first-row metal with Z=25 has charge−1. A controlled one-electron reduction in strong alkali preserves four oxygen atoms and gives charge−2. Which joint statement follows for the metal d counts and spin-only moments?

(A) Original: d^1; product: d^0; the original colour must be a d–d transition.

(B) Both are d^0 and both must be colourless.

(C) Original: d^0; product: d^2 with $\sqrt{8}$ BM; colour proves two-electron reduction.

(D) Original: d^0, zero spin-only moment; product: d^1, $\sqrt{3}$ BM. The original colour cannot be a d–d transition.

36. Yellow chromate becomes orange dichromate after adding acid; elemental analysis shows no chromium loss and a reducing-agent assay gives the same electron capacity per Cr before and after. Which explanation fits all observations?

(A) Acid reduces Cr(VI) to Cr(III); the unchanged capacity proves its d count is unchanged.

(B) Acid oxidises Cr(V) to Cr(VI); unchanged chromium amount forces equal oxidation states.

(C) The orange colour proves creation of unpaired d electrons; oxoanion association cannot alter colour.

(D) Acid changes oxoanion association while Cr stays +6 and d^0; colour change does not establish a chromium redox reaction.

37. A 6.95 g Fe–Al alloy dissolves completely in a nonoxidising-acid procedure, giving Fe(II), Al(III) and 0.175 mol H_2. With Fe=56, Al=27, aluminium mass fraction is

(A) $\frac{1}{3}$ (B) $\frac{27}{139}$

(C) $\frac{112}{139}$ (D) $\frac{1}{2}$

38. A dissolved Fe–Cr alloy contains 0.100 mol metal atoms as Fe(II), Cr(III). A validated complete oxidation converts them only to Fe(III), Cr(VI), releasing 0.180 mol electrons. With Fe=56, Cr=52, original alloy mass is

(A) 5.20 g (B) 5.44 g

(C) 5.36 g (D) 5.60 g

39. A 100 mL Cu(II)/Ni(II) solution has absorbances 0.500 and 0.400 at two wavelengths. Calibrated laws are $A_1 = 20[Cu] + 10[Ni]$, $A_2 = 10[Cu] + 20[Ni]$, concentrations in M. A validated selective procedure recovers 90% of Ni as a precipitate of molar mass 289, one Ni per formula. Precipitate mass is

(A) 0.2601 g (B) 0.2890 g

(C) 0.13005 g (D) 0.5202 g

40. A 100 mL Co(II)/Ni(II) solution obeys $A_1 = 10[Co] + 20[Ni]$, $A_2 = 30[Co] + 10[Ni]$. Both measured absorbances are 0.500; concentrations are in M. Selective separation recovers 80% of Co, which is converted entirely to Co_3O_4 (M=241). Oxide mass is

(A) 241/3750 g (B) 241/1875 g

(C) 241/3000 g (D) 241/7500 g

41–50 / Preserved hard practice

41. A 0.110 mol mixture of free high-spin Fe^{2+} and Fe^{3+} obeys the ideal Curie mixing rule: meanμeff^{2}is mole-fraction weighted mean of n(n+2). Measured meanμeff^2=29.0. Acidic MnO_4^- needed to oxidize all Fe^{2+} in mol is

(A) 0.014 (B) 0.022

(C) 0.01 (D) 0.012

42. A first-row M^{2+} ion has three unpaired electrons and more than five d-electrons in the free-ion ground state. Candidate atomic masses are V=51, Co=59, Ni=59, Cu=64. A 1.30 g sample of anhydrous MCl_2 is reduced completely to metal with 80% current efficiency. With Cl=35.5, the required charge in Faraday s is

(A) 0.02 (B) 0.1

(C) 0.05 (D) 0.025

43. A 0.100 mol mixture of MnO_4^- and $Cr_2O_7^{2-}$ accepts 0.550 mol electrons in acid. Moles of dichromate initially were

(A) 0.025 (B) 0.05

(C) 0.075 (D) 0.1

44. A 0.100 mol mixture of Sc^{3+} and Ti^{3+} has mean ideal spin-onlyμ^2=.75μB^{2}under Curie addition. Moles of Ti^{3+} are

(A) 0.025 (B) 0.0125

(C) 0.075 (D) 0.05

45. A 0.0200 mol $Cr_2O_7^{2-}$ sample is fully reduced to Cr^{3+}. Precipitating all chromium as $Cr(OH)_3$ requires OH^- amount in mol of

(A) 0.12 (B) 0.18

(C) 0.04 (D) 0.06

46. An oxide of a first-row metal has formula M_2O_3. A 3.20 g sample contains 0.0400 mol metal ions. Atomic mass of M in u is

(A) 24 (B) 40

(C) 56 (D) 64

47. A 0.100 mol mixture of high-spin Fe^{2+} and Ni^{2+} has 0.300 mol unpaired electrons under free-ion counting. Moles of Fe^{2+} are

(A) 0.075 (B) 0.025

(C) 0.05 (D) 0.1

48. A mixture of La_2O_3 and CeO_2 reacts completely with acidified peroxide. Use $La_2O_3 + 6H^+ \rightarrow 2La^{3+} + 3H_2O$ and $2CeO_2 + 6H^+ + H_2O_2 \rightarrow 2Ce^{3+} + 4H_2O + O_2$. The process consumes 0.120 mol H^+ and forms 0.0100 mol oxygen. With La=139, Ce=140, O=16, the original mixture mass in g is

(A) 3.26 (B) 6.7

(C) 10.14 (D) 3.44

49. A 24.4 g mixture contains a total of 0.100 mol of K_2CrO_4 and $K_2Cr_2O_7$ formula units. Their molar masses are 194 and 294 g/mol. After complete acidic reduction to Cr^{3+}, the moles of electrons accepted are

(A) 0.9 (B) 0.6

(C) 0.3 (D) 0.45

50. A first-row metal M has a free M^{3+} ion with d^5 configuration. A 7.68 g sample contains only MO and M_2O_3, and 0.100 mol total M atoms. Use Fe=56, Mn=55, Co=59, O=16. After acid dissolution, the M^{2+} alone is oxidized to M^{3+} by acidic permanganate. Required MnO_4^- amount in mol is

(A) 0.01

(B) 0.006

(C) 0.008

(D) 0.02

Answer key

d & f Block

C-DFB / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**C**	21	**A**	31	**D**	41	**D**
02	**A**	12	**C**	22	**A**	32	**A**	42	**D**
03	**D**	13	**D**	23	**B**	33	**B**	43	**B**
04	**A**	14	**C**	24	**B**	34	**C**	44	**A**
05	**D**	15	**A**	25	**C**	35	**D**	45	**A**
06	**A**	16	**B**	26	**C**	36	**D**	46	**C**
07	**B**	17	**D**	27	**C**	37	**B**	47	**C**
08	**B**	18	**B**	28	**C**	38	**B**	48	**B**
09	**A**	19	**D**	29	**B**	39	**A**	49	**D**
10	**C**	20	**B**	30	**D**	40	**A**	50	**C**

Chapter 19

Metallurgy

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Calcining 20.0 g $ZnCO_3$ gives ZnO. With $ZnCO_3$ =125 and ZnO =81 g/mol, oxide mass in g is

(A) 12.96 (B) 16.2

(C) 8.1 (D) 20

02. In Cu_2S +2 Cu_2O →6 Cu+ SO_2, 0.100 mol Cu_2S with oxide excess produces Cu in mol of

(A) 0.4 (B) 0.2

(C) 0.3 (D) 0.6

03. For Al^{3+} +3 e→Al, charge 0.300 F deposits Al mass in g of(Al=27)

(A) 5.4 (B) 0.9

(C) 8.1 (D) 2.7

04. A 100 kg ore contains 80% Fe_2O_3 and 20% inert gangue. Theoretical Fe mass in kg is(Fe_2O_3 =160, Fe=56)

(A) 40 (B) 80

(C) 70 (D) 56

05. An ore has 20.0% $ZnCO_3$ and 100 kg is processed with 90% overall metal recovery. Zn obtained in kg is(Zn=65, $ZnCO_3$ =125)

(A) 11.7 (B) 18

(C) 9.36 (D) 10.4

06. At a given temperature, 2 A+ O_2 →2 AO has ΔG° =-400 kJ, and 2 B+ O_2 →2 BO has ΔG° =-500 kJ. ΔG for B+ AO → BO +A in kJ is

(A) -50 (B) -100

(C) 100 (D) 50

07. For M+0.5 O_2 → MO, ΔH =-200 kJ /mol and ΔS =-100 J/mol/K. Neglecting temperature variation, ΔG at 1000 K in kJ /mol is

(A) -100 (B) -300

(C) 0 (D) 100

08. For M+0.5 O_2 → MO with ΔH =-200 kJ /mol and ΔS =-100 J/mol/K, the formal temperature where ΔG° is zero in K is

(A) 1000 (B) 3000

(C) 2000 (D) 4000

09. A molten Al_2O_3 electrolysis run uses 0.600 F charge at 75% current efficiency. Aluminium deposited in mol is

(A) 0.2 (B) 0.1

(C) 0.45 (D) 0.15

10. A 1.00 mol mixture of Cu_2S and Cu_2O reacts by Cu_2S +2 Cu_2O →6 Cu+ SO_2 with no starting material left. Initial Cu_2S amount in mol is

(A) 0.66666667 (B) 0.25

(C) 0.33333333 (D) 0.5

11–40 / New middle-to-hard practice

11. On a common per-mol-O_2 basis, an oxide formation line has $\Delta G^\circ = -300$ kJ at 1000 K and−200 at 1500 K, linear in T. The carbon oxidation line is $\Delta G^\circ = -200 - 0.100T$ kJ. For carbon reduction to be strictly favourable, the least allowed temperature from 900,1000,1100,1200 K is

(A) 1000 K (B) 900 K

(C) 1200 K (D) 1100 K

12. On a per-mol-O_2 basis, oxide-formation lines are $G_M = -600 + 0.200T$, $G_N = -400 + 0.0500T$, and carbon oxidation is $G_C = -200 - 0.150T$ (kJ). Width of the temperature interval where carbon reduces N oxide strictly but cannot reduce M oxide strictly is

(A) 2000/7 K (B) 1000 K

(C) 8000/7 K (D) 1000/7 K

13. At a fixed T, standard formation free energies are−300 kJ per M_2O_3, −80.0 per CO and−220 per CO_2. Products may be only metal plus CO, or metal plus CO_2; choose a strictly favourable standard route. With 0.100 mol oxide and 0.120 mol C, maximum mol M obtainable by such a route is

(A) 0.0800 mol (B) 0.160 mol

(C) 0.200 mol (D) 0.120 mol

14. At a fixed T, $\Delta G_f^\circ(MO_2) = -240$, $\Delta G_f^\circ(CO) = -120$, $\Delta G_f^\circ(CO_2) = -260$ kJ mol^{-1}. Only a route with strictly negative standard reduction free energy is allowed. With 0.100 mol MO_2 and 0.0800 mol carbon, maximum metal amount is

(A) 0.160 mol (B) 0.0400 mol

(C) 0.100 mol (D) 0.0800 mol

15. A 20.0 g ore contains only Fe_2O_3 and SiO_2. Complete reduction produces 11.2 g Fe. Acid gangue is removed as $CaSiO_3$ using limestone of 80.0% $CaCO_3$ purity, the rest inert. With Fe=56, Fe_2O_3=160, SiO_2=60, $CaCO_3$=100, limestone mass needed is

(A) 5.00 g (B) 20/3 g

(C) 25/3 g (D) 10.0 g

16. A 30.0 g Fe_3O_4–SiO_2 ore gives 16.8 g Fe on complete reduction. All SiO_2 becomes $CaSiO_3$ with sufficient pure CaO. With Fe=56, Fe_3O_4=232, SiO_2=60, $CaSiO_3$=116, slag mass is

(A) 986/75 g (B) 13.60 g

(C) 232/15 g (D) 6.80 g

17. A 10.70 g basic gangue contains CaO and Al_2O_3 only; complete acid dissolution takes 0.500 mol HCl. For this supplied slag model, $CaO+SiO_2 \rightarrow CaSiO_3$ and $Al_2O_3+SiO_2 \rightarrow Al_2SiO_5$. With masses 56,102,60, silica flux required is

(A) 10.70 g (B) 6.00 g

(C) 15.0 g (D) 9.00 g

18. A 10.70 g acidic gangue contains SiO_2 and P_2O_5 only; phosphorus assay gives 0.100 mol P. Remove it by CaO to form $CaSiO_3$ and $Ca_3(PO_4)_2$. With SiO_2=60, P_2O_5=142, CaO=56, CaO required is

(A) 8.40 g (B) 10.08 g

(C) 11.76 g (D) 5.60 g

19. Flotation of 100 kg feed gives 20.0 kg concentrate containing 60.0% valuable metal and 80.0 kg tailings containing 5.00% metal. No metal is otherwise lost. Fraction of feed metal recovered in concentrate is

(A) $\frac{3}{5}$ (B) $\frac{2}{3}$

(C) $\frac{3}{4}$ (D) $\frac{4}{5}$

20. A 100 kg ore has 15.0% metal. Flotation concentrate has 60.0% metal and tailings 3.00%; no metal is lost elsewhere. Fraction of the original metal in concentrate is

(A) $\frac{16}{19}$ (B) $\frac{15}{19}$

(C) $\frac{4}{5}$ (D) $\frac{3}{4}$

21. A bauxite leach dissolves only Al_2O_3 and SiO_2. Total NaOH consumption is 0.160 mol; dissolved-silicon assay gives 0.0200 mol Si. Use two NaOH per formula of either oxide. Recovered aluminium is 80% of dissolved aluminium. With Al=27, metal mass obtained is

(A) 5.184 g (B) 2.592 g

(C) 1.728 g (D) 3.240 g

22. A 4.00 g gangue fraction accompanying bauxite contains only Fe_2O_3 and SiO_2. Selective acid treatment consumes 0.0600 mol HCl on Fe_2O_3; silica is unaffected. The remaining silica is then dissolved by $SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$. With Fe_2O_3=160, SiO_2=60, NaOH required is

(A) 0.0600 mol (B) 0.120 mol

(C) 0.0800 mol (D) 0.0400 mol

23. A 14.36 g Au–Ag inventory is cyanide-leached quantitatively by $4M + 8CN^- + O_2 + 2H_2O \rightarrow 4[M(CN)_2]^- + 4OH^-$. Oxygen uptake is 0.0250 mol. Gold is separated and 90% recovered. With Au=197, Ag=108, gold mass recovered is

(A) 7.092 g (B) 12.924 g

(C) 6.480 g (D) 7.880 g

24. A 12.20 g Au–Ag sample is completely cyanide-leached with 0.0200 mol O_2, one O_2 per four metal atoms. Both metals are later recovered quantitatively. With Au=197, Ag=108, silver mass fraction among recovered metals is

(A) $\frac{108}{305}$ (B) $\frac{1}{2}$

(C) $\frac{197}{305}$ (D) $\frac{2}{5}$

25. A validated iodide separation feed contains 0.100 mol total ZrI_4 and TiI_4, mass 57.75 g. A stage deposits 80% of its Zr and 10% of its Ti as metals. With Zr=91, Ti=48, I=127, Zr mass fraction in deposited metal is

(A) $\frac{19}{20}$ (B) $\frac{91}{139}$

(C) $\frac{91}{97}$ (D) $\frac{4}{5}$

26. A 10.84 g Zr–Hf inventory consumes 0.200 mol I_2 on complete conversion to tetraiodides, two I_2 per metal atom. A specified separation then deposits 95% Zr and 20% Hf. With Zr=91, Hf=178, Zr mass fraction in the deposited metal is

(A) $\frac{91}{269}$ (B) $\frac{19}{20}$

(C) $\frac{4}{5}$ (D) $\frac{1729}{1907}$

27. A 0.100 mol $Ni(CO)_4$–$Fe(CO)_5$ gas feed releases 0.440 mol CO on complete decomposition. A purification removes the iron carbonyl, then 90% of the remaining nickel carbonyl yields Ni. With Ni=59, recovered nickel mass is

(A) 2.124 g (B) 3.186 g

(C) 5.310 g (D) 3.540 g

28. A gas feed of $Ni(CO)_4$ and $Fe(CO)_5$ weighs 18.10 g and releases 0.440 mol CO on complete decomposition. Molar masses are 171 and 196. Nickel atom fraction in the deposited metal is

(A) $\frac{3}{5}$ (B) $\frac{3}{4}$

(C) $\frac{2}{5}$ (D) $\frac{1}{2}$

29. In an electrorefiner, a Cu–Fe anode loses 6.00 g while$0.200F$ passes; both dissolve as +2 ions with full anodic efficiency. Cathodic current efficiency for Cu deposition is 90.0%; no other Cu sink exists. Initially 1.00 L solution has$[Cu^{2+}]$=0.100 M. With Cu=64, Fe=56, final $[Cu^{2+}]$ is

(A) 0.110 M (B) 0.0600 M

(C) 0.0100 M (D) 0.0500 M

30. A 10.0 g Cu–Ag anode is electrolysed until all Cu has dissolved as Cu^{2+}; all Ag stays as sludge. Total charge is$0.250F$; cathode gains 4.80 g Cu and all other cathodic charge forms H_2. With Cu=64, the pair (Ag sludge mass, H_2 amount) is

(A) $(2.00\text{ g}, 0.100\text{ mol})$ (B) $(2.00\text{ g}, 0.0500\text{ mol})$

(C) $(2.00\text{ g}, 0.0250\text{ mol})$ (D) $(5.20\text{ g}, 0.0500\text{ mol})$

31. A 100 g solid initially contains 1.00% impurity. Each idealised zone-refining pass retains 90% of host and a fixed fraction r of impurity. After one pass retained mass is 89.30 g. After two passes impurity mass fraction is

(A) $\frac{1}{100}$ (B) $\frac{4}{8019}$

(C) $\frac{4}{8023}$ (D) $\frac{2}{893}$

32. An idealised refining pass retains 90% of host metal and 50% of impurity. Feed has 95 g host and 5 g impurity. An integer number of passes must give at least 99.5% purity while retaining at least 70% of original host. Which is correct?

(A) Four passes satisfy both. (B) No integer number of passes satisfies both targets.

(C) Five passes satisfy both. (D) Three passes satisfy both.

33. A 10.70 g Pb–Ag alloy undergoes controlled cupellation: only Pb oxidises to PbO and 0.0250 mol O_2 is consumed. Oxide is quantitatively removed. With Pb=207, initial silver mass fraction is

(A) $\frac{7}{207}$ (B) $\frac{7}{214}$

(C) $\frac{1}{20}$ (D) $\frac{207}{214}$

34. An 18.46 g Pb–Sn–Ag alloy has 0.100 mol Pb plus Sn by an independent assay. Complete selective oxidation forms PbO and SnO_2, consuming 0.0750 mol O_2; Ag is unchanged and recovered quantitatively. With Pb=207, Sn=119, silver mass is

(A) 2.16 g (B) 3.24 g

(C) 1.08 g (D) 4.32 g

35. A ZnS–FeS_2 ore fraction contains 0.100 mol S. Complete roasting to ZnO, Fe_2O_3, SO_2 consumes 0.145 mol O_2. Use1.5 O_2 per ZnS and 2.75 per FeS_2. If90% of zinc is later recovered, zinc mass is (Zn=65)

(A) 1.17 g (B) 3.90 g

(C) 5.85 g (D) 3.51 g

36. A 12.02 g PbS–ZnS sample consumes 0.120 mol O_2 on full roasting to PbO, ZnO, SO_2,1.5 O_2 per sulfide. With PbS=239, ZnS=97, ZnO=81, zinc oxide formed is

(A) 6.48 g (B) 3.24 g

(C) 4.05 g (D) 2.43 g

37. A charge has 0.0500 mol PbS and 0.0600 mol PbO. First $PbS + 2PbO \rightarrow 3Pb + SO_2$ goes to its limiting feed. Remaining PbS then reacts with 0.0150 mol Fe by $PbS + Fe \rightarrow FeS + Pb$. With Pb=207, total Pb formed is

(A) 15.525 g (B) 21.735 g

(C) 18.630 g (D) 24.840 g

38. A 15.68 g mixture contains 0.100 mol Fe_2O_3 plus Cr_2O_3 formula units combined. Aluminium feed is 0.180 mol; both oxides reduce in the same fractional proportion by $M_2O_3 + 2Al \rightarrow Al_2O_3 + 2M$. With oxide masses 160,152 and Cr=52, chromium formed is

(A) 3.744 g (B) 2.808 g

(C) 7.488 g (D) 4.160 g

39. A 10.0 g Fe_2O_3–inert ore is completely reduced by a gas feed containing 0.300 mol CO and 0.100 mol CO_2. After reduction dry CO_2 mole fraction is 0.550; no carbon deposition or other gas reaction occurs. With Fe=56, metal mass produced is

(A) 3.36 g (B) 5.60 g

(C) 6.16 g (D) 4.48 g

40. A 12.0 g pure oxide M_2O_n yields 8.40 g metal on complete reduction. The metal has the same valency n in chloride MCl_n, whose molar mass is 162.5. With O=16, Cl=35.5, atomic molar mass of M is

(A) 56 (B) 27

(C) 52 (D) 40

41–50 / Preserved hard practice

41. A 10.0 g ore contains Fe_2O_3 and Fe_3O_4 only. Complete reduction gives 7.10 g Fe. With Fe_2O_3 =160, Fe_3O_4 =232, Fe=56, mass fraction Fe_3O_4 is

(A) 0.75 (B) 0.41428571

(C) 0.5 (D) 0.25

42. A 50.0 g ore contains 20% SiO_2 gangue. All silica is removed as $CaSiO_3$ using $CaCO_3$ flux that calcines completely. Required pure $CaCO_3$ mass in g is(SiO_2 =60, $CaCO_3$ =100)

(A) 10 (B) 20

(C) 16.666667 (D) 25

43. A 1.00 mol mixture of Cu_2S and Cu_2O has Cu_2S amount 0.400 mol. Itself-reduces by Cu_2S +2 Cu_2O →6 Cu+ SO_2. Moles Cu formed are

(A) 1.8 (B) 3.6

(C) 1.2 (D) 2.4

44. An Al_2O_3 electrolysis run passes 96500 Cat 80% current efficiency. Anode current efficiency is 100% for carbon oxidation to CO_2 with no CO. For F=96500, Al mass and carbon mass consumed in g are respectively

(A) $7.2, 2.4$ (B) $7.2, 3.0$

(C) $9.0, 3.0$ (D) $9.0, 2.4$

45. At a temperature T, 2 M+ O_2 →2 M O has ΔG =-300 kJ and 2 C+ O_2 →2 CO has ΔG =-400 kJ. For MO +C→M+ CO, the equilibrium constant obeys ln K=a/(RT). What is a in J/mol?

(A) -100000 (B) 50000

(C) 100000 (D) -50000

46. A 1.00 g Ag -containing ore leach contains $Ag(CN)_2^-$. Zinc displacement consumes 0.00100 mol Zn and recovers all Ag. With Ag=108, mass percentage Ag in the ore is

(A) 10.8 (B) 21.6

(C) 54 (D) 43.2

47. A mixture contains 0.100 mol ZnS and 0.100 mol $ZnCO_3$. Both are converted fully to ZnO. In roasting and calcination combined, total gas moles SO_2 + CO_2 produced are

(A) 0.2 (B) 0.3

(C) 0.15 (D) 0.1

48. A 1.00 g impure Ni sample forms 0.0100 mol $Ni(CO)_4$ with CO in excess; inert impurity does not react. Using Ni=59, CO =28, carbonyl mass in g is

(A) 0.59 (B) 1.71

(C) 1.12 (D) 2.3

49. A copper refining cell passes 0.100 F charge.Cathode efficiency is 90%. Dissolving anode Cu is 100% efficient and Cu=64. The mass of copper that stays dissolved in excess of cathode deposit in g is

(A) 3.2 (B) 0.64

(C) 0.32 (D) 0.16

50. A 1.87 g Ti/Zr metal mixture contains 0.0300 mol total metal atoms. Use Ti=48, Zr=91. Both metals form MI_4 quantitatively. During transfer, 20% of TiI_4 is lost, but all ZrI_4 reaches the decomposition stage. Each transferred MI_4 gives M+2 I_2. The iodine amount recovered, in mol, is

(A) 0.048 (B) 0.06

(C) 0.04 (D) 0.052

Answer key

Metallurgy

C-METL / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**B**	31	**C**	41	**B**
02	**D**	12	**D**	22	**C**	32	**B**	42	**C**
03	**D**	13	**B**	23	**A**	33	**B**	43	**A**
04	**D**	14	**D**	24	**A**	34	**A**	44	**B**
05	**C**	15	**C**	25	**C**	35	**D**	45	**B**
06	**A**	16	**A**	26	**D**	36	**C**	46	**B**
07	**A**	17	**D**	27	**B**	37	**B**	47	**A**
08	**C**	18	**C**	28	**A**	38	**A**	48	**B**
09	**D**	19	**C**	29	**B**	39	**D**	49	**C**
10	**C**	20	**A**	30	**B**	40	**A**	50	**D**

Chapter 20

Coordination Compounds

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The moles AgCl formed immediately from 0.10 mol $[Co(NH_3)_5Cl]Cl_2$ with $AgNO_3$ excess, without ligand substitution, are

(A) 0.5 (B) 0.1

(C) 0.2 (D) 0.3

02. For an ideal octahedral d^3 ion, CFSE in units Δ_o is

(A) -1.2 (B) -0.6

(C) 0 (D) -1.8

03. For an ideal high-spin octahedral d^5 ion, CFSE excluding pairing energy in units Δ_o is

(A) 0 (B) -1.2

(C) -2 (D) 1.2

04. For an ideal low-spin octahedral d^6 ion, CFSE excluding pairing energy in units Δ_o is

(A) -2.4 (B) 0

(C) -1.2 (D) -0.4

05. Number of unpaired electrons in high-spin octahedral Fe^{2+} (d^6) is

(A) 0 (B) 2

(C) 6 (D) 4

06. For an ideal tetrahedral d^3 ion, CFSE in units Δt is

(A) 0 (B) -1.2

(C) 0.8 (D) -0.8

07. A central ion has d^5 configuration. Strong-field octahedral pairing leaves how many unpaired electrons?

(A) 3 (B) 1

(C) 5 (D) 0

08. The total number of donor atoms in one hexadentate EDTA and two monodentate H_2O ligands, if they all bind same centre, is

(A) 10 (B) 8

(C) 6 (D) 3

09. An octahedral cobalt(III) complex contains three identical bidentate ethylenediamine ligands, $[Co(en)_3]^{3+}$. Counting optical isomers separately but excluding conformers, its number of distinct stereoisomers is

(A) 1 (B) 4

(C) 3 (D) 2

10. A mononuclear octahedral Cr(III) complex has two neutral bidentate en ligands and only chloride as its other ligand and counterion. Excess $AgNO_3$ immediately precipitates all outer-sphere chloride. Complete destruction of the complex then makes all its chloride available. Ratio of total AgCl after destruction to immediate AgCl, for equal amounts of the salt, is

(A) 2 (B) 4

(C) 3 (D) 1

11–40 / New middle-to-hard practice

11. A mixture of $[Co(NH_3)_6]Cl_3$ and $[Co(NH_3)_5Cl]Cl_2$ contains 0.100 mol Co. Immediate quantitative AgCl assay of outer-sphere chloride gives 0.260 mol AgCl; coordinated Cl does not exchange during the test. Total coordinated NH_3 amount is

(A) 0.600 mol (B) 0.560 mol

(C) 0.500 mol (D) 0.520 mol

12. A mixture of $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2]Cl{\cdot}2H_2O$ contains 0.100 mol Cr and gives 0.160 mol immediate AgCl from outer chloride only. Water lost on selectively removing only crystal water is (H_2O=18)

(A) 1.44 g (B) 2.52 g

(C) 1.26 g (D) 3.60 g

13. A 0.100 mol Co mixture contains ionisation isomers $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$. Immediate $BaSO_4$ assay gives 0.0400 mol from outer sulfate only. In an equal sample, immediate outer-sphere AgBr mass is (M=188)

(A) 7.52 g (B) 18.80 g

(C) 15.04 g (D) 11.28 g

14. A 0.100 mol Co mixture contains $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5Cl]Cl(NO_2)$. Immediate outer-chloride assay gives 0.140 mol AgCl. A validated selective assay oxidises only initially outer NO_2^- to NO_3^-, two electrons per nitrite. Electron amount released is

(A) 0.200 mol (B) 0.120 mol

(C) 0.0800 mol (D) 0.0600 mol

15. Coordination isomer A=$[Co(NH_3)_6][Fe(CN)_6]$, B=$[Fe(NH_3)_6][Co(CN)_6]$. Both metals are +3. Use Co(III) low-spin in both; Fe(III) low-spin with CN and high-spin with NH_3. A 0.100 mol salt mixture has mean sum of spin-only μ^2 per formula 19 BM^2. Amount of B is

(A) 0.0250 mol (B) 0.0750 mol

(C) 0.100 mol (D) 0.0500 mol

16. For A=$[Co(NH_3)_6][Fe(CN)_6]$ and B=$[Fe(NH_3)_6][Co(CN)_6]$, use Co(III) low-spin throughout, Fe(III) low-spin with CN and high-spin with NH_3. A 0.100 mol mixture has mean formula spin-only $\mu^2 = 11$ BM^2. After selectively isolating the cobalt-cyanide anion fraction, total coordinated CN^- in that fraction is

(A) 0.600 mol (B) 0.450 mol

(C) 0.0750 mol (D) 0.150 mol

17. A linkage-isomer mixture contains 0.100 mol nitro-N isomer X plus nitrito-O isomer Y. A calibrated infrared signal is $I = x + 3y$ for amounts x, y in mol; initial I=0.180. Irradiation converts half of X to Y without loss or reverse conversion. Final I is

(A) $\frac{27}{100}$ (B) $\frac{6}{25}$

(C) $\frac{3}{10}$ (D) $\frac{21}{100}$

18. A 0.100 mol linkage-isomer mixture has nitro-N amount x, nitrito-O amount y, with calibrated $I = x + 3y$. I falls from 0.260 to 0.140 during conversion of Y to X only. Fraction of original Y converted is

(A) $\frac{3}{4}$ (B) $\frac{3}{5}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

19. A 0.100 mol cis/trans mixture of square-planar PtA_2B_2 has mean squared permanent dipole $0.600\mu_b^2$. In the supplied equal-bond model cis has $2\mu_b^2$ and trans zero. Then 25.0% of trans converts to cis. Final cis amount is

(A) 0.0475 mol (B) 0.0550 mol

(C) 0.0750 mol (D) 0.0300 mol

20. A 0.100 mol $[Co(en)_2Cl_2]^+$ sample is 80.0% cis and 20.0% trans; cis has two enantiomers and trans is achiral. Calibrated optical rotation gives 0.0200 mol excess of right- over left-rotating cis. After quantitative removal of trans only, left-rotating fraction is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{3}{8}$ (D) $\frac{3}{10}$

21. An octahedral M(III) complex contains only coordinated NH_3 and ethane-1,2-diamine (en). Carbon assay gives four carbon atoms per M; en has two C and is bidentate. For the resulting ligand set, total stereoisomers, counting optical isomers separately, is

(A) 1 (B) 2

(C) 3 (D) 4

22. An octahedral Cr(III) complex contains only en and oxalate, each bidentate. Nitrogen assay gives two N atoms per Cr; only en contains N, two each. The neutral potassium salt has what molar mass? (Cr=52, en=60, $C_2O_4^{2-}$=88, K=39)

(A) 366 $g\,mol^{-1}$ (B) 327 $g\,mol^{-1}$

(C) 300 $g\,mol^{-1}$ (D) 288 $g\,mol^{-1}$

23. A d^6 metal forms octahedral and tetrahedral complexes. Octahedral splitting is 240 kJ mol^{-1}, tetrahedral splitting is 4/9 of that, and pairing energy 180 kJ mol^{-1}. Apply the usual weak/strong-field filling. A 0.0800 mol mixture has mean spin-only $\mu^2 = 6.00$ BM^2. Tetrahedral-complex amount is

(A) 0.0100 mol (B) 0.0200 mol

(C) 0.0400 mol (D) 0.0600 mol

24. A d^4 octahedral sample initially has $\Delta_o = 150$ and P=200 kJ mol^{-1}. Treatment converts some ions to a new complex with $\Delta_o = 300$, same P. Only old and new complexes remain; mean spin-only $\mu^2 = 12$ BM^2. With t_{2g} energy$-0.4\Delta_o$, e_g energy$+0.6\Delta_o$ and P per pair, mean field-plus-pair energy per mol metal is

(A) -280 kJ mol^{-1} (B) -210 kJ mol^{-1}

(C) -232.5 kJ mol^{-1} (D) -137.5 kJ mol^{-1}

25. A 0.100 mol Ni mixture contains tetrahedral $[NiCl_4]^{2-}$ and square-planar $[Ni(CN)_4]^{2-}$. Mean spin-only $\mu^2 = 2.40$ BM^2. Excess CN^- then converts half of the chlorocomplex into cyanocomplex, four Cl^- released per converted complex. Chloride released is

(A) 0.0600 mol (B) 0.0300 mol

(C) 0.140 mol (D) 0.120 mol

26. A 0.100 mol Ni inventory contains $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$. Selective complete carbonyl decomposition gives 0.240 mol CO; cyanide complex is unchanged. Formal electron amount needed to reduce all initially ionic Ni to Ni(0) is

(A) 0.120 mol (B) 0.200 mol

(C) 0.0800 mol (D) 0.0400 mol

27. A 0.100 mol Fe mixture contains $[Fe(H_2O)_5NO]SO_4$ and $Na_2[Fe(CN)_5NO]$. For this formal bookkeeping treat NO as NO^+ in both. Oxidising every Fe formally to Fe(III) releases 0.160 mol electrons. Total Na amount in the original sample is

(A) 0.0800 mol (B) 0.120 mol

(C) 0.0400 mol (D) 0.200 mol

28. A 0.100 mol total-metal mixture contains $K[Co(CO)_4]$ and $Na_2[Fe(CO)_4]$. Oxidising the metal centres formally to oxidation state zero releases 0.160 mol electrons; CO is neutral. Total sodium amount in the original mixture is

(A) 0.0600 mol (B) 0.0800 mol

(C) 0.200 mol (D) 0.120 mol

29. A 0.100 mol $[Cr(en)_3]^{3+}$ sample contains only its two enantiomers. Calibrated rotation is proportional to molar difference $n_\Delta - n_\Lambda$, initially 0.0200 mol and finally 0.00500 mol. A fraction f of every initial population becomes exactly racemic, with no complex loss. Pair (f, final n_Λ) is

(A) $(0.250, 0.0475$ mol) (B) $(0.500, 0.0250$ mol)

(C) $(0.750, 0.0475$ mol) (D) $(0.750, 0.0450$ mol)

30. A 0.100 mol $[Cr(en)_3]^{3+}$ sample has $n_\Delta - n_\Lambda = 0.0200$ mol. Selective crystallisation removes 50.0% of initial Delta and 10.0% of initial Lambda. In the remaining solution, enantiomeric excess and majority form are

(A) 1/10, Delta (B) 1/11, Delta

(C) 1/11, Lambda (D) 1/5, Lambda

31. A 0.100 mol octahedral MA_3B_3 sample contains fac and mer isomers only. A calibrated structural probe counts 0.0600 mol trans A–A pairs; fac has none and mer one per complex. Half the mer is then converted to fac. Final fac amount is

(A) 0.0400 mol (B) 0.0700 mol

(C) 0.0300 mol (D) 0.0800 mol

32. A 0.100 mol octahedral MA_3B_3 sample has fac and mer only. In a selective fast substitution, only A ligands trans to B are released; total fast A release is 0.180 mol. fac has all three A trans to B, while mer has one. Initial mer amount is

(A) 0.0600 mol (B) 0.0800 mol

(C) 0.0200 mol (D) 0.0400 mol

33. A $CoCl_3$ complex contains only en, NH_3 and chloride. Co amount is 0.100 mol; carbon assay gives 0.400 mol C, and complete ligand release gives 0.200 mol NH_3 (excluding en nitrogen). Assume Co coordination number 6 and bidentate en with two C. Immediate outer-sphere AgCl amount is

(A) 0.200 mol (B) 0.400 mol

(C) 0.100 mol (D) 0.300 mol

34. An octahedral Cr(III) chloride contains en, NH_3, Cl only. Elemental ratios are C:Cr=2 and N:Cr=4; en has two C, two N and occupies two sites. Total Cl:Cr=3. Number of ions from complete outer-sphere dissociation of one formula unit is

(A) 3 (B) 4

(C) 2 (D) 1

35. A fixing bath initially contains 0.100 mol $S_2O_3^{2-}$. It dissolves AgBr as $[Ag(S_2O_3)_2]^{3-}$, two ligands per Ag. A validated separation isolates only unused free thiosulfate, which consumes 0.0100 mol I_2 to form tetrathionate (two thiosulfates per I_2). With AgBr=188, film AgBr dissolved was

(A) 9.40 g (B) 7.52 g

(C) 15.04 g (D) 3.76 g

36. A 17.02 g AgCl–AgBr film is completely fixed by thiosulfate, two ligands per Ag. Initial ligand is 0.220 mol; validated separation measures 0.0200 mol unused free ligand. With AgCl=143.5, AgBr=188, original AgBr formula-unit fraction is

(A) $\frac{3}{5}$ (B) $\frac{2}{5}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

37. A $CoCl_3 \cdot nNH_3$ octahedral salt gives two mol immediate outer-sphere AgCl per mol Co. Its idealised osmotic factor is 2.50. Assume fraction alpha undergoes complete outer-sphere dissociation and the rest stays as a single particle; no intermediate ion pairs. Pair (n, alpha) is

(A) $(5, 0.750)$ (B) $(4, 0.750)$

(C) $(5, 0.500)$ (D) $(6, 0.500)$

38. A $CrCl_3 \cdot 6H_2O$ hydrate has coordination number 6. A 0.000100 mol Cr sample gives 0.01435 g immediate AgCl from outer chloride only (M=143.5). Its unperturbed osmotic factor is 1.80; use a model of intact particles or complete outer-sphere dissociation only. Pair (crystal-water molecules per formula, dissociated fraction) is

(A) $(2, 0.400)$ (B) $(2, 0.800)$

(C) $(3, 0.800)$ (D) $(1, 0.800)$

39. An unknown neutral iron carbonyl contains two Fe per formula. A sample has 0.0200 mol Fe and releases 0.0900 mol CO on complete decomposition. An independent structural probe counts 0.120 mol Fe–C contacts. Each CO is terminal (one contact) or bridging (two), with no other mode. Number of bridging CO per dimer is

(A) 6 (B) 4

(C) 3 (D) 2

40. A neutral dicobalt carbonyl isomer has 0.0200 mol Co in a sample,0.0800 mol total CO and 0.100 mol Co–C contacts. Each molecule has two Co; CO is only terminal or two-metal bridging. Terminal CO ligands per dicobalt molecule in this observed isomer number

(A) 2 (B) 4

(C) 8 (D) 6

41–50 / Preserved hard practice

41. A 2.685 g sample of a coordination compound contains 0.0100 mol Co and has only NH_3, Cl^-, and crystal water. It releases 0.0200 mol Cl^- immediately with $AgNO_3$. Co is+3 and coordination number 6; exactly one chloride ligand is bound. Molar amount of crystal water per mole compound is(Co=59, NH_3 17, Cl=35.5, H_2O =18)

(A) 3 (B) 0

(C) 1 (D) 2

42. A 0.100 mol mixture of $[Fe(CN)_6]^{4-}$ and $[FeF_6]^{3-}$ has 0.300 mol unpaired electrons under ideal low-spin CN^- /high-spin F^- models. Moles of $[FeF_6]^{3-}$ are

(A) 0.06 (B) 0.05

(C) 0.08 (D) 0.04

43. For an octahedral d^6 ion, high-spin CFSE =-.4 Δ_o with one pair and low-spin CFSE =-2.4 Δ_o with three pairs. If Δ_o =200 kJ /mol and P=150 kJ per pair, E low-E high in kJ /mol is

(A) -400 (B) -100

(C) 100 (D) 300

44. A 0.0100 mol $[Co(NH_3)_6]Cl_3$ sample and 0.0100 mol $[Co(NH_3)_5Cl]Cl_2$ sample are titrated together with $AgNO_3$ without ligand substitution. Mass of AgCl formed in g is(AgCl =143.5)

(A) 4.305 (B) 7.175

(C) 2.87 (D) 8.61

45. A 1.30 g anhydrous MCl_2 sample forms 0.0100 mol $[M(CN)_4]^{2-}$. The complex is squareplanar d^8 and diamagnetic. Using Cl=35.5, atomic mass of M from sample assay is

(A) 64 (B) 56

(C) 52 (D) 59

46. A 0.100 mol mixture of $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2$ requires 0.250 mol Ag^+ to precipitate all immediate outer Cl^-. Amount of hexaaqua compound in mol is

(A) 0.05 (B) 0.075

(C) 0.1 (D) 0.025

47. A low-spin d^6 complex has no unpaired electrons. Oxidation removes one electron without changing strong-field octahedral geometry. Spin-only moment of product in Bohr magnetons is

(A) 2.8284271 (B) 0

(C) 3.8729833 (D) 1.7320508

48. A 0.100 mol complex[M(en)2 Cl_2]Cl contains M^{3+} and is six-coordinate. On complete breakdown, all chloride ions are released. Extra AgCl formed after breakdown beyond the immediate outer-sphere precipitate in mol is

(A) 0.4 (B) 0.1

(C) 0.3 (D) 0.2

49. Two complexes contain identical M^{3+} d^5 centres but one is strong-field low-spin, and the other weak-field high-spin. Under ideal spin-only Curie addition, ratio of high-spin$\mu^2$to low-spin$\mu^2$is

(A) 11.666667 (B) 5

(C) 7 (D) 25

50. A six-coordinate entity contains M^{3+}, one neutral bidentate en ligand, and four chlorido ligands. The required net charge on the coordination entity is

(A) 0 (B) 1

(C) -1 (D) 3

Answer key

Coordination Compounds

C-COCO / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**C**	31	**B**	41	**C**
02	**A**	12	**B**	22	**B**	32	**A**	42	**A**
03	**A**	13	**D**	23	**B**	33	**D**	43	**B**
04	**A**	14	**B**	24	**C**	34	**C**	44	**B**
05	**D**	15	**D**	25	**A**	35	**B**	45	**D**
06	**D**	16	**D**	26	**C**	36	**A**	46	**A**
07	**B**	17	**B**	27	**A**	37	**A**	47	**D**
08	**B**	18	**A**	28	**D**	38	**B**	48	**D**
09	**D**	19	**A**	29	**C**	39	**C**	49	**A**
10	**C**	20	**C**	30	**C**	40	**D**	50	**C**

Chapter 21

GOC

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. The number of alpha $C-H$ bonds available for hyperconjugation with the positive centre in $(CH_3)_3C^+$ is

(A) 9

(B) 3

(C) 12

(D) 6

02. The number of constitutional alcohol isomers with formula C_3H_8O is

(A) 4

(B) 1

(C) 3

(D) 2

03. Which conjugate base is stabilized most directly by resonance across two equivalent oxygen atoms?

(A) $CH_3CH_2^-$

(B) CH_3COO^-

(C) $CH_3CH_2O^-$

(D) NH_2^-

04. Which is the strongest acid among ethane, ethene, and ethyne under the usual intrinsic hybridization trend?

(A) Ethane

(B) All equal

(C) Ethyne

(D) Ethene

05. Which simple carbanion is most stable under the usual alkyl+I trend, without resonance?

(A) $(CH_3)_2CH^-$

(B) CH_3^-

(C) $CH_3CH_2^-$

(D) $(CH_3)_3C^-$

06. Why is a benzyl cation more stable than an unsubstituted primary ethyl cation?

(A) It has no sigma bonds

(B) Its positive charge is resonance-delocalized into the aromatic ring

(C) It has no positive charge

(D) Benzene supplies extra neutrons

07. The degree of unsaturation of an acyclic hydrocarbon C_5H_8 is

(A) 2 (B) 3

(C) 1 (D) 4

08. A neutral compound has formula C_4H_7Cl. Treat Cl as equivalent to H. Degree of unsaturation is

(A) 0 (B) 2

(C) 1 (D) 3

09. Number of constitutional ether isomers with formula $C_4H_{10}O$ is

(A) 3 (B) 4

(C) 2 (D) 1

10. Total number of constitutional alcohol and ether isomers with formula C_3H_8O is

(A) 3 (B) 4

(C) 5 (D) 2

11–40 / New middle-to-hard practice

11. Two monocarboxylic acids A and B have calibrated pK_a values 3.00 and 5.00. A sample has 0.110 mol acids combined. At buffered pH4.00, a selective species assay counts 0.0730 mol carboxylate groups. Use $[A^-]/[HA] = 10^{pH-pK_a}$. Amount of the weaker acid is

(A) 0.0770 mol (B) 0.0550 mol

(C) 0.0370 mol (D) 0.0330 mol

12. Acids A and B have pK_a4.00 and 5.00. Total amount is 0.100 mol; at buffered pH4.00, carboxylate assay gives37/1100 mol. At buffered pH3.00, total carboxylate amount in an equal sample is

(A) 13/2222 mol (B) 1/110 mol

(C) 37/11000 mol (D) 1/101 mol

13. Two monobasic organic bases A, B have calibrated conjugate-acid pK_a values 6.00,8.00. At buffered pH7.00, a 0.110 mol sample has 0.0730 mol protonated base. Amount of the intrinsically stronger base is

(A) 0.0550 mol (B) 0.0330 mol

(C) 0.0370 mol (D) 0.0770 mol

14. A pyridine/aniline model has conjugate-acid pK_a values$4.60 + \log 4$ and 4.60, respectively. At buffered pH4.60, a 0.100 mol sample has 0.0680 mol total protonated base. Neutral pyridine amount is

(A) 0.0120 mol (B) 0.0300 mol

(C) 0.0200 mol (D) 0.0480 mol

15. For a supplied nitronate resonance model, contributor X is $H_2C = N^+(O^-)_2$; contributors Y, Z have $H_2C^- - N^+(=O) - O^-$ with the double bond on either O. Y, Z have equal weights. Measured C–N bond order is 1.40. Pair (mean N–O bond order, mean formal charge on C) is

(A) $(1.70, -0.600)$ (B) $(1.30, -0.600)$

(C) $(1.30, -0.400)$ (D) $(1.40, -0.400)$

16. Two amide species A, B are described by neutral $O = C - N$ and ionic $O^- - C = N^+$ contributors. Their independently calibrated ionic weights are 0.200 and 0.500. A 0.100 mol mixture has mean C–O bond order 1.68. Amount of B is

(A) 0.0200 mol (B) 0.0600 mol

(C) 0.0400 mol (D) 0.0800 mol

17. A 0.100 mol primary-alkyl-halide inventory is split between KCN and AgCN routes. Under the stated selective conditions KCN gives only nitrile RCN, AgCN only isocyanide RNC. Product hydrolysis yields 0.0600 mol carboxylic acid from nitrile; isocyanide yields primary amine RNH_2 and formic acid. Initial isocyanide amount is

(A) 0.100 mol (B) 0.0600 mol

(C) 0.0400 mol (D) 0.0200 mol

18. A 0.100 mol nitrile/isocyanide mixture has R=C_2H_5. Complete hydrolysis gives 0.140 mol total carboxylic acid molecules, counting both propanoic acid from nitrile and formic acid from isocyanide, and a separately measured 0.0400 mol ethylamine. Which diagnosis is correct?

(A) KCN must have attacked only through nitrogen; total acid count fixes that mechanism.

(B) The sample contains0.0400 mol nitrile and0.0600 mol isocyanide, and every assay is consistent.

(C) The sample contains0.0600 mol nitrile and0.0400 mol isocyanide, and every assay is consistent.

(D) The stated acid assay is inconsistent: both routes give one acid molecule per starting molecule.

19. A trapped carbocation ensemble contains 0.100 mol total tert-butyl and isopropyl cations. Count one classical C–H hyperconjugative contributor per alpha H. Ensemble contributor count is 0.780 mol (summed over ions). Quantitative water trapping of tert-butyl ions gives alcohol of M=74. Mass of that alcohol is

(A) 2.96 g (B) 4.44 g

(C) 5.92 g (D) 7.40 g

20. A 0.100 mol $C_4H_9^+$ ensemble contains only straight-chain primary and secondary butyl cations, trapped before rearrangement. Its mean count of alpha C–H hyperconjugative contributors is 3.20 per ion. Each ion traps water once. Butan-2-ol amount is

(A) 0.0800 mol (B) 0.0600 mol

(C) 0.0400 mol (D) 0.0200 mol

21. Two fully conjugated planar carbon rings A and B each contribute one p electron per neutral carbon; charges remove or add electrons conventionally. A is neutral and has n carbon atoms; B has n+2 carbons and charge+2. Combined carbon count is 14. Their pi-electron counts and aromatic classification are

(A) Both have eight pi electrons and are antiaromatic.
(B) A has four,B six; only B satisfies the rule.
(C) Both have six pi electrons and satisfy the $4k+2$ rule.
(D) A has six,B eight; only A satisfies the rule.

22. Assume every ring listed is planar and continuously conjugated: cyclopentadienyl anion, cyclopentadienyl cation, cycloheptatrienyl cation, cycloheptatrienyl anion. Removing one electron from the anion of each ring gives the corresponding neutral radical. Which pair of ionic rings is aromatic before removal?

(A) Cyclopentadienyl cation and cycloheptatrienyl anion.
(B) Both cations.
(C) Cyclopentadienyl anion and cycloheptatrienyl cation.
(D) Both anions.

23. A neutral organic compound contains only C, H, N and one nitrogen atom per molecule. Molar mass is 67;0.0100 mol gives 0.0400 mol CO_2 on complete combustion. Its hydrogen deficiency index is

(A) 4
(B) 1
(C) 2
(D) 3

24. A compound contains only C, H, O, Cl, has one O and one Cl per molecule, and molar mass 122.5. A 0.0100 mol sample gives 0.0500 mol CO_2 on combustion. With C=12, H=1, O=16, Cl=35.5, hydrogen deficiency index is

(A) 2
(B) 1
(C) 0
(D) 3

25. A mixture contains 0.100 mol total ethanol and ethane-1,2-diol. Reaction with excess sodium releases 0.0700 mol H_2; all OH hydrogens react and no other hydrogen does. Initial diol amount is

(A) 0.0700 mol
(B) 0.0600 mol
(C) 0.0200 mol
(D) 0.0400 mol

26. A 9.00 g phenol–ethanol mixture releases 0.0750 mol H_2 with sodium, one half H_2 per molecule. With molar masses 94 and 46, amount of phenol neutralised by aqueous NaOH under ordinary conditions is

(A) 7/160 mol
(B) 0.0900 mol
(C) 0.0600 mol
(D) 0.0750 mol

27. A primary alkyl chloride has formula $C_5H_{11}Cl$. Ionisation gives a primary carbocation whose adjacent carbon bears no H and is attached to three methyl groups. The favourable first rearrangement and resulting tertiary cation are

(A) No rearrangement; the primary cation is already tertiary by hyperconjugation.
(B) One 1,2-methyl shift; 2-methylbutan-2-yl cation.
(C) One 1,2-hydride shift; 2-methylbutan-2-yl cation.
(D) One 1,2-methyl shift; tert-butyl cation.

28. Ionisation of 3-methylbutan-2-ol is modelled as water loss to $CH_3 - CH^+ - CH(CH_3) - CH_3$, followed by one shift before nucleophilic capture. Which shift increases carbocation degree while preserving the five-carbon skeleton?

(A) Moving only the positive-charge symbol to C3 without an electron-pair shift.
(B) Loss of a methyl radical from C3, leaving the same five-carbon skeleton.
(C) 1,2-hydride transfer from C1 to C2, leaving a tertiary charge at C1.
(D) 1,2-hydride transfer from C3 to C2, leaving a tertiary charge at C3.

29. Two tautomers K, E equilibrate with E/K=1/4. A selective trapping step converts 80% of E and 10% of K without interconversion during trapping. Starting from 0.100 mol equilibrated material, trapped product derived from E is what fraction of total trapped product?

(A) $\frac{1}{2}$
(B) $\frac{2}{3}$
(C) $\frac{4}{5}$
(D) $\frac{1}{5}$

30. A tautomer mixture at equilibrium has 0.100 mol total. Selective trapping converts 90% of E and 20% of K, giving 0.0480 mol total product. No interconversion occurs during trapping. Original E/K ratio is

(A) $\frac{2}{3}$
(B) $\frac{3}{5}$
(C) $\frac{3}{2}$
(D) $\frac{2}{5}$

31. A neutral acyclic C_4H_5N compound contains exactly one nitrile and no ring or other heteroatom. The remaining unsaturation is only a C=C bond. Among carbon atoms only, the pair (sp, sp^2 counts) is

(A) $(1, 3)$
(B) $(2, 2)$
(C) $(2, 1)$
(D) $(1, 2)$

32. An acyclic C_5H_6O compound has exactly one aldehyde, one C=C andone C≡C, with no other multiple bond. Is this structural inventory compatible with its formula, and if not what hydrogen count would match these features?

(A) Incompatible; C_5H_4O is required.
(B) Incompatible; the oxygen atom adds one to the hydrogen deficiency index.
(C) Incompatible; C_5H_8O is required.
(D) Compatible; C_5H_6O already has four unsaturation units.

33. For 3-methylphenol, number OH carbon 1 and methyl carbon 3. Treat each substituent as ortho/para directing. Exclude occupied carbons; also exclude the site ortho to both substituents as sterically blocked. Number of available sites directed by both is

(A) 0
(B) 2
(C) 1
(D) 3

34. Compare 3-methylphenol and 4-methylphenol, with OH at carbon 1. Before steric exclusions, count unoccupied ring carbons simultaneously ortho/para to both OH and CH_3. The ordered pair of counts is

(A) $(1, 1)$
(B) $(2, 2)$
(C) $(3, 2)$
(D) $(3, 0)$

35. A 0.100 mol acetone–benzophenone sample undergoes validated complete alpha-H/D exchange without side reaction. It incorporates 0.240 mol D; aromatic H does not exchange. With molar masses 58 and 182, original sample mass is

(A) 13.24 g (B) 5.80 g

(C) 15.72 g (D) 11.60 g

36. An 8.12 g ethanal–benzaldehyde mixture incorporates 0.120 mol D in a validated alpha-H/D exchange with no side reactions. Ethanal has three such H; benzaldehyde none. With molar masses 44,106 and Ag=108, complete subsequent Tollens reaction gives silver mass

(A) 8.64 g (B) 21.6 g

(C) 10.8 g (D) 12.96 g

37. An 8.80 g mixture contains butanoic acid and methyl propanoate, both M=88. A bicarbonate test gives 0.0400 mol CO_2 from acid only. Hydrolysis of all original ester gives methanol (M=32). Methanol mass is

(A) 2.56 g (B) 1.92 g

(C) 3.20 g (D) 1.28 g

38. A 4.60 g ethanol–dimethyl-ether mixture contains only isomers of C_2H_6O (M=46). Reaction with sodium releases 0.0200 mol H_2; ether is inert and each alcohol molecule gives half H_2. Ether mole fraction is

(A) $\frac{1}{2}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{1}{5}$

39. An organic substitution has empirical $r = k[S]^m[Nu]^n$. Doubling S and tripling Nu multiplies rate by 6; tripling S and doubling Nu also multiplies it by 6. Halving both concentrations changes rate by what factor, and which minimal kinetic description fits?

(A) 1/4; bimolecular dependence on substrate and nucleophile. (B) 1/2; substrate-only first order.

(C) 1/6; termolecular dependence. (D) 1; zero order in both.

40. An organic substitution rate is unchanged when Nu concentration is quadrupled at fixed S. Increasing both S and Nu threefold multiplies rate by 3. Under one power-law rate model, rate after reducing S to one quarter and doubling Nu is

(A) 2 times the original. (B) 1/4 of the original.

(C) 1/8 of the original. (D) 1/2 of the original.

41–50 / Preserved hard practice

41. A neutral acyclic compound has formula C_4H_8O. It reacts with 2, 4-DNP but not Tollens reagent. Number of constitutional carbonyl isomers satisfying these data is

(A) 4 (B) 1

(C) 2 (D) 3

42. A $C_4H_{10}O$ compound reacts with Na and exhibits no carbonyl group. Complete oxidation gives a ketone without $C-C$ cleavage. Number of constitutional structures possible is

(A) 3 (B) 1

(C) 2 (D) 4

43. A 0.100 mol acyclic monoalkene consumes 0.100 mol Br_2. Its bromine addition product has molar mass 216 g/mol(Br=80). Number of carbon atoms in starting alkene is

(A) 3 (B) 2

(C) 4 (D) 5

44. A 0.100 mol mixture of acyclic C_4H_8 alkene and C_4H_{10} alkane consumes 0.0600 mol H 2 on complete hydrogenation. Total number of pi bonds in the original sample, expressed as multiple of N_A, is

(A) 0.1 (B) 0.12

(C) 0.06 (D) 0.04

45. A compound contains C, H, O only and has molar mass 88 g/mol. It contains one carboxyl group and no other unsaturation or ring. Number of carbon atoms is

(A) 5 (B) 6

(C) 3 (D) 4

46. A primary monoamine contains one N atom and has formula $C_4H_{11}N$. Number of acyclic constitutional primary amines with this formula is

(A) 4 (B) 3

(C) 2 (D) 5

47. A monochloroalkane has formula C_4H_9Cl. Number of constitutional structures giving 2-methylpropene as the sole carbon skeleton alkene on dehydrohalogenation is

(A) 3 (B) 1

(C) 4 (D) 2

48. An acyclic organic compound C_4H_6 contains no triple bond and shows conjugation. Number of $C-C$ sigma bonds is

(A) 2 (B) 4

(C) 5 (D) 3

49. A 0.0500 mol monocarboxylic acid HA with one acidic proton is neutralized completely. If the normal carboxylate resonance model assigns equivalent C – O bonds, total moles of such C – O bonds are

(A) 0.1 (B) 0.05

(C) 0.025 (D) 0.2

50. A compound C_3H_6O has an aldehyde group. After complete hydrogenation of the carbonyl, the product has how many sp^3-hybridized carbon atoms per molecule?

(A) 4 (B) 2

(C) 3 (D) 1

Answer key

GOC

C-GOC / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**D**	21	**C**	31	**D**	41	**B**
02	**D**	12	**A**	22	**C**	32	**A**	42	**B**
03	**B**	13	**D**	23	**D**	33	**B**	43	**C**
04	**C**	14	**A**	24	**C**	34	**D**	44	**C**
05	**B**	15	**B**	25	**D**	35	**A**	45	**D**
06	**B**	16	**C**	26	**A**	36	**B**	46	**A**
07	**A**	17	**C**	27	**B**	37	**B**	47	**D**
08	**C**	18	**D**	28	**D**	38	**C**	48	**D**
09	**A**	19	**B**	29	**B**	39	**A**	49	**A**
10	**A**	20	**C**	30	**A**	40	**B**	50	**C**

Chapter 22

Purification & Characterisation

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. If 0.200 g compound gives 0.287 g AgCl in Carius analysis, chlorine mass percentage is(AgCl =143.5, Cl=35.5)

(A) 53.25 (B) 35.5

(C) 17.75 (D) 71

02. If 0.100 g compound gives 0.233 g $BaSO_4$ in sulfur estimation, sulfur mass percentage is($BaSO_4$ =233, S=32)

(A) 24 (B) 16

(C) 32 (D) 64

03. A 0.140 g compound gives 0.00100 mol N_2 in Dumas analysis. Nitrogen mass percentage is

(A) 28 (B) 10

(C) 20 (D) 14

04. A 100 g solution has 20% solute. Evaporating 25 g solvent with no solute loss changes mass percentage solute to

(A) 40 (B) 20

(C) 26.666667 (D) 25

05. A compound contains 40% C, 6.667% H, and remainder O. Using exact mole ratio 1: 2: 1, the empirical formula mass is(C=12, H=1, O=16)

(A) 30 (B) 90

(C) 15 (D) 60

06. In a Kjeldahl assay, NH_3 from a sample consumes 25.0 mL 0.100 M HCl. Nitrogen mass in g is

(A) 0.35 (B) 0.0175

(C) 0.035 (D) 0.07

07. A 0.500 g sample gives 0.00250 mol NH_3 in Kjeldahl estimation. Nitrogen mass percentage is

(A) 3.5 (B) 28

(C) 7 (D) 14

08. A compound contains one Cl atom per molecule. A 0.250 g sample forms 0.287 g AgCl. Its molar mass in g/mol is(AgCl =143.5)

(A) 125 (B) 62.5

(C) 358.75 (D) 250

09. An immiscible organic-water mixture contains 10.0 g solute. One extraction removes 60% of the currently remaining amount. After two identical extractions, total extracted mass in g is

(A) 6 (B) 8.4

(C) 7.2 (D) 9.6

10. At fixed T, a steam distillation model has p_{water}=600 torr and p organic=160 torr. Vapor molar ratio organic/water is

(A) 0.26666667 (B) 0.21052632

(C) 3.75 (D) 0.78947368

11–40 / New middle-to-hard practice

11. A 0.500 g organic sample gives 0.3315 g AgCl+AgBr by a validated Carius assay. Complete reduction of that precipitate gives 0.2160 g Ag. With Ag=108, Cl=35.5, Br=80, chlorine mass percentage in the sample is

(A) 14.2% (B) 7.10%

(C) 16.0% (D) 23.1%

12. A 0.500 g organic sample gives 0.3785 g AgCl+AgI. Quantitative reduction of precipitate yields 0.2160 g Ag. With Ag=108, Cl=35.5, I=127, iodine mass percentage is

(A) 50.8% (B) 7.10%

(C) 12.7% (D) 25.4%

13. A 0.300 g compound of molar mass 300 gives 0.233 g $BaSO_4$ on complete sulfur assay and 0.00100 mol N_2 in a separate Dumas assay of an equal sample. With $BaSO_4$=233, pair (S atoms, N atoms) per molecule is

(A) $(2, 1)$ (B) $(1, 2)$

(C) $(1, 1)$ (D) $(2, 2)$

14. A 0.250 g compound of molar mass 250 gives 0.466 g $BaSO_4$ and 22.4 mL dry N_2 at STP from equal sample portions. With $BaSO_4$=233, STP molar volume 22.4 L, total N+S atoms per compound molecule is

(A) 3 (B) 6

(C) 4 (D) 2

15. A 0.200 g sample containing only C, H, N, O yields 50.0 mL wet CO_2+N_2 at 300 K,780 mmHg. Water vapour is 30.0 mmHg. After quantitative CO_2 absorption, dry N_2 volume is 25.0 mL at 300 K,750 mmHg. With R=62.5 L mmHg mol^{-1} K^{-1}, pair (N%, C%) is

(A) $(14.0, 6.00)$ (B) $(14.0, 12.0)$

(C) $(14.56, 6.24)$ (D) $(7.00, 6.00)$

16. A 0.0900 g CHNO compound of molar mass 90 gives 67.2 mL dry CO_2+N_2 at STP; after CO_2 absorption 22.4 mL N_2 remains. A separate equal combustion sample gives 0.0540 g H_2O. With C=12, H=1, N=14, O=16, molecular formula is

(A) $C_3H_6N_2O$ (B) $C_2H_6N_2O$

(C) CH_3NO (D) $C_2H_6N_2O_2$

17. A 1.20 g compound of molar mass 120 is completely Kjeldahl-digested to 250 mL. A 25.0 mL aliquot yields NH_3 trapped in 0.00500 mol HCl; excess acid requires 0.00200 mol NaOH. Number of Kjeldahl-detectable N atoms per compound molecule is

(A) 2 (B) 3

(C) 6 (D) 1

18. A 7.44 g urea–methylurea mixture is completely Kjeldahl-digested; each molecule gives two NH_3. NH_3 is trapped in 0.250 mol HCl and excess acid needs 0.0300 mol NaOH. With molar masses 60 and 74, original urea mass fraction is

(A) $\frac{37}{62}$ (B) $\frac{5}{11}$

(C) $\frac{1}{2}$ (D) $\frac{25}{62}$

19. A 0.116 g CHO compound gives 0.264 g CO_2. Water trap gain is 0.112 g, but an identical blank contributes 0.00400 g. Molar mass is 58. With standard integer atomic masses, hydrogen deficiency index is

(A) 3 (B) 1

(C) 2 (D) 0

20. A 0.217 g CHOCl compound of molar mass 108.5 gives 0.264 g CO_2,0.0900 g H_2O and 0.287 g AgCl in independent complete assays. With AgCl=143.5, pair (O atoms per molecule, hydrogen deficiency index) is

(A) $(2, 2)$ (B) $(2, 1)$

(C) $(3, 1)$ (D) $(1, 2)$

21. An aqueous 100 mL solution contains 10.0 g solute. One equilibrium extraction with 50.0 mL organic solvent transfers 6.00 g. Assume constant $K = C_{org}/C_{aq}$ and unchanged phase volumes. For an equal fresh aqueous sample extracted twice with fresh 25.0 mL portions, solute left in water is

(A) 40/7 g (B) 100/49 g

(C) 4.00 g (D) 160/49 g

22. A 100 mL aqueous solution with 10.0 g solute is equilibrated with 50.0 mL organic solvent;6.00 g enters the organic layer. That isolated organic layer is then equilibrated with 100 mL fresh water. Constant $K = C_{org}/C_{aq}$, no volume change. Solute transferred into the fresh water is

(A) 2.40 g (B) 1.20 g

(C) 3.60 g (D) 4.00 g

23. A chromatogram has solvent-front travel 8.00 cm and two spots at 2.00,6.00 cm, matching standards A($R_f = 0.250$), B($R_f = 0.750$). Integrated signals at the two spots are 60.0,40.0 units; calibrated responses are 3.00 units/mg A,2.00 units/mg B. A mass fraction is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$

(C) $\frac{1}{2}$ (D) $\frac{3}{4}$

24. Standard (R_f in solvent I, solvent II) pairs are A:(0.40,0.20), B:(0.40,0.60), C:(0.20,0.60), D:(0.20,0.20). Unknown travels 2.40 cm with a 6.00 cm front in I and 4.80 cm with an 8.00 cm front in II. Its identity is

(A) C (B) A

(C) D (D) B

25. An immiscible water–organic system steam-distils at total pressure 760 mmHg; water vapour pressure is 700 mmHg. Distillate contains 6.00 g organic and 14.0 g water. Assuming ideal vapour and separate liquid phases, organic molar mass is (H_2O=18)

(A) 90.0 $\mathrm{g\,mol^{-1}}$ (B) 60.0 $\mathrm{g\,mol^{-1}}$

(C) 75.0 $\mathrm{g\,mol^{-1}}$ (D) 120 $\mathrm{g\,mol^{-1}}$

26. Three separate immiscible liquid phases water, A, B have vapour pressures 600,100,50 mmHg at the distillation temperature. A collected equilibrium distillate contains 18.0 g water and 30.0 g organics, with organic mass ratio A:B=2:1. With water M=18, molar mass of A is

(A) 90.0 $\mathrm{g\,mol^{-1}}$ (B) 180 $\mathrm{g\,mol^{-1}}$

(C) 120 $\mathrm{g\,mol^{-1}}$ (D) 60.0 $\mathrm{g\,mol^{-1}}$

27. An ideal equimolar A–B liquid has equilibrium vapour A fraction 0.750 at a fixed T. Add pure B equal to the initial total liquid-mole amount, then re-establish equilibrium at that T. New vapour A fraction is

(A) $\frac{3}{8}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

28. For an ideal binary liquid at one T, total pressure is 180 mmHg at liquid $x_A = 0.200$ and 420 mmHg at $x_A = 0.800$. Liquid x_A needed for equilibrium vapour $y_A = 0.750$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{8}$

(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

29. A 12.0 g crude solid contains 2.00 g impurity insoluble in the hot solvent, removed by hot filtration. Product solubilities are 20.0 g per 100 g solvent hot and 2.00 g per 100 g cold. Using minimum hot solvent and neglecting evaporation, maximum dry product crystals recovered are

(A) 10.8 g (B) 9.00 g

(C) 10.0 g (D) 8.80 g

30. A crude solid contains 10.0 g A and 2.00 g B. Independent hot solubilities are 20.0,10.0 g per 100 g solvent; cold values 2.00,1.00. Use the minimum hot solvent dissolving both and cool without evaporation. Assume independent solubilities. A mass fraction in recovered crystals is

(A) $\frac{5}{6}$ (B) $\frac{9}{10}$

(C) $\frac{4}{5}$ (D) $\frac{6}{7}$

31. A 10.0 g A–B solid undergoes a specified sublimation separation retaining 90.0% of A and 10.0% of B in the collector. Collected mass is 5.80 g. Repeating the identical separation on that collected material gives A mass fraction

(A) $\frac{243}{245}$ (B) $\frac{24}{25}$

(C) $\frac{9}{10}$ (D) $\frac{27}{29}$

32. A sublimation collector retains unknown fractions p of A, q of B. Calibration feeds (8.00 g A,2.00 g B) and (2.00 g A,8.00 g B) give 6.60 and 2.40 g collected. An unknown 10.0 g feed gives 3.80 g under identical conditions. Initial A mass fraction is

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$

(C) $\frac{1}{2}$ (D) $\frac{2}{5}$

33. A neutral organic liquid gives H_2 with sodium, negative neutral $FeCl_3$ phenol test, and mixes extensively with water. A drying trial with anhydrous $CaCl_2$ retains part of the organic liquid as a complex. Under the usual qualitative assumptions, which pair is best supported?

(A) A phenol; the negative $FeCl_3$ result proves $CaCl_2$ is inert.

(B) An ether; the sodium gas arises from an ether O-H bond.

(C) A hydrocarbon; loss with $CaCl_2$ proves chemical oxidation.

(D) An ordinary alcohol; prefer a compatible drying agent such as anhydrous K_2CO_3 rather than $CaCl_2$.

34. A 3.24 g mixture contains benzoic acid, phenol, anisole. Selective $NaHCO_3$ extraction gives 0.0100 mol CO_2 from benzoic acid; subsequent NaOH extraction removes 0.0100 mol phenol. Both extracted components are quantitatively removed. With molar masses 122,94,108, anisole remaining is

(A) 2.02 g (B) 0.940 g

(C) 1.08 g (D) 1.22 g

35. For a calibrated dilute-impurity liquidus model $T = T_0 - kx$, standards with impurity mole fractions 0.0200 and 0.0600 melt at 118 and 110 degrees C. An unknown obeying the same model melts at 116 degrees C. Impurity mole fraction is

(A) $\frac{1}{25}$ (B) $\frac{1}{20}$

(C) $\frac{3}{100}$ (D) $\frac{1}{50}$

36. Use $T = T_0 - kx$ for impurity mole fraction x. Standards x=0.0100,0.0400 melt at 120,114 degrees C. A 100 mmol sample melts at 116 degrees C. Recrystallisation retains 90% of host and 50% of impurity. Impurity mole fraction in retained crystals is

(A) $\frac{5}{296}$ (B) $\frac{5}{291}$

(C) $\frac{3}{100}$ (D) $\frac{1}{60}$

37. A CHO acid has empirical formula CH_2O by complete elemental assay and true vapour molar mass 60. In a nonpolar solvent its apparent colligative molar mass is 100. Assume only monomers and dimers and fraction alpha of original acid molecules enters dimers. Pair (molecular formula, alpha) is

(A) $(C_2H_4O_2, 0.400)$ (B) $(CH_2O, 0.800)$

(C) $(C_2H_4O_2, 0.800)$ (D) $(C_3H_6O_3, 0.600)$

38. A compound contains only C, H and one halogen per molecule. Mean molar mass 104.5; molecular-ion isotope peaks at 104 and 106 have intensity 3:1, identifying one chlorine (mean 35.5). Carbon assay gives 6.00 g C in 0.100 mol compound. Molecular formula is

(A) C_4H_9Cl (B) C_5H_9Cl

(C) $C_5H_{11}Cl$ (D) C_5H_9Br

39. A 0.500 g moist organic sample is 10.0% water by an independent moisture assay. Sulfur assay gives 0.300 g $BaSO_4$, including 0.0670 g from the reagent blank. With $BaSO_4$=233, S=32, sulfur percentage on a dry-sample basis is

(A) 32/9% (B) 6.40%

(C) 80/9% (D) 64/9%

40. A 0.161 g CHOCl compound of molar mass 80.5 gives 0.220 g CO_2 including 0.0440 g blank,0.0900 g H_2O after its blank correction, and 0.287 g AgCl. With AgCl=143.5 and standard atomic masses, molecular formula is

(A) C_2H_5ClO (B) C_2H_5Cl

(C) C_3H_5ClO (D) CH_3ClO_2

41–50 / Preserved hard practice

41. A 0.590 g sample containing only C, H, N produces 0.00500 mol N_2 in Dumas analysis and 1.320 g CO_2 on complete combustion. Molar mass is 59 g/mol. Number of hydrogen s per molecule is

(A) 5 (B) 9

(C) 7 (D) 11

42. A 0.300 g compound containing C, H, O only gives 0.440 g CO_2 and 0.180 g water. Its molar mass is 60 g/mol. Number of constitutional monocarboxylic acid isomers with this formula is

(A) 3 (B) 1

(C) 4 (D) 2

43. A compound contains only C, H, and one Cl atom per molecule. A 0.185 g sample gives 0.287 g AgCl in Carius analysis and contains 0.0960 g carbon. With AgCl =143.5, Cl=35.5, C=12, H=1, the number of constitutional monochloroalkane structures consistent with the data is

(A) 3 (B) 2

(C) 5 (D) 4

44. A solute has organic/water concentration distribution ratio D=4.0. A 100 mL water phase initially contains 10.0 g solute. One extraction with 25.0 mL organic phase reaches partition equilibrium with constant volumes. Solute mass in g inorganic phase is

(A) 5 (B) 8

(C) 2 (D) 4

45. A 0.500 gamine sample is processed by Kjeldahl. Evolved NH_3 is absorbed in 0.0100 mol HCl; remaining acid needs 0.00750 mol NaOH. Nitrogen percentage is

(A) 7 (B) 28

(C) 14 (D) 3.5

46. A compound contains one S atom each molecule. A 0.150 g sample gives 0.233 g $BaSO_4$. Independently, molar mass is 150 g/mol. Carbon mass fraction 48%. Number of C atoms per molecule is

(A) 8 (B) 3

(C) 4 (D) 6

47. A 5.00 g sample of a solid is crystallized from 50.0 g water. Solubilities are 20.0 g per 100 g water hot and 4.00 g per 100 g water cold. Assuming complete initial dissolution and no solvent loss, crystal mass in g is

(A) 3 (B) 1

(C) 2 (D) 4

48. A 0.100 mol gas mixture consists of N_2 and CO_2 and has mass 3.60 g. Passing through an absorber removes all CO_2 but no N_2. Remaining gas amount in mol is

(A) 0.075 (B) 0.05

(C) 0.1 (D) 0.025

49. A sample contains a 1: 1 molar mixture of NaCl and NaBr. Precipitating all halide as AgCl and AgBr gives 3.315 g solid. With AgCl =143.5, AgBr =188, moles of each original salt are

(A) 0.01 (B) 0.005

(C) 0.015 (D) 0.02

50. A 0.100 mol sample of a monochloroalkane undergoes partial hydrolysis to a monohydric alcohol. All inorganic chloride is removed. Carius analysis of the entire remaining organic mixture gives 5.74 g AgCl. All alcohol is then reacted with excess Na. With AgCl =143.5, the hydrogen amount formed in mol is

(A) 0.06 (B) 0.04

(C) 0.02 (D) 0.03

Answer key

Purification & Characterisation

C-PUCH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**B**	21	**D**	31	**A**	41	**B**
02	**C**	12	**D**	22	**A**	32	**D**	42	**B**
03	**C**	13	**B**	23	**C**	33	**D**	43	**D**
04	**C**	14	**C**	24	**D**	34	**C**	44	**A**
05	**A**	15	**A**	25	**A**	35	**C**	45	**A**
06	**C**	16	**D**	26	**C**	36	**A**	46	**D**
07	**C**	17	**B**	27	**C**	37	**C**	47	**A**
08	**A**	18	**D**	28	**B**	38	**B**	48	**B**
09	**B**	19	**B**	29	**B**	39	**D**	49	**A**
10	**A**	20	**B**	30	**D**	40	**A**	50	**D**

Chapter 23

Practical Chemistry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 25.0 mL 0.100 M Na_2CO_3 sample is completely neutralized by 0.100 M HCl. Required volume in mL is

(A) 12.5 (B) 25

(C) 50 (D) 100

02. The phenolphthalein endpoint for 25.0 mL 0.100 M Na_2CO_3 with 0.100 M HCl requires volume in mL of

(A) 25 (B) 50

(C) 12.5 (D) 75

03. A 0.0100 mol Cu^{2+} sample forming $[Cu(NH_3)_4]^{2+}$ requires NH_3 amount in mol of

(A) 0.04 (B) 0.02

(C) 0.01 (D) 0.08

04. A 0.0100 mol aldehyde sample reduces Tollens reagent with 2 Ag^+ per aldehyde. Silver mass in g is(Ag=108)

(A) 4.32 (B) 10.8

(C) 1.08 (D) 2.16

05. A chloride bromide mixture contains 0.0100 mol of each ion. Excess $AgNO_3$ gives precipitate mass in g of(AgCl =143.5, AgBr =188)

(A) 1.435 (B) 3.315

(C) 1.88 (D) 6.63

06. A 0.0200 mol NH_4^+ sample is converted quantitatively to NH_3. The ammonia needs 0.100 M HCl in volume L of

(A) 0.4 (B) 2

(C) 0.2 (D) 0.1

07. A 0.0100 mol Cu^{2+} sample is reduced to Cu^+ by I^-, forming 0.00500 mol I_2. The iodine needs $S_2O_3^{2-}$ in mol of

(A) 0.01 (B) 0.005

(C) 0.02 (D) 0.04

08. A 0.0100 mol $[Co(NH_3)_6]Cl_3$ sample gives immediate AgCl precipitate mass in g of(AgCl =143.5)

(A) 2.87 (B) 8.61

(C) 1.435 (D) 4.305

09. A barium test sample contains 0.0100 mol Ba^{2+} and 0.0150 mol SO_4^{2-}. Assuming complete precipitation, $BaSO_4$ formed in mol is

(A) 0.01 (B) 0.005

(C) 0.025 (D) 0.015

10. In $Fe^{3+} + SCN^- \rightleftharpoons$red complex, adding SCN^- at fixed temperature initially shifts the equilibrium toward product. If 1: 1 binding is assumed, maximum complex from 0.010 mol Fe^{3+} and 0.004 mol SCN^- in mol is

(A) 0.006 (B) 0.004

(C) 0.002 (D) 0.01

11–40 / New middle-to-hard practice

11. Acidic permanganate is standardised:0.134 g $Na_2C_2O_4$ (M=134, two electrons per oxalate) uses 20.0 mL. A 0.100 g iron-containing sample, all its iron as Fe(II), then uses 5.00 mL of this permanganate with no interference. With Fe=56, iron mass percentage is

(A) 56.0% (B) 7.00%

(C) 28.0% (D) 14.0%

12. Acidic $KMnO_4$ uses 25.0 mL for 0.126 g $H_2C_2O_4 \cdot 2H_2O$ (M=126, two electrons per oxalate). A 10.0 mL aliquot of a 100 mL peroxide sample uses 10.0 mL of the same permanganate. With H_2O_2=34 and two electrons donated per peroxide, total peroxide mass is

(A) 0.0680 g (B) 0.340 g

(C) 1.36 g (D) 0.136 g

13. A thiosulfate solution is standardised with 0.214 g KIO_3 (M=214): one iodate liberates three I_2, each needing two thiosulfates. Titration volume 30.0 mL. A 0.500 g bleach sample then requires 15.0 mL of that thiosulfate after iodide treatment. Available Cl_2 mass percentage is (Cl_2=71;one Cl_2 gives one I_2)

(A) 10.65% (B) 14.2%

(C) 42.6% (D) 21.3%

14. KIO_3 standardisation:0.107 g (M=214) requires 15.0 mL thiosulfate, six thiosulfates per iodate. A Cu(II) sample liberates iodine by $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$ and needs 20.0 mL of that thiosulfate. With Cu=64, copper mass is

(A) 0.256 g (B) 0.512 g

(C) 0.0640 g (D) 0.128 g

15. A 4.00 g alkaline sample may contain NaOH, Na_2CO_3, $NaHCO_3$ plus inert solid; equilibrated OH^- and bicarbonate do not coexist. Acid uptake to phenolphthalein endpoint is 0.0200 mol and total uptake to methyl-orange endpoint 0.0600 mol. With carbonate M=106, bicarbonate M=84, active-salt mass percentage is

(A) 84.0% (B) 95.0%

(C) 79.0% (D) 53.0%

16. A 100 mL alkaline solution may contain NaOH, Na_2CO_3, $NaHCO_3$ but no equilibrated OH^-/HCO_3^- coexistence. Acid uptake to phenolphthalein is 0.0400 mol, to full methyl-orange endpoint 0.0600 mol. Initial NaOH molarity is

(A) 0.600 M (B) 0.400 M

(C) 0.200 M (D) 0.100 M

17. A 0.0300 mol total Cu(II)/Fe(III) sample gives 0.955 g CuS in a validated selective acidic sulfide separation (CuS=95.5); Fe remains dissolved. All Fe is then precipitated as $Fe(OH)_3$ (M=107). Precipitate mass is

(A) 3.21 g (B) 1.07 g

(C) 2.14 g (D) 0.535 g

18. A 0.0200 mol Cu(II)/Cd(II) mixture gives 2.400 g total CuS+CdS. Molar masses are 95.5,144.5. A subsequent validated separation recovers all copper as $Cu(OH)_2$ (M=97.5), with cadmium excluded. Copper hydroxide mass is

(A) 0.4875 g (B) 0.975 g

(C) 1.445 g (D) 1.950 g

19. For a calibrated sulfide separation, $[S^{2-}] = 10^{-30}/[H^+]^2$. Initially Cd^{2+} and Zn^{2+} are 0.0100 M each; K_{sp} values are10^{-28},10^{-22}. Require at least 99.9% Cd removal and no onset of ZnS. Allowed pH interval is

(A) $[2.00, 5.00)$ (B) $[5.00, 6.50)$

(C) $[3.50, 5.00)$ (D) $[3.50, 6.00)$

20. Fe^{3+} and Mg^{2+} are initially 0.0100 M. Given $K_{sp}(Fe(OH)_3) = 10^{-38}$,$K_{sp}(Mg(OH)_2) = 10^{-11}$,$pK_w = 14$, require 99.9% Fe removal and no Mg precipitation. An ammonia buffer obeys $pH = 9.25 + \log([NH_3]/[NH_4^+])$ and is additionally restricted to its useful range $|pH - 9.25| \leq 1$. Ignore complex formation. Allowed interval for that logarithm is

(A) $[-1.00, 1.00]$ (B) $[-0.250, 1.00]$

(C) $[-6.25, 0.250)$ (D) $[-1.00, 0.250)$

21. Borax standardisation uses 0.381 g $Na_2B_4O_7 \cdot 10H_2O$ (M=381), requiring 20.0 mL HCl, two HCl per borax. A 1.00 g NH_4Cl–NaCl sample releases all NH_3 into 50.0 mL of this acid; excess acid needs 20.0 mL0.100 M NaOH. With NH_4Cl=53.5, ammonium-chloride mass percentage is

(A) 16.05% (B) 10.70%

(C) 32.10% (D) 26.75%

22. KHP (M=204, monoprotic)0.204 g requires 20.0 mL NaOH. A 0.110 g ester–inert sample is completely saponified with 30.0 mL of this NaOH; excess base requires 10.0 mL0.0500 M HCl. Ester M=88, one OH^- per ester. Ester mass percentage is

(A) 60.0% (B) 100%

(C) 40.0% (D) 80.0%

23. For a fixed-time Fe(II) permanganate assay, validated matrix background volume is proportional to chloride concentration. The same Fe standard uses 12.0 mL at 0.100 M Cl^- and 16.0 mL at 0.300 M Cl^-. An unknown at 0.200 M Cl^- uses 20.0 mL0.0200 M $KMnO_4$. True Fe(II) amount is

(A) 0.00160 mol (B) 0.00200 mol

(C) 0.000320 mol (D) 0.00120 mol

24. An Fe assay obeys $A = kC$ with no blank. A 10.0 mL sample gives A=0.400. Adding 2.00 mL0.0100 M Fe standard to that same aliquot gives A=0.500; volumes add and chemistry is unchanged. Original Fe concentration is

(A) 0.00500 M (B) 0.00800 M

(C) 0.00400 M (D) 0.00200 M

25. A sodium-fusion extract contains only CN^- and SCN^- as nitrogen-bearing anions. Total N is 0.100 mol and total S0.0300 mol. After a validated selective separation of thiocyanate, Fe(III) gives its blood-red complex. Fraction of extract nitrogen originally in the blood-red-test precursor is

(A) $\frac{1}{2}$ (B) $\frac{3}{10}$

(C) $\frac{7}{10}$ (D) $\frac{3}{7}$

26. A 100 mL fusion extract contains 0.0200 mol N as CN^-/SCN^-. A validated selective SCN assay follows $A = 20[SCN^-]$ and gives A=0.800. Remaining CN is quantitatively converted to $Fe_4[Fe(CN)_6]_3$ (M=860),18 CN per formula. Product mass is

(A) 43/225 g (B) 86/225 g

(C) 86/45 g (D) 172/225 g

27. Acidification of a carbonate–sulfite sample gives 0.100 mol total CO_2+SO_2. In a separate equal sample oxidation of sulfite to sulfate releases 0.0800 mol electrons; carbonate is inert. Remove SO_2 quantitatively from the gas, then pass CO_2 into enough limewater to form carbonate without overcarbonation. $CaCO_3$ mass is (M=100)

(A) 8.00 g (B) 10.0 g

(C) 4.00 g (D) 6.00 g

28. A Na_2CO_3–Na_2SO_3 sample gives 0.100 mol total acid gas. A separate equal sample consumes 0.0600 mol I_2 in quantitative oxidation of sulfite to sulfate, one I_2 per sulfite. Oxidation is kept alkaline with sufficient added base to retain all inorganic carbon as carbonate. Before acidification, excess $BaCl_2$ then precipitates both carbonate and newly formed sulfate. With $BaCO_3$=197, $BaSO_4$=233, total precipitate mass is

(A) 13.98 g (B) 21.86 g

(C) 23.30 g (D) 7.88 g

29. A mixture of $(NH_4)_2CO_3$ and NH_4HCO_3 decomposes quantitatively to NH_3, CO_2, water. Validated collection gives 0.100 mol NH_3 and 0.0600 mol CO_2. Molar masses are 96 and 79. Original sample mass is

(A) 7.90 g (B) 4.74 g

(C) 5.42 g (D) 6.00 g

30. A sample contains anhydrous $(NH_4)_2C_2O_4$ and NH_4Cl. Alkaline ammonia assay gives 0.100 mol NH_3. A separate equal sample loses 0.0600 mol electrons on complete oxidation of oxalate to CO_2; chloride is excluded from the validated redox assay. With molar masses 124,53.5, initial mass is

(A) 7.44 g (B) 5.35 g

(C) 3.72 g (D) 5.86 g

31. A 0.100 mol mixture contains ethanal, acetone, benzaldehyde. Tollens assay gives 0.120 mol Ag; iodoform assay gives 0.0700 mol CHI_3. Assume quantitative standard selectivity: both aldehydes give 2Ag, ethanal and acetone give 1CHI_3. With benzaldehyde=106, its mass is

(A) 6.36 g (B) 3.18 g

(C) 4.24 g (D) 2.12 g

32. A 0.100 mol methanol–ethanol–ethanal mixture has 0.0600 mol O-H groups by a validated spectral assay. Quantitative iodoform assay gives 0.0700 mol CHI_3 from ethanol and ethanal, one each. With molar masses 32,46,44, mean molar mass is

(A) 40.0 $g\,mol^{-1}$ (B) 42.0 $g\,mol^{-1}$

(C) 41.0 $g\,mol^{-1}$ (D) 43.0 $g\,mol^{-1}$

33. In an idealised Mohr chloride titration, $K_{sp}(AgCl) = 10^{-10}$ and $K_{sp}(Ag_2CrO_4) = 10^{-12}$. Free chromate is held 0.0100 M. Initially chloride is 0.0100 M; volumes are treated constant. Percentage chloride precipitated when chromate indicator just starts precipitating is

(A) 99.9% (B) 90.0%

(C) 99.0% (D) 99.99%

34. Use $K_{sp}(AgCl) = 10^{-10}$,$K_{sp}(Ag_2CrO_4) = 10^{-12}$. For initial 0.0100 M chloride at unchanged volume, choose free chromate so indicator precipitation starts exactly after 99.99% chloride removal. Required chromate concentration is

(A) 10^{-3} M (B) 10^{-4} M

(C) 10^{-2} M (D) 10^{-6} M

35. A 2.11 g sample contains MCl_2 and MCO_3, with one divalent metal M. Complete selective chloride assay gives 0.0200 mol AgCl; acidification of an equal sample gives 0.0100 mol CO_2. With Cl=35.5, C=12, O=16, atomic molar mass M is

(A) 88 (B) 56

(C) 24 (D) 40

36. A 2.43 g sample contains MCl_2 and $M(NO_3)_2$, with one divalent metal. Validated assays give 0.0200 mol AgCl and 0.0100 mol N_2 from complete nitrate-nitrogen analysis. With Cl=35.5, N=14, O=16, atomic molar mass M is

(A) 24 (B) 56

(C) 40 (D) 65

37. A 10.0 g Zn–ZnO–Cu sample reacts completely with nonoxidising acid except for Cu. H_2 evolved is 0.0600 mol; total H^+ consumed is 0.180 mol. With Zn=65, ZnO=81, copper mass percentage is

(A) 61.0% (B) 36.7%

(C) 24.3% (D) 39.0%

38. An 8.00 g Mg–MgO–Cu sample evolves 0.100 mol H_2 and consumes 0.300 mol H^+ on complete nonoxidising-acid treatment; Cu is inert. With Mg=24, MgO=40, copper mass is

(A) 3.60 g (B) 2.40 g

(C) 2.00 g (D) 5.60 g

39. An 8.00 g technical carbide contains CaC_2, Ca_3P_2 and inert matter. Complete hydrolysis yields 0.100 mol total C_2H_2+PH_3; validated gas analysis finds 0.0200 mol PH_3. CaC_2 gives one acetylene and Ca_3P_2 gives two phosphines per formula. With CaC_2=64, carbide mass percentage is

(A) 80.0% (B) 32.0%

(C) 16.0% (D) 64.0%

40. A dry sample contains Ca(OCl)Cl and $CaCl_2$ only, total 0.100 mol Ca. Iodometry measures 0.0600 mol I_2, one per Ca(OCl)Cl; $CaCl_2$ is inert. With molar masses 127,111 and Cl_2=71, available-chlorine mass fraction as Cl_2 equivalents is

(A) $\frac{3}{5}$ (B) $\frac{71}{201}$

(C) $\frac{37}{67}$ (D) $\frac{127}{201}$

41–50 / Preserved hard practice

41. A 1.00 g sample contains NaCl and NaBr plus inert material. Total halide is 0.0100 mol and Ag precipitate mass is 1.6575 g. Moles of bromide are(AgCl =143.5, AgBr =188)

(A) 0.01
(B) 0.0075
(C) 0.0025
(D) 0.005

42. A mixture contains a total of 0.100 mol of Na_2CO_3 and $NaHCO_3$ and requires 0.150 mol H^+ for complete acidification. With formula masses 106 and 84, the mass fraction of Na_2CO_3 is

(A) 0.75
(B) 0.5
(C) 0.55789474
(D) 0.44210526

43. A 0.0100 mol mixture of ethanal and propanone produces 0.0120 mol Ag with Tollens reagent. Both react with 2, 4-DNP. Moles of propanone are

(A) 0.004
(B) 0.008
(C) 0.006
(D) 0.002

44. A 0.100 mol mixture of ethanol and ethanoic acid gives 0.0400 mol CO_2 with excess bicarbonate. In a fresh identical sample, all ethanol is selectively converted quantitatively to ethanal, with acid unchanged. Subsequent Tollens test produces Ag in mol of

(A) 0.04
(B) 0.12
(C) 0.06
(D) 0.08

45. A 0.0100 mol Cu^{2+} sample liberates I_2 with I^- then requires 0.100 M thiosulfate. Expected volume in mL is

(A) 100
(B) 400
(C) 200
(D) 50

46. A 0.0100 mol Al^{3+} sample forms $Al(OH)_3$ then redissolves as $Al(OH)_4{}^-$. Net OH^- consumed through complete complex formation in mol is

(A) 0.01
(B) 0.04
(C) 0.03
(D) 0.07

47. A 0.100 mol mixture of Fe^{2+} and Fe^{3+} needs 0.0120 mol acid $MnO_4{}^-$ to oxidize all Fe^{2+}. In a fresh identical sample before oxidation, oxygen is excluded and iron is precipitated quantitatively as $Fe(OH)_2$ and $Fe(OH)_3$. Required OH^- amount in mol is

(A) 0.18
(B) 0.3
(C) 0.24
(D) 0.12

48. A sulfate precipitation assay gives 0.466 g $BaSO_4$ from 20.0 mL sample. Sulfate concentration in M is($BaSO_4$ =233)

(A) 0.2
(B) 0.05
(C) 0.1
(D) 0.02

49. A total 0.100 mol mixture of Na_2CO_3 and NaOH requires 0.140 mol HCl for complete neutralization. With Na_2CO_3 formula mass 106, the mass of carbonate in g is

(A) 2.12 (B) 4.24

(C) 10.6 (D) 6.36

50. A nitrogen assay produces NH_3 captured by 0.0100 mol HCl. Backtitrating residual acid needs 20.0 mL 0.100 M NaOH. Nitrogen mass in g is

(A) 0.112 (B) 0.028

(C) 0.224 (D) 0.14

Answer key

Practical Chemistry

C-PRCH / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**C**	21	**A**	31	**B**	41	**D**
02	**A**	12	**D**	22	**D**	32	**C**	42	**C**
03	**A**	13	**D**	23	**A**	33	**A**	43	**A**
04	**D**	14	**A**	24	**C**	34	**B**	44	**B**
05	**B**	15	**B**	25	**B**	35	**D**	45	**A**
06	**C**	16	**C**	26	**D**	36	**A**	46	**B**
07	**A**	17	**C**	27	**D**	37	**B**	47	**C**
08	**D**	18	**B**	28	**B**	38	**A**	48	**C**
09	**A**	19	**C**	29	**C**	39	**D**	49	**B**
10	**B**	20	**D**	30	**D**	40	**B**	50	**A**

Chapter 24

Hydrocarbons

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Complete combustion of 1 mol benzene consumes O_2 in mol of

(A) 9 (B) 6

(C) 7 (D) 7.5

02. Number of pi electrons in a cyclopentadienyl anion with full conjugation is

(A) 6 (B) 4

(C) 8 (D) 5

03. The number of distinct constitutional monochlorination products of neopentane is

(A) 4 (B) 2

(C) 1 (D) 3

04. The number of distinct constitutional monochlorination products of n-butane is

(A) 3 (B) 4

(C) 2 (D) 1

05. The number of constitutional acyclic monoalkenes with formula C_4H_8 is

(A) 3 (B) 4

(C) 2 (D) 5

06. An acyclic monoalkene has molar mass 70 g/mol. Number of C atoms is

(A) 5 (B) 7

(C) 6 (D) 4

07. An acyclic alkyne with one triple bond has M 68 g/mol. Carbon count is

(A) 4 (B) 6

(C) 7 (D) 5

08. Number of terminal acyclic alkyne constitutional structures with formula C_5H_8 is

(A) 1 (B) 4

(C) 2 (D) 3

09. For 1, 3-butadiene adding one HBr through 1, 4-addition, the remaining C = C is between original carbons

(A) $2, 3$ (B) $1, 4$

(C) $1, 2$ (D) $3, 4$

10. A benzene ring bears one methyl group. Number of constitutionally distinct ring-mono nitration products is

(A) 2 (B) 5

(C) 4 (D) 3

11–40 / New middle-to-hard practice

11. A 0.100 mol ethane–ethene mixture consumes 0.320 mol O_2 on complete combustion. With ethane=30, ethene=28, original ethene mass is

(A) 2.80 g (B) 2.24 g

(C) 1.12 g (D) 1.68 g

12. A 0.100 mol methane–propane mixture yields 0.180 mol CO_2 on complete combustion. Oxygen used is

(A) 0.400 mol (B) 0.360 mol

(C) 0.320 mol (D) 0.280 mol

13. A 0.100 mol ethene–ethyne mixture completely hydrogenates to ethane using 0.160 mol H_2. Original ethyne mass fraction is (ethene=28, ethyne=26)

(A) $\frac{1}{2}$ (B) $\frac{39}{67}$

(C) $\frac{3}{5}$ (D) $\frac{28}{67}$

14. A 4.06 g mixture contains ethane, ethene, ethyne, total 0.150 mol. Complete hydrogenation uses 0.180 mol H_2. With masses 30,28,26, can all stated data be consistent?

(A) No; the first two amount assays require total mass4.14 g.

(B) No; hydrogenation must increase the number of hydrocarbon molecules.

(C) Yes; ethane,ethene,ethyne are0.0300,0.0600,0.0600 mol.

(D) Yes; any triple with total0.150 mol can have mass4.06 g.

15. Propene and but-2-ene, total 0.100 mol, undergo complete reductive ozonolysis. A validated selective assay finds 0.0400 mol formaldehyde. Remaining aldehyde product is ethanal. With ethanal=44, total ethanal mass is

(A) 7.04 g (B) 8.80 g

(C) 5.28 g (D) 4.40 g

16. A 0.100 mol mixture of 1-butene and 2-methylpropene undergoes reductive ozonolysis. Gas-chromatographic assay gives 0.0300 mol acetone; the other non-formaldehyde product is propanal. With propanal=58, its mass is

(A) 4.06 g (B) 1.74 g

(C) 3.48 g (D) 5.80 g

17. A 4.98 g propyne–but-1-yne–but-2-yne mixture requires 0.200 mol H_2 for complete hydrogenation. A validated terminal-alkyne assay counts 0.0500 mol terminal C-H groups. With propyne=40, butynes=54, original but-1-yne amount is

(A) 0.0500 mol (B) 0.0200 mol

(C) 0.0300 mol (D) 0.0700 mol

18. A 0.100 mol mixture contains acetylene, propyne, but-2-yne. A validated terminal-C-H assay counts 0.120 mol terminal H; selective hydration gives 0.0400 mol ethanal from acetylene only, other components give ketones. Original propyne amount is

(A) 0.0200 mol (B) 0.0800 mol

(C) 0.0400 mol (D) 0.0600 mol

19. An 8.00 g benzene–cyclohexene mixture quantitatively consumes 0.0500 mol Br_2 under selective dark conditions: cyclohexene adds one Br_2, benzene does not. With masses 78,82, complete catalytic hydrogenation including benzene rings requires

(A) 0.100 mol H_2 (B) 0.200 mol H_2

(C) 0.150 mol H_2 (D) 0.250 mol H_2

20. A 0.100 mol benzene–cyclohexene–cyclohexane mixture consumes 0.0300 mol Br_2 selectively in the dark and 0.150 mol H_2 on complete hydrogenation. Cyclohexene uses one of each, benzene three H_2, cyclohexane neither. With cyclohexane=84, its mass is

(A) 1.68 g (B) 2.52 g

(C) 5.88 g (D) 3.36 g

21. For statistical monochlorination, rate contribution is(number of equivalent H)$\times$(per-H reactivity). Propane gives 1-chloropropane:2-chloropropane=3:2. Assume transferable primary and secondary per-H reactivities. Fraction of monochlorinated n-butane giving 2-chlorobutane is

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$

(C) $\frac{4}{7}$ (D) $\frac{1}{2}$

22. An ideal transferable-reactivity model gives $k_s/k_p = 2$. Isobutane has 9 primary H,1 tertiary H; its tertiary monochlorination fraction is 0.400. For 2-methylbutane (9 primary,2 secondary,1 tertiary H), secondary-substitution fraction is

(A) $\frac{4}{15}$ (B) $\frac{2}{15}$

(C) $\frac{2}{9}$ (D) $\frac{4}{19}$

23. An aromatic hydrocarbon has molar mass 106 and gives only one symmetry-distinct ring-mononitration product. Strong side-chain oxidation gives a benzene dicarboxylic acid. Which hydrocarbon fits all observations?

(A) o-Xylene (B) p-Xylene

(C) Mesitylene (D) Ethylbenzene

24. A C_8H_{10} aromatic hydrocarbon gives a $C_8H_6O_4$ product on complete benzylic oxidation and exactly two symmetry-distinct ring-mononitro derivatives. It is

(A) p-Xylene (B) o-Xylene

(C) Ethylbenzene (D) m-Xylene

25. A 0.100 mol propene inventory reacts with HBr in two separately controlled modes: ordinary ionic addition or peroxide-initiated radical addition. After hydrolysis of all alkyl bromide, quantitative iodoform assay gives 0.0400 mol CHI_3. Fraction of propene assigned to radical mode was

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{2}{5}$ (D) $\frac{3}{5}$

26. A 0.100 mol 1-butene/2-methylpropene mixture gives 0.0300 mol acetone on selective reductive ozonolysis of an equal sample. Another equal sample undergoes ordinary ionic HBr addition then complete hydrolysis. CHI_3 amount is

(A) 0.0400 mol (B) 0.0300 mol

(C) 0.0700 mol (D) 0.100 mol

27. A 0.100 mol cis/trans-but-2-ene mixture has mean squared dipole 0.360 times that of pure cis; trans dipole is zero. Hydrogenation releases 120 kJ/mol cis and 116 kJ/mol trans under supplied conditions. Total heat evolved is

(A) 11.856 kJ (B) 11.600 kJ

(C) 11.744 kJ (D) 12.000 kJ

28. A cis/trans-but-2-ene sample releases 11.744 kJ on full hydrogenation (cis 120, trans 116 kJ/mol). Its calibrated total squared-dipole response equals 0.0360 mol of pure cis response; trans contributes zero. Trans mole fraction is

(A) $\frac{1}{2}$ (B) $\frac{16}{25}$

(C) $\frac{4}{5}$ (D) $\frac{9}{25}$

29. A 5.60 g cyclobutane–but-1-ene mixture (both M=56) consumes 0.0400 mol H_2 under validated mild conditions that hydrogenate only C=C, without ring opening. Cyclobutane mass is

(A) 3.36 g (B) 1.12 g

(C) 2.24 g (D) 4.48 g

30. A 7.00 g cyclopentane–pent-1-ene–pent-2-ene mixture (all M=70) consumes 0.0600 mol Br_2 under conditions selective for C=C. An equal sample gives 0.0200 mol formaldehyde on reductive ozonolysis, only from pent-1-ene. Initial pent-2-ene mole fraction is

(A) $\frac{1}{5}$ (B) $\frac{1}{2}$

(C) $\frac{3}{5}$ (D) $\frac{2}{5}$

31. A styrene–ethylbenzene inventory gives 0.800 mol CO_2 on complete combustion. Under selective mild hydrogenation, only styrene side-chain C=C reacts, using 0.0500 mol H_2. On a fresh equal sample, full hydrogenation including benzene rings requires

(A) 0.200 mol H_2 (B) 0.300 mol H_2

(C) 0.350 mol H_2 (D) 0.400 mol H_2

32. A styrene–phenylacetylene sample gives 0.800 mol CO_2 on complete combustion anduses 0.160 mol H_2 to saturate only side-chain multiple bonds. Ring is unchanged. Each terminal phenylacetylene forms one C_8H_5Ag equivalent in a validated terminal-alkyne assay (M=209). Product mass is

(A) 6.27 g (B) 20.90 g

(C) 12.54 g (D) 8.36 g

33. An acetylene–propyne mixture is completely hydrated to ethanal and acetone, respectively. The products give 0.100 mol CHI_3 and 0.0800 mol Ag in validated assays (one iodoform per carbonyl, only ethanal gives two Ag). With acetylene=26, propyne=40, original gas mass is

(A) 3.44 g (B) 4.00 g

(C) 3.16 g (D) 2.60 g

34. A but-1-yne/but-2-yne mixture gives 0.100 mol iodoform after complete hydration; each gives one methyl ketone and one iodoform. A validated terminal-C-H assay finds 0.0400 mol but-1-yne. Complete oxidative cleavage of an equal sample gives propanoic acid+CO_2 from terminal alkyne and two ethanoic acids from internal. After removing inorganic CO_2, organic acids consume bicarbonate to evolve

(A) 0.160 mol CO_2 (B) 0.120 mol CO_2

(C) 0.100 mol CO_2 (D) 0.200 mol CO_2

35. In an ideal random Wurtz-coupling model, methyl-fragment fraction is p, ethyl 1-p; ethane, propane, butane fractions are $p^2, 2p(1-p), (1-p)^2$. Propane fraction is 0.480 and butane exceeds ethane. For0.100 mol total products, butane mass is (M=58)

(A) 0.928 g (B) 2.784 g

(C) 1.392 g (D) 2.088 g

36. In an ideal random methyl/ethyl Wurtz-coupling model, ethane, propane, butane fractions are$p^2, 2p(1-p), (1-p)^2$ and masses 30,44,58. A 0.100 mol product mixture has mean mass 46.8. Selective collection recovers 75% of propane. Propane mass collected is

(A) 2.112 g (B) 1.584 g

(C) 3.168 g (D) 1.056 g

37. A 0.100 mol ethylbenzene–tert-butylbenzene mixture is vigorously oxidised under stated selective benzylic-H conditions: only ethylbenzene gives benzoic acid,0.0600 mol. tert-Butylbenzene lacks benzylic H and is unchanged. With molar masses 106,134, initial mass is

(A) 12.28 g (B) 10.60 g

(C) 13.40 g (D) 11.72 g

38. A 12.0 g cumene–mesitylene mixture (both M=120) oxidises completely at every benzylic side chain. Cumene gives monobasic benzoic acid;mesitylene gives tribasic trimesic acid. Product neutralisation needs 0.220 mol OH^-. With trimesic acid=210, its mass is

(A) 21.0 g (B) 8.40 g

(C) 4.20 g (D) 12.6 g

39. A 0.100 mol buta-1,3-diene sample adds exactly one HBr per molecule, giving only 1,2-product $CH_3CHBrCH=CH_2$ and 1,4-product $CH_3CH=CHCH_2Br$. After hydrolysis, a validated iodoform assay responds only to the 1,2-derived secondary alcohol and gives 0.0300 mol CHI_3. Fraction of initial addition products that were 1,4 is

(A) $\frac{3}{10}$ (B) $\frac{1}{2}$

(C) $\frac{3}{7}$ (D) $\frac{7}{10}$

40. A 0.100 mol 1,2/1,4 HBr-adduct mixture from butadiene is probed by the validated hydrolysis/iodoform test: only 1,2-adduct gives one CHI_3. Initial response 0.0600 mol falls to 0.0200 after a controlled process converting only some 1,2 to 1,4. Pair (fraction of initial 1,2 converted, final 1,4 amount) is

(A) $(2/3, 0.0600\ \text{mol})$ (B) $(2/3, 0.0800\ \text{mol})$

(C) $(1/3, 0.0800\ \text{mol})$ (D) $(1/2, 0.0400\ \text{mol})$

41–50 / Preserved hard practice

41. A 0.100 mol mixture of ethene and ethyne consumes 0.160 mol H 2 on full hydrogenation. Ethyne amount in mol is

(A) 0.06 (B) 0.02

(C) 0.08 (D) 0.04

42. An alkene on yield-preserving reductive ozonolysis gives only propanone. Its molar mass in g/mol is(C=12, H=1)

(A) 56 (B) 70

(C) 84 (D) 98

43. A 0.100 mol mixture of propane and propene consumes 0.480 mol O 2 on complete combustion. Hydrogen needed to fully saturate afresh equal sample in mol is

(A) 0.04 (B) 0.08

(C) 0.02 (D) 0.06

44. A 0.100 mol mixture of benzene and cyclohexene consumes 0.200 mol H_2 to fully saturate all rings, to cyclohexane. Under complete benzene hydrogenation, benzene uses 3 H_2 and cyclohexene 1 H_2. Benzene amount in mol is

(A) 0.025 (B) 0

(C) 0.05 (D) 0.075

45. A terminal alkyne with formula C_4H_6 is converted quantitatively to a silver acetylide by replacement of one acidic H with Ag. Molar mass of product ing/mol is(Ag=108, C=12, H=1)

(A) 216 (B) 161

(C) 162 (D) 108

46. A 0.0500 mol alkene produces 0.100 mol total carbonyl molecules on reductive ozonolysis. One product is 0.0500 mol methanal and the other 0.0500 mol butan-2-one. Carbon count in starting alkene is

(A) 5 (B) 6

(C) 7 (D) 4

47. An acyclic alkene has formula C_5H_{10} and one hydrogen on one sp 2 carbon but none on the other. Number of structural isomers satisfying this is

(A) 4 (B) 1

(C) 2 (D) 3

48. A 0.100 mol mixture of but-1-ene and but-2-ene on reductive ozonolysis gives 0.0400 mol methanal. Total ethanal formed in mol is

(A) 0.06 (B) 0.12

(C) 0.08 (D) 0.16

49. A 0.100 mol mixture of propyne and propene is fully hydrogenated to propane. It consumes 0.140 mol H_2. Complete combustion of fresh equal sample requires O_2 in moles of

(A) 0.42 (B) 0.45

(C) 0.44 (D) 0.43

50. A 0.100 mol mixture of n-butane and isobutane is radically monochlorinated under a hypothetical model with all primary H reactivity 1 and all secondary/tertiary H reactivity 0. Only total 0.780 mol primary H atoms are present initially. Moles of isobutane are

(A) 0.08 (B) 0.06

(C) 0.02 (D) 0.04

Answer key

Hydrocarbons

C-HYDR / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**D**	21	**C**	31	**C**	41	**A**
02	**A**	12	**C**	22	**D**	32	**C**	42	**C**
03	**C**	13	**B**	23	**B**	33	**A**	43	**A**
04	**C**	14	**A**	24	**B**	34	**A**	44	**C**
05	**A**	15	**A**	25	**D**	35	**D**	45	**B**
06	**A**	16	**A**	26	**C**	36	**B**	46	**A**
07	**D**	17	**B**	27	**C**	37	**D**	47	**B**
08	**C**	18	**C**	28	**B**	38	**D**	48	**B**
09	**A**	19	**B**	29	**A**	39	**D**	49	**D**
10	**D**	20	**B**	30	**D**	40	**B**	50	**B**

Chapter 25

Haloalkanes & Haloarenes

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. Number of constitutionally distinct monobromo alkanes with formula C_3H_7Br is

(A) 3 (B) 1

(C) 2 (D) 4

02. Complete Wurtz coupling of 0.200 mol bromoethane gives n-butane in mol of

(A) 0.05 (B) 0.2

(C) 0.4 (D) 0.1

03. The degree of unsaturation of C_6H_5Cl, treating Cl as equivalent to H, is

(A) 3 (B) 4

(C) 1 (D) 2

04. In ideal SN2 rate=k [RX] [Nu-], doubling both concentrations changes rate by factor

(A) 2 (B) 4

(C) 8 (D) 0.5

05. In ideal SN1 rate=k [RX], tripling only nucleophile concentration changes rate by factor

(A) 3 (B) 0.33333333

(C) 1 (D) 9

06. If an optically active sample has specific rotation+20 and its mirror image-20 in matching conditions, a 50: 50 mixture has average rotation of

(A) -20 (B) 0

(C) 20 (D) 10

07. In a stated linear rotation model, a mixture has 75% ofthe+20 enantiomer and 25% ofthe-20 enantiomer. Average rotation is

(A) 5 (B) 0

(C) 15 (D) 10

08. An SN1 substrate has first-order half-life 10 min. After 20 min, the unreacted fraction is

(A) 0.125 (B) 0.25

(C) 0.75 (D) 0.5

09. An alkyl chloride has molar mass 92.5 $\mathrm{g\,mol^{-1}}$ and formula $\mathrm{C_nH_{2n+1}Cl}$. Its carbon count is (Cl=35.5)

(A) 4 (B) 5

(C) 3 (D) 6

10. For the overall oxidative change $\mathrm{2CHCl_3 + O_2 \rightarrow 2COCl_2 + 2HCl}$, the oxygen amount consumed per mole of chloroform is

(A) 1 (B) 0.75

(C) 0.5 (D) 0.25

11–40 / New middle-to-hard practice

11. For parallel substitution paths of one substrate,$k_{obs} = k_1 + k_2[Nu]$, where first term is unimolecular and second bimolecular. Measured k values are 0.0300 $\mathrm{s^{-1}}$ at Nu=0.100 M and 0.0700 at Nu=0.300 M. At Nu=0.200 M, fraction reacting through the bimolecular path is

(A) $\frac{1}{5}$ (B) $\frac{4}{5}$

(C) $\frac{2}{3}$ (D) $\frac{3}{5}$

12. A single chiral substrate reacts by exactly racemising SN1 and exactly inverting SN2. At Nu=0.100 M,$k_{obs} = 0.0500\ \mathrm{s^{-1}}$ and product is 80.0% inverted,20.0% retained. Assume SN1 contribution independent of Nu and SN2 proportional to Nu. At doubled Nu, inverted fraction is

(A) $\frac{4}{5}$ (B) $\frac{3}{4}$

(C) $\frac{9}{10}$ (D) $\frac{7}{8}$

13. A substrate has unimolecular E1 plus bimolecular E2 and SN2 pathways. $k_{obs} = k_1 + (k_E + k_S)[B]$. At B=0.100,0.300 M, total k is 0.0800,0.160 $\mathrm{s^{-1}}$; alkene fraction at 0.100 M is 0.750. At0.200 M, alkene fraction is

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$

(C) $\frac{1}{2}$ (D) $\frac{4}{5}$

14. A 0.100 mol 2-bromo-2-methylbutane sample eliminates HBr; only 2-methylbut-1-ene and 2-methylbut-2-ene form. Reductive ozonolysis detects 0.0700 mol formaldehyde, only from terminal alkene. The other carbonyls are butan-2-one from terminal, and acetone plus ethanal from internal. Each listed methyl ketone/ethanal gives one iodoform. Total $\mathrm{CHI_3}$ is

(A) 0.100 mol (B) 0.130 mol

(C) 0.160 mol (D) 0.0700 mol

15. A C_4H_9Br isomer gives exactly three stereochemically distinct alkene products on beta elimination, including one cis/trans pair. Which substrate fits, and what alcohol is obtained on ordinary unrearranged hydrolysis?

(A) 1-Bromobutane; butan-1-ol.
(B) 2-Bromobutane; butan-2-ol.
(C) Isobutyl bromide; 2-methylpropan-1-ol.
(D) tert-Butyl bromide; tert-butanol.

16. A $C_5H_{11}Br$ substrate has a tertiary C–Br carbon. Only 2-methylbut-1-ene and 2-methylbut-2-ene form on elimination. Treat per-beta-H reactivities as k_t for the terminal-alkene route,k_i for internal. Terminal product fraction is 0.900. Ratio k_t/k_i is

(A) 3
(B) 9
(C) 1
(D) $\frac{1}{3}$

17. A 9.12 g mixture contains ethyl chloride and ethyl bromide. Quantitative iodide exchange gives 0.0400 mol NaCl and converts both substrates to ethyl iodide. Molar masses are 64.5,109,156. Ethyl iodide mass formed is

(A) 6.24 g
(B) 9.36 g
(C) 14.4 g
(D) 15.6 g

18. A mixture RCl/RBr has the same acyclic saturated monovalent alkyl group R. Quantitative iodide exchange gives 0.0300 mol chloride and 0.0200 mol bromide ions, plus 8.50 g RI. With I=127, C=12, H=1, carbon count in R is

(A) 4
(B) 5
(C) 3
(D) 2

19. A mixture contains 1,2-dichloroethane and 1,1,2-trichloroethane. Complete combustion gives 0.100 mol CO_2; complete halogen assay gives 0.120 mol AgCl. With molar masses 99,133.5, original mass is

(A) 5.64 g
(B) 4.95 g
(C) 6.675 g
(D) 5.985 g

20. A chlorobenzene–benzyl-chloride mixture gives 0.100 mol AgCl only after complete destructive halogen assay. A validated mild hydrolysis releases 0.0600 mol chloride from benzyl chloride only; chlorobenzene remains unchanged. With chlorobenzene=112.5, its mass is

(A) 11.25 g
(B) 7.59 g
(C) 4.50 g
(D) 6.75 g

21. A 0.100 mol feed contains ethyl bromide,2-bromopropane and tert-butyl bromide. Complete conversion gives only the corresponding alcohols and alkenes; total alkene is 0.0600 mol. Controlled oxidation of the separated alcohols gives 0.0300 mol carbonyl compounds; the isolated aldehyde deposits 0.0400 mol Ag with Tollens reagent. Mass of the original separated alcohol fraction is (M ethanol 46, propan-2-ol60, tert-butanol 74)

(A) 2.54 g
(B) 2.26 g
(C) 1.84 g
(D) 2.40 g

22. A 0.100 mol ethyl-bromide/2-bromopropane feed converts only to the corresponding alcohols and alkenes. Total alkene is 0.0400 mol. After separating all alcohols, complete iodoform processing gives only sodium formate and sodium acetate as organic sodium salts, total 4.50 g (M68,82). Ethanol originally present in the isolated alcohol fraction was

(A) 0.0400 mol (B) 0.0200 mol

(C) 0.0300 mol (D) 0.0600 mol

23. A 0.100 mol para-/meta-nitrochlorobenzene mixture is treated under calibrated aromatic addition-elimination conditions. Conversion per pass is 96.0% para,2.00% meta. First-pass total nitrophenol is 0.0584 mol. Original para-isomer mole fraction is

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$

(C) $\frac{3}{5}$ (D) $\frac{2}{5}$

24. A 0.100 mol para-/meta-nitrochlorobenzene feed produces 0.0584 mol nitrophenol in one pass that converts 96.0% para,2.00% meta. Unreacted material is isolated and passed once more under identical conditions. Final unreacted meta-isomer amount is

(A) 0.0016 mol (B) 0.0392 mol

(C) 0.038416 mol (D) 0.003088 mol

25. In an ideal random phenyl/ethyl coupling model, phenyl fraction p gives biphenyl, ethylbenzene, butane fractions$p^2, 2p(1-p), (1-p)^2$ with masses 154,106,58. A 0.100 mol product mixture has mean molar mass 106. Ethylbenzene mass is

(A) 7.95 g (B) 5.30 g

(C) 10.60 g (D) 2.65 g

26. In an ideal random phenyl/methyl coupling model, biphenyl, toluene, ethane fractions are$p^2, 2p(1-p), (1-p)^2$, masses 154,92,30. A 0.200 mol product mixture has mean mass 79.6. Collection recovers 75% of toluene. Collected mass is

(A) 6.624 g (B) 4.416 g

(C) 8.832 g (D) 13.248 g

27. Two equal aliquots contain the same CH_3MgBr/C_2H_5MgBr mixture. Hydrolysis of one gives 0.100 mol gas of mean molar mass 21.6 (CH_4=16, C_2H_6=30). The other is completely carbonated then acidified to ethanoic and propanoic acids (M=60,74). Total acid mass is

(A) 6.56 g (B) 5.60 g

(C) 6.00 g (D) 7.40 g

28. From 0.100 mol ethyl bromide,80.0% becomes C_2H_5MgBr. A water impurity destroys 0.0100 mol Grignard reagent before 0.0500 mol acetone is added; carbonyl addition is complete for the limiting feed. During final aqueous workup only unused Grignard evolves ethane. Ethane formed in this final workup is

(A) 0.0200 mol (B) 0.0500 mol

(C) 0.0300 mol (D) 0.0100 mol

29. A 0.100 mol benzyl-chloride/tert-butyl-chloride/chlorobenzene mixture gives 0.0600 mol AgCl in a validated short solvolysis assay; only the first two react. Selective benzylic oxidation of an equal sample gives 0.0200 mol benzoic acid from benzyl chloride only. With masses 126.5,92.5,112.5, initial total mass is

(A) 12.65 g (B) 10.73 g

(C) 11.25 g (D) 10.09 g

30. A 10.65 g allyl-chloride/benzyl-chloride mixture gives 0.100 mol total chloride on complete assay. Hydrolysis replaces every Cl by OH without rearrangement. With allyl chloride=76.5, benzyl chloride=126.5, benzyl alcohol=108, benzyl-alcohol mass formed is

(A) 8.80 g (B) 4.32 g

(C) 10.80 g (D) 6.48 g

31. An optically active secondary bromide gives substitution with inversion. Doubling substrate alone doubles initial rate;doubling nucleophile alone also doubles it. No alkene is found. Which mechanistic conclusion is warranted, without assuming rate law alone uniquely determines mechanism?

(A) First order in nucleophile proves a unimolecular rate-limiting ionisation.

(B) Inversion proves a freely equilibrating planar carbocation formed first.

(C) Bimolecular kinetics by itself proves SN2 even if an alkene were formed.

(D) The substitution product and inversion support SN2; the bimolecular rate law alone would also be compatible with an E2 rate form.

32. A tertiary chloride ionises in a rate-limiting step independent of nucleophile concentration. Its carbocation then gives substitution at rate proportional to Nu, or elimination at constant competing capture rate. Initially products are 50:50. Tripling Nu changes neither ionisation rate nor elimination capture constant. Pair (overall-rate factor, substitution fraction) is

(A) $(3, 1/2)$ (B) $(1, 3/4)$

(C) $(1, 1/2)$ (D) $(3, 3/4)$

33. A 0.100 mol 1,1-/1,2-dichloroethane mixture hydrolyses completely. Geminal isomer gives ethanal, one net water per molecule;vicinal gives ethane-1,2-diol, two waters. Validated carbonyl assay finds 0.0400 mol ethanal. Net water consumed is

(A) 0.120 mol (B) 0.100 mol

(C) 0.200 mol (D) 0.160 mol

34. A 0.100 mol mixture contains 1,1-,1,2-,2,2-dibromopropane. Complete hydrolysis gives respectively propanal, propane-1,2-diol, acetone. Carbonyl assay counts 0.0700 mol aldehyde+ketone; Tollens gives 0.0800 mol Ag from propanal, two Ag each. Original 2,2-dibromopropane mole fraction is

(A) $\frac{3}{10}$ (B) $\frac{1}{5}$

(C) $\frac{2}{5}$ (D) $\frac{7}{10}$

35. A 0.100 mol $CHCl_3$–CCl_4 mixture is selectively oxidised by $2CHCl_3 + O_2 \rightarrow 2COCl_2 + 2HCl$;$CCl_4$ is unchanged. A calibrated endpoint converts 60.0% of the initial chloroform and gives 0.0300 mol HCl. After separating all $COCl_2$ and HCl, mass of unchanged original chlorinated compounds is (M119.5,154)

(A) 10.09 g (B) 9.24 g

(C) 8.47 g (D) 13.675 g

36. A $CHCl_3$–CCl_4 sample gives 0.400 mol total chloride after complete destruction. Under a separately validated complete selective oxidation, only chloroform becomes $COCl_2$+HCl and produces 0.0200 mol HCl. With masses 119.5,154, initial CCl_4 mass fraction is

(A) $\frac{4}{5}$ (B) $\frac{239}{1548}$

(C) $\frac{1309}{1548}$ (D) $\frac{17}{21}$

37. A calibrated E2 model has equivalent beta sites carrying 3H and 2 D;per-bond rates arek_H, k_D. D-loss product fraction is2/17. For another substrate with 9 eligible H and 1 D at otherwise equivalent beta sites, the predicted D-loss fraction is

(A) $\frac{1}{10}$ (B) $\frac{1}{16}$

(C) $\frac{1}{46}$ (D) $\frac{2}{17}$

38. An E2/SN2 model has 3 eligible beta H,2 D and$k_H/k_D = 5$. Initially 85.0% of reaction gives alkenes;the rest is SN2. A separate nonbasic nucleophile doubles only the SN2 channel, leaving E2 per-bond rates fixed. Final fraction giving the D-loss alkene is

(A) $\frac{2}{23}$ (B) $\frac{2}{17}$

(C) $\frac{1}{10}$ (D) $\frac{1}{23}$

39. Two calibrated ether routes are run separately: A methyl iodide+tert-butoxide gives 90% MTBE (no alkene);B tert-butyl bromide+methoxide gives 10% MTBE,80% isobutene. Combined assays find 0.0500 mol MTBE and 0.0320 mol isobutene. Feed methyl iodide amount in route A was

(A) 0.0400 mol (B) 0.0500 mol

(C) 1/18 mol (D) 23/450 mol

40. Route A MeI+tert-butoxide gives 90% MTBE and 10% unchanged MeI. Route B tert-BuBr+methoxide gives 10% MTBE,80% isobutene,10% unchanged halide. Total halide feed 0.100 mol;MTBE mass 3.696 g (M=88). Released Br^-:I^- mole ratio is

(A) 3 (B) $\frac{3}{2}$

(C) $\frac{2}{3}$ (D) 1

41–50 / Preserved hard practice

41. A 0.100 mol mixture of ethyl chloride and ethyl bromide is fully hydrolyzed. Released halides give 16.575 g AgX precipitate(AgCl =143.5, AgBr =188). Ethyl bromide amount in mol is

(A) 0.025 (B) 0.075

(C) 0.05 (D) 0.1

42. A 0.100 mol mixture of 1-bromopropane and 2-bromopropane hydrolyzes with experimentally measured pseudo-first-order half-lives of 10 and 5 min, respectively. After 10 min, total Br^- released is 0.0650 mol. Initial 2-bromopropane amount in mol is

(A) 0.02 (B) 0.08

(C) 0.04 (D) 0.06

43. A 0.100 mol 2-bromobutane sample undergoes 80% elimination and 20% substitution with no double counting. Elimination products consume Br_2 quantitatively. Required Br_2 in mol is

(A) 0.04 (B) 0.08

(C) 0.1 (D) 0.02

44. A 0.100 mol mixture of two mirror-image halides has observed rotation+8 in the stated linear model with pure enantiomer rotations+20 and-20. Amount of positive enantiomer in mol is

(A) 0.07 (B) 0.08

(C) 0.03 (D) 0.04

45. A 0.100 mol C_4H_9Cl sample converts completely to alcohol. If all chloride is collected as AgCl and all alcohol then reacts with Na to give H_2, the AgCl: H 2 mole ratio is

(A) 4 (B) 1

(C) 0.5 (D) 2

46. A 0.100 mol mixture of methylmagnesium bromide and ethyl magnesium bromide is quenched quantitatively with D_2O. The collected organic products weigh 2.54 g. Use CH_3D =17 and C_2H_5D =31 g/mol. Amount of C_2H_5D in mol is

(A) 0.02 (B) 0.04

(C) 0.06 (D) 0.08

47. An alkyl chloride RCl contains one Cl and has M=92.5. Full Wurtz coupling of 0.100 mol RCl forms R − R. With Cl=35.5, product mass in g is

(A) 5.7 (B) 11.4

(C) 9.25 (D) 2.85

48. A first-order SN1 substrate has initial amount 0.100 mol.After 10 min, 0.0250 mol remains. After 15 min, total alcohol product amount in mol is

(A) 0.09375 (B) 0.05

(C) 0.075 (D) 0.0875

49. A benzyl chloride $C_6H_5CH_2Cl$ sample of 0.0500 mol forms benzyl alcohol quantitatively. Subsequent selective oxidation to aldehyde followed by Tollens reaction deposits Ag mass in g of(Ag=108)

(A) 10.8 (B) 21.6

(C) 5.4 (D) 16.2

50. A 0.100 mol mixture of ethyl chloride and ethyl bromide weighs 8.675 g. Their formula masses are 64.5 and 109 g/mol. After complete hydrolysis, the bromide alone is quantitatively isolated as AgBr. With AgBr =188, the AgBr mass in g is

(A) 4.7 (B) 18.8

(C) 14.1 (D) 9.4

Answer key

Haloalkanes & Haloarenes

C-HAHA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**B**	31	**D**	41	**C**
02	**D**	12	**D**	22	**C**	32	**B**	42	**D**
03	**B**	13	**A**	23	**C**	33	**D**	43	**B**
04	**B**	14	**B**	24	**C**	34	**A**	44	**A**
05	**C**	15	**B**	25	**B**	35	**A**	45	**D**
06	**B**	16	**A**	26	**A**	36	**C**	46	**C**
07	**D**	17	**D**	27	**A**	37	**C**	47	**A**
08	**B**	18	**C**	28	**A**	38	**A**	48	**D**
09	**A**	19	**A**	29	**B**	39	**D**	49	**A**
10	**C**	20	**C**	30	**D**	40	**B**	50	**D**

Chapter 26

Alcohols Phenols & Ethers

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. One mole of ethane-1, 2-diol reacts completely with Na. H_2 formed in mol is

(A) 1.5 (B) 0.5

(C) 2 (D) 1

02. One mole phenol yields 2, 4, 6-tribromophenol with excess Br 2 water. Br 2 consumed in mol is

(A) 3 (B) 2

(C) 6 (D) 1

03. Intermolecular dehydration 2 ethanol→diethyl ether+ H_2O forms ether from 0.100 mol ethanol in mol of

(A) 0.2 (B) 0.1

(C) 0.05 (D) 0.025

04. Number of constitutional alcohol isomers with formula $C_4H_{10}O$ is

(A) 2 (B) 3

(C) 4 (D) 5

05. Among constitutional $C_4H_{10}O$ alcohols, number of secondary alcohols is

(A) 1 (B) 0

(C) 2 (D) 3

06. Among constitutional $C_4H_{10}O$ alcohols, number of primary alcohols is

(A) 4 (B) 2

(C) 1 (D) 3

07. One mole primary alcohol oxidized to aldehyde releases formal redox electron amount in mol of

(A) 3 (B) 2

(C) 4 (D) 1

08. Complete oxidation of ethanol to ethanoic acid releases formal redox electron amount per mole in mol of

(A) 4 (B) 3

(C) 2 (D) 6

09. Number of distinct constitutional ortho, meta, and para dihydroxybenzene isomers is

(A) 1 (B) 4

(C) 3 (D) 2

10. A 0.100 mol tert-butanol sample dehydrates fully to 2-methylpropene. The product consumes Br_2 in mol of

(A) 0.1 (B) 0.3

(C) 0.05 (D) 0.2

11–40 / New middle-to-hard practice

11. A 0.100 mol mixture contains methanol, ethanol and propan-2-ol. Its mass is 4.46 g. Complete iodoform reaction on an equal sample gives 0.0700 mol CHI_3; one mole of either responsive alcohol gives one CHI_3. With molar masses 32,46,60, original ethanol amount is

(A) 0.0700 mol (B) 0.0200 mol

(C) 0.0500 mol (D) 0.0300 mol

12. A mixture of methanol, ethanol and propan-2-ol has 0.100 mol molecules and gives 0.0800 mol iodoform. Separate calibrated oxidation converts all methanol to formic acid, ethanol to ethanoic acid, and propan-2-ol to acetone. The two isolated acids require 0.0700 mol NaOH. Initial mixture mass is (M=32,46,60)

(A) 4.74 g (B) 4.60 g

(C) 4.46 g (D) 5.02 g

13. A 0.100 mol mixture contains butan-1-ol, butan-2-ol and tert-butanol. Controlled oxidation gives carbonyl compounds only from the first two; their total is 0.0700 mol. Isolated aldehyde deposits 0.0800 mol Ag with Tollens reagent. Ketone mass formed is (butan-2-one=72)

(A) 1.44 g (B) 5.04 g

(C) 2.16 g (D) 2.88 g

14. A $C_5H_{12}O$ alcohol is oxidised without carbon-skeleton rearrangement to a ketone that gives iodoform. The alcohol itself is optically active. Which pair is consistent?

(A) Pentan-2-ol and 3-methylbutan-2-ol are both possible; pentan-3-ol is excluded.

(B) Only pentan-3-ol is possible because secondary alcohols are all iodoform-positive.

(C) Pentan-1-ol and 3-methylbutan-1-ol are both possible because primary alcohols give ketones.

(D) Only 2-methylbutan-2-ol is possible because tertiary centres are optically active.

15. A 0.100 mol anisole/tert-butyl-methyl-ether mixture undergoes the specified first cleavage only: anisole gives phenol+MeI; MTBE gives tert-BuI+MeOH with cold HI. Isolated MeI is 0.0400 mol. Under a subsequent validated excess-HI treatment, all methanol also becomes MeI while phenol remains phenol. Total HI consumed over both stages is

(A) 0.100 mol (B) 0.200 mol

(C) 0.140 mol (D) 0.160 mol

16. Anisole and ethyl methyl ether are completely cleaved using excess HI under calibrated conditions: anisole yields phenol+MeI; ethyl methyl ether ultimately yields MeI+EtI+water. A sample gives 0.100 mol MeI and 0.0300 mol EtI. Isolated phenol mass is (M=94)

(A) 6.58 g (B) 2.82 g

(C) 4.70 g (D) 9.40 g

17. A 0.100 mol mixture contains phenol, ethanol and ethanoic acid. A bicarbonate assay releases 0.0200 mol CO_2;complete neutralisation by NaOH consumes 0.0600 mol. After quantitatively removing both the carboxylic acid and phenol by validated extraction, ethanol mass left is (M46)

(A) 2.76 g (B) 4.60 g

(C) 0.92 g (D) 1.84 g

18. A phenol/ethanol/ethanoic-acid mixture gives 0.0200 mol CO_2 with $NaHCO_3$, consumes 0.0700 mol NaOH on complete acid-base neutralisation, and evolves 0.0600 mol H_2 with excess Na in a separate dry equal sample. After extracting all carboxylic acid only, mass left is (phenol=94, ethanol=46)

(A) 6.54 g (B) 7.00 g

(C) 8.20 g (D) 5.76 g

19. A 0.0100 mol mixture of ethane-1,2-diol and glycerol weighs 0.770 g (M62,92). Each molecule forms exactly one cyclic acetonide, masking two OH groups and retaining any additional OH. After isolating these protected products, complete acetylation adds 42 g per mole of residual OH. Acetylation mass gain is

(A) 0.420 g (B) 0.210 g

(C) 0.630 g (D) 1.050 g

20. A mixture contains glycerol and ethane-1,2-diol, 0.100 mol molecules total. Complete acetylation adds 10.50 g; each OH adds 42 g/mol. After complete hydrolysis of those esters, recovered polyol mass is (M=92,62)

(A) 6.20 g (B) 8.00 g

(C) 7.70 g (D) 9.20 g

21. A 0.100 mol mixture contains cyclohexanol and cyclohexanone. Initial carbonyl assay is 0.0300 mol. Only the alcohol dehydrates; 80.0% converts quantitatively to cyclohexene. A separate bromination of the isolated unsaturated product consumes Br_2 amount

(A) 0.0560 mol (B) 0.0240 mol

(C) 0.0800 mol (D) 0.0700 mol

22. A butan-2-ol sample forms only but-1-ene and but-2-ene on complete dehydration. Total alkene is 0.100 mol. Reductive ozonolysis gives 0.160 mol ethanal (only but-2-ene gives ethanal, two each). Carbonyl products that give iodoform total

(A) 0.100 mol (B) 0.160 mol

(C) 0.200 mol (D) 0.180 mol

23. Two routes are proposed for ethyl tert-butyl ether. Route I uses EtBr+tert-BuO^-; route II uses tert-BuBr+EtO^-. Calibrated yields are 70% ether/20% alkene for I, and 10% ether/80% alkene for II. Equal substrate feeds give 0.0400 mol ether in total. Total alkene is

(A) 0.0200 mol (B) 0.0800 mol

(C) 0.0400 mol (D) 0.0500 mol

24. An optically active secondary alcohol is converted to its alkoxide without breaking its C–O bond, then reacted with methyl iodide. Alternatively, it is converted to the corresponding secondary bromide with inversion, then displaced by methoxide with inversion. Assume both substitutions are SN2 and no side reaction. Relative to the original C–O stereochemical arrangement, the two ether products have

(A) Retention in the alkoxide route and inversion after two inversions.

(B) Retention in both routes; their stereochemical arrangement at the original centre is the same.

(C) Inversion in both routes because every ether synthesis inverts the alcohol carbon.

(D) Racemisation in the alkoxide route and retention after the bromide route.

25. A purified ethanolamine mixture contains mono-, di-, triethanolamine only,0.100 mol N atoms. Total OH is 0.220 mol; independently,0.0300 mol molecules have two N–H bonds. Triethanolamine amount is

(A) 0.0300 mol (B) 0.0600 mol

(C) 0.0200 mol (D) 0.0500 mol

26. Ammonia incorporates 0.240 mol ethylene oxide to give only mono-, di-, triethanolamine,0.100 mol total molecules. Triethanolamine is 0.0600 mol. Amount of monoethanolamine is

(A) 0.0600 mol (B) 0.0300 mol

(C) 0.0400 mol (D) 0.0200 mol

27. A phenol conversion gives salicylaldehyde by Reimer–Tiemann and salicylic acid by Kolbe processing in separate quantified branches. Combined isolated products total 0.100 mol. Tollens deposits 0.0800 mol Ag from the aldehyde only. Complete neutralisation of all phenolic and carboxylic OH by strong NaOH consumes

(A) 0.160 mol (B) 0.120 mol

(C) 0.100 mol (D) 0.200 mol

28. A 0.100 mol salicylaldehyde/salicylic-acid mixture consumes 0.0600 mol $NaHCO_3$. It is completely reduced under calibrated conditions that convert CHO to CH_2OH but leave COOH unchanged. The phenolic-alcohol product is then separated quantitatively from the remaining carboxylic acid. Its mass is (salicyl alcohol M124)

(A) 4.88 g (B) 4.96 g

(C) 12.40 g (D) 7.44 g

29. A nitrophenol mixture contains only ortho and para isomers. A calibrated steam-distillation pass transfers 80.0% of ortho and 5.00% of para;0.100 mol feed transfers 0.0500 mol. A second identical pass on the residue transfers ortho amount

(A) 0.0120 mol (B) 0.0480 mol

(C) 0.00200 mol (D) 0.00960 mol

30. A 0.100 mol ortho-/para-nitrophenol mixture is steam-distilled in a calibrated stage transferring 80% ortho and 5% para. The distillate contains 0.0500 mol total nitrophenol. If that distillate is crystallised with 90% ortho and 50% para recovery, ortho mole fraction among collected crystals is

(A) $\frac{108}{113}$ (B) $\frac{24}{25}$

(C) $\frac{3}{5}$ (D) $\frac{216}{221}$

31. A mixture contains methanol labelled only at oxygen as $CH_3{}^{18}OH$ and unlabelled ethanol. Complete esterification with unlabelled acetic acid is calibrated to retain the alcohol oxygen in each ester. Total esters 0.100 mol contain 0.0600 mol ^{18}O atoms. Subsequent ester hydrolysis cleaves the acyl–O bond and retains ester alkoxy oxygen in the recovered alcohol. Recovered labelled methanol mass is (M=34)

(A) 1.92 g (B) 3.40 g

(C) 1.36 g (D) 2.04 g

32. A 0.100 mol mixture of labelled anisole Ph–^{18}O–Me and unlabelled MTBE undergoes one cold-HI cleavage: anisole gives phenol+MeI;MTBE gives tert-BuI+MeOH. Methanol formed is 0.0400 mol. All phenol is isolated with no oxygen exchange. Labelled phenol mass is (C=12, H=1,^{18}O=18)

(A) 5.76 g (B) 9.60 g

(C) 3.84 g (D) 5.64 g

33. A 0.100 mol mixture contains a secondary and tertiary alcohol. A calibrated Lucas assay has independent first-order conversions, $f_i = 1 - e^{-k_i t}$. At time t, the tertiary is 75% converted and secondary 50%;total chloride formed is 0.0650 mol. At time 2t, total chloride is

(A) 0.09375 mol (B) 0.0900 mol

(C) 0.08625 mol (D) 0.0800 mol

34. A calibrated first-order Lucas assay of a 0.100 mol secondary/tertiary alcohol mixture gives 0.0640 mol chloride at t. At t, the tertiary conversion is 80.0%, secondary 40.0%. After selectively removing all chloride, another interval t is applied to the residual alcohols. Additional chloride is

(A) 0.01920 mol (B) 0.0160 mol

(C) 0.0288 mol (D) 0.0256 mol

35. A 0.100 mol mixture contains ethane-1,2-diol and propane-1,2-diol. Complete periodate cleavage gives formaldehyde from both, two per ethylene glycol and one per propylene glycol;the latter also gives ethanal. Formaldehyde is 0.140 mol. Iodoform from isolated ethanal is

(A) 0.140 mol (B) 0.0600 mol

(C) 0.100 mol (D) 0.0400 mol

36. A mixture contains ethane-1,2-diol and propane-1,3-diol,0.100 mol total. Under validated conditions periodate cleaves only the vicinal diol and consumes 0.0400 mol. Complete acetylation of the unchanged nonvicinal fraction, after separation, adds mass

(A) 8.40 g (B) 5.04 g

(C) 3.36 g (D) 2.52 g

37. An acyclic saturated $C_4H_{10}O$ compound X neither evolves H_2 with Na nor gives a carbonyl on mild alcohol oxidation. Excess HI completely cleaves X to exactly one alkyl-iodide constitutional type. X is

(A) 1-Methoxypropane. (B) Ethoxyethane.

(C) Butan-2-ol. (D) 2-Methoxypropane.

38. A $C_5H_{12}O$ ether gives methanol plus a tertiary alkyl iodide in the specified first cold-HI cleavage. Its synthesis is most directly achieved by SN2 without attacking a tertiary carbon using

(A) 1-Iodopropane and sodium ethoxide. (B) Ethyl iodide and sodium propoxide.

(C) Methyl iodide and potassium tert-butoxide. (D) tert-Butyl iodide and sodium methoxide.

39. A 0.100 mol phenol/para-cresol mixture undergoes calibrated exhaustive ring bromination at all vacant ortho/para positions relative to OH, without side-chain reaction. Br_2 uptake is 0.240 mol. An equal original sample is quantitatively deoxygenated by zinc dust to benzene/toluene. Hydrocarbon mass is (M78,92)

(A) 7.80 g (B) 9.20 g

(C) 8.64 g (D) 8.08 g

40. A phenol/para-cresol sample weighs 8.36 g and consumes 0.180 mol Br_2 in exhaustive substitution of every vacant ortho/para site, without side-chain reaction. An equal original sample is fully acetylated. With starting molar masses 94,108 and 42 g/mol mass added per OH, acetylated-product mass is

(A) 15.92 g (B) 10.88 g

(C) 11.72 g (D) 12.14 g

41–50 / Preserved hard practice

41. A 0.100 mol mixture of ethanol and ethane-1, 2-diol reacts with Na to give 0.0700 mol H_2. Moles of diol were

(A) 0.04 (B) 0.02

(C) 0.06 (D) 0.08

42. A 0.100 mol mixture of phenol and ethanol consumes 0.0400 mol NaOH quantitatively under conditions where only phenol reacts. A fresh identical sample consumes Br_2 in mol of on exhaustive phenol tribromination

(A) 0.18 (B) 0.04

(C) 0.06 (D) 0.12

43. A 6.00 g mixture of propan-1-ol and propan-2-ol is fully oxidized under aldehyde/ketone-selective conditions. Tollens assay of products deposits 10.8 g Ag. Moles of propan-2-ol initially were

(A) 0.1 (B) 0.05

(C) 0.075 (D) 0.025

44. A 0.100 mol mixture of ethanol and dimethylether has formula C_2H_6O for each component. Na treatment releases 0.0300 mol H_2 and ether does not react. Amount of dimethylether in mol is

(A) 0.04 (B) 0.08

(C) 0.02 (D) 0.06

45. A 0.0500 mol anisole sample is cleaved quantitatively by HI to phenol and MeI. All phenol is then brominated with Br 2 water. Number of moles Br 2 required are

(A) 0.05 (B) 0.1

(C) 0.15 (D) 0.2

46. A 0.100 mol mixture of phenol and catechol releases 0.0700 mol hydrogen with excess Na. With phenol=94 and catechol=110 g/mol, the catechol mass fraction is

(A) 0.56175299 (B) 0.43824701

(C) 0.4 (D) 0.6

47. A 0.100 mol primary alcohol sample is 80% converted to aldehyde with remaining alcohol unchanged. Tollens reaction of aldehyde deposits Ag mass in g of(Ag=108)

(A) 8.64 (B) 10.8

(C) 21.6 (D) 17.28

48. A monohydric acyclic saturated alcohol of mass 3.70 g produces 0.0250 mol H_2 with Na. Number of carbon atoms per molecule is

(A) 2 (B) 5

(C) 3 (D) 4

49. A 0.100 mol mixture of butan-1-ol and butan-2-ol is oxidized completely without C – C cleavage: primary alcohol to carboxylic acid and secondary alcohol to ketone. The oxidant accepts 0.280 mol electrons. Mole fraction of butan-1-ol initially is

(A) 0.8 (B) 0.2

(C) 0.4 (D) 0.6

50. A mixture of phenol and ethanoic acid consumes 0.100 mol NaOH but gives 0.0300 mol CO_2 with $NaHCO_3$. Moles of phenol were

(A) 0.1 (B) 0.07

(C) 0.03 (D) 0.04

Answer key

Alcohols Phenols & Ethers

C-APE / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**C**	21	**A**	31	**D**	41	**A**
02	**A**	12	**A**	22	**B**	32	**A**	42	**D**
03	**C**	13	**C**	23	**D**	33	**C**	43	**B**
04	**C**	14	**A**	24	**B**	34	**A**	44	**A**
05	**A**	15	**D**	25	**D**	35	**B**	45	**C**
06	**B**	16	**A**	26	**D**	36	**B**	46	**B**
07	**B**	17	**D**	27	**A**	37	**B**	47	**D**
08	**A**	18	**B**	28	**B**	38	**C**	48	**D**
09	**C**	19	**B**	29	**D**	39	**C**	49	**C**
10	**A**	20	**C**	30	**D**	40	**C**	50	**B**

Chapter 27

Aldehydes Ketones & Carboxylic Acids

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. For 2 benzaldehyde→benzyl alcohol+benzoate(Cannizzaro), 0.100 mol benzaldehyde can form benzyl alcohol in mol of

(A) 0.05
(B) 0.025
(C) 0.2
(D) 0.1

02. The number of constitutional aldehydes with formula C_4H_8O is

(A) 1
(B) 4
(C) 3
(D) 2

03. The number of constitutional ketones with formula $C_5H_{10}O$ is

(A) 3
(B) 1
(C) 2
(D) 4

04. One mole ethanoic acid esterifies with 0.80 mol ethanol. Maximum ester amount in mol is

(A) 1
(B) 0.2
(C) 1.8
(D) 0.8

05. A 0.100 mol aldehyde sample deposits Ag mass in g of through quantitative Tollens reaction(Ag=108)

(A) 43.2
(B) 10.8
(C) 21.6
(D) 32.4

06. For butan-2-one, total number of alpha H atoms is

(A) 2
(B) 5
(C) 3
(D) 6

07. A 0.0500 mol methyl ketone sample needs I_2 in mol of for complete haloform reaction

(A) 0.1 (B) 0.05

(C) 0.2 (D) 0.15

08. One mole but-2-enal has molar formula C_4H_6O. Complete hydrogenation of both C = C and C = O consumes H_2 in mol of

(A) 1 (B) 2

(C) 4 (D) 3

09. An ester has molar mass 88 g/mol and formula $C_4H_8O_2$. Number of constitutional acyclic saturated esters possible is

(A) 3 (B) 4

(C) 2 (D) 5

10. Complete hydrolysis of one cyclic an hydride containing two acyl groups produces carboxylic acid groups in mol per mol substrate of

(A) 3 (B) 1

(C) 2 (D) 4

11–40 / New middle-to-hard practice

11. A 0.100 mol mixture contains ethanal, acetone and benzaldehyde. Complete Tollens assay deposits 0.140 mol Ag;iodoform assay on an equal sample gives 0.0600 mol CHI_3. Initial mass is (M44,58,106)

(A) 7.30 g (B) 6.82 g

(C) 5.80 g (D) 8.26 g

12. A 0.100 mol ethanal/acetone/benzaldehyde mixture gives 0.0800 mol Ag with Tollens and 0.0700 mol iodoform in separate quantitative assays. Selective complete oxidation of only benzaldehyde in another equal sample gives benzoic acid (M122). Its mass is

(A) 3.66 g (B) 4.88 g

(C) 1.22 g (D) 7.32 g

13. A 0.100 mol benzaldehyde/ethanal feed weighs 6.88 g (M106,44). A calibrated crossed aldol process gives only cinnamaldehyde from one molecule of each aldehyde, with 80.0% yield based on the limiting component. Product mass is (M132)

(A) 10.56 g (B) 5.280 g

(C) 4.224 g (D) 6.336 g

14. For 2,6-heptanedione, $CH_3COCH_2CH_2CH_2COCH_3$, intramolecular aldol addition followed by loss of water gives the favoured six-membered conjugated enone. Which identification includes the correct carbon skeleton and formula?

(A) 3-Methylcyclohexan-1-one; $C_7H_{12}O$, because aldol dehydration hydrogenates the ring.

(B) Cyclohex-2-en-1-one; C_6H_8O, because one terminal carbon is lost.

(C) 3-Methylcyclohex-2-en-1-one; $C_7H_{10}O$.

(D) 3-Methylcyclobut-2-en-1-one; C_5H_6O, because the inner alpha carbon must attack.

15. A 0.100 mol benzaldehyde/formaldehyde feed weighs 8.32 g (M106,30). A calibrated two-stage Cannizzaro model first transfers one hydride from each available HCHO to one benzaldehyde until one is exhausted;then remaining benzaldehyde self-disproportionates completely. On acidification, the total benzoic-acid+formic-acid mass is (M122,46)

(A) 2.76 g (B) 3.82 g

(C) 6.10 g (D) 5.04 g

16. A 0.100 mol benzaldehyde/formaldehyde feed has mass 6.04 g (M106,30). In a calibrated sequence, all possible 1:1 crossed Cannizzaro occurs with HCHO oxidised;excess HCHO then self-disproportionates. Total isolated alcohol mass is (benzyl alcohol 108, methanol 32)

(A) 4.64 g (B) 5.60 g

(C) 6.48 g (D) 4.32 g

17. An acyclic saturated methyl ketone $RCOCH_3$ gives 0.0500 mol iodoform in complete haloform reaction. After acidifying and isolating only RCOOH, the acid mass is 4.40 g. With C12, H1, O16, molar mass of the original ketone is

(A) 86 $g\,mol^{-1}$ (B) 88 $g\,mol^{-1}$

(C) 102 $g\,mol^{-1}$ (D) 72 $g\,mol^{-1}$

18. A 0.100 mol acetophenone/propiophenone mixture gives 0.0400 mol iodoform on complete assay. Original mass is (acetophenone 120, propiophenone 134)

(A) 13.40 g (B) 12.00 g

(C) 12.56 g (D) 12.84 g

19. An ethanoic-acid/oxalic-acid mixture weighs 6.60 g and consumes 0.140 mol NaOH on complete neutralisation (M60,90). Acidic permanganate oxidises oxalic acid only, by 2 electron equivalents per molecule;MnO_4^- accepts 5 each. Permanganate amount needed is

(A) 0.0280 mol (B) 0.0600 mol

(C) 0.0120 mol (D) 0.0240 mol

20. A succinic-acid/fumaric-acid mixture requires 0.200 mol NaOH and consumes 0.0400 mol Br_2 by addition across C=C in a validated complete assay. With M118,116, initial mixture mass is

(A) 11.72 g (B) 11.68 g

(C) 11.80 g (D) 23.44 g

21. A 7.12 g ethyl-ethanoate/methyl-benzoate sample is completely saponified with 0.100 mol NaOH;back-titration finds 0.0300 mol unused base. With ester masses 88,136 and alcohol masses 46,32, total liberated alcohol mass is

(A) 3.22 g (B) 2.94 g

(C) 4.60 g (D) 2.24 g

22. Methyl and ethyl esters of the same saturated acyclic monocarboxylic acid have total mass 10.76 g. Complete saponification gives 0.100 mol carboxylate and 0.0400 mol ethanol, the remaining alcohol being methanol. Carbon count in the original acid is

(A) 3 (B) 6

(C) 4 (D) 5

23. A 0.100 mol acetyl-chloride/benzoyl-chloride feed weighs 10.33 g (M78.5,140.5). Complete Rosenmund reduction is calibrated to stop at aldehydes. Quantitative iodoform assay of the isolated aldehydes gives CHI_3 mass (M394)

(A) 39.40 g (B) 15.76 g

(C) 7.88 g (D) 23.64 g

24. A 0.100 mol acetyl-/benzoyl-chloride feed gives 0.0600 mol iodoform after quantitative Rosenmund reduction. A separate equal feed is completely reduced with $LiAlH_4$ to the corresponding primary alcohols. Isolated alcohol mass is (M46,108)

(A) 8.32 g (B) 7.08 g

(C) 6.24 g (D) 10.80 g

25. A C_4H_8O aldehyde gives a negative iodoform test. Its self-aldol addition product retains an alpha-H at the donor carbon and readily loses water to a conjugated carbonyl. Which candidate satisfies both the addition and dehydration constraints?

(A) Butan-2-one; being a ketone makes it an aldehyde under aldol conditions.

(B) Butanal; its original alpha-CH_2 leaves one alpha-H after C–C bond formation.

(C) 2-Methylpropanal; its alpha-CH leaves an alpha-H after the new C–C bond forms.

(D) 2-Methylpropanal; ordinary dehydration does not require any donor alpha-H.

26. An acyclic saturated $C_6H_{12}O$ ketone gives iodoform. Its isolated monocarboxylic-acid haloform product is optically active as a pure enantiomer. Which ketone structure fits?

(A) Hexan-3-one. (B) Hexan-2-one.

(C) 3-Methylpentan-2-one. (D) 4-Methylpentan-2-one.

27. Two monoprotic substituted benzoic acids A, B have$pK_a = 3,5$. At pH4.00, a 0.100 mol analytical mixture has 0.0500 mol total carboxylate, with no other ions contributing to that measured inventory. Use$[A^-]/[HA] = 10^{pH-pK_a}$. At pH5.00, the ratio of neutral HA to neutral HB amounts is

(A) $\frac{1}{10}$ (B) $\frac{2}{101}$

(C) $\frac{20}{101}$ (D) $\frac{1}{101}$

28. Acids A, B have$pK_a = 4,6$. A 0.100 mol mixture at pH5.00 has total anion amount16/275 mol. At pH4.00, total anion amount is (use ideal Henderson–Hasselbalch speciation)

(A) 3/100 mol (B) 16/275 mol

(C) 307/10100 mol (D) 7/100 mol

29. An 8.90 g sodium-ethanoate/sodium-propanoate mixture is completely decarboxylated with soda lime. The isolated methane/ethane gas has mean molar mass 23.0 (M16,30);salt masses are 82,96. Complete combustion of all this gas gives CO_2 amount

(A) 0.150 mol (B) 0.200 mol

(C) 0.100 mol (D) 0.0750 mol

30. A calibrated complete thermal treatment of an oxalic-/malonic-acid mixture yields 0.0400 mol CO and 0.100 mol CO_2. Only oxalic acid gives CO+CO_2+water;malonic acid gives ethanoic acid+CO_2, which is isolated before further heating. Original mixture mass is (M90,104)

(A) 9.84 g (B) 9.36 g

(C) 10.40 g (D) 13.76 g

31. A 0.0500 mol carbonyl compound gives a single dioxime by complete reaction with hydroxylamine;the isolated dioxime mass is 5.80 g. Each C=O to C=NOH conversion adds 15 g/mol. Parent molecular formula is $C_nH_{2n-2}O_2$. Carbon count n is

(A) 4 (B) 5

(C) 3 (D) 6

32. A 0.100 mol glyoxal/methylglyoxal mixture is selectively assayed under calibrated conditions: each CHO gives 2 Ag with Tollens, while ketone groups do not react. Total Ag is 0.280 mol. Both carbonyls of each molecule form oximes on complete hydroxylamine treatment. Total isolated dioxime mass is (parent M58,72;each oximation adds 15)

(A) 10.20 g (B) 6.64 g

(C) 8.80 g (D) 9.64 g

33. A mixture of ethanoic anhydride and benzoyl chloride is completely hydrolysed. Liberated HCl is 0.0400 mol;after separately removing HCl, the carboxylic acids require 0.100 mol NaOH. Original mass is (anhydride 102, benzoyl chloride 140.5)

(A) 8.68 g (B) 7.66 g

(C) 5.62 g (D) 10.20 g

34. A glyoxylic-acid/pyruvic-acid mixture requires 0.100 mol NaOH for its COOH groups. Under calibrated aldehyde-only Tollens conditions, it deposits 0.0800 mol Ag. Initial mass is (glyoxylic HOOCCHO=74, pyruvic $CH_3COCOOH$=88)

(A) 8.80 g (B) 8.24 g

(C) 7.96 g (D) 7.40 g

35. An ester $C_6H_{10}O_3$ gives ethanol on complete hydrolysis. The resulting beta-keto acid loses CO_2 to acetone on heating. Which identification also predicts the original ester iodoform response?

(A) Methyl levulinate,$CH_3COCH_2CH_2COOMe$; positive,but it gives ethanol on hydrolysis.

(B) Diethyl oxalate,EtOOC–COOEt; positive because all carbonyl compounds give iodoform.

(C) Ethyl pyruvate,$CH_3COCOOEt$; negative because every ester prevents iodoform reaction.

(D) Ethyl acetoacetate,CH_3COCH_2COOEt; positive because it contains CH_3CO–.

36. Methyl and ethyl acetoacetate are completely hydrolysed;the separated beta-keto acids then each decarboxylate once. CO_2 is 0.100 mol and ethanol recovered from hydrolysis is 0.0400 mol, the remaining alcohol being methanol. Initial ester mass is (C12, H1, O16)

(A) 13.00 g (B) 12.16 g

(C) 10.76 g (D) 11.60 g

37. A 0.100 mol ethanal/ethanal-dimethyl-acetal mixture is partly hydrolysed. Hydrolysis releases 0.0800 mol methanol;the final free ethanal amount is 0.0700 mol. No other process changes aldehyde. Initial mixture mass is (ethanal 44, acetal 90)

(A) 5.78 g (B) 6.24 g

(C) 7.62 g (D) 9.00 g

38. A 0.100 mol benzaldehyde/cyclic-ethylene-glycol-acetal mixture is partially hydrolysed. Liberated ethane-1,2-diol is 0.0200 mol;final free benzaldehyde is 0.0600 mol. Molar masses are 106,150. Initial sample mass was

(A) 13.24 g (B) 12.36 g

(C) 14.12 g (D) 15.00 g

39. A $C_3H_6O_2$ compound gives a carbonyl derivative with 2,4-DNP, iodoform, and a positive reducing test characteristic of an alpha-hydroxy carbonyl. It gives no CO_2 with bicarbonate. Which structure satisfies all these observations?

(A) 3-Hydroxypropanal,$HOCH_2CH_2CHO$. (B) Methyl ethanoate,CH_3COOCH_3.

(C) Propanoic acid,CH_3CH_2COOH. (D) Hydroxyacetone,CH_3COCH_2OH.

40. A compound $C_3H_6O_2$ is bicarbonate-negative and 2,4-DNP-negative. Complete saponification gives one mole methanol and one mole a monocarboxylate per mole compound. Which compound and acidification product are consistent?

(A) Propanoic acid; propan-1-ol. (B) Methyl ethanoate; ethanoic acid.

(C) Ethyl methanoate; ethanoic acid. (D) Hydroxyacetone; methanoic acid.

41–50 / Preserved hard practice

41. A mixture of ethanal and propanone consumes 0.100 mol HCN on total cyanohydrin formation. A fresh sample deposits 12.96 g Ag with Tollens. Moles of propanone were

(A) 0.06 (B) 0.08

(C) 0.04 (D) 0.02

42. A 0.100 mol mixture of benzaldehyde and anisole undergoes quantitative Cannizzaro conversion of the aldehyde while the ether remains unchanged. Benzyl alcohol isolated is 0.0200 mol. Initial anisole amount in mol is

(A) 0.08 (B) 0.04

(C) 0.02 (D) 0.06

43. A mixture of ethanoic acid and ethyl ethanoate consumes 0.0600 mol NaOH in a rapid acid-selective neutralization where ester hydrolysis is negligible. A fresh equal sample consumes 0.100 mol NaOH after complete neutralization and saponification. Initial ester amount in mol is

(A) 0.06 (B) 0.08

(C) 0.04 (D) 0.02

44. An acyclic monoaldehyde has M 72 g/mol. A 3.60 g sample is reduced to alcohol then reacted with Na. Hydrogen formed in mol is

(A) 0.0125 (B) 0.025

(C) 0.05 (D) 0.1

45. A 0.100 mol mixture of propanone and pentan-3-one consumes 0.180 mol I_2 in quantitative iodoform testing. Moles of pentan-3-one were

(A) 0.08 (B) 0.06

(C) 0.04 (D) 0.02

46. A 0.0500 mol acyclic alkene produces only ethanal on reductive ozonolysis. Completely oxidizing all carbonyl products to acid requires formal oxidant electron capacity in mol of

(A) 0.2 (B) 0.1

(C) 0.05 (D) 0.4

47. A mixture of ethanal and propanal deposits 0.200 mol Ag in a quantitative Tollens assay. The original sample weighs 5.10 g. With formula masses 44 and 58 g/mol, the propanal amount in mol is

(A) 0.025 (B) 0.15

(C) 0.075 (D) 0.05

48. A 0.100 mol monocarboxylic acid is saturated and acyclic. It weighs 8.80 g and forms ester with methanol. Molar mass of ester in g/mol is

(A) 88 (B) 74

(C) 102 (D) 116

49. A 0.100 mol mixture of ethanal and ethanol requires 0.300 mol electron capacity for complete oxidation to ethanoic acid. Moles of ethanol were

(A) 0.075 (B) 0.05

(C) 0.1 (D) 0.025

50. A 5.04 g sample of an acyclic compound containing only C, H, O has one aldehyde and no other oxygen group. It deposits 0.120 mol Ag with Tollens reagent and consumes 0.120 mol H 2 on complete hydrogenation of all multiple bonds. The carbon count per molecule is(C=12, H=1, O=16)

(A) 7 (B) 6

(C) 4 (D) 5

Answer key

Aldehydes Ketones & Carboxylic Acids

C-AKCA / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**A**	21	**B**	31	**A**	41	**C**
02	**D**	12	**A**	22	**C**	32	**D**	42	**D**
03	**A**	13	**C**	23	**D**	33	**A**	43	**C**
04	**D**	14	**C**	24	**B**	34	**B**	44	**B**
05	**C**	15	**B**	25	**B**	35	**D**	45	**C**
06	**B**	16	**A**	26	**C**	36	**B**	46	**A**
07	**D**	17	**A**	27	**B**	37	**C**	47	**D**
08	**B**	18	**D**	28	**C**	38	**A**	48	**C**
09	**B**	19	**D**	29	**A**	39	**D**	49	**B**
10	**C**	20	**A**	30	**A**	40	**B**	50	**D**

Chapter 28

Amines

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. One mole propanamide undergoes Hofmann bromamide degradation. Carbon count of the amine product is

(A) 2 (B) 3

(C) 4 (D) 1

02. One mole propanenitrile is fully reduced to amine. Carbon count of product is

(A) 3 (B) 2

(C) 1 (D) 4

03. Formal reduction of one mole-nitro group from N+3 to N-3 requires mol electrons of

(A) 2 (B) 6

(C) 4 (D) 3

04. Brominating 0.100 mol aniline to tribromoaniline consumes Br_2 in mol of

(A) 0.6 (B) 0.3

(C) 0.1 (D) 0.2

05. A saturated acyclic primary monoamine has M 45 g/mol. Carbon count is(C=12, H=1, N=14)

(A) 1 (B) 4

(C) 2 (D) 3

06. The number of constitutional primary amines with formula C_3H_9N is

(A) 2 (B) 4

(C) 3 (D) 1

07. The number of constitutional secondary amines with formula C_3H_9N is

(A) 3 (B) 0

(C) 1 (D) 2

08. Total number of constitutional acyclic amines with formula C_3H_9N is

(A) 4 (B) 3

(C) 5 (D) 2

09. An amine buffer has pK_b 5 and [BH^+] / [B] =10. At 25°C, pH is

(A) 6 (B) 10

(C) 9 (D) 8

10. For an amide $RCONH_2$ containing 4 carbons including carbonyl, Hofmann product contains carbon atoms per molecule of

(A) 4 (B) 2

(C) 5 (D) 3

11–40 / New middle-to-hard practice

11. A 0.100 mol methylamine/dimethylamine/trimethylamine mixture undergoes a validated Hinsberg separation. Sulfonamide that dissolves in excess alkali then precipitates on acidification is 0.0400 mol;alkali-insoluble sulfonamide is 0.0300 mol. Original mass is (M31,45,59)

(A) 5.20 g (B) 4.36 g

(C) 3.94 g (D) 4.50 g

12. A 0.100 mol methylamine/diethylamine/trimethylamine mixture incorporates 9.80 g mass on complete Hinsberg derivatisation;each reacting amine adds 140 g/mol, and unchanged tertiary amine is retained. A separate carbylamine assay identifies 0.0400 mol responsive molecules. Initial mass is (M31,73,59)

(A) 4.36 g (B) 5.20 g

(C) 6.04 g (D) 5.62 g

13. Bases A, B have conjugate-acid$pK_a = 5, 9$. At pH7.00, a 0.100 mol analytical mixture has 0.0500 mol total protonated base. Use$[B]/[BH^+] = 10^{pH-pK_a}$. At pH5.00, the ratio of neutral A to neutral B amounts is

(A) 100 (B) 10001

(C) $\frac{10001}{2}$ (D) $\frac{101}{2}$

14. Anilinium and methylamine have conjugate-acid$pK_a = 5, 11$. Equal initial concentrations 0.100 M of anilinium and methylamine react in a fixed volume, with no products initially. For proton transfer to methylamine, ignore other equilibria. Fraction of original anilinium remaining is

(A) $\frac{1}{2}$ (B) $\frac{1}{1000001}$

(C) $\frac{1}{1000}$ (D) $\frac{1}{1001}$

15. A 0.100 mol aniline/benzylamine mixture gives 0.0400 mol N_2 during a validated cold nitrous-acid treatment:the aliphatic primary amine deaminates, while aniline forms stable diazonium at that stage. All diazonium then couples with excess phenol at 80.0% isolated yield. Azo-product mass is (M198)

(A) 7.92 g (B) 6.336 g

(C) 11.88 g (D) 9.504 g

16. A 0.100 mol methylamine/aniline mixture weighs 6.82 g (M31,93). Cold diazotisation is followed by warming in water until every aromatic diazonium ion hydrolyses. Isolated phenol mass at quantitative yield is (M94)

(A) 3.76 g (B) 9.40 g

(C) 5.64 g (D) 6.82 g

17. A 0.100 mol aniline/N-methylaniline mixture weighs 10.14 g (M93,107). In quantitative carbylamine processing, only responsive molecules consume one $CHCl_3$ each. Chloroform amount required is

(A) 0.0200 mol (B) 0.100 mol

(C) 0.0400 mol (D) 0.0600 mol

18. A 0.100 mol ethylamine/diethylamine mixture weighs 5.62 g (M45,73). Quantitative carbylamine conversion gives ethyl isocyanide from the responsive fraction. Product mass is (C12, H1, N14)

(A) 3.30 g (B) 5.50 g

(C) 2.20 g (D) 4.50 g

19. A 9.59 g n-butyl-bromide/tert-butyl-bromide mixture undergoes a calibrated Gabriel route:only primary halide substitutes, and all substituted material is released as butan-1-amine. Isolated amine is 2.92 g (M73);both bromides have M137. Original tertiary-halide mole fraction is

(A) $\frac{4}{7}$ (B) $\frac{1}{2}$

(C) $\frac{2}{7}$ (D) $\frac{3}{7}$

20. An 8.38 g acetamide/benzamide mixture consumes 0.100 mol Br_2 in complete Hofmann bromamide rearrangement, one Br_2 per amide under the stated quantitative conditions. Total isolated amine mass is (amide M59,121;amine M31,93)

(A) 4.34 g (B) 8.38 g

(C) 6.14 g (D) 5.58 g

21. A 0.100 mol methylamine/dimethylamine/trimethylamine mixture consumes 0.240 mol MeI in exhaustive conversion to quaternary ammonium iodides. A calibrated carbylamine assay gives 0.0600 mol responsive molecules. Initial mass is (M31,45,59)

(A) 4.50 g (B) 4.22 g

(C) 3.94 g (D) 5.06 g

22. A 0.100 mol methylamine/dimethylamine/trimethylamine mixture needs 0.200 mol MeI for exhaustive quaternisation. Primary amine is independently 0.0200 mol. In a subsequent separate Hinsberg assay, amount forming sulfonamides in total is

(A) 0.0800 mol
(B) 0.0200 mol
(C) 0.100 mol
(D) 0.0600 mol

23. A 6.74 g mixture contains acetamide and one saturated acyclic primary monoamine RNH_2. Total nitrogen assay gives 0.100 mol N atoms. Under calibrated aqueous titration, only the amine consumes HCl,0.0600 mol;acetamide is unprotonated in the analytical endpoint window. Carbon count in the amine is (acetamide 59)

(A) 3
(B) 4
(C) 2
(D) 5

24. An ethane-1,2-diamine/piperazine mixture takes 0.200 mol HCl for full protonation. A validated nitrous-acid assay counts only primary NH_2 groups as N_2 and releases 0.0800 mol. Both compounds contain two basic N atoms;M60,86. Original mixture mass is

(A) 6.52 g
(B) 6.00 g
(C) 8.60 g
(D) 7.56 g

25. Aniline is converted through diazonium to Ph–^{13}CN, then hydrolysed to benzoic acid and converted to benzamide. Compare Hofmann rearrangement with $LiAlH_4$ reduction of this labelled amide. Where does the label go?

(A) Hofmann: carbonate carbon; $LiAlH_4$: the CH_2 carbon of benzylamine.
(B) Hofmann: nitrogen; $LiAlH_4$: carbonate carbon.
(C) Hofmann: aniline ring carbon; $LiAlH_4$: nitrogen.
(D) Both routes: the same aniline ring carbon.

26. A 0.100 mol aniline inventory is converted to isolated diazonium at 80.0% yield. Cyanation of that isolated material is 75.0% efficient;complete nitrile hydrolysis is 90.0% efficient in isolated benzoic acid. The acid is quantitatively converted to benzamide then Hofmann rearranged. Final aniline mass is (M93)

(A) 9.300 g
(B) 5.022 g
(C) 6.696 g
(D) 5.580 g

27. A 0.100 mol aniline/acetanilide mixture consumes 0.180 mol Br_2 in calibrated reactions yielding 2,4,6-tribromoaniline from aniline and para-bromoacetanilide from acetanilide. Initial mass is (M93,135)

(A) 11.82 g
(B) 9.30 g
(C) 13.50 g
(D) 10.98 g

28. Aniline is acetylated, then nitrated mainly para, and the amide protecting group is hydrolysed. Which product formula and mechanistic purpose are consistent with the complete sequence?

(A) Para-nitroaniline,$C_6H_6N_2O_2$; protection moderates NH_2 activation and avoids treating anilinium as an ortho/para director.

(B) Meta-nitroaniline,$C_6H_6N_2O_2$; acetanilide is intrinsically meta-directing.

(C) Para-nitrotoluene,$C_7H_7NO_2$; amide hydrolysis converts $NHCOCH_3$ to CH_3.

(D) Para-nitroacetanilide,$C_8H_8N_2O_3$; hydrolysis permanently retains the acetyl carbons.

29. A 0.100 mol methylamine/dimethylamine/trimethylamine mixture gives 0.0300 mol N_2 in a validated nitrous-acid assay. Secondary-amine nitrosation product is isolated,4.44 g (N-nitrosodimethylamine M74);tertiary amine produces no such neutral N-nitrosamine. Initial mass is (M31,45,59)

(A) 4.36 g (B) 3.94 g

(C) 4.50 g (D) 4.22 g

30. A 0.100 mol aniline/N-methylaniline/N, N-dimethylaniline mixture contains 0.0400 mol carbylamine-positive molecules. In a separately calibrated secondary-N-H nitrosation assay,4.08 g N-nitroso-N-methylaniline (M136) is isolated quantitatively;other components do not enter that isolated derivative. Original mass is (M93,107,121)

(A) 10.98 g (B) 9.72 g

(C) 10.56 g (D) 12.10 g

31. A hydrated hydrochloride of a saturated acyclic primary monoamine weighs 12.75 g. Drying removes 1.80 g water;the released anhydrous free base gives 0.100 mol N_2 on quantitative deamination. The hydrated salt formula is (HCl36.5, water 18)

(A) $C_3H_9N.HCl.2H_2O$. (B) $C_5H_{13}N.HCl.H_2O$.

(C) $C_4H_{11}N.HCl.H_2O$. (D) $C_4H_{11}N.2HCl.H_2O$.

32. An unknown saturated acyclic primary monoamine gives 0.0500 mol N_2 in complete quantitative deamination and has original mass 2.95 g. It is optically inactive but can be either straight-chain or branched. Which statement follows?

(A) Formula C_3H_9N is fixed; only trimethylamine is primary.

(B) Formula $C_4H_{11}N$ is fixed; only butan-1-amine is possible.

(C) Only propan-1-amine is possible because every branched amine is tertiary.

(D) Formula C_3H_9N is fixed; propan-1-amine and propan-2-amine remain possible.

33. A quaternary ammonium hydroxide contains one n-propyl and three methyl groups on N. Under a calibrated complete Hofmann elimination, n-propyl alone has eligible beta-H. If0.0600 mol salt is present and 80.0% eliminates, complete reductive ozonolysis of the isolated alkene gives ethanal amount

(A) 0.0960 mol (B) 0.0600 mol

(C) 0.0240 mol (D) 0.0480 mol

34. A calibrated quaternary-ammonium elimination has eligible ethyl and n-propyl groups. Ethyl loss gives ethene, n-propyl loss propene;total alkene 0.100 mol. Complete reductive ozonolysis gives 0.0600 mol ethanal. Total formaldehyde is

(A) 0.140 mol (B) 0.160 mol

(C) 0.100 mol (D) 0.200 mol

35. A 0.0600 mol aniline/phenol mixture is diazotised at 0–5 C with 0.0600 mol $NaNO_2$;only aniline reacts, and phenol does not consume nitrite under the validated assay. Residual nitrite is 0.0200 mol. Every resulting diazonium couples with the available phenol 1:1 at 90.0% isolated yield. Azo-product mass is (M198)

(A) 7.920 g (B) 7.128 g

(C) 3.564 g (D) 10.692 g

36. A 0.100 mol aniline/phenol mixture gives 0.0600 mol diazonium quantitatively. Without adding phenol, the same batch couples 1:1 wherever a partner is available;all such coupling is complete. Remaining diazonium is then hydrolysed to phenol. Final total free phenol amount is

(A) 0.0400 mol (B) 0.0200 mol

(C) 0.0600 mol (D) 0.100 mol

37. A nitrobenzene/dinitrobenzene mixture produces 0.600 mol CO_2 on combustion of an equal sample. Complete reduction to aniline/phenylenediamine consumes 0.420 mol Fe, each Fe giving 2 electrons;each NO_2 to NH_2 needs 6. Isolated amine mass is (M93,108)

(A) 9.90 g (B) 10.80 g

(C) 10.20 g (D) 9.30 g

38. A 0.100 mol nitrobenzene/nitrosobenzene mixture requires 0.260 mol Fe for complete reduction to aniline, with Fe ending as Fe^{2+}. Reduction requires 6 electrons per nitro and 4 per nitroso group. Original mass is (M123,107)

(A) 11.66 g (B) 10.70 g

(C) 11.02 g (D) 12.30 g

39. An 8.46 g aniline/pyridine mixture consumes 0.100 mol HCl quantitatively, one per molecule. Complete carbylamine reaction of an equal sample gives phenyl isocyanide from the responsive fraction only. Product mass is (aniline 93, pyridine 79, PhNC103)

(A) 10.30 g (B) 6.18 g

(C) 4.12 g (D) 5.15 g

40. Compare a pyramidal tertiary amine N(Me)(Et)(n-Pr) with quaternary N^+(Me)(Et)(n-Pr)(n-Bu). Ignore chiral carbon centres and assume ordinary rapid nitrogen inversion for the neutral amine. Which conclusion follows?

(A) Neither can be chiral because nitrogen always has a planar geometry.

(B) The ammonium ion inverts rapidly through a lone pair exactly as the neutral amine does.

(C) The neutral amine rapidly interconverts nitrogen configurations; the four-distinct-substituent ammonium ion can have persistent nitrogen-centred enantiomers.

(D) Both are configurationally stable merely because their alkyl groups differ.

41–50 / Preserved hard practice

41. A mixture of primary aromatic and tertiary monoamines consumes 0.100 mol HCl. Diazotization of primary aromatic amine part quantitatively releases 0.0400 mol N 2 after hydrolysis; tertiary component does not release N_2. Tertiary amine amount in mol is

(A) 0.08 (B) 0.06

(C) 0.04 (D) 0.02

42. A 0.0100 mol nitrobenzene sample is reduced completely to aniline, and then 80% of the aniline is converted into diazonium salt. Quantitative hydrolysis of that salt releases N 2 volume at 22.4 L/mol, in L, of

(A) 0.224 (B) 0.112

(C) 0.1792 (D) 0.3584

43. A primary acyclic monoamine has molar mass 59 g/mol. A sample of 1.18 g is converted quantitatively to isocyanide by carbylamine chemistry. The mass of isocyanide product obtained in g is

(A) 1.38 (B) 1.46

(C) 1.42 (D) 1.18

44. A primary amide of molar mass 73 g/mol undergoes Hofmann degradation. A 0.730 g sample gives the amine in 80% yield. Product amount in mol is

(A) 0.004 (B) 0.008

(C) 0.01 (D) 0.02

45. A 0.100 mol mixture of C_3H_9N constitutional primary and tertiary amines contains 0.140 mol N – H bonds. Amount of tertiary amine in mol is

(A) 0.05 (B) 0.02

(C) 0.07 (D) 0.03

46. A 0.100 mol mixture of aniline and nitrobenzene weighs 10.80 g. Formula masses are 93 and 123. Before any reduction, aniline alone consumes HCl in mol of

(A) 0.1 (B) 0.05

(C) 0.025 (D) 0.075

47. A 1.10 g sample of an acyclic saturated mononitrile contains C, H, N only. Complete hydrogenation of its single CN group consumes 0.0400 mol H_2. The number of carbons per molecule is(C=12, H=1, N=14)

(A) 2 (B) 5

(C) 4 (D) 3

48. A 0.100 mol mixture of NH_3 and CH_3NH_2 has mass 2.40 g. Molar masses are 17 and 31. Moles of methylamine are

(A) 0.1 (B) 0.05

(C) 0.075 (D) 0.025

49. A 0.100 mol weak amine is 50% protonated by acid. At 25°C, the buffer pH is 9.00. The K_b of neutral amine is

(A) 1×10^{-9} (B) 1×10^{-5}

(C) 0.0001 (D) 1×10^{-10}

50. A mixture of benzenediazonium chloride and NaCl releases 0.0600 mol N 2 on quantitative diazonium hydrolysis. All chloride in the original sample gives 14.35 g AgCl. With AgCl =143.5, the original NaCl amount in mol is

(A) 0.1 (B) 0.04

(C) 0.02 (D) 0.06

Answer key

Amines

C-AMIN / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**A**	11	**B**	21	**C**	31	**C**	41	**B**
02	**A**	12	**B**	22	**A**	32	**D**	42	**C**
03	**B**	13	**C**	23	**B**	33	**D**	43	**A**
04	**B**	14	**D**	24	**D**	34	**A**	44	**B**
05	**C**	15	**D**	25	**A**	35	**C**	45	**D**
06	**A**	16	**C**	26	**B**	36	**B**	46	**B**
07	**C**	17	**C**	27	**A**	37	**A**	47	**D**
08	**A**	18	**A**	28	**A**	38	**A**	48	**B**
09	**D**	19	**D**	29	**D**	39	**C**	49	**B**
10	**D**	20	**D**	30	**C**	40	**C**	50	**B**

Chapter 29

Biomolecules

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A 0.100 mol mixture contains glucose and a disaccharide only. Complete hydrolysis gives 0.160 mol hexoses, with two hexoses formed per disaccharide. Use glucose=180 and disaccharide=342. The original sample mass in g is

(A) 18

(B) 28.8

(C) 25.2

(D) 27.72

02. A 0.0500 mol mixture contains sucrose and lactose. Before hydrolysis it deposits 0.0600 mol Ag in an ideal quantitative Tollens assay. Assume one reducing carbonyl equivalent per lactose and none per sucrose, with two Ag per equivalent. The mole fraction of sucrose is

(A) 0.7

(B) 0.6

(C) 0.3

(D) 0.4

03. Complete hydrolysis of 0.0200 mol identical linear glycine peptides consumes 0.0800 mol water. Each peptide has one free amino end and one free carboxyl end. Use glycine=75 and water=18. Peptide molar mass in $\mathrm{g\,mol^{-1}}$ is

(A) 303

(B) 285

(C) 246

(D) 375

04. A 0.100 mol mixture contains only linear tetrapeptides and cyclic tetrapeptides with no side-chain ionizable groups. Complete hydrolysis consumes 0.350 mol water. Count one free amino and one free carboxyl end per linear chain, and none per cycle. Total amount of these free end groups before hydrolysis is (mol)

(A) 0.2

(B) 0.1

(C) 0.15

(D) 0.05

05. A linear double-stranded DNA fragment contains 1000 bases in total, including 120 adenine residues across both strands. Assume only canonical A–T and G–C pairing. With two H bonds per A–T pair and three per G–C pair, its hydrogen-bond count is

(A) 1260 (B) 1120

(C) 1500 (D) 1380

06. A double-stranded DNA fragment has 60 base pairs and 160 canonical base-pair hydrogen bonds. Use two per A–T pair and three per G–C pair. Cytosine is what fraction of all bases?

(A) 0.33333333 (B) 0.25

(C) 0.16666667 (D) 0.66666667

07. Identical linear glycine peptides bind 0.0200 mol HCl quantitatively at their free amino ends only. Complete hydrolysis of the same sample consumes 0.100 mol water. Use glycine=75. Total glycine mass recovered in g is

(A) 9 (B) 7.5

(C) 1.5 (D) 10.8

08. A linear double-stranded DNA fragment contains 90 cytosine and 60 adenine residues across both strands. Assume canonical pairing, equal strand lengths, and no circular closure. How many phosphodiester linkages join successive nucleotides along the two backbones?

(A) 300 (B) 299

(C) 150 (D) 298

09. A 0.100 mol amino-acid mixture contains glycine (75) and alanine (89) and weighs 8.20 g. All residues are incorporated, without loss of residues, into 0.0100 mol linear peptide chains. The mean number of alanine residues per chain is

(A) 10 (B) 6

(C) 5 (D) 4

10. A DNA duplex has 100 base pairs and 240 canonical pair hydrogen bonds. A–T has two H bonds and G–C three. For this count, A, G and C each carry one amino group, while T carries none. Total amino-group count on the bases is

(A) 140 (B) 160

(C) 120 (D) 100

11–40 / New middle-to-hard practice

11. A 0.100 mol maltose/sucrose mixture gives 0.0800 mol Ag under a calibrated reducing-disaccharide Tollens assay,2 Ag per reducing molecule. Both disaccharides then hydrolyse completely in a separate equal sample. Glucose:fructose mole ratio is

(A) $\frac{3}{2}$ (B) $\frac{7}{3}$

(C) $\frac{2}{3}$ (D) $\frac{4}{3}$

12. A 0.100 mol maltose/lactose mixture is completely hydrolysed. A selective glucose assay gives 0.160 mol glucose. Fraction of all monosaccharide molecules that are galactose is

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$

(C) $\frac{1}{5}$ (D) $\frac{2}{5}$

13. A 0.100 mol glucose/fructose mixture consumes 0.0400 mol Br_2 in validated neutral bromine-water oxidation, which oxidises aldose only without base-catalysed isomerisation. Oxidised acid is removed quantitatively. Mass of unchanged sugar is (both 180)

(A) 14.40 g (B) 7.20 g

(C) 10.80 g (D) 18.00 g

14. A 0.100 mol glucose/sucrose mixture weighs 27.72 g (M180,342). All glucose is selectively oxidised by neutral bromine water and removed as gluconic acid. Remaining sucrose is completely hydrolysed. In a calibrated alkaline reducing assay, each resulting monosaccharide deposits 2 Ag. Total silver amount is

(A) 0.240 mol (B) 0.400 mol

(C) 0.120 mol (D) 0.0800 mol

15. A 0.0500 mol glucose/galactose mixture forms osazones completely. A calibrated separation isolates 0.0300 mol osazone identical to that from pure glucose. The remaining isomeric osazone has the same molar mass 358. Its mass is

(A) 7.16 g (B) 3.58 g

(C) 17.90 g (D) 10.74 g

16. A 0.100 mol glucose/mannose/fructose mixture consumes 0.0700 mol Br_2 in neutral aldose-selective oxidation. A separate validated glucose-specific assay counts 0.0400 mol glucose. Original mannose:fructose mole ratio is

(A) $\frac{7}{3}$ (B) 1

(C) $\frac{4}{3}$ (D) $\frac{3}{4}$

17. A 0.0100 mol pure peptide hydrolyses completely to 0.0600 mol amino-acid molecules while consuming 0.0500 mol water, one per cleaved peptide bond. Assume one unbranched molecule and no side-chain amide hydrolysis. The peptide is

(A) A cyclic pentapeptide with six peptide bonds. (B) A cyclic hexapeptide with six peptide bonds.

(C) A linear hexapeptide with five peptide bonds. (D) A linear pentapeptide with five peptide bonds.

18. A mixture contains linear diglycine and cyclic diglycine. Complete hydrolysis gives 0.200 mol glycine and consumes 0.140 mol water. Linear dimer has one peptide bond, M132;cyclic dimer has two, M114. Original mass is

(A) 18.48 g (B) 12.48 g

(C) 13.20 g (D) 11.40 g

19. Tetrapeptides use Gly, Ala, Ser, Val once each. C-terminal residue must be Ala, N-terminal residue must be chiral, and Gly and Ser must be adjacent in either order. Number of allowed primary sequences is

(A) 4 (B) 3

(C) 2 (D) 6

20. Pentapeptides use Gly, Ala, Ser, Val, Leu once each. C-terminus is Ala;N-terminus must be chiral;Gly must occur immediately before Ser. Number of allowed sequences is

(A) 6 (B) 8

(C) 4 (D) 12

21. A linear tetrapeptide contains Gly, Ser, Val, Ala once each. C-terminus is Ala;partial hydrolysis establishes Gly–Ser and Ser–Val dipeptides. Which complete order is forced?

(A) Gly–Ser–Val–Ala. (B) Ser–Gly–Val–Ala.

(C) Val–Ser–Gly–Ala. (D) Ala–Gly–Ser–Val.

22. A pentapeptide uses Val, Gly, Ser, Leu, Ala once each. N-terminus is Val, C-terminus Ala;partial hydrolysis detects Val–Gly and Ser–Leu. Which additional adjacent pair must be present?

(A) Gly–Leu. (B) Leu–Ser.

(C) Ala–Val. (D) Gly–Ser.

23. An ideal 20-residue alpha-helical model permits each C=O at i to H-bond only to N–H at i+4 within the chain. Proline occupies positions 2 and 10 and has no N–H. Keep all geometry unchanged and ignore any helix disruption. Number of these model H-bonds is

(A) 14 (B) 16

(C) 18 (D) 15

24. A 40-residue model protein has ideal alpha-helical segments 1–12 and 17–40 only. H-bonds are i to i+4 inside the same segment. Proline at 10 and 20 removes N–H only;all other geometry is fixed. Number of permitted bonds is

(A) 26 (B) 32

(C) 27 (D) 28

25. A double-stranded DNA molecule has 200 base pairs and 520 interbase H-bonds. A–T pairs have 2, G–C3. Fraction of all nucleotides that are guanine is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$

(C) $\frac{3}{10}$ (D) $\frac{1}{5}$

26. A double-stranded DNA fragment has 300 base pairs. Guanine is 20.0% of all its nucleotides. Total interbase H-bonds are (A–T2, G–C3)

(A) 780
(B) 720
(C) 840
(D) 660

27. A coding DNA strand is 5-prime ATG CCA TAA3-prime. RNA has the coding sequence with U replacing T. Given AUG=Met, CCA=Pro, UAA=stop, which pair is correct?

(A) Template3-prime ATG CCA TAA5-prime; peptide Tyr–Gly.
(B) Template3-prime TAC GGT ATT5-prime; translated peptide Met–Pro.
(C) Template3-prime TAC GGT ATT5-prime; peptide Pro–Met.
(D) Template5-prime TAC GGT ATT3-prime; peptide Met–Pro–stop amino acid.

28. A coding DNA segment is 5-prime ATG GAA TTC TAA3-prime. Given AUG=Met, GAA=Glu, UUC=Phe, UAA=stop, a mutation replaces the third coding triplet TTC by TAA. Which result follows without a frameshift?

(A) Original peptide Met–Glu; mutant Met–Glu–Phe,with an extra residue.
(B) Both have three amino acids because stop is itself an amino acid.
(C) Original peptide Met–Glu–Phe; mutant Met–Glu,with a premature stop.
(D) The mutation reverses the DNA template direction and changes every codon.

29. A purified enzyme loses activity and its alpha-helical spectroscopic signal on heating. Cooling restores both;peptide-bond count and terminal-group assays remain unchanged throughout. Which interpretation fits all observations?

(A) Complete peptide hydrolysis followed by spontaneous resynthesis on cooling.

(B) A change in amino-acid sequence without any peptide-bond change.

(C) Permanent conversion of every peptide bond to a disulfide bond.

(D) Reversible loss and recovery of higher-order structure,with primary peptide connectivity retained.

30. Two DNA populations A, B have equal absorbance per unit mass when intact. At T1, only A melts, giving 40% hyperchromicity;whole-sample absorbance rises 24%. At T2 both melt, with A 40% and B30% hyperchromicity. For initial absorbance 1.000, the final value is (assume additive absorbance)

(A) $\frac{13}{10}$
(B) $\frac{67}{50}$
(C) $\frac{7}{5}$
(D) $\frac{34}{25}$

31. A mixture of starch and sucrose yields 0.150 mol glucose and 0.0500 mol fructose on complete hydrolysis. Each sucrose gives one of each;each starch anhydroglucose unit gives one glucose. Ignoring chain-end corrections, initial starch mass is (repeat unit 162)

(A) 24.30 g
(B) 8.10 g
(C) 18.00 g
(D) 16.20 g

32. A pure unbranched linear amylose sample has 0.00100 mol reducing ends, one per chain. Complete hydrolysis gives 0.100 mol glucose. A chain of n glucose units has molar mass$162n + 18$. Original dry amylose mass is

(A) 16.218 g (B) 16.380 g

(C) 16.200 g (D) 18.000 g

33. A calibrated amino-acid ampholyte has only two ionisation steps, cation to zwitterion to anion. Its isoelectric pH is 6.00 and its second half-titration pH is 9.00. Use$pI = (pK_{a1} + pK_{a2})/2$ and ideal speciation. Mean molecular charge at pH7.00, in elementary-charge units, is

(A) 0 (B) $-\frac{33}{3367}$

(C) $-\frac{1}{101}$ (D) $\frac{33}{3367}$

34. A 0.100 mol glycine/aspartic-acid mixture is in the specified dominant-form model at pH7:glycine has net charge 0, aspartate net charge minus 1. Measured total charge is minus 0.0400 mol elementary-charge equivalents. Initial neutral amino-acid mass is (M75,133)

(A) 13.30 g (B) 9.82 g

(C) 8.66 g (D) 10.98 g

35. A 5.00 g mixture of alpha- and beta-D-glucose is made to 100 mL. In a 2.00 dm polarimeter, the observed angle is+5.62 degrees. Given$\alpha_{obs} = lc[\alpha]$, c in g/mL, and pure specific rotations 112,19 degrees, amount of alpha form is (M180;ignore mutarotation during measurement)

(A) 1/45 mol (B) 1/60 mol

(C) 1/90 mol (D) 1/180 mol

36. Consider D-aldohexoses whose C3, C4, C5 configurations match D-glucose;C2 is allowed to vary. For each resulting open-chain sugar, count its alpha and beta pyranose anomers only. Number of cyclic stereochemical forms in this restricted set is

(A) 4 (B) 8

(C) 2 (D) 16

37. A 0.0100 mol single-chain protein has 0.0200 mol free thiol groups before reduction and 0.0800 mol after quantitative disulfide reduction. Each S–S gives two thiols. Pair (cysteine residues per chain, disulfide bonds per chain) is

(A) $(8, 3)$ (B) $(8, 4)$

(C) $(6, 3)$ (D) $(2, 6)$

38. A mixture contains protein A with one disulfide and protein B with three, with no free thiols. Total chains 0.00100 mol;complete reduction consumes 0.00220 mol H_2, one per disulfide. Chain masses are 10000,20000 g/mol. Initial protein mass is

(A) 20.0 g (B) 18.0 g

(C) 16.0 g (D) 14.0 g

39. A 0.100 mol glycine/arginine mixture gives 0.220 mol NH_3 on a calibrated total-nitrogen digestion. Glycine has 1 N, arginine 4 N;M75,174. Initial amino-acid mass is

(A) 11.46 g (B) 17.40 g

(C) 13.44 g (D) 10.47 g

40. A glycine/lysine mixture consumes 0.100 mol base for all its carboxylic-acid groups and gives 0.140 mol NH_3 in quantitative total-N digestion. Each has one COOH;glycine 1 N, lysine 2 N. Initial mass is (M75,146)

(A) 9.17 g (B) 10.34 g

(C) 11.76 g (D) 14.60 g

41–50 / Preserved hard practice

41. A 0.100 mol mixture of glucose and sucrose is assayed assuming one Tollens -reducing carbonyl per glucose and sucrose nonreducing. It deposits 0.120 mol Ag before hydrolysis. After complete hydrolysis and assuming both hexoses are quantitatively reducing, total reducing monosaccharides in mol are

(A) 0.16 (B) 0.14

(C) 0.1 (D) 0.12

42. A linear peptide formed from 5 glycine units(M=75)has one free amine and one free carboxyl end. Its molar mass in g/mol is

(A) 285 (B) 321

(C) 375 (D) 303

43. A double-stranded DNA fragment has 200 base pairs and total 500 canonical hydrogen bonds. Number of G-C pairs is

(A) 100 (B) 50

(C) 150 (D) 200

44. A 6.06 g sample of a pure linear peptide made only from glycine gives 0.100 mol glycine on complete hydrolysis. Use glycine=75 and water=18 g/mol. Number of glycine residues per peptide is

(A) 6 (B) 5

(C) 3 (D) 4

45. A sample contains cytidine (a nucleoside) and adenosine monophosphate (one phosphate per molecule). Complete hydrolysis gives 0.140 mol total bases and 0.0400 mol phosphate. Cytosine contains 3 N atoms and adenine 5. Total nitrogen-atom amount in the released bases, in mol, is

(A) 0.7 (B) 0.4

(C) 0.3 (D) 0.5

46. A 0.100 mol mixture of glucose(M=180) and ribose(M=150)weighs 16.2 g. Moles of glucose are

(A) 0.04 (B) 0.08

(C) 0.06 (D) 0.02

47. A 0.0100 mol mixture contains linear, unbranched glucose polymers with either 40 or 100 residues per chain. There are no crosslinks or rings. Complete hydrolysis consumes 0.690 mol water. With glucose=180 and water=18, the original dry polymer mass in g is

(A) 112.32 (B) 126

(C) 113.58 (D) 124.2

48. A nonreducing disaccharide sample of 3.42 g is fully hydrolyzed to two hexoses(M=180 each). Total monosaccharide mass in g is

(A) 3.24 (B) 3.6

(C) 3.78 (D) 3.42

49. An ideal neutral-side-chain amino acid has only two ionizable groups, with $pK_{a1} = 2.00$ and $pK_{a2} = 10.00$. A sample initially contains 0.100 mol zwitterion. Add 0.0250 mol HCl and then 0.0500 mol NaOH, with negligible volume change. Neglect minor species. Using $\log_{10} 3 = 0.4771$, the final pH is

(A) 10.4771 (B) 2.4771

(C) 6 (D) 9.5229

50. A 6.42 g mixture contains only linear diglycine and triglycine. Complete hydrolysis gives 0.100 mol glycine. Use glycine=75 and water=18 g/mol. Initial triglycine amount in mol is

(A) 0.03 (B) 0.04

(C) 0.02 (D) 0.01

Answer key

Biomolecules

C-BIOM / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**D**	11	**B**	21	**A**	31	**D**	41	**B**
02	**D**	12	**C**	22	**D**	32	**A**	42	**D**
03	**A**	13	**C**	23	**D**	33	**B**	43	**A**
04	**B**	14	**A**	24	**C**	34	**B**	44	**B**
05	**D**	15	**A**	25	**C**	35	**C**	45	**D**
06	**A**	16	**B**	26	**B**	36	**A**	46	**A**
07	**A**	17	**C**	27	**B**	37	**A**	47	**C**
08	**D**	18	**B**	28	**C**	38	**C**	48	**B**
09	**C**	19	**B**	29	**D**	39	**A**	49	**D**
10	**A**	20	**C**	30	**D**	40	**B**	50	**C**

Chapter 30

Polymers

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A polymer blend contains only PE and PVC repeat residues, with negligible end corrections. A 9.05 g sample contains 3.55 g chlorine. Use PE repeat=28, PVC repeat=62.5, Cl=35.5. Mole fraction of PVC repeat residues is

(A) 0.25 (B) 0.75

(C) 0.5 (D) 0.66666667

02. A PE sample has chains with 100 or 200 repeat residues only. Its number-average chain mass is 4200 u; end corrections are negligible and repeat mass is 28 u. Its weight-average chain mass in u is

(A) 4200 (B) 3500

(C) 5600 (D) 4666.6667

03. A linear nylon-6 oligomer formed from 6-aminohexanoic acid has molar mass $583\,\mathrm{g\,mol^{-1}}$. Use amino acid=131, water=18, with ordinary amino and carboxyl ends. Complete hydrolysis of 0.0200 mol oligomer yields amino-acid mass in g of

(A) 13.1 (B) 2.62

(C) 11.66 (D) 15.72

04. A linear nylon-6, 6 chain has 20 amide linkages and two amino end groups. It forms only from adipic acid (146) and hexamethylenediamine (116), losing water (18) per linkage. Chain molecular mass in u is

(A) 2376 (B) 2394

(C) 2260 (D) 2620

05. An OH -ended linear PET oligomer contains five terephthalic-acid residues and six ethylene-glycol residues per chain, with no acid ends. Use monomer masses 166 and 62 and water=18. Formation of 20.44 g oligomer releases water in amount (mol)

(A) 0.2 (B) 0.22

(C) 0.1 (D) 0.02

06. A chain mixture contains masses 10000 u and 20000 u only and has number-average mass 12500 u. Mass fraction of the 20000 u chains is

(A) 0.6 (B) 0.25

(C) 0.5 (D) 0.4

07. A polymer has chain masses M and 2 M only. Its dispersity is 10/9, and the number fraction of 2 M chains is less than 1/3. The mass fraction of 2 M chains is

(A) 0.66666667 (B) 0.2

(C) 0.5 (D) 0.33333333

08. All pendant nitrile groups of an ideal PAN sample are hydrolyzed to neutral carboxylic-acid groups. Each original repeat C_3H_3N becomes $C_3H_4O_2$ and gives one ammonia equivalent. All released nitrogen is separately recovered as dry ammonia after separation. The dry hydrolyzed polymer weighs 7.20 g. Use C=12, H=1, N=14, O=16. Ammonia volume at conventional STP is (L)

(A) 3.36 (B) 2.24

(C) 4.48 (D) 1.12

09. An ideal PMMA specimen has repeat $C_5H_8O_2$, including one pendant methyl ester per repeat. Complete hydrolysis converts each ester to a carboxylic acid and methanol, without backbone cleavage. Starting with 10.0 g polymer, dry acid-polymer mass in g is (end corrections negligible; C=12, H=1, O=16)

(A) 8.6 (B) 7.2

(C) 10 (D) 9.8

10. An ideal step-growth feed has 0.100 mol A–A and 0.100 mol B–B monomer molecules. One water is lost per A–B link. The product contains 0.0100 mol acyclic chains, with every monomer incorporated and no rings. Water released in g is (water=18)

(A) 3.42 (B) 0.18

(C) 3.6 (D) 1.8

11–40 / New middle-to-hard practice

11. A dry butadiene/acrylonitrile copolymer weighs 16.0 g and contains 2.80 g nitrogen. Repeat masses are 54 and 53;only acrylonitrile contains one N. Each incorporated butadiene retains one C=C. Complete selective backbone hydrogenation needs H_2 amount

(A) 0.0500 mol (B) 0.100 mol

(C) 0.200 mol (D) 0.300 mol

12. A styrene/butadiene copolymer weighs 10.6 g. A validated selective assay adds 0.100 mol Br_2 to residual backbone double bonds without aromatic substitution. Each butadiene repeat 54 retains one C=C;styrene repeat 104 retains no backbone C=C. Styrene repeat mole fraction is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) $\frac{2}{3}$ (D) $\frac{1}{3}$

13. For an ideal linear nylon-6,6 batch,0.100 mol adipic acid(M146) and 0.100 mol hexamethylenediamine(M116) form 0.180 mol amide bonds, each releasing one water. No rings or branching occur;each intermolecular bond reduces molecule count by one. Number-average molecular mass is

(A) 2260 $\mathrm{g\,mol^{-1}}$ (B) 574 $\mathrm{g\,mol^{-1}}$

(C) 1275 $\mathrm{g\,mol^{-1}}$ (D) 1148 $\mathrm{g\,mol^{-1}}$

14. A calibrated nylon-6 ring-opening batch incorporates 0.100 mol caprolactam(M113) into linear chains. Exactly 0.00100 mol water initiates one chain each and is incorporated in terminal groups;there is no water loss or chain transfer. Number-average chain mass is

(A) 11318 $\mathrm{g\,mol^{-1}}$ (B) 11118 $\mathrm{g\,mol^{-1}}$

(C) 11300 $\mathrm{g\,mol^{-1}}$ (D) 22618 $\mathrm{g\,mol^{-1}}$

15. A 9.60 g sample contains PET and inert filler. Complete saponification consumes 0.0800 mol NaOH;each PET repeat $\mathrm{C_{10}H_8O_4}$(M192) contains two ester links. Ignore chain ends. PET mass fraction is

(A) $\frac{4}{5}$ (B) $\frac{2}{5}$

(C) $\frac{9}{10}$ (D) $\frac{3}{5}$

16. A PHBV copolyester has 0.100 mol repeat units and mass 9.16 g. Hydroxybutyrate-derived repeats have formula $\mathrm{C_4H_6O_2}$(86);hydroxyvalerate-derived $\mathrm{C_5H_8O_2}$(100). Carbon mass in the polymer is

(A) 5.28 g (B) 5.52 g

(C) 4.80 g (D) 6.00 g

17. A 10.0 g polymer mixture contains monodisperse chains of masses 10000 and 30000 g/mol. Total chain amount is 0.000500 mol. Weight-average mass$M_w = \sum n_i M_i^2 / \sum n_i M_i$ is

(A) 20000 $\mathrm{g\,mol^{-1}}$ (B) 15000 $\mathrm{g\,mol^{-1}}$

(C) 25000 $\mathrm{g\,mol^{-1}}$ (D) 30000 $\mathrm{g\,mol^{-1}}$

18. A 12.5 g polymer mixture has 0.00100 mol total chains, with only 5000 and 20000 g/mol chain species. Using$M_w = \sum n_i M_i^2 / \sum n_i M_i$, the polydispersity ratioM_w/M_n is

(A) $\frac{17}{10}$ (B) 1

(C) $\frac{34}{25}$ (D) $\frac{5}{4}$

19. An ideal bifunctional step-growth model has$X_n = (1+r)/(1+r-2rp)$, r<=1 the functional-group ratio. Complete conversion gives$X_n = 99$. At p=0.990, the number-average degree is

(A) 100 (B) 50

(C) 99 (D) 25

20. Two ideal equimolar bifunctional batches each contain 0.100 mol incorporated monomer units. Their conversions are 0.980 and 0.990, and$X_n = 1/(1-p)$. When batches are mixed without further reaction, the combined number-average degree is

(A) 75 (B) 50

(C) 100 (D) $\frac{200}{3}$

21. A 10.0 g PVC/PTFE mixture contains 3.80 g fluorine. Repeat formulas are C_2H_3Cl(M62.5) and C_2F_4(100). Chlorine mass is (F19, Cl35.5)

(A) 1.42 g
(B) 5.68 g
(C) 3.55 g
(D) 2.84 g

22. A PVC/polyvinylidene-chloride mixture contains 0.100 mol repeat units and 0.160 mol chlorine atoms. Repeat formulas are C_2H_3Cl and $C_2H_2Cl_2$. A validated elemental hydrogen assay collects every two polymer H atoms as one H_2O, with halogen interference removed and blank-corrected. Water amount is

(A) 0.120 mol
(B) 0.0800 mol
(C) 0.100 mol
(D) 0.160 mol

23. In a soluble acyclic rubber-linking model,0.200 mol C_5H_8 repeats(M68) initially form chains of 1000 repeats each. Added sulfur 0.00640 g forms only two-S bridges, each connecting two previously separate components. S32;ignore all other mass changes. Number-average final component mass is

(A) 68032 $\mathrm{g\,mol^{-1}}$
(B) 136000 $\mathrm{g\,mol^{-1}}$
(C) 136064 $\mathrm{g\,mol^{-1}}$
(D) 272128 $\mathrm{g\,mol^{-1}}$

24. A model rubber sample initially has 0.00100 mol separate chains. After acyclic intermolecular linking, it has 0.000400 mol components. Bridges contain either one or two S atoms;total sulfur is 0.000800 mol atoms. Fraction of bridges containing two sulfur atoms is

(A) $\frac{1}{3}$
(B) $\frac{1}{4}$
(C) $\frac{2}{3}$
(D) $\frac{1}{2}$

25. A radical-polymerisation batch begins with 0.00200 mol growing radical chains. Only pairwise combination or disproportionation terminates them;no transfer. Final dead polymer molecules total 0.00150 mol. Fraction of termination events that were combination is

(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{3}{4}$
(D) $\frac{2}{3}$

26. A polystyrene batch incorporates 0.100 mol styrene units(M104) into 0.00200 mol initially growing radical chains. Termination events are 75% combination,25% disproportionation;no transfer or other loss, and ignore end-group mass. Number-average polymer mass is

(A) 6944 $\mathrm{g\,mol^{-1}}$
(B) 8320 $\mathrm{g\,mol^{-1}}$
(C) 5200 $\mathrm{g\,mol^{-1}}$
(D) 10400 $\mathrm{g\,mol^{-1}}$

27. Material A repeatedly softens and can be reshaped on heating, and dissolves as separate polymer molecules in a suitable solvent. Material B only swells in that solvent and does not melt before decomposition. Which structural inference fits both tests?

(A) Both tests identify the monomer uniquely without any chemical analysis.
(B) A is consistent with a thermoplastic;B with a covalently cross-linked network.
(C) A must be a fully cross-linked network because it dissolves.
(D) B must be a linear thermoplastic because swelling proves molecular dissolution.

28. A 10.0 g sample contains polyethylene and PHB polyester only. A validated selective hydrolysis leaves PE unchanged and gives 3.12 g isolated 3-hydroxybutanoic acid(M104) from PHB;each PHB repeat(M86) gives one acid. PE mass fraction is

(A) $\frac{43}{52}$ (B) $\frac{43}{125}$

(C) $\frac{371}{500}$ (D) $\frac{13}{50}$

29. A polyester feed contains 0.100 mol alcohol molecules, only ethylene glycol(two OH) and glycerol(three OH). It is exactly OH/COOH balanced with 0.125 mol pure diacid(two COOH). Glycerol mole fraction among alcohol molecules is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$

(C) $\frac{1}{4}$ (D) $\frac{1}{3}$

30. A 0.100 mol ethylene-glycol/glycerol mixture weighs 7.10 g (M62,92). It reacts with 0.100 mol diacid, two COOH each. Every available COOH forms an ester, one OH consumed per COOH, with no other reaction. Free OH amount remaining is

(A) 0.0500 mol (B) 0.0100 mol

(C) 0.100 mol (D) 0.0300 mol

31. An ideal phenol-formaldehyde model incorporates 0.100 mol phenol rings(M94). Every retained formaldehyde makes one methylene bridge between two ortho/para ring sites and loses one water;there are no hydroxymethyl end groups. Dry resin mass 10.12 g. Vacant ortho/para site amount is

(A) 0.0600 mol (B) 0.180 mol

(C) 0.240 mol (D) 0.120 mol

32. An ideal unbranched acyclic novolac model has 0.100 mol phenol rings and 0.0800 mol methylene bridges. Each bridge comes from HCHO minus H_2O and joins separate components. Phenol M94. Number-average resin component mass is

(A) 647.5 $\mathrm{g\,mol^{-1}}$ (B) 470 $\mathrm{g\,mol^{-1}}$

(C) 1036 $\mathrm{g\,mol^{-1}}$ (D) 518 $\mathrm{g\,mol^{-1}}$

33. One linear copolymer chain has 100 A and 100 B units. A sequence probe finds all 199 adjacent dyads are A–B or B–A, with no A–A or B–B. Which conclusion uses both composition and adjacency?

(A) It must have the same unit type at both ends because its length is even.

(B) The chain alternates A and B and has different terminal unit types.

(C) It is a physical blend of A and B homopolymers,each with no heterodyads.

(D) It is a two-block chain because equal composition always implies blocks.

34. A single open linear chain has 40 A and 60 B units. Arrange it to maximise unlike adjacent pairs A–B/B–A. Pair(maximum unlike dyads, minimum B–B dyads) is

(A) $(40, 59)$ (B) $(99, 0)$

(C) $(80, 19)$ (D) $(79, 20)$

35. A 10.0 g polymer/glucose sample in 1.00 L initially has$\pi/(RT) = 0.0110$ mol/L. A membrane passes glucose completely but retains all polymer;after exhaustive dialysis back to 1.00 L,$\pi/(RT) = 0.00100$ mol/L. Glucose M180. Polymer number-average mass is

(A) 909.09 $\mathrm{g\,mol^{-1}}$ (B) 8200 $\mathrm{g\,mol^{-1}}$

(C) 1800 $\mathrm{g\,mol^{-1}}$ (D) 10000 $\mathrm{g\,mol^{-1}}$

36. A 2.00 g polymer sample has one measurable end group per covalent chain,0.000200 mol ends total. Osmometry gives 0.000150 mol particles. Under an explicit monomer/two-chain-aggregate model, pair(true number-average covalent-chain mass, fraction of covalent chains in dimers)is

(A) $(10000\ \mathrm{g\,mol^{-1}}, 1/4)$ (B) $(13333.3\ \mathrm{g\,mol^{-1}}, 1/3)$

(C) $(13333.3\ \mathrm{g\,mol^{-1}}, 1/2)$ (D) $(10000\ \mathrm{g\,mol^{-1}}, 1/2)$

37. A linear PET sample initially has 0.00100 mol chains and mass 19.218 g. Controlled hydrolysis cleaves 0.00900 mol internal ester bonds, each adding one water and splitting one chain, with no cycles or fragments lost. Final number-average chain mass is

(A) 1921.8 $\mathrm{g\,mol^{-1}}$ (B) 1938 $\mathrm{g\,mol^{-1}}$

(C) 2135.3 $\mathrm{g\,mol^{-1}}$ (D) 19218 $\mathrm{g\,mol^{-1}}$

38. A linear polyester has 9.609 g dry mass and 0.000500 mol chains. After controlled internal-ester hydrolysis, all products retained, there are 0.00200 mol chains. One water enters per scission. Final number-average mass is

(A) 4804.5 $\mathrm{g\,mol^{-1}}$ (B) 6406 $\mathrm{g\,mol^{-1}}$

(C) 19218 $\mathrm{g\,mol^{-1}}$ (D) 4818 $\mathrm{g\,mol^{-1}}$

39. A 10.4 g polystyrene sample is selectively sulfonated;each new SO_3H group replaces one ring H and adds 80 g/mol net. Acid-form resin mass 14.4 g. Complete conversion of every SO_3H to SO_3Na changes H1 to Na23. Dry sodium-form mass is

(A) 14.4 g (B) 11.5 g

(C) 15.5 g (D) 18.4 g

40. A pure polyvinyl-acetate (PVAc) sample weighs 8.60 g, repeat M86. Partial hydrolysis replaces $OCOCH_3$ by OH, lowering polymer mass 42 per converted repeat. Dry final polymer mass 6.50 g. NaOH needed to hydrolyse all remaining acetate groups is

(A) 0.0750 mol (B) 0.0250 mol

(C) 0.0500 mol (D) 0.100 mol

41–50 / Preserved hard practice

41. A single linear nylon-6 oligomer forms from 10 molecules of 6-aminohexanoic acid(M=131), with one water lost per each amide bond. Its molecular mass in u is

(A) 1292 (B) 1130

(C) 1148 (D) 1310

42. An equimolar chain feed contains 10 adipic acid and 10 hexamethylenediamine molecules and forms one unbranched chain with amide bonds. With M=146 and 116, its molecular mass in u is

(A) 2620 (B) 2278

(C) 2260 (D) 2440

43. A PET chain has 10 terephthalic acid and 10 ethyleneglycol residues with one COOH and one OH end. Monomer masses 166 and 62. Chain molecular mass in u is

(A) 1938 (B) 1920

(C) 2100 (D) 2280

44. A polymer mixture contains twice as many 10000 u chains as 20000 u chains. M_w / M_n is

(A) 1 (B) 1.25

(C) 1.125 (D) 1.5

45. A 0.0100 mol mixture contains OH -ended linear PET chains with either 10 or 20 ester linkages per chain and no COOH end groups. Complete saponification consumes 0.140 mol OH^-. All ester-derived glycol is recovered quantitatively. The amount of ethylene glycol recovered, in mol, is

(A) 0.06 (B) 0.07

(C) 0.1 (D) 0.08

46. A copolymer contains 60 mole%styrene units(M=104) and 40 mole%butadiene units(M=54). Its total mass for 1.00 mol repeat residues in g is

(A) 94 (B) 84

(C) 104 (D) 74

47. A 0.100 mol nylon-6 repeat unit sample is hydrolyzed to the corresponding amino acid, ignoring end corrections. Product mass in g is(amino acid 131)

(A) 26.2 (B) 11.3

(C) 13.1 (D) 9.5

48. A mixture of PE and PVC contains 0.100 mol of total repeat units and 1.775 g chlorine. With Cl=35.5, PE 28, PVC 62.5, total polymer mass in g is

(A) 2.8 (B) 6.25

(C) 4.525 (D) 3.6625

49. A monodisperse oligomer built from glycine only has M 303 g/mol. Complete hydrolysis of 0.0100 mol oligomer gives glycine amount in mol of(M gly 75, H_2O =18)

(A) 0.05 (B) 0.06

(C) 0.04 (D) 0.03

50. A linear polyamide chain contains 3 glycine(M=75) and 3 aminohexanoic acid(M=131)units with one NH_2 and one COOH end. Molecular mass in u is

(A) 546 (B) 528

(C) 510 (D) 618

Answer key

Polymers

C-POLY / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**C**	11	**B**	21	**D**	31	**B**	41	**C**
02	**D**	12	**D**	22	**A**	32	**D**	42	**B**
03	**A**	13	**D**	23	**C**	33	**B**	43	**A**
04	**A**	14	**A**	24	**A**	34	**C**	44	**C**
05	**A**	15	**A**	25	**B**	35	**B**	45	**D**
06	**D**	16	**A**	26	**B**	36	**D**	46	**B**
07	**D**	17	**C**	27	**B**	37	**B**	47	**C**
08	**B**	18	**C**	28	**C**	38	**D**	48	**C**
09	**A**	19	**B**	29	**A**	39	**C**	49	**A**
10	**A**	20	**D**	30	**D**	40	**C**	50	**B**

Chapter 31

Environmental Chemistry

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. A water sample contains 2.00 mg pollutant in 1.00 kg solution. Mass concentration in ppm is

(A) 200 (B) 2

(C) 20 (D) 0.2

02. Rainwater pH changes from 5 to 3. H^+ activity increases by factor

(A) 2 (B) 100

(C) 1000 (D) 10

03. A 100 mL sample loses 4.00 mg/L dissolved O 2 during the stated BOD test with no dilution correction. Oxygen mass consumed in mg is

(A) 40 (B) 4

(C) 0.04 (D) 0.4

04. In Cl+ O_3 → ClO +O 2 followed by ClO +O→Cl+ O_2, net O_2 molecules formed per cycle is

(A) 4 (B) 2

(C) 1 (D) 3

05. For $CaCO_3$ → CaO + CO_2, atom economy if CaO is the desired product is($CaCO_3$ =100, CaO =56)

(A) 56 (B) 100

(C) 44 (D) 50

06. A gas mixture contains 40 micro mol CO_2 per 0.100 mol total gas. Mole fraction in ppm is

(A) 400 (B) 4000

(C) 0.4 (D) 40

07. A first-order pollutant decay model has 10 day half-life. Remaining fraction after 30 days is

(A) 0.5 (B) 0.75

(C) 0.125 (D) 0.25

08. An ideal reaction has stoichiometric reactant mass sum 200 g per reaction and desired product mass 150 g. Atom economy percentage is

(A) 25 (B) 150

(C) 50 (D) 75

09. A process produces 20 g product and 80 g discarded waste under a stated mass-accounting boundary. Waste/product mass ratio is

(A) 0.25 (B) 4

(C) 1 (D) 5

10. A model pollutant moves from 10 ppm at a low trophic level to 1000 ppm at higher level. Concentration magnification factor is

(A) 0.01 (B) 1000

(C) 10 (D) 100

11–40 / New middle-to-hard practice

11. A dry emission contains only SO_2 and NO, total 0.100 mol. Complete oxidation to SO_3 and NO_2 followed by conversion to H_2SO_4 and HNO_3 yields acids consuming 0.140 mol OH^-. With both protons of sulfuric acid neutralised, SO_2 mass in the emission is (M=64)

(A) $\frac{128}{25}$ (B) $\frac{96}{25}$

(C) $\frac{64}{25}$ (D) $\frac{32}{25}$

12. A dry 0.200 mol SO_2/NO emission is completely converted to sulfuric and nitric acids. Its acids require 0.300 mol NaOH for complete neutralisation. Minimum O_2 needed for both oxidation sequences, excluding oxygen already in water, is

(A) $\frac{1}{5}$ (B) $\frac{3}{20}$

(C) $\frac{1}{10}$ (D) $\frac{1}{8}$

13. Equal volumes of two ideal, unbuffered strong-acid rain samples have pH 3 and 5. The mixture is diluted until its pH is 4. Ignoring water ionisation, final volume divided by the volume just after mixing is

(A) 5 (B) $\frac{101}{10}$

(C) $\frac{101}{20}$ (D) $\frac{11}{2}$

14. A 100 mL unbuffered strong-acid rain sample of pH 3 is mixed with 100 mL of pH 4. After adding 0.0900 mmol NaOH in negligible volume, residual $[H^+]$ is, in mol L^{-1}

(A) $\frac{1}{20000}$ (B) $\frac{1}{10000}$

(C) $\frac{11}{20000}$ (D) $\frac{1}{2000}$

15. In a 300 mL BOD bottle, 15.0 mL wastewater is diluted to volume. Dissolved O_2 falls from 8.0 to 3.0 mg/L; an otherwise identical blank falls by 0.50 mg/L. Treat the blank correction as additive at bottle scale. The wastewater BOD is, in mg/L

(A) 100 (B) 90

(C) 95 (D) 9

16. A wastewater has blank-corrected BOD 120 mg/L. A 300 mL test bottle starts at 8.0 mg/L O_2; its blank consumes 0.50 mg/L. What maximum wastewater aliquot, in mL, leaves at least 2.0 mg/L after the test under the additive-demand model?

(A) 15 (B) $\frac{55}{2}$

(C) $\frac{55}{4}$ (D) $\frac{25}{2}$

17. Two oxygen-demanding water streams flow at 2 and 3 L/min, with ultimate demands 40 and 10 mg O_2/L. Treatment removes 50% of this incoming demand and then adds 20 mg/min oxygen. In a closed oxygen-balance model, residual unmet demand per litre of final stream is

(A) 11 (B) 14

(C) 18 (D) 7

18. A 4 L/min water stream carries demand 30 mg O_2/L. It mixes with x L/min clean water containing 8 mg/L available O_2; the original stream has none. In this closed balance, at least 2 mg/L oxygen must remain after all demand is met. Minimum x is, in L/min

(A) $\frac{64}{3}$ (B) 15

(C) 20 (D) 24

19. A model Cl radical destroys one O_3 per completed catalytic cycle. After each cycle it is permanently trapped with probability 0.001; every radical starts active. Treating cycles as independent, expected O_3 molecules destroyed by 0.0100 mol initially active radicals is, in moles

(A) 10 (B) $\frac{1}{10}$

(C) $\frac{1}{100}$ (D) 100

20. Each completed Cl cycle consumes one O_3 and one O and makes two O_2. A model batch initially contains 0.0800 mol O_3, 0.0500 mol O and 0.00100 mol radicals, each capped at 100 cycles. Greatest newly formed O_2 amount is

(A) $\frac{4}{25}$ (B) $\frac{1}{20}$

(C) $\frac{1}{10}$ (D) $\frac{1}{5}$

21. For the simplified sunlit $NO/NO_2/O_3$ cycle at steady state, $J[NO_2] = k[NO][O_3]$. In one air parcel $[NO_2]/[NO]$=3. On doubling J and reducing total NO_x to half, the ratio becomes 2 while k is unchanged. New ozone concentration divided by old is

(A) 2 (B) $\frac{2}{3}$

(C) $\frac{4}{3}$ (D) 3

22. A simplified steady photochemical cycle obeys $[O_3] = (J/k)[NO_2]/[NO]$. Two observations at the same J/k give $[NO_2]$ fractions 1/4 and 2/5 of total NO_x. Ozone in the second observation divided by the first is

(A) 2 (B) $\frac{8}{5}$

(C) $\frac{1}{2}$ (D) $\frac{5}{2}$

23. Assess these statements: I, stratospheric ozone can shield damaging UV; II, ozone cannot act as a greenhouse gas because it contains only one element; III, ozone near the ground can be an oxidising secondary pollutant; IV, in the Cl/ClO catalytic ozone-loss cycle, each destroyed ozone molecule permanently consumes one Cl radical. Which set is correct?

(A) I, III and IV only. (B) I, II and III only.

(C) I and III only. (D) II and IV only.

24. A simplified sunlit cycle contains $NO_2 \rightarrow NO + O$, $O + O_2 \rightarrow O_3$ and $RO_2 + NO \rightarrow RO + NO_2$. Ignore all other reactions. Which net chemical conversion and nitrogen-oxide role follow after one complete cycle?

(A) $RO_2 + O_3 \rightarrow RO + 2O_2$; nitrogen oxides are regenerated.

(B) $RO_2 + O_2 \rightarrow RO + O_3$; nitrogen oxides are regenerated.

(C) $RO_2 + O_2 \rightarrow RO + O_3$; one nitrogen dioxide is permanently consumed.

(D) $RO_2 + NO_2 \rightarrow RO + NO + O_2$; nitrogen dioxide is consumed.

25. In a calibrated lake model, algal growth requires 16 mol available N per mol P incorporated; its future decomposition needs 20 mol O_2 per mol incorporated P. Initially available N and P are 0.160 and 0.0200 mol. The closed lake has 0.100 mol O_2 and must retain at least 0.0200 mol after decomposition. Minimum fraction of P to remove before growth is

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$

(C) $\frac{9}{10}$ (D) $\frac{1}{2}$

26. A calibrated algal model uses 16 mol N per mol P incorporated and has later oxygen demand proportional to biomass. A lake starts with 0.0960 mol N and 0.0100 mol P. Treatment A removes half the P before growth. Treatment B removes no nutrients but exports a fraction of the mature biomass. Minimum exported fraction under B that leaves no more later oxygen demand than A is

(A) $\frac{1}{6}$ (B) $\frac{1}{5}$

(C) $\frac{1}{3}$ (D) $\frac{1}{2}$

27. A 1.00 mol dry gas feed contains 400 ppm SO_2 by mole. A scrubber removes 75% of SO_2 and no other gas. SO_2 in the dry outlet, in ppm by mole, is exactly

(A) 100 (B) 75

(C) $\frac{1000000}{9997}$ (D) 25

28. A 0.100 mol gas sample contains CO and N_2 only and has mass 2.80 g. Oxidising all CO increases mass by 0.160 g without losing gas. The initial CO content, in ppm by mole, is

(A) 10000 (B) 200000

(C) 50000 (D) 100000

29. In a stated food-chain concentration model, predator tissue receives contributions proportional to dietary mass fractions. Prey A contains 20 ppm persistent pollutant, prey B 80 ppm. The predator eats 75% A and 25% B and has a tissue/diet concentration factor 4. Its tissue concentration is, in ppm

(A) 200 (B) 100

(C) 140 (D) 320

30. A predator tissue/diet factor is 3 in a model with prey concentrations 10 and 50 ppm. Its tissue concentration is 90 ppm. Fraction of higher-concentration prey by consumed mass is

(A) $\frac{2}{3}$ (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{1}{4}$

31. A filter removes 90% of pollutant from the gas that passes through it; 20% of the feed bypasses the filter and rejoins the outlet. To halve the present total pollutant emission while retaining filter efficiency, the bypass fraction must become

(A) $\frac{1}{20}$ (B) $\frac{1}{5}$

(C) $\frac{2}{45}$ (D) $\frac{1}{10}$

32. A water purifier treats 80% of feed at 75% removal; untreated bypass rejoins it. A second unit then removes half the pollutant from the entire mixed stream. If final concentration is 6 mg/L and volumes are unchanged, feed concentration was, in mg/L

(A) 40 (B) 24

(C) 15 (D) 30

33. Pollutants A and B initially have concentration ratio 4:1 and decay independently with half-lives 2 and 4 days. First time their concentrations become equal is, in days

(A) 8 (B) 6

(C) 4 (D) 16

34. A pollutant follows first-order loss. Dilution immediately halves its concentration, and after a further 6 days concentration is one-eighth its original pre-dilution value. Its kinetic half-life is, in days

(A) 3 (B) $\frac{3}{2}$

(C) 6 (D) 2

35. A model wastewater contains methanol and ethanol only, total 0.100 mol. Complete oxidation to CO_2 and water consumes 0.240 mol O_2. Initial ethanol mass is (M=46)

(A) $\frac{46}{25}$ (B) $\frac{23}{5}$

(C) $\frac{23}{25}$ (D) $\frac{69}{25}$

36. In a closed oxygen-demand model, 0.0100 mol glucose and an unknown amount of acetic acid are completely oxidised to CO_2 and water. Oxygen demand is 0.100 mol O_2. With glucose=180 and acetic acid=60, total original organic mass is

(A) 3 (B) $\frac{9}{5}$

(C) $\frac{18}{5}$ (D) $\frac{12}{5}$

37. A process forms a theoretical 120 g desired product from 200 g reactants. Actual desired-product recovery is 75%; every other gram is discarded, with no solvent or recycle. Its atom economy is 60%. The actual waste/product mass ratio is

(A) $\frac{5}{3}$ (B) $\frac{11}{9}$

(C) $\frac{2}{3}$ (D) $\frac{4}{9}$

38. A process consumes 200 g reactants and 300 g solvent per batch and recovers 100 g product. It recycles 80% of solvent; all remaining output is waste and no other material enters. Waste/product ratio on this boundary is

(A) $\frac{3}{5}$ (B) $\frac{8}{5}$

(C) 1 (D) 4

39. A flue gas contains 0.100 mol SO_2. A limestone scrubber captures 80% as $CaSO_3$, using one $CaCO_3$ per captured SO_2. Limestone is 80% $CaCO_3$ by mass, and only half of its $CaCO_3$ reacts. Minimum limestone mass is ($CaCO_3$=100)

(A) 16 (B) 20

(C) 25 (D) 10

40. A 10.0 g limestone sample is 80% $CaCO_3$; the rest is inert. It treats 0.100 mol SO_2 by the one-to-one $CaCO_3/SO_2$ reaction, but $CaCO_3$ conversion stops at 75%. Fraction of original SO_2 left uncaptured is

(A) $\frac{1}{4}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

41–50 / Preserved hard practice

41. A 10.0 g mixture of $CaCO_3$ and inert solid neutralizes 0.120 mol H^+ in acid-rain-treatment model. With $CaCO_3$ =100, the calcium carbonate mass percentage is

(A) 80 (B) 60

(C) 40 (D) 100

42. A 0.300 mol gas mixture contains only SO_2 and CO_2 and weighs 15.20 g. All SO_2 is selectively oxidized and captured as pure H_2SO_4, with CO_2 unchanged. Use SO_2 =64, CO_2 =44, H_2SO_4 =98 g/mol. The acid mass obtained in g is

(A) 19.6 (B) 14.7

(C) 4.9 (D) 9.8

43. Two water streams flow at 10.0 and 20.0 L/min with pollutant concentrations 2.00 and 5.00 mg/L. Mixing is instant and no reaction occurs. Final concentration in mg/L is

(A) 4 (B) 4.5

(C) 3 (D) 3.5

44. A 0.100 mol mixture of SO_2 and NO_2 is fully oxidized to sulfate and nitrate. Oxidant accepts 0.150 mol electrons. Moles of SO_2 initially were

(A) 0.025 (B) 0.075

(C) 0.1 (D) 0.05

45. A 0.200 mol gas mixture contains CO and H_2 only and weighs 2.48 g. It burns completely and water is condensed away. With CO =28, H_2 =2, CO_2 =44 g/mol, the dry CO_2 mass in g is

(A) 1.76 (B) 8.8

(C) 5.28 (D) 3.52

46. A 20.0 mL polluted-water aliquot is diluted to 200.0 mL for a stated BOD test. Dissolved oxygen falls from 8.00 to 5.00 mg/L. The corresponding dilution-water blank loses 0.200 mg/L. Using BOD =(sample DO loss-blank DO loss)times dilution factor, the original sample BOD in mg/L is

(A) 32 (B) 30

(C) 2.8 (D) 28

47. A polluted-water model contains only a monoprotic weak acid HA with $K_a = 10^{-5}$ and has pH 3.00. Neglect water autoionization and activity corrections. For 10.0 L, the OH^- amount required for complete stoichiometric neutralization of the total acid, in mol, is

(A) 0.01 (B) 1

(C) 1.01 (D) 0.1

48. A first-order pollutant decay model has initial concentration 80 mg/Land 20 mg/L after 20 days. At 30 days, remaining concentration in mg/L is

(A) 10 (B) 5

(C) 15 (D) 20

49. A green-process model has A+B→P+Q, with stoichiometric masses 100+80→150+30 g. One such feed batch reaches 80% conversion. It isolates 75% of the P formed. All unreacted A/B are recovered. Of 500 g solvent, 95% is recovered. Count Q, lost P, and un recovered solvent as waste. Waste/product mass ratio is

(A) 0.87777778 (B) 0.6

(C) 1.2777778 (D) 0.27777778

50. An idealized ozone-cycle model starts with 0.00100 mol chlorine atoms; 40% are inactive. Of the active atoms, 10% terminate after 500 cycles and the rest complete 1000 cycles. Each cycle has net O_3 +O→2 O_2. The O_2 amount formed by these cycles, in mol, is

(A) 1.14 (B) 0.6

(C) 1.2 (D) 0.57

Answer key

Environmental Chemistry

C-ENVI / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**C**	31	**C**	41	**B**
02	**B**	12	**D**	22	**A**	32	**D**	42	**D**
03	**D**	13	**C**	23	**C**	33	**A**	43	**A**
04	**B**	14	**B**	24	**B**	34	**A**	44	**D**
05	**A**	15	**B**	25	**B**	35	**D**	45	**D**
06	**A**	16	**C**	26	**A**	36	**A**	46	**D**
07	**C**	17	**D**	27	**C**	37	**B**	47	**C**
08	**D**	18	**A**	28	**D**	38	**B**	48	**A**
09	**B**	19	**A**	29	**C**	39	**B**	49	**A**
10	**D**	20	**C**	30	**B**	40	**C**	50	**A**

Chapter 32

Chemistry in Everyday Life

50 questions / Single correct / Four options

01–10 / Selected middle practice

01. One mole triglyceride undergoes complete saponification with NaOH. Sodiumsoap amount formed in mol is

(A) 6 (B) 3

(C) 1 (D) 2

02. For a sodium soap RCOONa, precipitation as $Ca(RCOO)_2$ consumes how many moles of soap per mole of Ca^{2+}?

(A) 1 (B) 4

(C) 3 (D) 2

03. A sodium carboxylate soap with M 306 g/mol has parent acid M in $g\,mol^{-1}$ of(Na=23, H=1)

(A) 285 (B) 284

(C) 283 (D) 328

04. A solution contains 0.0100 mol benzoic acid and 0.0100 mol benzoate. Using pK_a 4.20, buffer pH is

(A) 8.4 (B) 4.2

(C) 7 (D) 2.1

05. Hydrolysis of one mole aspirin forms one mole salicylic acid and one mole ethanoic acid. Total acid molecule amount in mol is

(A) 4 (B) 2

(C) 1 (D) 3

06. A 0.100 mol Mg^{2+} sample precipitates all soap carboxylate as $Mg(RCOO)_2$. Soap carboxylate consumed in mol is

(A) 0.2 (B) 0.1

(C) 0.3 (D) 0.05

07. A micelle model has 100 surfactant molecules, each with one anionic head. Magnitude of total formal head charge in elementary-charge units is

(A) 50 (B) 200

(C) 100 (D) 1000

08. A 0.100 mol stearic acid sample is neutralised to sodium stearate without loss. Soap mass, in g, is (sodium stearate molar mass=306 g/mol)

(A) 30.6 (B) 61.2

(C) 28.4 (D) 15.3

09. Complete hydrolysis of one aspartame molecule has one peptide bond and one methyl-ester bond available. Water molecules required for cleaving both are

(A) 2 (B) 3

(C) 4 (D) 1

10. A fatty acid soap has 18 carbons per formula unit. A 0.0100 mol soap contains C atoms in mol of

(A) 0.18 (B) 0.09

(C) 0.36 (D) 0.01

11–40 / New middle-to-hard practice

11. A 0.100 mol mixture of stearic acid (M=284) and oleic acid (M=282) consumes 0.0400 mol I_2 quantitatively at C=C only. Complete neutralisation with NaOH is followed by isolation of all sodium soaps. With sodium replacing acidic H, final soap mass is, in g

(A) $\frac{761}{25}$ (B) $\frac{153}{5}$

(C) $\frac{763}{25}$ (D) $\frac{709}{25}$

12. A 30.52 g sodium-stearate/sodium-oleate mixture has molar masses 306 and 304. Acidification quantitatively gives 0.100 mol total fatty acids. Hydrogenating all C=C in those acids requires H_2, in mol, of

(A) $\frac{1}{50}$ (B) $\frac{1}{25}$

(C) $\frac{1}{10}$ (D) $\frac{3}{50}$

13. A laboratory sample contains a triglyceride and a free monocarboxylic fatty acid only. Complete alkaline hydrolysis consumes 0.0700 mol NaOH; isolated glycerol is 0.0200 mol. Free fatty-acid amount initially present is

(A) $\frac{3}{100}$ (B) $\frac{1}{50}$

(C) $\frac{1}{25}$ (D) $\frac{1}{100}$

14. A sample contains a monofunctional methyl ester of a fatty acid and the free parent acid only. Total sample amount is 0.100 mol. Selective neutralisation before ester hydrolysis uses 0.0300 mol NaOH. Complete treatment of a fresh equal sample with excess aqueous alkali gives methanol, in mol, of

(A) $\frac{7}{100}$ (B) $\frac{3}{100}$

(C) $\frac{7}{200}$ (D) $\frac{1}{10}$

15. A hard-water sample contains Ca^{2+} and Mg^{2+}, total 0.00600 mol. Selective removal of all Ca^{2+} leaves Mg^{2+} consuming 0.00400 mol monocarboxylate soap in complete precipitation. An excess of sodium stearate is added to a fresh equal untreated sample. Mass of the calcium-stearate component of its precipitate is, in g ($Ca(stearate)_2$=606)

(A) $\frac{303}{125}$ (B) $\frac{303}{250}$

(C) $\frac{909}{250}$ (D) $\frac{606}{125}$

16. A hard-water sample consumes 0.0120 mol monocarboxylate soap in complete precipitation. Selective Ca^{2+} removal eliminates 0.00400 mol cations. Subsequent addition of 0.00500 mol soap to the Ca-free sample leaves dissolved soap, in mol, of

(A) $\frac{1}{200}$ (B) $\frac{1}{1000}$

(C) $\frac{3}{1000}$ (D) $\frac{1}{500}$

17. Four surfactants are $RCOO^-Na^+$, $ROSO_3^-Na^+$, $R_4N^+Br^-$ and a long-chain alcohol ethoxylate with no ionisable group. Which pair is classified correctly and also distinguishes soap from a synthetic anionic detergent?

(A) Both sodium compounds are soaps because both contain Na.

(B) $ROSO_3Na$ is cationic; RCOONa is nonionic.

(C) RCOONa is an anionic soap; $ROSO_3Na$ is an anionic synthetic detergent.

(D) R_4NBr is nonionic; the alcohol ethoxylate is cationic.

18. A long-chain surfactant remains usable in Ca^{2+}-rich water and has an uncharged polyoxyethylene head. A second has a quaternary ammonium head and bromide counterion. Their correct paired classes are

(A) Nonionic soap; anionic detergent.

(B) Cationic detergent; anionic soap.

(C) Anionic soap; nonionic detergent.

(D) Nonionic detergent; cationic detergent.

19. An ideal micelle model has constant monomer concentration c above a threshold and fixed aggregation number N. At total surfactant concentrations 4 and 7 mmol/L, micelle concentrations are 0.0200 and 0.0500 mmol/L. The inferred aggregation number N is

(A) 150 (B) 200

(C) 50 (D) 100

20. Above its threshold, an ideal surfactant has free monomer concentration 2 mmol/L and N=100 molecules per micelle. A 1.00 L solution contains 7 mmol total surfactant. Dilute to 2.00 L without changing threshold or N. Final micelle amount divided by its initial amount is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$

(C) $\frac{7}{10}$ (D) $\frac{1}{2}$

21. A model mixture contains a monovalent anionic detergent and a monovalent cationic detergent, total 0.100 mol. They pair quantitatively into neutral insoluble 1:1 units. After pairing, 0.0200 mol free anions remain. Initial cationic-detergent amount is

(A) $\frac{1}{20}$ (B) $\frac{1}{50}$

(C) $\frac{1}{25}$ (D) $\frac{3}{50}$

22. A model mixture contains monocationic and dicationic surfactants, total 0.0500 mol molecules. Adding 0.0900 mol monovalent anionic surfactant precipitates all positive heads as neutral units and leaves 0.0100 mol free anions. Initial dicationic mole fraction is

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

23. A 19.50 g sample contains only aspirin (M=180) and salicylic acid (M=138). Complete ester hydrolysis gives 0.120 mol salicylic acid, and no material is lost. Initial aspirin amount is

(A) $\frac{7}{100}$ (B) $\frac{11}{200}$

(C) $\frac{3}{25}$ (D) $\frac{13}{200}$

24. A sample contains aspirin and salicylic acid only. Complete hydrolysis gives 0.0800 mol ethanoic acid and 0.100 mol salicylic acid. Treat each product carboxyl group as one titratable acid and exclude phenolic OH. Carboxylic-acid equivalents after hydrolysis divided by before are

(A) $\frac{8}{5}$ (B) $\frac{5}{9}$

(C) 2 (D) $\frac{9}{5}$

25. In a quantitative model, p-aminophenol (M=109) is mono-N-acetylated to paracetamol (M=151). A 10.90 g feed is 80% p-aminophenol and 20% inert material. Conversion is 75% and product-isolation recovery 80%. Paracetamol isolated is, in g

(A) $\frac{453}{50}$ (B) $\frac{604}{125}$

(C) $\frac{906}{125}$ (D) $\frac{151}{25}$

26. A p-aminophenol/paracetamol mixture contains 0.100 mol total molecules. A validated selective free-amino assay counts 0.0300 mol p-aminophenol. Paracetamol is its mono-N-acetyl derivative, adding C_2H_2O to each precursor molecule. With p-aminophenol=109 g/mol, initial mixture mass is, in g

(A) $\frac{304}{25}$ (B) $\frac{151}{10}$

(C) $\frac{346}{25}$ (D) $\frac{109}{10}$

27. A model preservative buffer contains 0.0200 mol benzoic acid and 0.0300 mol sodium benzoate. After adding 0.0100 mol HCl in negligible volume, the undissociated-acid fraction of total benzoate species is (assume complete strong-acid reaction first)

(A) $\frac{1}{2}$ (B) $\frac{2}{5}$

(C) $\frac{3}{5}$ (D) $\frac{4}{5}$

28. For an ideal benzoic-acid preservative, $pK_a = 4.20$. At pH 5.20, 10.0 mmol total benzoate species are present. A selective extraction removes half the undissociated acid, then a fixed-pH reservoir re-establishes equilibrium without adding benzoate. Total benzoate amount left after this one extraction is, in mmol

(A) $\frac{100}{11}$ (B) 5

(C) $\frac{105}{11}$ (D) $\frac{109}{11}$

29. Sucralose is obtained conceptually by replacing three OH groups of sucrose ($C_{12}H_{22}O_{11}$, M=342) with Cl. Using Cl=35.5, H=1 and O=16, chlorine mass fraction in the resulting formula is

(A) $\frac{71}{265}$ (B) $\frac{71}{228}$

(C) $\frac{71}{342}$ (D) $\frac{71}{297}$

30. For a three-chlorine sweetener molecule, assume each Cl independently has probability 3/4 of the lighter isotope and 1/4 of the isotope two mass units heavier. Among molecules heavier than the all-light species, fraction exactly two mass units heavier is

(A) $\frac{27}{37}$ (B) $\frac{9}{16}$

(C) $\frac{9}{37}$ (D) $\frac{27}{64}$

31. A 0.0500 mol mixture contains aspartame, represented as Asp–Phe–OMe, and the corresponding unesterified dipeptide Asp–Phe–OH. Complete hydrolysis yields 0.0300 mol methanol. Net water consumed in cleaving every peptide and methyl-ester bond is

(A) $\frac{3}{50}$ (B) $\frac{1}{20}$

(C) $\frac{1}{10}$ (D) $\frac{2}{25}$

32. A mixture of aspartame (Asp–Phe–OMe) and free phenylalanine is completely hydrolysed. Methanol formed is 0.0200 mol and total final free amino-acid amount is 0.0600 mol. In a validated assay counting carboxyl groups only, Asp contributes two and Phe one per molecule. Total carboxyl-group amount is

(A) $\frac{1}{25}$ (B) $\frac{3}{50}$

(C) $\frac{1}{10}$ (D) $\frac{2}{25}$

33. In a calibrated food-oxidation model, oxygen uptake is zero until antioxidant traps all initiating radicals, then is constant q. Each antioxidant molecule traps two radicals; initiation rate j is constant. At t=20 min, samples with 1 and 2 mmol antioxidant have taken up 30 and 20 mmol O_2, both after induction. With 3 mmol antioxidant, uptake by 20 min is, in mmol

(A) 10 (B) 15

(C) 0 (D) 20

34. In the same induction model, each antioxidant traps two radicals. At 20 min, 1 and 2 mmol antioxidant give 30 and 20 mmol oxygen uptake; after induction uptake rate q is constant. Initiating-radical production j, in mmol/min, is

(A) $\frac{2}{5}$ (B) 4

(C) 2 (D) $\frac{1}{5}$

35. A laboratory antacid model contains $Mg(OH)_2$ and $NaHCO_3$ only. Complete neutralisation consumes 0.100 mol HCl and releases 0.0200 mol CO_2. With molar masses 58 and 84, original mixture mass is, in g

(A) 4 (B) $\frac{58}{25}$

(C) $\frac{16}{5}$ (D) $\frac{126}{25}$

36. A laboratory mixture contains $CaCO_3$ and $Al(OH)_3$ only, total mass 8.68 g. Complete acid treatment releases 0.0400 mol CO_2. With molar masses 100 and 78, total HCl consumed is

(A) $\frac{1}{5}$ (B) $\frac{13}{50}$

(C) $\frac{11}{50}$ (D) $\frac{7}{25}$

37. In a stated competitive receptor model, drug-bound fraction is $\theta_D = d/(1+d+\ell)$ with $d = [D]/K_D$ and $\ell = [L]/K_L$. At fixed $\ell = 2$, measured $\theta_D = 1/2$. Double free drug concentration without changing free ligand. New θ_D is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

38. A model competitive inhibitor obeys $v/V_{max} = S/[K_m(1 + I/K_i) + S]$. With no inhibitor, S=2 mmol/L gives $v/V_{max} = 1/2$; with I=6 mmol/L at the same S, the fraction is 1/5. At S=4 and I=6 mmol/L, predicted v/V_{max} is

(A) $\frac{2}{5}$ (B) $\frac{1}{3}$

(C) $\frac{1}{2}$ (D) $\frac{1}{4}$

39. A candidate is a non-narcotic analgesic obtained by O-acetylating the phenolic group of salicylic acid. A second is the N-acetyl derivative of p-aminophenol. Which pair also gives the correct first candidate hydrolysis products?

(A) Paracetamol and aspirin; the first gives phenol and methanol.

(B) Aspirin and saccharin; aspirin gives benzoic acid and ethanol.

(C) Aspirin and paracetamol; aspirin gives salicylic acid and ethanoic acid.

(D) Morphine and paracetamol; the first gives salicylic acid and water.

40. A formulation contains sodium stearate, BHA and sodium benzoate. One component is a carboxylate soap, one inhibits food oxidation, and one is used as a food preservative. Which ordering assigns all three roles correctly without treating every sodium salt as a soap?

(A) Soap: sodium stearate; antioxidant: BHA; preservative: sodium benzoate.

(B) Soap: sodium stearate; antioxidant: sodium benzoate; preservative: BHA.

(C) Soap: sodium benzoate; antioxidant: sodium stearate; preservative: BHA.

(D) Soap: BHA; antioxidant: sodium benzoate; preservative: sodium stearate.

41–50 / Preserved hard practice

41. A 1.00 g model base contains $Mg(OH)_2$ and inert matter. It consumes 0.0200 mol HCl in a chemical assay. With $Mg(OH)_2$ =58, the base purity percentage is

(A) 100 (B) 29

(C) 40 (D) 58

42. A 0.100 mol mixture of sodium palmitate(M=278) and sodium stearate(M=306)weighs 28.92 g. Moles of sodium stearate are

(A) 0.06 (B) 0.04

(C) 0.08 (D) 0.02

43. A 0.0100 mol tristearin sample(M=890) is fully saponified with NaOH. Total Na^- stearate mass in g is(M=306)

(A) 18.36 (B) 3.06

(C) 9.18 (D) 8.9

44. A benzoate preservative model has pK_a 4.20 and pH 3.20. Fraction of all benzoic-species present as undissociated HA is

(A) 0.99 (B) 0.5

(C) 0.090909091 (D) 0.90909091

45. A 0.0100 mol aspirin sample is hydrolyzed fully. It produces one carboxylic acid group in salicylic acid and one in ethanoic acid per aspirin. NaOH needed to form only carboxylates, with phenol left protonated, in mol is

(A) 0.04 (B) 0.03

(C) 0.01 (D) 0.02

46. A 2.00 L soap model initially contains 0.0100 mol Ca^{2+} and 0.00500 mol Mg^{2+}. Add 0.0400 mol monovalent soap. Assume both divalent ions precipitate completely as M(soap)2. The remaining soluble soap has monomer concentration fixed at its CMC of 0.00200 M; all excess forms 100-molecule micelles. Amount of micelles in mol is

(A) 2×10^{-5} (B) 0.0006

(C) 0.0001 (D) 6×10^{-5}

47. A model-additive mixture contains A with one basic amine and one ester per molecule, and B with two basic amines and no ester. One aliquot consumes 0.140 mol HCl for complete protonation. A fresh equal aliquot consumes 0.0600 mol NaOH for complete ester saponification, with no other base consumption. Mole fraction of B is

(A) 0.25 (B) 0.6

(C) 0.4 (D) 0.5

48. A 0.100 mol mixture of sodium benzoate(M=144) and benzoic acid(M=122)weighs 13.08 g. At pK_a 4.20, using Henderson and log(2/3)=-.1761, model pH is

(A) 4.3761 (B) 4.0239

(C) 4.2 (D) 4.8

49. A 0.0100 mol mixture of quaternary-ammonium chloride and bromide has the same cation of formula mass 284 in both salts. It weighs 3.373 g. Bromide alone is quantitatively isolated as AgBr. With Cl=35.5, Br=80, AgBr =188, the AgBr mass in g is

(A) 1.88 (B) 0.376

(C) 0.752 (D) 1.128

50. A 0.0100 mol sample of a nonionic surfactant weighs 8.02 g. Each molecule has an end group of mass 186 u plus either 10 or 20 ethylene oxide units of 44 u each, with no condensation loss. Mass fraction of the 20-unit molecules is

(A) 0.6

(B) 0.46832918

(C) 0.4

(D) 0.53167082

Answer key

Chemistry in Everyday Life

C-CEVL / Questions 1–50

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
01	**B**	11	**C**	21	**C**	31	**D**	41	**D**
02	**D**	12	**B**	22	**B**	32	**D**	42	**B**
03	**B**	13	**D**	23	**A**	33	**A**	43	**C**
04	**B**	14	**A**	24	**D**	34	**A**	44	**D**
05	**B**	15	**A**	25	**C**	35	**A**	45	**D**
06	**A**	16	**B**	26	**C**	36	**B**	46	**D**
07	**C**	17	**C**	27	**C**	37	**B**	47	**C**
08	**A**	18	**D**	28	**C**	38	**B**	48	**B**
09	**A**	19	**D**	29	**A**	39	**C**	49	**C**
10	**A**	20	**B**	30	**A**	40	**A**	50	**D**